

APPENDIX C

SPECIAL TOOLS AND GROUND SUPPORT EQUIPMENT

C-1 SPECIAL TOOLS AND GROUND SUPPORT EQUIPMENT.

C-1.1 Introduction.

NOTE

New support equipment purchased from Sikorsky Aircraft has passed all quality requirements and should be considered calibrated upon delivery. The recommended calibration requirements specified within this appendix are recommended for new support equipment that have been inducted into each customer's inventory. Specific calibration requirements may vary depending on the internal process that have been established by each customer for support equipment calibration.

This appendix includes special tools and ground support equipment required to support and maintain the helicopter. Individual illustrations for these items include simplified line drawings and supporting text containing item description and calibration procedures if an item requires calibration.

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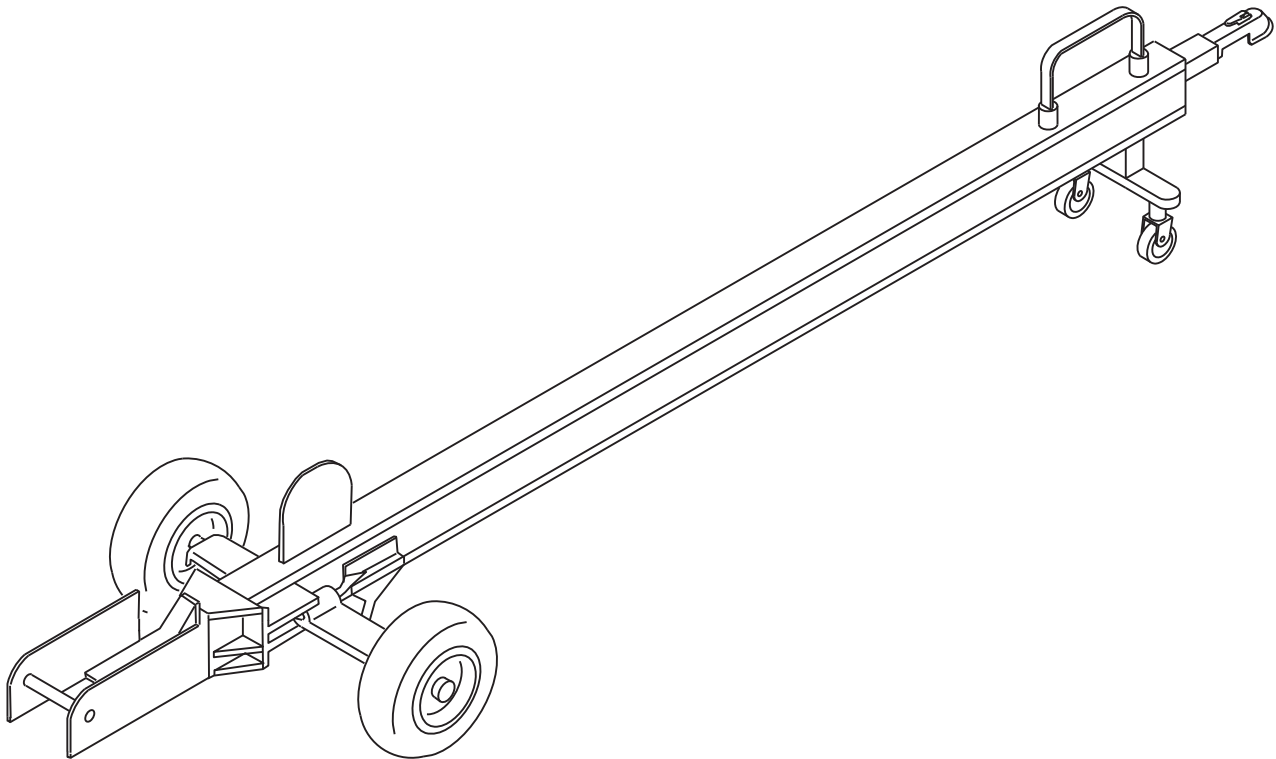
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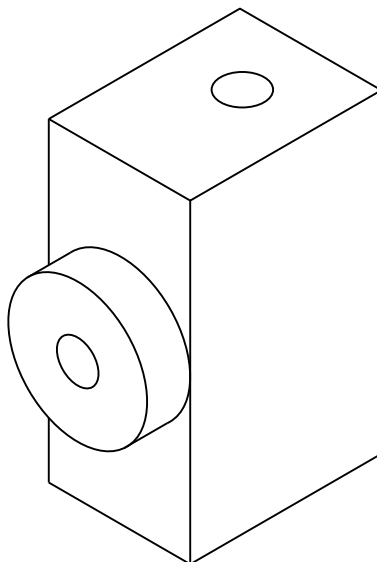
DESCRIPTION: The tow bar is designed as a single leg aluminum tubular structure having a yoke at one end for attachment to the helicopter tail wheel. It is designed to accept loads imposed by a fully loaded helicopter.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:Not Applicable

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FIGURE C-1. AIRCRAFT TOW BAR, 01-1185-0000 OR 01-1185-0010



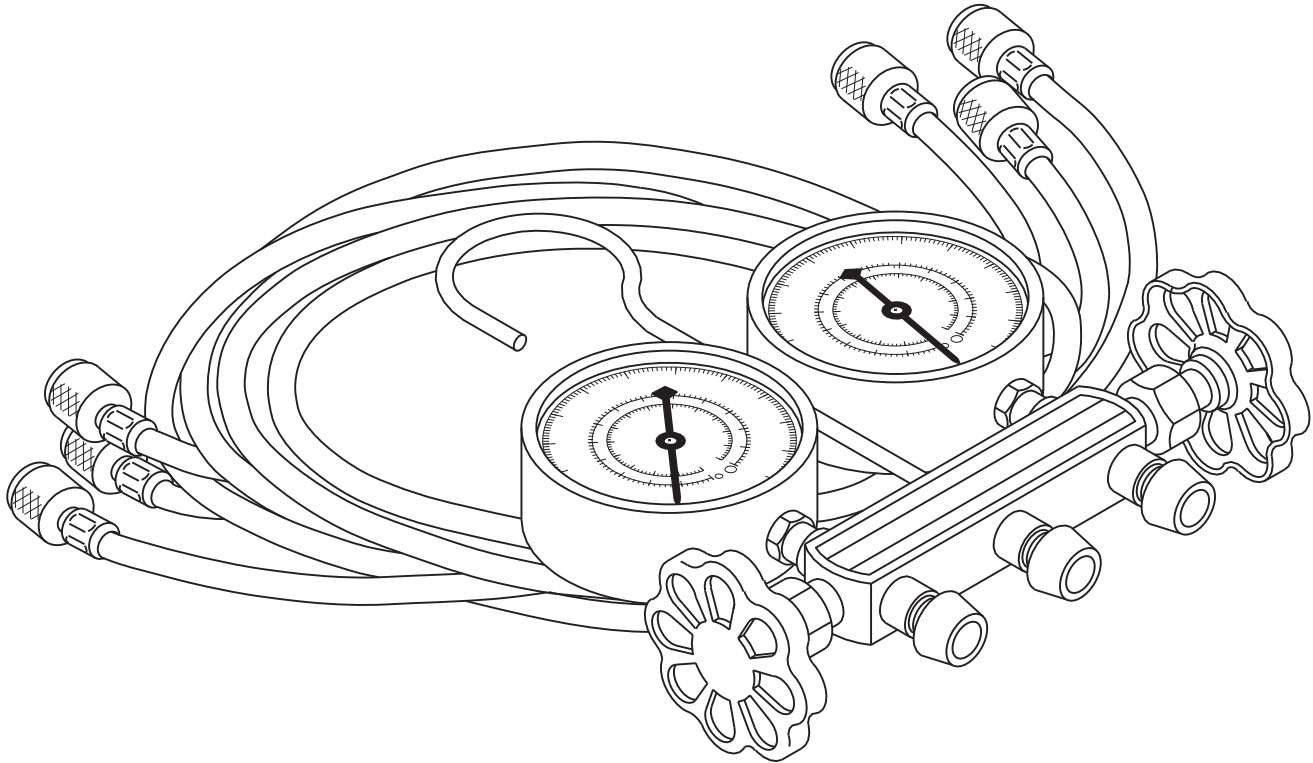
DESCRIPTION: The accelerometer bracket is mounted to the airframe and is used as a base for the track and balance accelerometer. When track and balance of the helicopter is complete, the bracket is removed from the airframe.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

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FIGURE C-2. ACCELEROMETER BRACKET, 70351-08408-102



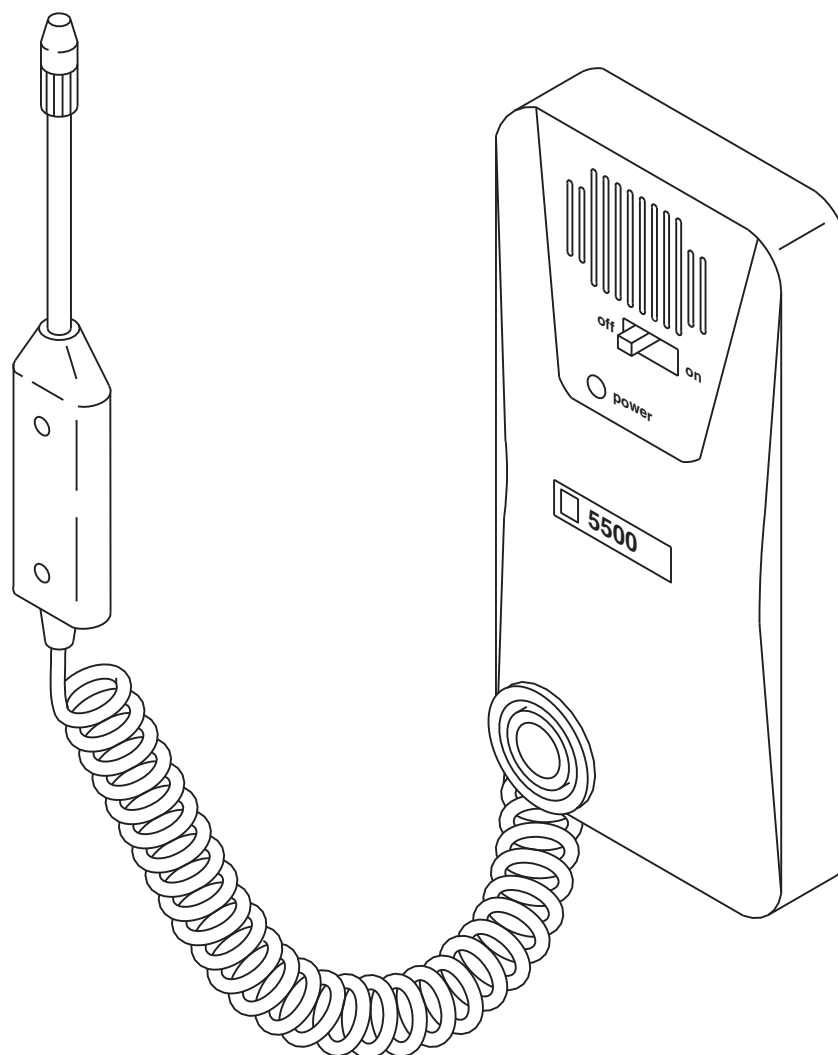
DESCRIPTION: The manifold gage is used to check and service the gas pressure in the air-conditioning system. It consists of two gages, hoses, and a regulating valve to control the pressure while servicing.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

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SA

FIGURE C-3. AIR CONDITIONING GAGE AND MANIFOLD, 41213J



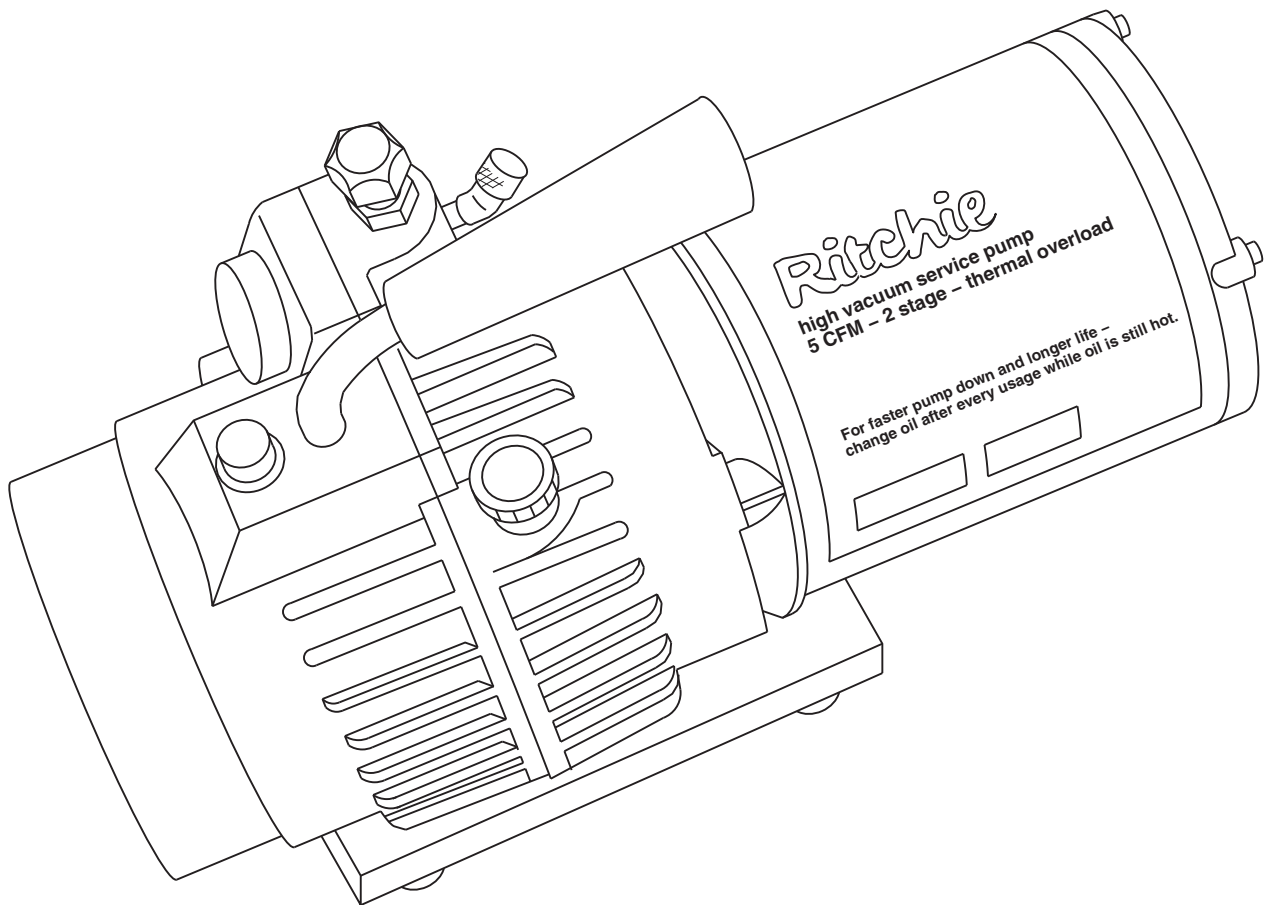
DESCRIPTION: The leak detector is designed to detect all HFC, CFC, and HCFC refrigerants. It is a hand held unit with 36-inch cord for hard to reach places. The detector responds to minute traces of halogen gas with a leak rate accuracy of better than 0.40 ounces per year.

POWER REQUIREMENTS: Battery powered, two C-Cell 3 Volt DC.

CALIBRATION REQUIREMENTS: Not Applicable

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FIGURE C-4. AIR CONDITIONING LEAK DETECTOR, TIF MODEL 5650



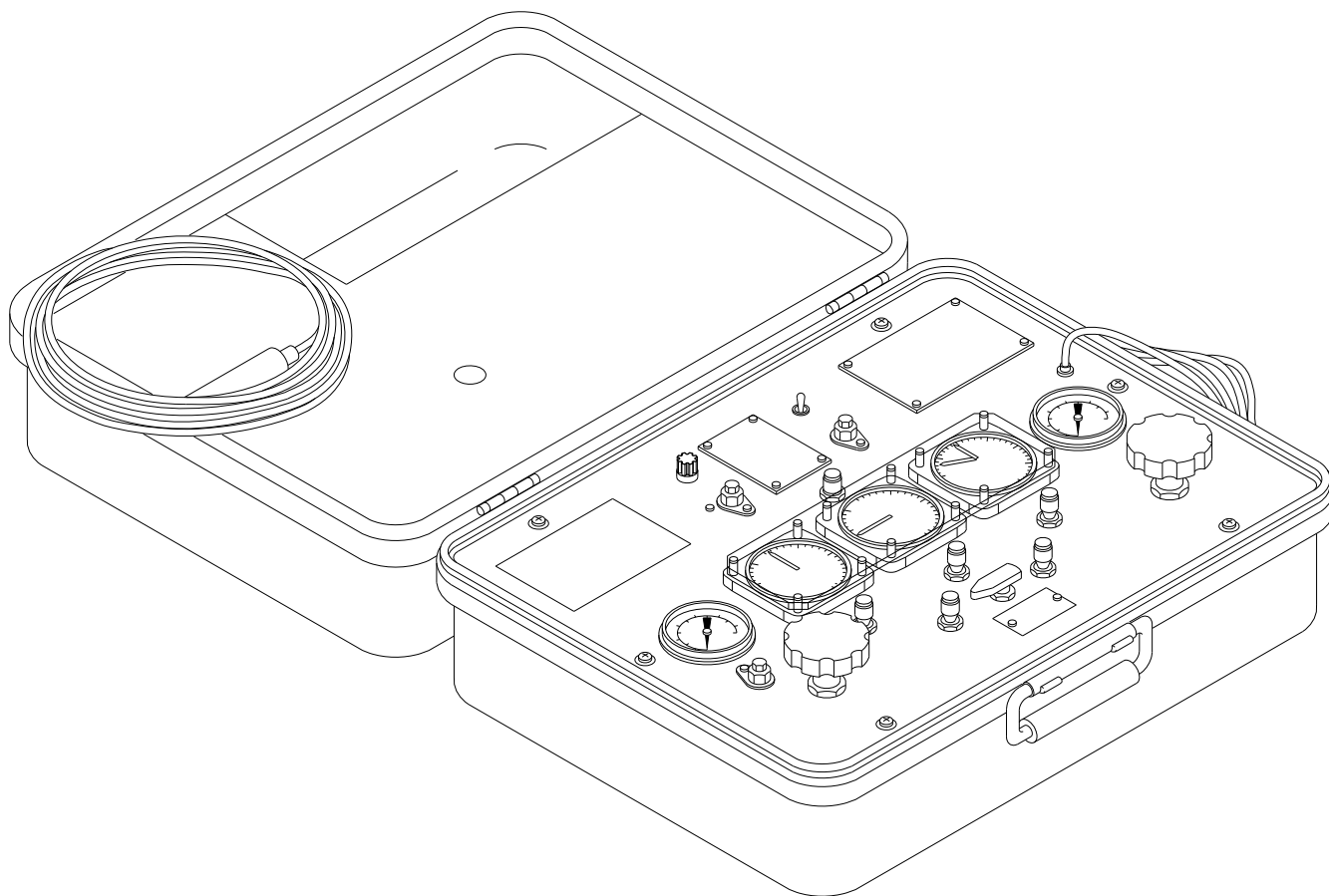
DESCRIPTION: The air conditioning vacuum pump is used when removing and reinstalling components of the helicopter air conditioning system.

POWER REQUIREMENTS: 120 / 220 volts, 50 / 60 Hz.

CALIBRATION REQUIREMENTS: Not Applicable

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FIGURE C-5. AIR CONDITIONING VACUUM PUMP, 93462



DESCRIPTION: The air data test set is used to test the helicopter pitot static system. It is also used in testing the operation and calibration of airspeed, engine pressure ratio, manifold pressure indicators, and other vacuum or low pressure systems.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Air data test set must be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Mensor Precision Quartz Pressure Gage or equivalent
- Micron Vacuum Gage and Sensor
- Vacuum Pump
- Stop Watch

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FIGURE C-6. AIR DATA TEST SET, 1811H-665 (SHEET 1 OF 5)

CALIBRATION REQUIREMENTS:(CONT)

CALIBRATION PROCEDURE

Inspect

1. Check test set for excessive oil, dirt, and foreign matter on unit. Check connectors, hoses, external pressure and vacuum ports for obstructions.
2. Check for broken valves, connectors, and instruments. Check pump and selector positions by observing tank gages.

Function and Leak Check

1. Close valves, cap test ports on static and pitot ports on test set, to check pitot side for leaks.
2. Turn on test set pump to achieve 5 psi in pressure tank as shown on gage. Slowly open pressure valve and observe airspeed indicator. Close valve at 200 knots on airspeed indicator. Airspeed shall not fall more than 2 knots in one minute. If airspeed falls 2 knots, repair system and repeat check.
3. Release pressure by slowly opening pressure vent valve.
4. Make sure all valves are closed, pitot and static test ports are capped. Fully open cross bleed valve to check static for leaks.
5. Turn on test set pump to achieve 20 in. on vacuum tank gage. Slowly open vacuum valve and observe altimeter. Close vacuum valve when altitude reads 20000 feet. Altimeter shall not drop more than 100 feet in one minute. If altimeter drops more than 100 feet, repair system and repeat check.
6. Release pressure by slowly opening pressure vent valve.
7. Make sure all valves are closed, pitot and static ports are capped, for overall functional check of tester. Turn on tester pump to achieve 5 psi on pressure tank gage and 20 in. on vacuum tank gage.
8. Open cross bleed valve and slowly open vacuum valve. Open vertical speed valve and observe rate of climb / descent indicator. Rate is controlled by vacuum valve. DO NOT EXCEED full scale range.

NOTE

Airspeed indicator may indicate temporary drop in airspeed below 0-knots. This condition will correct itself when vacuum valve is closed.

9. Vertical speed indicator should show some upscale movement with vacuum valve open. If pointer does not move, check and repair system.
10. Close cross bleed valve and slowly open pressure valve to achieve 100 knots on airspeed indicator. Close pressure valve and observe altimeter for drop in altitude, any drop beyond 100 feet is possible static system leak. Any drop in altitude and a drop in airspeed beyond 2 knots is possible leak between pitot and static. Any rise in airspeed is sign of leak in pitot. Repair any leaks and repeat check.
11. With checks complete, open cross bleed valve to bring airspeed down. Slowly open pressure vent to bring down altitude. Vent vacuum by means of pressure vent to avoid reverse airspeed.

Preliminary Measurement

1. Insert quartz gage capsule into gage. Turn on power at gage and allow to warmup.
2. Check capsule for proper temperature. Connect for differential as shown in view "A" and for absolute measurement in view "B". Connect as shown with both vent valves on tester open.
3. Turn on vacuum pump and observe vacuum gage. Gage should read 200 microns or less. Adjust gage containing absolute capsule to correspond with vacuum gage.
4. Close zero valve for differential. Stop vacuum pump for absolute. Vent gage and connect to testers static port, see view B. The gage with absolute capsule will read ambient atmospheric pressure.

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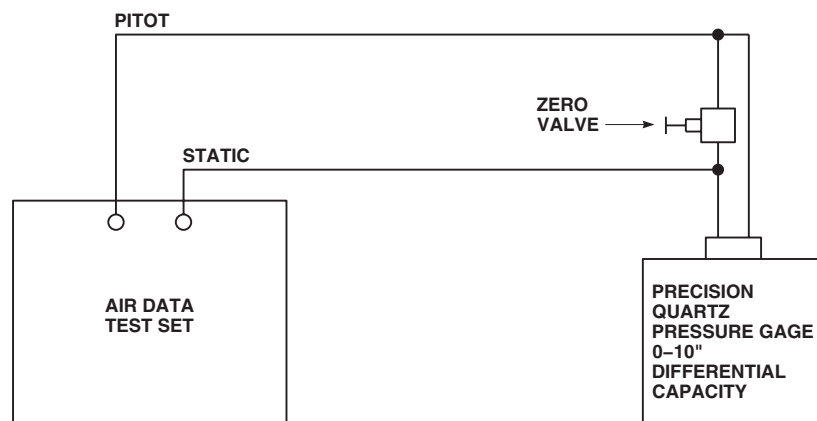
FIGURE C-6. AIR DATA TEST SET, 1811H-665 (SHEET 2 OF 5)

CALIBRATION REQUIREMENTS:(CONT)**CALIBRATION PROCEDURE (CONT)****Altimeter Check**

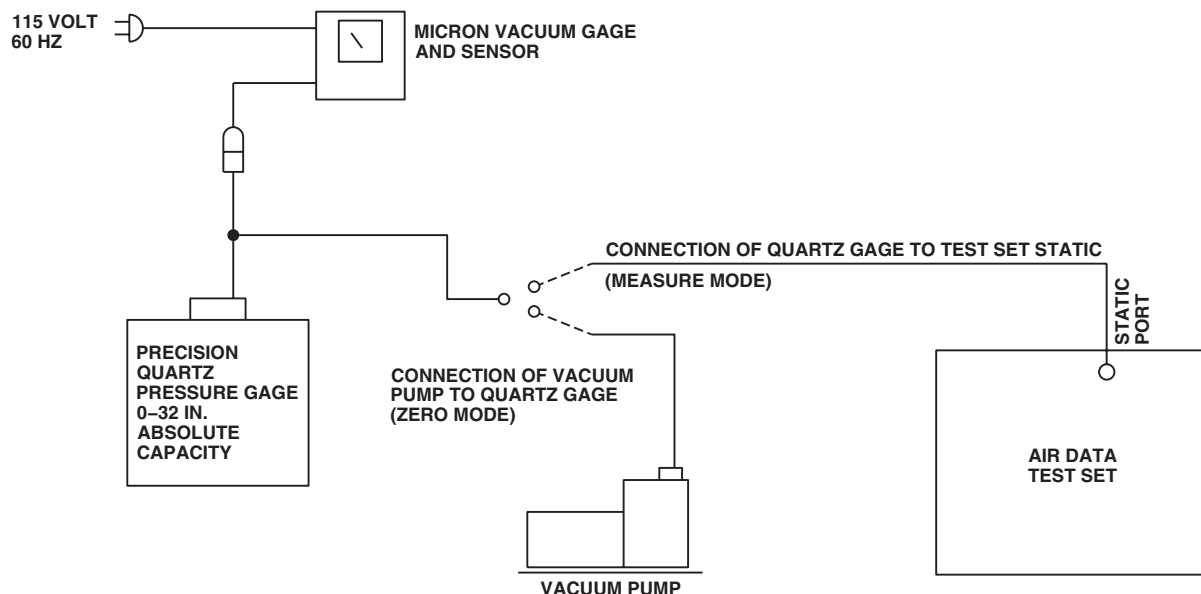
5. Connect test set as shown in view B, and open cross bleed and close pressure vent valve on tester. Make sure vacuum valve and vacuum vent valve are closed. Turn on tester pump to achieve 25 in. of vacuum on tank gage.
6. Slowly open vacuum valve and observe altimeter, also rate of climb indicator on tester indicators will rise up scale and quartz gage will go down scale. Close vacuum valve when altimeter is at 20000 feet. If altitude drops, check for leaks and repair as required.
7. Open pressure vent valve and slowly bring system back to atmospheric pressure.

NOTE

Altimeter reading may not be at zero with pressure at ambient due to work area most likely not being at sea-level pressure of 29.92 in.



**AIRSPEED MEASUREMENT
VIEW A**



**ALTITUDE MEASUREMENT
VIEW B**

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FIGURE C-6. AIR DATA TEST SET, 1811H-665 (SHEET 3 OF 5)

CALIBRATION REQUIREMENTS:(CONT)**CALIBRATION PROCEDURE (CONT)****Altimeter Check (cont)**

8. Repeat tests in steps 5 through 7.
9. Adjust barometric reading on test set altimeter to read 29.92 in. and close pressure vent valve.
10. Depending on ambient room pressure, open either vacuum or pressure valve on test set to achieve 29.92 in. on altimeter. Lightly tap altimeter and record reading. Repeat measurements at remaining points as listed in Table 1.

TABLE 1 (ALTIMETER)

ALTITUDE	PRESSURE	INDICATOR	STANDARD	TOTAL
0 ft	29.9213 "HgA	± 50 ft	± 4.3 ft	± 54.3 ft
+1000	28.8557	75	4.5	79.5
2000	27.8210	100	4.7	104.7
5000	24.8959	100	5.1	105.1
10000	20.5770	150	5.8	155.8
15000	16.8858	200	6.9	206.9
18000	14.9421	200	7.8	207.8
20000	13.7501	200	8.1	208.1
10000	20.5770	150	5.8	155.8
5000	24.8959	100	5.1	105.1
0	29.9213	50	4.3	54.3
-1000	31.0185	50	4.3	54.3

11. If tolerance in Table 1 can not be achieved, adjust and repeat check. Open vacuum vent before disconnecting system.

Airspeed Indicator Check

1. Connect airspeed indicator to test set as shown in view A. Make sure vacuum vent is open and zero valve is closed.
2. Make sure pressure valve, cross bleed valve, and pressure vent valve are closed. Turn on tester pump to achieve 5 psi on tank pressure gage.
3. Slowly open pressure valve and observe reading on airspeed indicator going upscale. Quartz gage shall also go upscale. Close pressure valve when airspeed indicator reaches 200 knots and check for leaks.
4. Open pressure vent valve to bring system back to ambient pressure and repeat step 3.
5. With pitot side of airspeed indicator at ambient pressure, open cross bleed valve and measure zero. Read and record indications. Repeat measurements for remaining points as listed in Table 2.

TABLE 2 (AIRSPEED)

AIRSPEED	PRESSURE	INDICATOR	STANDARD	TOTAL
20 Knots	.0192 "HgD	± 2.5 Knots	± 2.5 Knots	± 4.8 Knots
40	.0767	2.0	1.2	3.2
60	.1729	2.0	.8	2.8
80	.3079	2.0	.7	2.7
100	.4821	2.0	.6	2.6
120	.6959	2.0	.5	2.5
140	.9501	2.0	.4	2.4
160	1.2452	2.5	.4	2.9
200	1.9616	4.0	.4	4.4
100	.4821	2.0	.6	2.6
60	.1721	2.0	.8	2.8

6. If tolerance in Table 2 can not be achieved, adjust and repeat check. Open pressure vent valve and cross bleed valve before disconnecting system.

Rate of Climb / Descent Indicator Check

1. Close vent valves, make sure pitot and static output ports are capped, and cross bleed valve is open. Make sure vertical speed indicator control valve is open on tester.

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SA**FIGURE C-6. AIR DATA TEST SET, 1811H-665 (SHEET 4 OF 5)**

CALIBRATION REQUIREMENTS:(CONT)**CALIBRATION PROCEDURE (CONT)****Rate of Climb / Descent Indicator Check (cont)**

2. Close vacuum valve, pressure vent, and vacuum vent valves. Turn on tester pump to achieve 25 in. on vacuum tank gage.

NOTE

The following check requires the altimeter be operable and calibrated.

3. Slowly open vacuum valve while observing altimeter and vertical speed indicator. Using stop watch, measure the rate of climb at several points on the scale. Turn on tester pump if needed for more vacuum.
4. Check descent scale in same manner as step 3 except slowly open pressure vent valve to bleed vacuum. With checks completed, bring system back to ambient pressure.
5. Check suggested points and tolerances shown in Table 3.

TABLE 3 (RATE OF CLIMB / DESCENT)

INDICATED	TOLERANCE*	
500 ft / min	109.1	TO 133.3 sec
1000	54.6	66.7
2000	27.3	33.3
3000	18.2	26.2
**4000	13.7	16.7

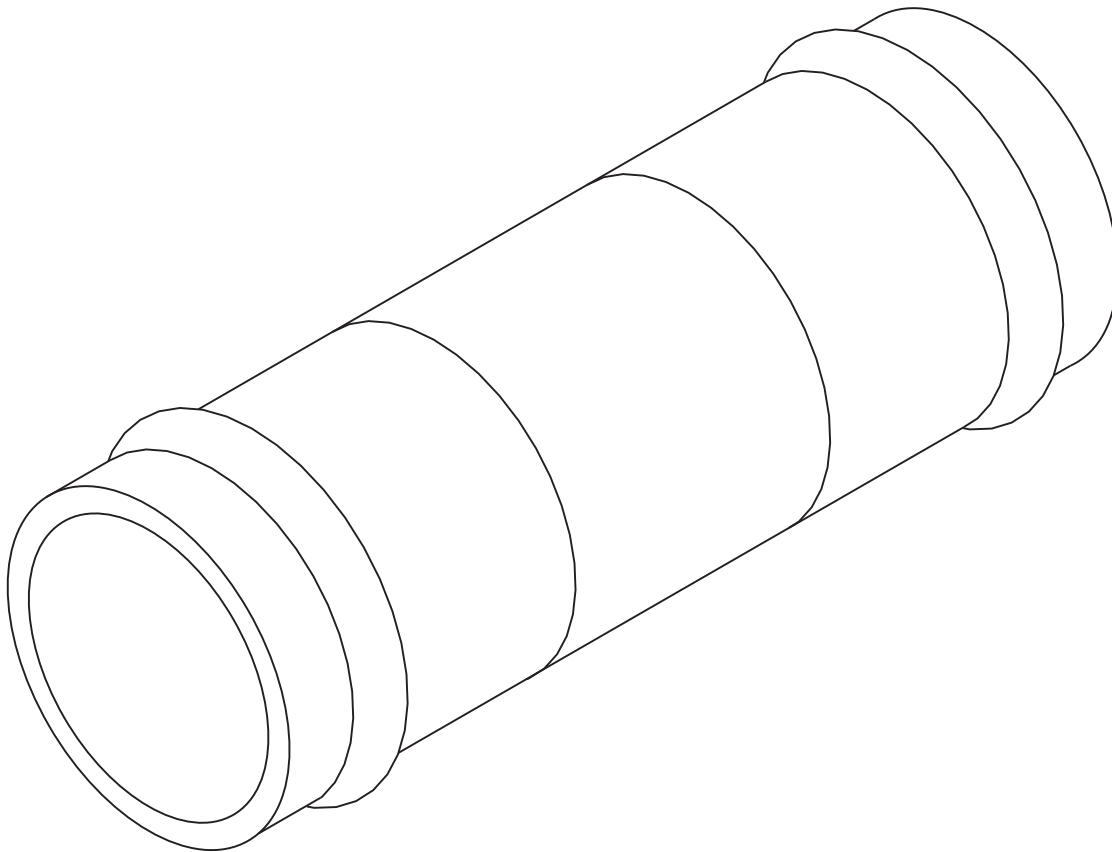
* Net altitude of 1000 ft and $\pm 10\%$ of indicator+ tol. of altimeter.

** Only on a 6000 ft / min indicator.

6. When checks are complete on tester, disconnect equipment and install calibration sticker.

FB0998_5A
SA

FIGURE C-6. AIR DATA TEST SET, 1811H-665 (SHEET 5 OF 5)



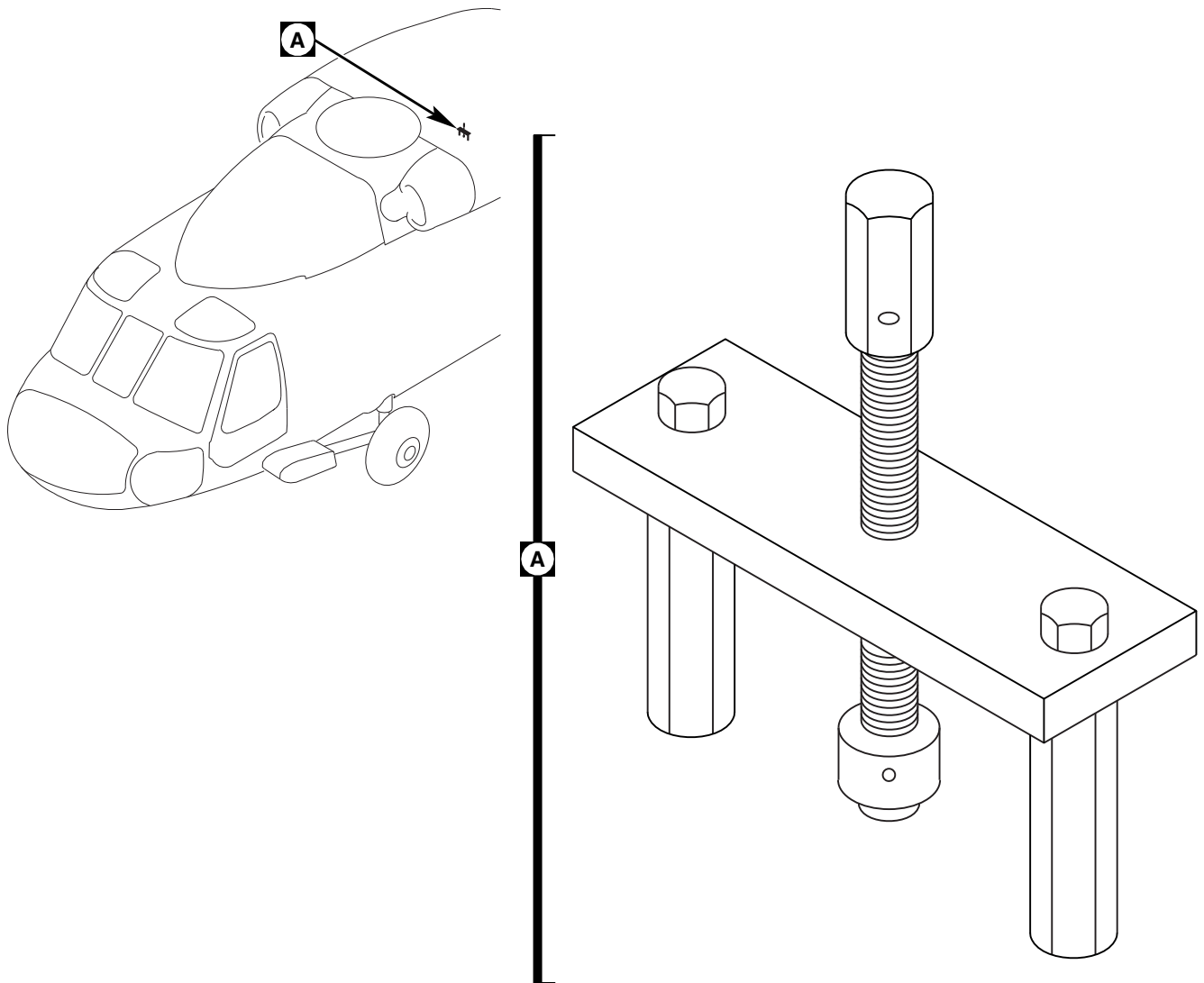
DESCRIPTION: The air inlet plug is used to block the inlet anti-ice bleed air tubes when the engine air inlets are removed to balance the engine output shaft.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB0999
SA

FIGURE C-7. AIR INLET PLUG, 70700-20114-109



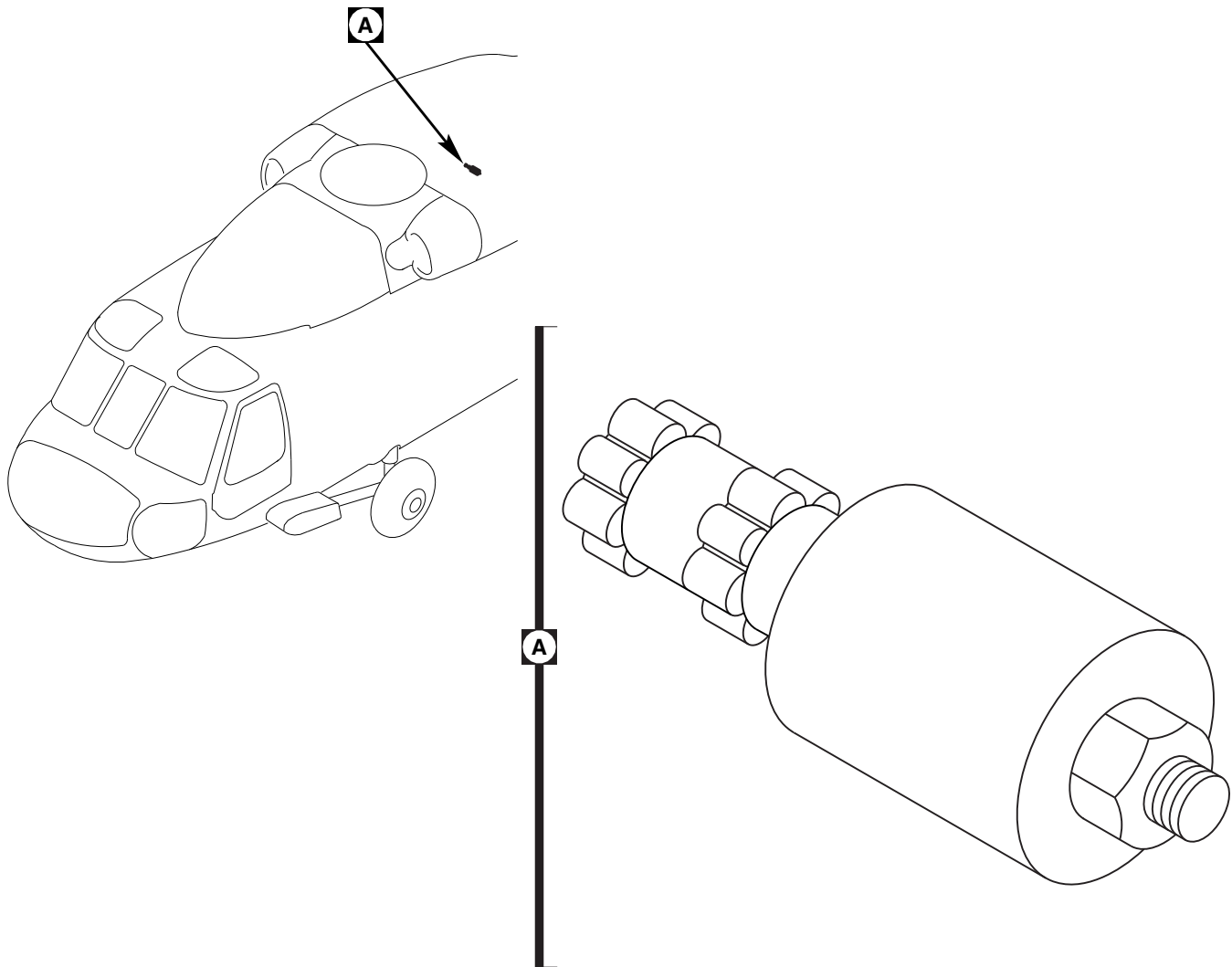
DESCRIPTION: The APU spline adapter installer consists of an externally threaded shaft, plug, plate, two externally threaded posts, and two hex head screws. It must be used when installing the vespel spline adapter into the APU generator recess.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1124
SA

FIGURE C-8. APU SPLINE ADAPTER INSTALLER, 1106841-1



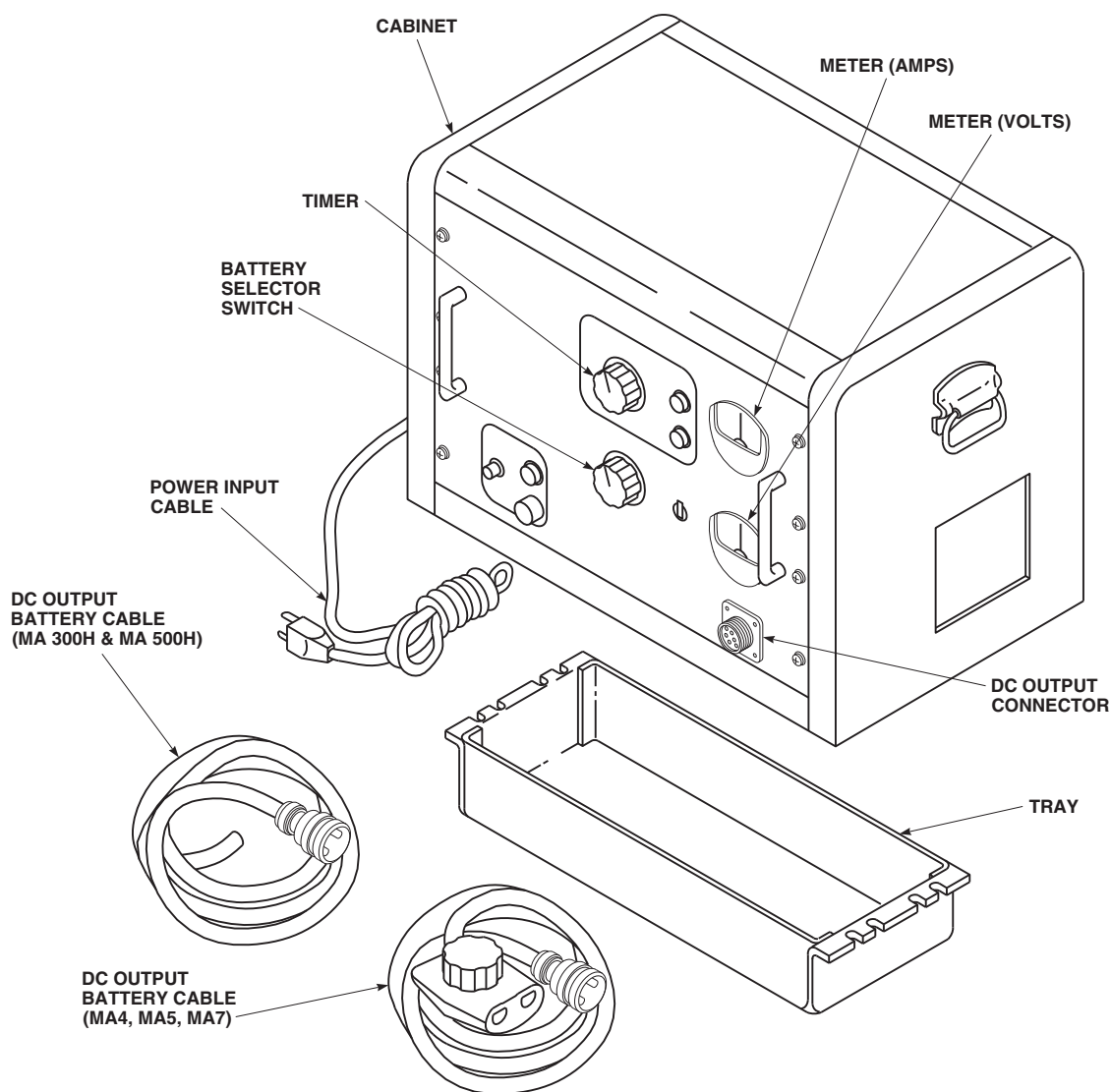
DESCRIPTION: The APU spline adapter puller is used to remove the vespel spline adapter and threaded shaft from the APU generator drive recess.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1125
SA

FIGURE C-9. APU SPLINE ADAPTER PULLER, 110679-4



DESCRIPTION: The battery charger / analyzer consists of a transformer, rectifier, passing regulator circuit, timer programmer, and test discharge load bank. The unit is designed to charge and analyze aircraft batteries of the nickel-cadmium (NiCad) type automatically.

POWER REQUIREMENTS: This unit operates from a source of either 110, 117, 220, or 234 volts, 50-60 Hz, single phase (nominal $\pm 10\%$).

CALIBRATION REQUIREMENTS:

FREQUENCY

As required due to abnormal or erratic operation.

EQUIPMENT REQUIRED

- Digital Multimeter
- TS-1037 Charger Interface
- 19 or 20 cell NiCad Battery

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SA

FIGURE C-10. BATTERY CHARGER/ANALYZER, PCA-131 50/60 (SHEET 1 OF 2)

CLEANING MATERIALS

- Source of dry air under pressure.

CALIBRATION PROCEDURES

Cleaning / Inspection

1. Clean battery charger / analyzer with clean dry air.
2. Check condition of fuse, exposed resistors, and wiring.

Calibration Voltmeter

1. Make sure air temperature is within 65° – 85°F (18° – 29°C) and relative humidity is less than 85% during calibration.
2. Apply power to battery charger / analyzer and digital multimeter. Allow 10 minutes for warmup.
3. Set selector switch on battery charger to type of NiCad battery being used in this test.
4. Connect the battery charger / analyzer to the NiCad battery with the charger interface in series with the battery charger cabling.
5. Set charger interface switch to VOLT position and connect digital multimeter input to charger interface binding post.
6. Verify battery charger / analyzer voltmeter and digital multimeter indications agree to within ± 1 volt DC.
7. Do not disconnect at this time, leave set-up for ammeter test.

Calibration Ammeter

1. Operate battery charger / analyzer in charge (+) and discharge (–) modes to apply and withdraw current from NiCad battery.
2. Monitor digital multimeter and battery charger / analyzer ammeter for current indications.

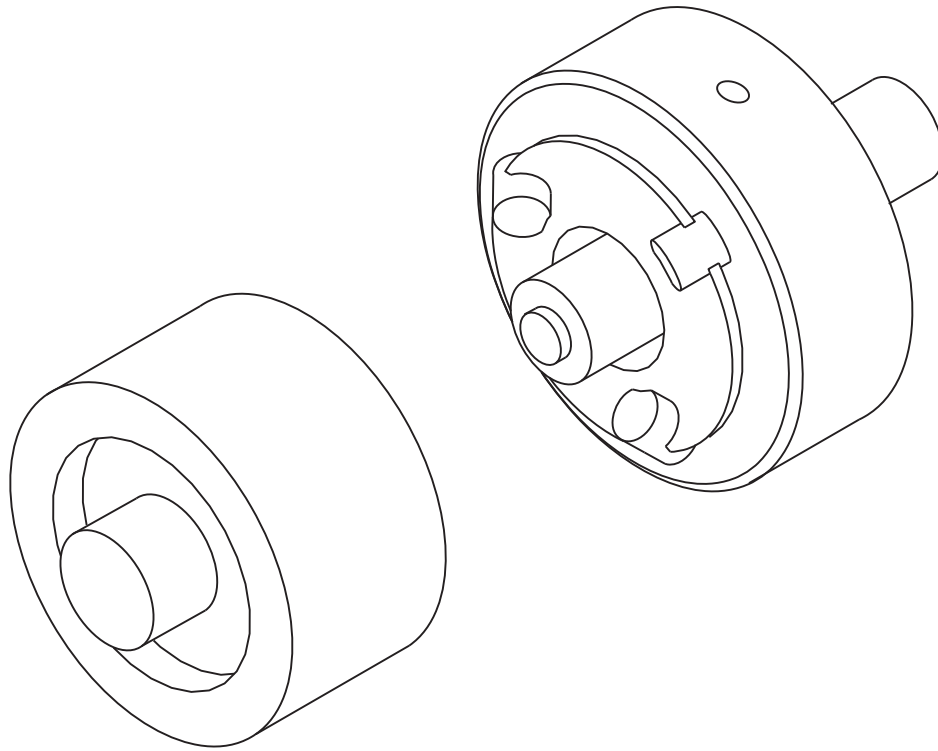
NOTE

Digital multimeter indication in millivolts is actual current reading in amps.

3. Check amp readings by comparing at least one positive (+) and one negative (–) indication. Make sure readings on ammeter and digital multimeter are within two amps of each other.
4. Turn off battery charger / analyzer and digital multimeter.
5. Disconnect battery, charger interface, and digital multimeter.
6. Install calibration label on battery charger / analyzer.

FB1142_2B
SA

FIGURE C-10. BATTERY CHARGER/ANALYZER, PCA-131 50/60 (SHEET 2 OF 2)



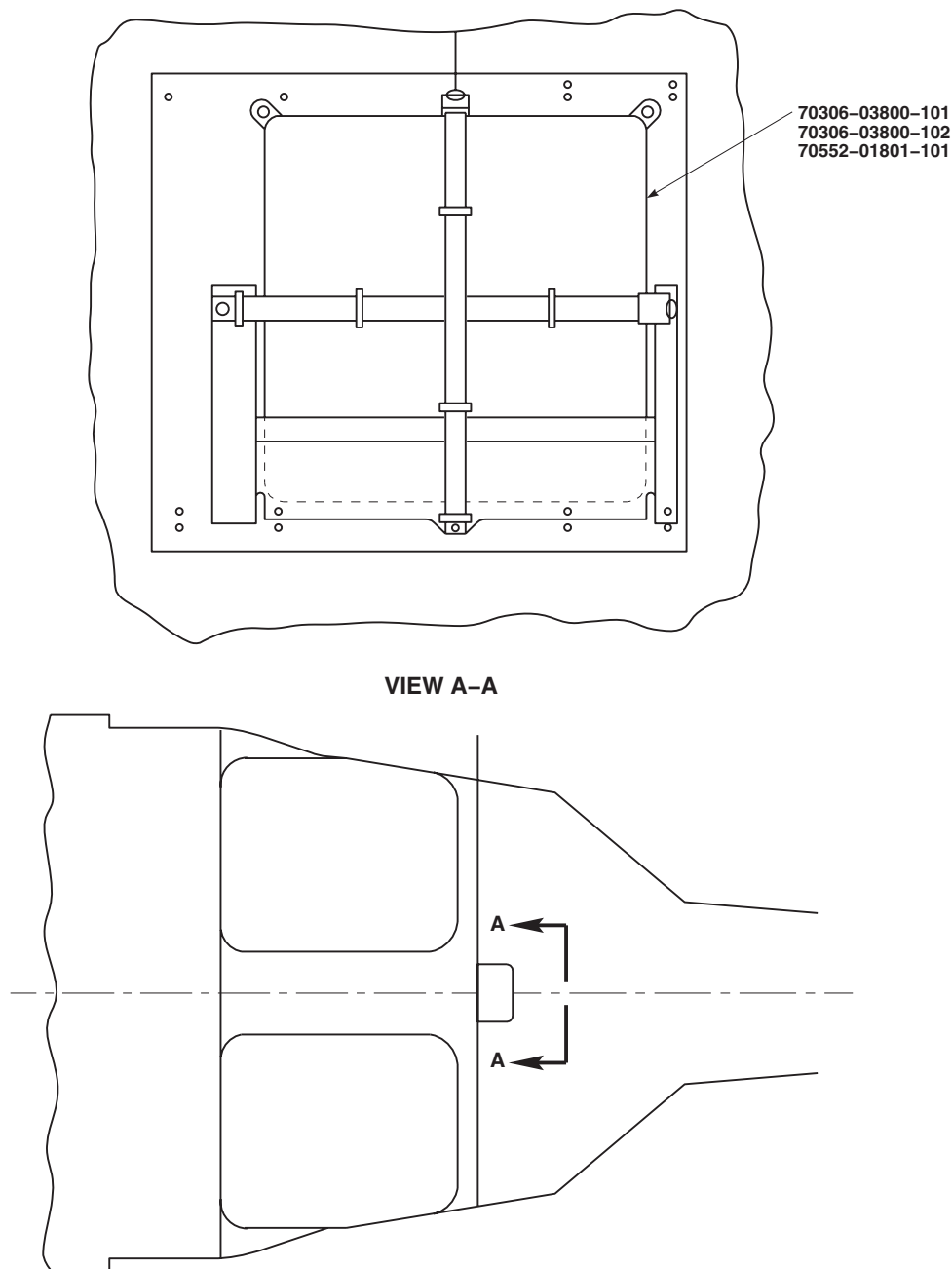
DESCRIPTION: This bearing staking tool is used to stake bearing, SB5302, into the rod end of the tail rotor pitch control rod 70103-11007-041.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1000A
SA

FIGURE C-11. BEARING STAKING TOOL, RST2153



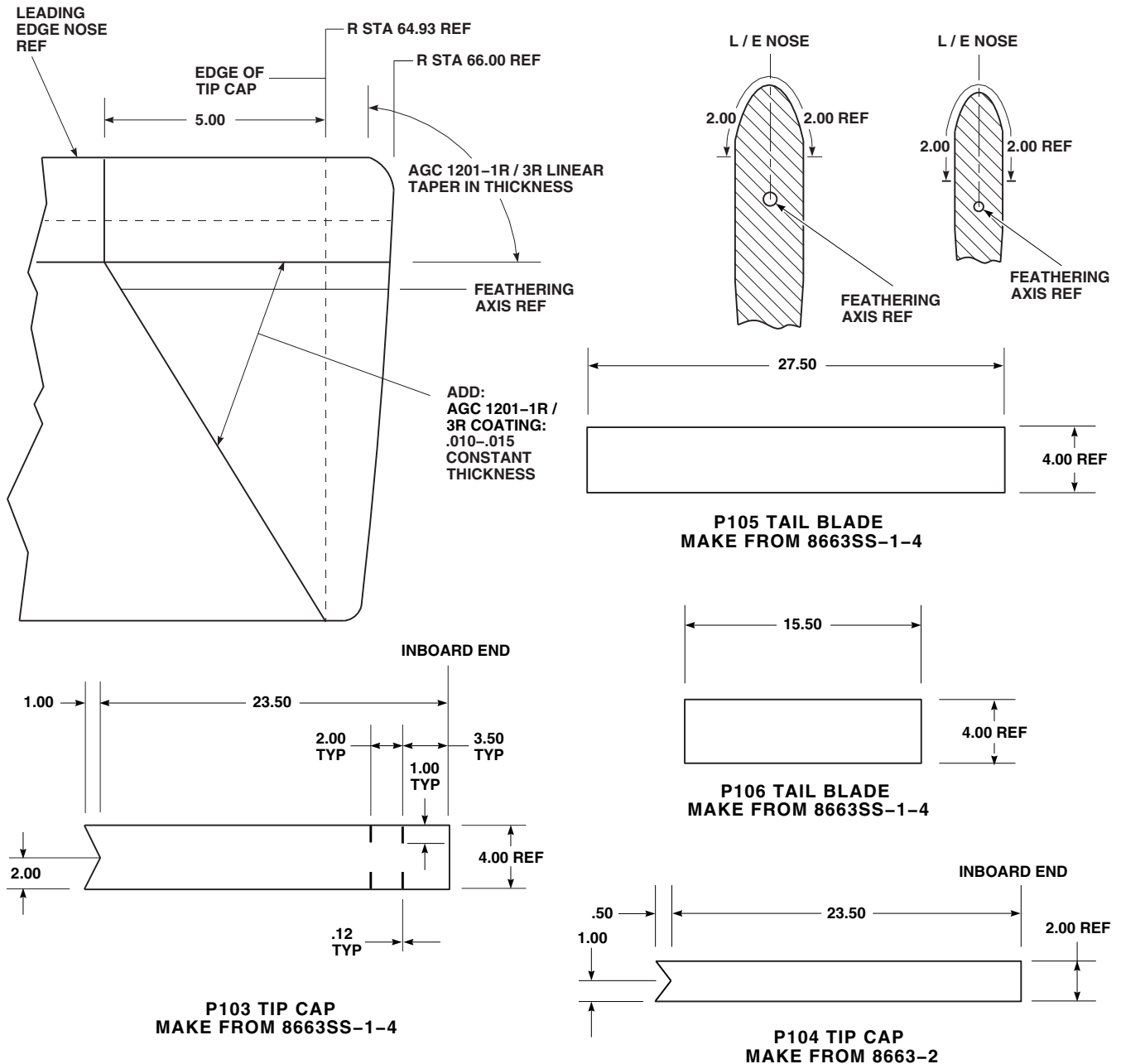
DESCRIPTION: The buddy start system is used to supply power from one helicopter to another for engine starts. When the second helicopter has an inoperable auxiliary power unit (APU) or no APU installed.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1001A
SA

FIGURE C-12. BUDDY START SYSTEM, 70073-30001-011



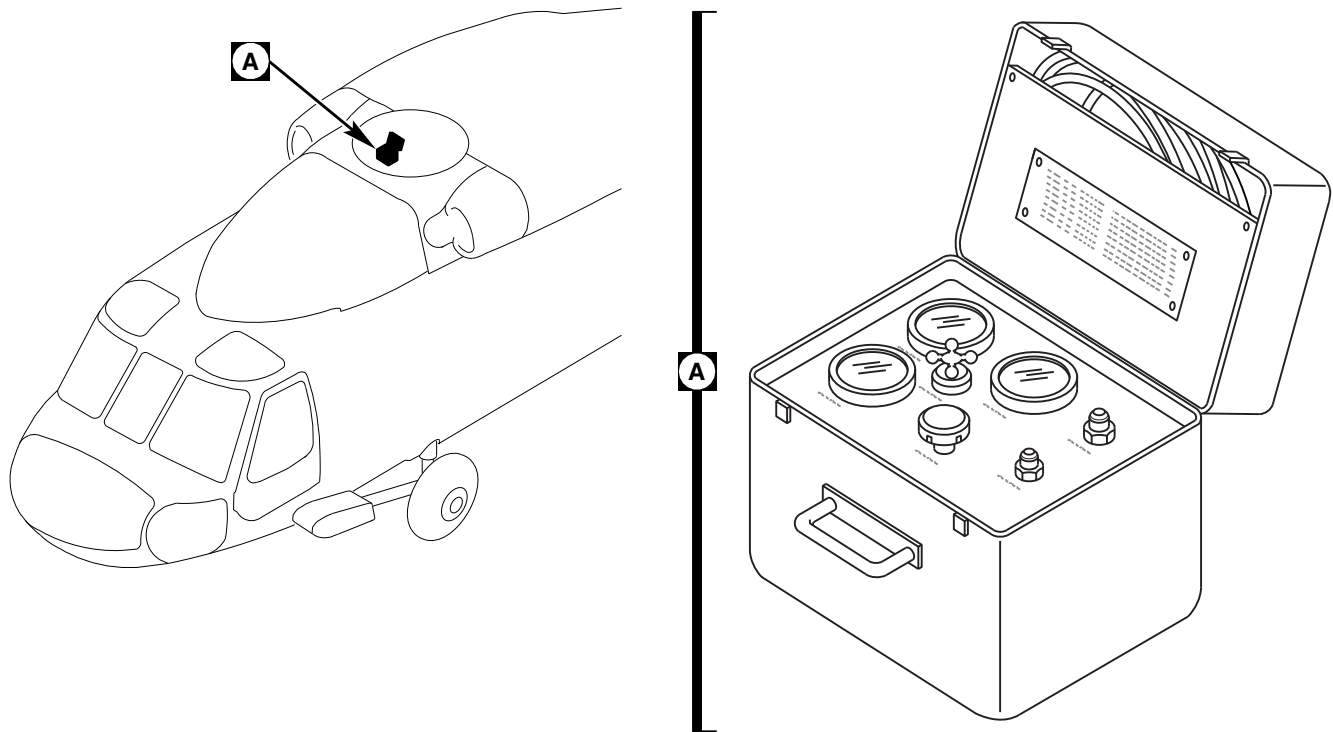
DESCRIPTION: The mission kit is used for repair of leading edge of main and tail rotor blades, and for repair of main rotor blade tip caps.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1002
23

FIGURE C-13. BLADE REPAIR MISSION KIT, 70073-15000-011



DESCRIPTION: The check and fill unit provides the capability to transfer nitrogen from 3000 psi source into the main rotor blade spar, at required 8 to 14 psi and check loss of spar pressure.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

EQUIPMENT REQUIRED

- Nitrogen Bottle, Item 221, Appendix D
- Vacuum Pump
- Ruska Pressure Controller
- Micron Vacuum Gage
- Differential Pressure Gage, 0 to 200 psi range
- Soap and Water
- Machinery Towel, Item 211, Appendix D

FK2211_1
SA

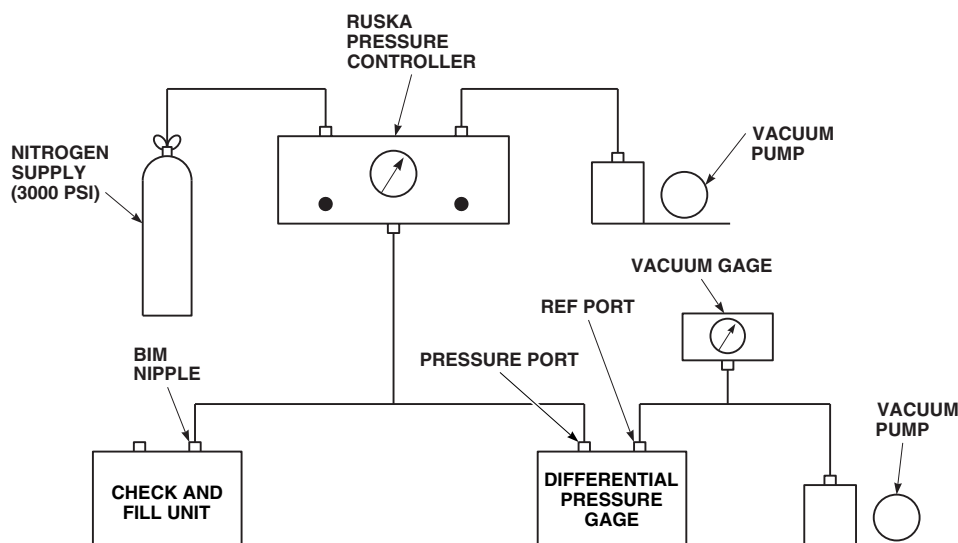
FIGURE C-14. CHECK AND FILL UNIT, BIM® SYSTEM, 70700-77669-041 (SHEET 1 OF 3)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES****Cleaning / Inspection**

1. Clean case and gage glass using soap and water. Wipe dry using machinery towel, Item 211, Appendix D.
2. Inspect check and fill unit for broken gage glass, bent pointers, damaged or unreadable dial face and any damage to case and fittings.

Calibration

1. Make sure thermometer is calibrated.
2. Connect check and fill unit to differential pressure gage and nitrogen bottle, 3000 psi in accordance with view A-A.
3. Connect Ruska Pressure Controller and vacuum pumps in accordance with view A-A.

**VIEW A-A**

4. Disconnect Ruska Pressure Controller. Using switches on front panel adjust gage to zero (PSIA). Reconnect pressure controller.
5. Turn on vacuum pump. Evacuate pressure in differential pressure gage at reference port. Pressure shall be 100 millitorr or less.

FK2211_2B
SA**FIGURE C-14. CHECK AND FILL UNIT, BIM® SYSTEM, 70700-77669-041 (SHEET 2 OF 3)**

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES (CONT)****Calibration (cont)**

6. Check calibration points of gage by increasing pressure from absolute zero as shown in Table 1.

NOTE

To ensure proper readings on gage, tap gage lightly prior to each reading.

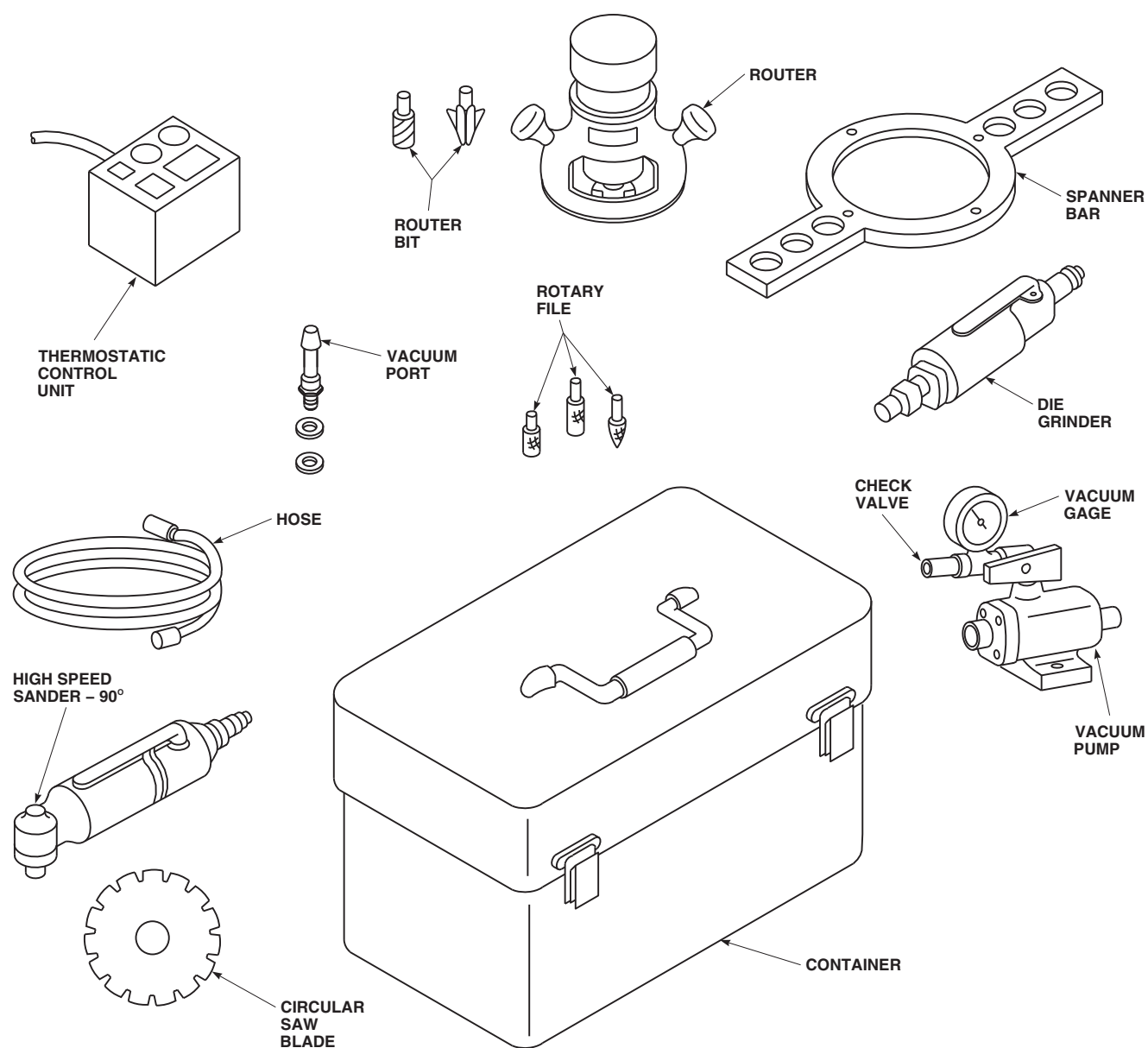
TABLE 1

CHECK AND FILL UNIT – SCALE	DIFFERENTIAL PRESSURE GAGE TOLERANCE (PSIA)
10.0 in Hg	4.783 to 5.012
20.0 in Hg	9.681 to 9.910
29.9 in Hg	14.585 to 14.815
5.0 psig	19.585 to 19.815
10.0 psig	24.585 to 24.815
15.0 psig	29.585 to 29.815

7. Adjust pointer on gage if all readings are off by same value over entire range. Adjustment is limited to 10 degree movement of pointer. If readings are off after adjustment of pointer, replace gage.
8. To adjust absolute pressure gage, remove gage from panel and disconnect fittings from back of gage.
9. Remove glass face from gage, adjust gage to zero with adjusting screw on face of gage.
10. Install glass face on gage and fittings on back of gage. Install gage into panel.
11. To check accuracy of gage adjustment, repeat step 6. If requirements are not met, replace gage.
12. Disconnect the test setup from check and fill unit.
13. Apply calibration label.

FB1126_3C
SA

FIGURE C-14. CHECK AND FILL UNIT, BIM® SYSTEM, 70700-77669-041 (SHEET 3 OF 3)



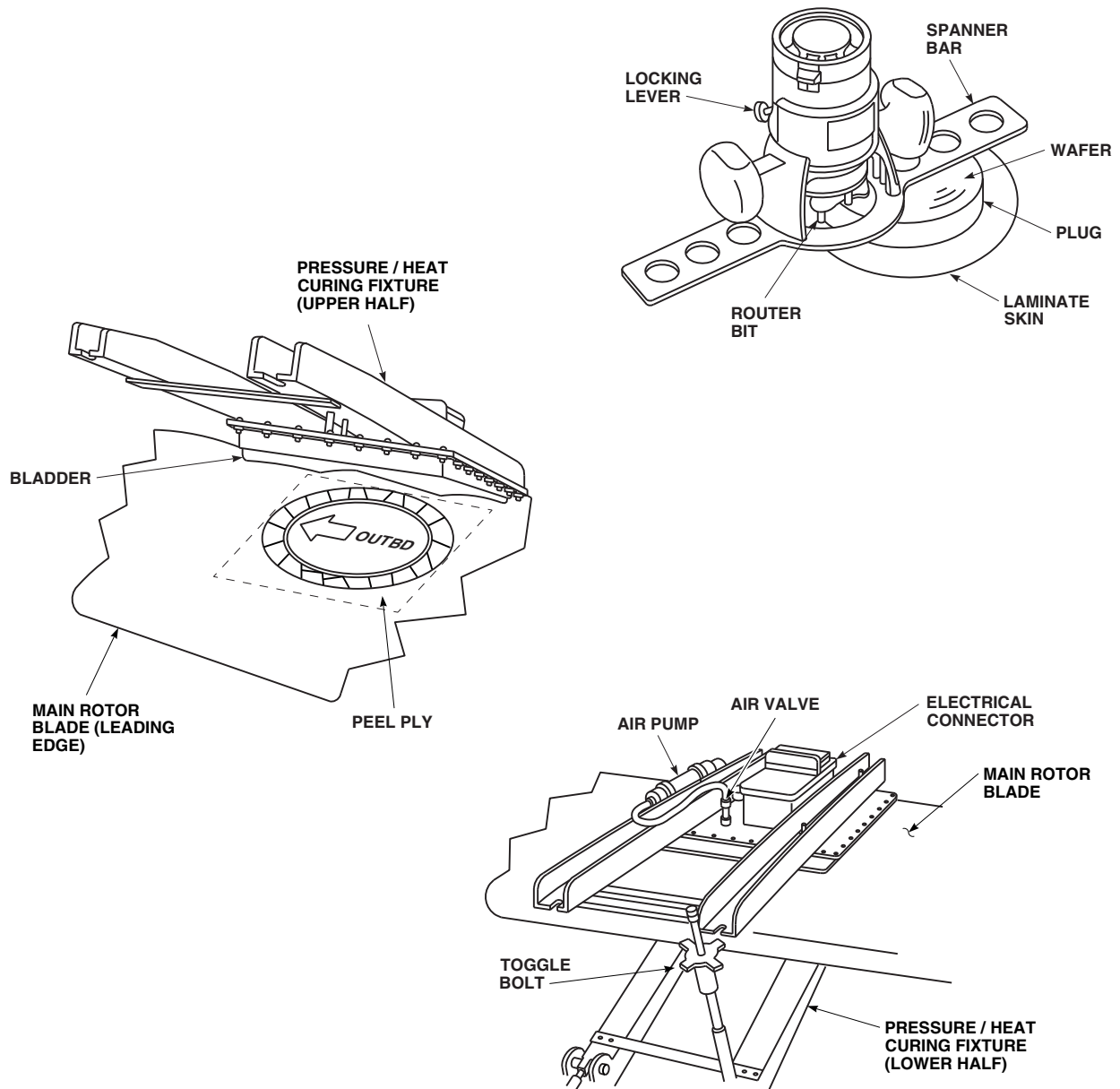
DESCRIPTION: The composite repair kit is required for repairs of components made of composite materials. Items include main and tail rotor blades and various helicopter components.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1003
SA

FIGURE C-15. COMPOSITES REPAIR KIT, 70700-77325-041



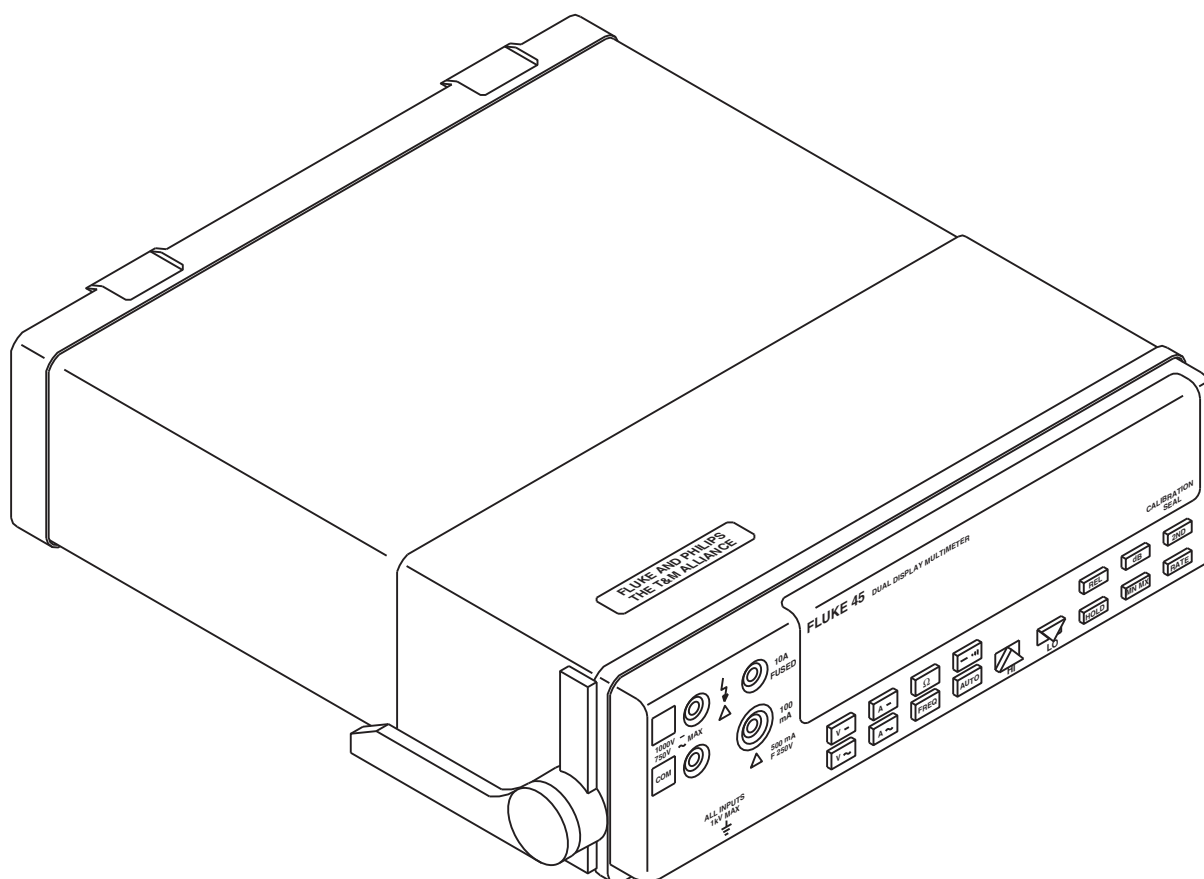
DESCRIPTION: The rotary wing blade repair kit is used in repairing main and tail rotor blade composite area, and main rotor blade tip cap.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1004B
SA

FIGURE C-16. ROTARY WING BLADE REPAIR KIT, 70072-15001-011



DESCRIPTION: The Fluke 45 is a 5 digit, 100,000 count multimeter, with a multifunction dual display allowing measurement of two signal parameters from a single test connection. This test set is used on electrical systems during installation and troubleshooting on or off the helicopter.

POWER REQUIREMENTS: 90 to 264 VAC, 50 Hz and 60 Hz
Battery: lead acid, rechargeable within unit, with unit turned on.

CALIBRATION REQUIREMENTS:

FREQUENCY

Fluke 45 multimeter must be calibrated whenever unit fails, or every 48 weeks.

EQUIPMENT REQUIRED

- Fluke 5700A Multifunction Calibrator (or Equivalent)
- Fluke 5725A Amplifier (or Equivalent)

NOTE:

All measurements listed in the performance test tables are made in the medium reading rate unless otherwise noted.

FB1005_1A
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FIGURE C-17. DIGITAL MULTIMETER, FLUKE 45 (SHEET 1 OF 3)

CALIBRATION REQUIREMENTS:(CONT)**CALIBRATION PROCEDURES****Cleaning / Inspection**

1. Inspect Fluke 45 for signs of physical damage, loose or broken parts. Clean and repair as required.

Calibration

1. Turn on multimeter and allow it to stabilize for one hour.
2. Connect a cable from output VA HI and LO connectors of 5700A to $V\Omega \rightarrow$ and COM connectors on Fluke 45.
3. Select function and range on Fluke 45 and input level from 5700A using values listed in Table 1. The display should read between minimum and maximum values listed in table 1.
4. Connect a cable from Output VA HI and LO connectors of 5700A to the 100 mA and COM connectors on the Fluke 45. Select function and range on Fluke 45 and input level from 5700A using the values in Table 2. The display should read between minimum and maximum readings listed in table 2.

NOTE

The following tests require a Fluke 5725A Amplifier (or equivalent) to be used with 5700A Multifunction Calibrator. The input level for the performance test will be set on 5700A but will be output from 5725A Amplifier.

5. Connect a cable from Output VA HI and LO connectors of 5725A to 10 A and COMM connectors on Fluke 45. Select function and range on Fluke 45 and input level from 5700A using values contained in Table 3. The display should read between the minimum and maximum readings listed in table 3.

Table 1. Performance Tests for Volts, Diode Test, Ohms, and Frequency, Functions

FUNCTION	RANGE (RATE)	INPUT LEVEL	FREQUENCY	DISPLAY	
				MIN	MAX
$V \text{ ---}$	100 mV (slow)	short	—	−0.006	0.006
		+90 mV	—	89.971	90.029
	1000 mV (slow)	+900 mV	—	899.71	900.29
	300 mV	short	—	−0.02	0.02
		300 mV	—	299.90	300.10
	3V	+3V	—	2.9990	3.0010
		−3V	—	−3.0010	−2.9990
	30V	+30V	—	29.990	30.010
	300V	+300V	—	299.90	300.10
	1000V	+1000V	—	999.5	1000.5
$\rightarrow $	—	short	—	−0.0008	0.0008
	—	open	—	(tone) OL	(tone) OL
$V \sim$	300 mV	short		—	< 75
		15 mV	1 kHz	14.87	15.13
			100 kHz	13.75	16.25
		300 mV	1 kHz	299.30	300.70
			100 kHz	284.50	315.50
	3V	3.0V	1 kHz	2.9930	3.0070
	30V	30V	1 kHz	29.930	30.070
	300V	300V	1 kHz	299.30	300.70
	750V	750V	1 kHz	747.5	752.5

FB1005_2A
SA**FIGURE C-17. DIGITAL MULTIMETER, FLUKE 45 (SHEET 2 OF 3)**

Table 1. Performance Tests for Volts, Diode Test, Ohms, and Frequency, Functions (cont)

FUNCTION	RANGE (RATE)	INPUT LEVEL	FREQUENCY	DISPLAY	
				MIN	MAX
Ω	Using decades of 1.9:				
	300 ohm	short	—	0.00	0.04
		190 ohms	—	189.88	190.14
	3 kohm	short	—	0.0000	0.0002
		1.9 kohms	—	1.8988	1.9012
	30 kohm	19 kohms	—	18.988	19.012
	300 kohm	190 kohms	—	189.88	190.12
	3 Mohm	1.9 Mohms	—	1.8987	1.9013
	30 Mohm*	19 Mohms	—	18.949	19.051
	300 Mohm*	190 Mohms	—	186.2	193.8
	Using decades of 1:				
	300 ohm	short	—	0.00	0.04
		100 ohms	—	99.93	100.09
	3 kohm	short	—	0.0000	0.0002
	3 kohm	1 kohms	—	0.9993	1.0007
	30 kohm	10 kohms	—	9.993	10.007
	300 kohm	100 kohms	—	99.93	100.07
	3 Mohm	1 Mohms	—	0.9992	1.0008
	30 Mohm*	10 Mohms	—	9.972	10.028
	300 Mohm*	100 Mohms	—	98.0	102.0
FREQ	10 / 100 kHz	0.1 to 10V	10 kHz	9.9949	10.006
* Optional test points that can be used if standards are available.					

Table 2. Performance Tests for mA Current Functions

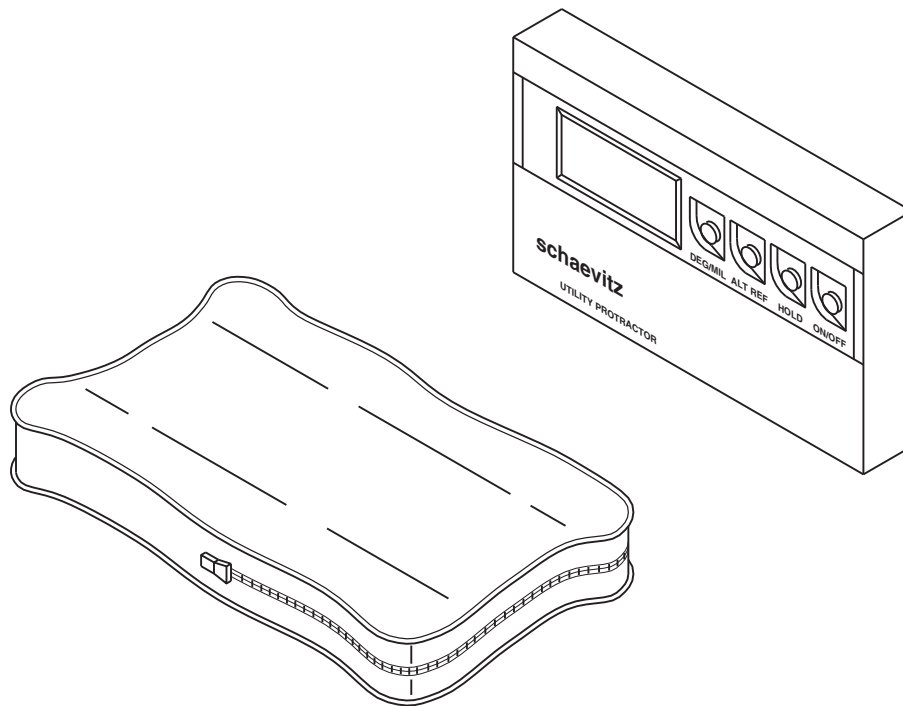
FUNCTION	RANGE (RATE)	INPUT LEVEL	FREQUENCY	DISPLAY	
				MIN	MAX
A $\square\square\square$ (mA)	30 mA	+30 mA	—	29.982	30.018
	100 mA	+100 mA	—	99.93	100.07
A $\sim$ (mA)	30 mA	30 mA	1 kHz	29.840	30.160
	100 mA	100 mA	1 kHz	99.40	100.60

Table 3. Performance Tests for A Current Functions

FUNCTION	RANGE (RATE)	INPUT LEVEL	FREQUENCY	DISPLAY	
				MIN	MAX
A $\square\square\square$	10A	+10A	—	9.975	10.025
A $\sim$	10A	10A	1 kHz	9.890	10.110

FB1005_3
SA

FIGURE C-17. DIGITAL MULTIMETER, FLUKE 45 (SHEET 3 OF 3)



DESCRIPTION: The digital prop protractor is a universal digital unit, which provides maintenance personnel with immediate indication of level, angle, or tilt of main rotor head at time of rigging. The protractor is handheld and battery powered.

POWER REQUIREMENTS: Standard 9 volt battery

CALIBRATION REQUIREMENTS:

FREQUENCY

Digital protractor must be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Precision Level
- Precision Leveling Plate
- Angle Gage Blocks 15° and 45°

CLEANING MATERIALS

- Isopropyl Alcohol, Item 72, Appendix D
- Machinery Towel, Item 211, Appendix D

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SA

FIGURE C-18. DIGITAL PROPELLER PROTRACTOR, 02500-01 (SHEET 1 OF 2)

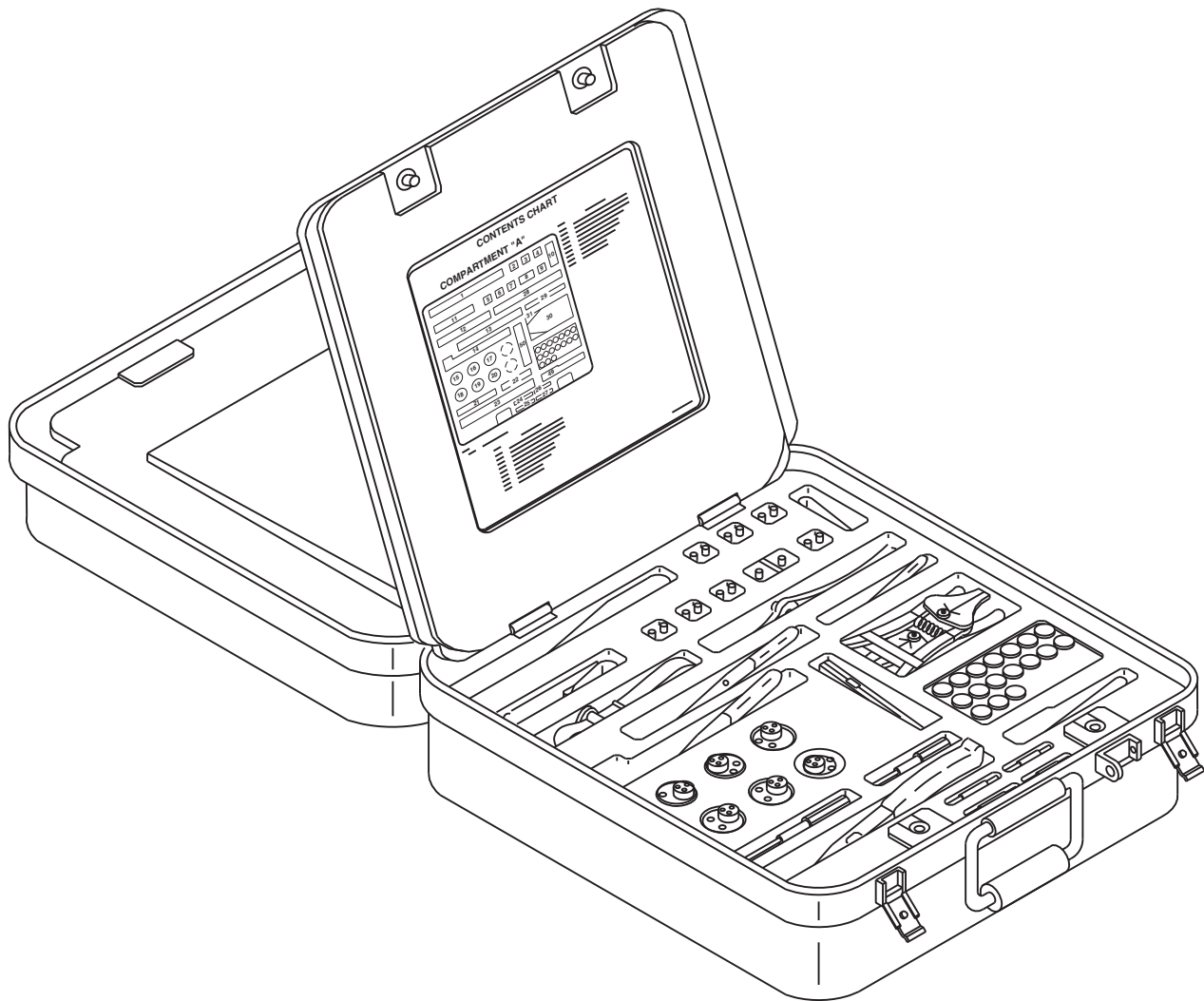
CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES****Cleaning / Inspection**

1. Clean protractor using isopropyl alcohol, Item 72, Appendix D and wipe dry using machinery towel, Item 211, Appendix D.
2. Inspect for signs of damage, abuse, nicks, and dents.
3. Make sure a fresh battery is installed.

Calibration

1. Make sure leveling plate and precision level are clean before starting calibration.
2. Adjust leveling plate and check for level using precision level to achieve centered bubble.
3. Place protractor on leveling plate. Turn on protractor and allow five minutes for warmup, allow for reading to stabilize.
4. Check protractor angle reading at zero, reading shall be $0^{\circ} \pm .05^{\circ}$.
5. Place 15° angle gage block on leveling plate, place protractor on angle block, allow reading to stabilize, record reading. Rotate protractor 180° and record reading. Readings shall be $15^{\circ} \pm .05^{\circ}$. Remove 15° angle block.
6. Place 45° angle gage block on leveling plate, place protractor on angle block, allow reading to stabilize, record reading. Rotate protractor 180° and record reading. Readings shall be $45^{\circ} \pm .1^{\circ}$. Remove 45° angle block.
7. If protractor does not meet specified readings, refer to manufacturer's calibration procedure.

FK2212_2A
SA**FIGURE C-18. DIGITAL PROPELLER PROTRACTOR, 02500-01 (SHEET 2 OF 2)**



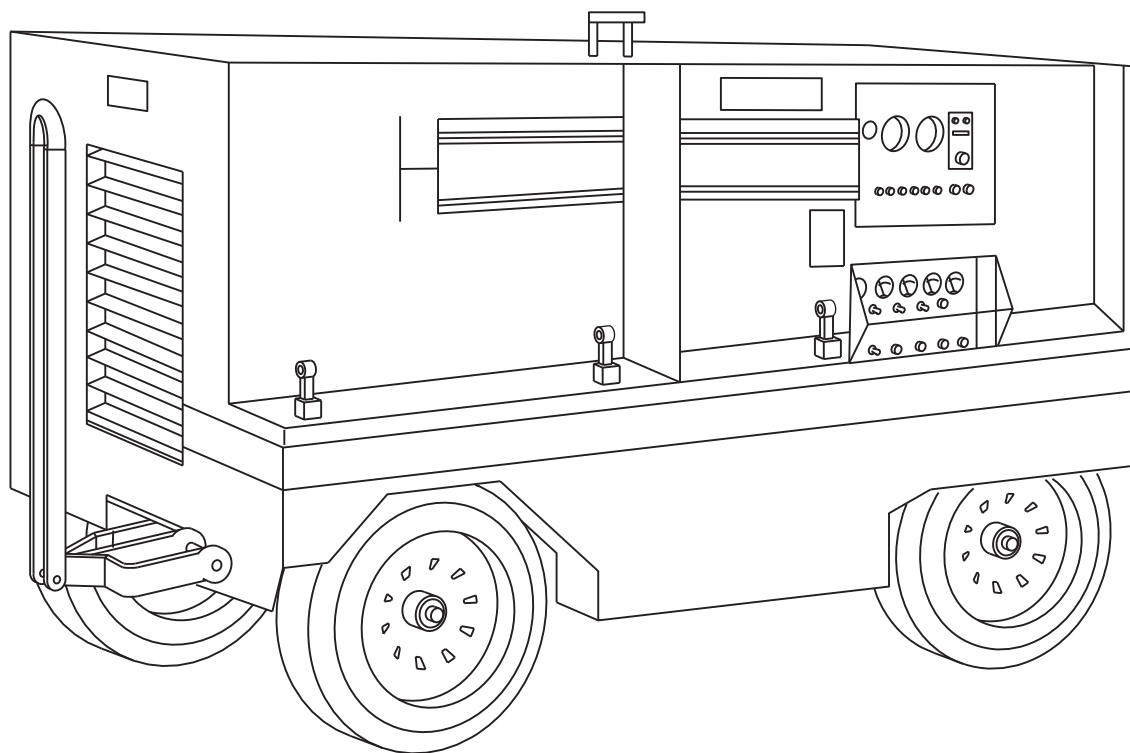
DESCRIPTION: The electronic maintenance / repair kit is used for installation, repair and splicing of wires to maintain helicopter electrical systems.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1007
SA

FIGURE C-19. ELECTRICAL CONNECTOR MAINTENANCE/REPAIR KIT, DMC 787-ELEC



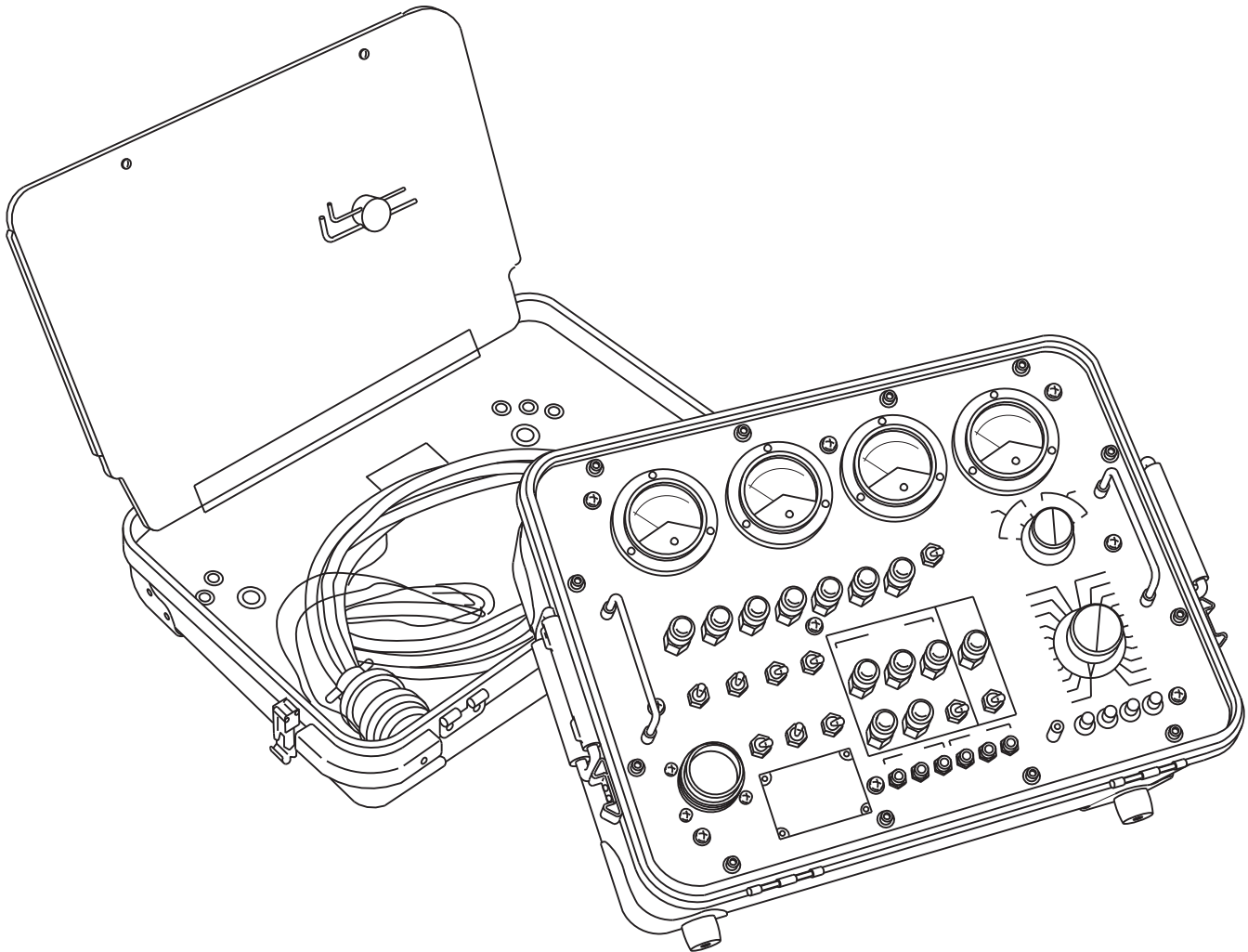
DESCRIPTION: The Generator unit will provide all AC electrical power needed to perform system checks and troubleshooting of various electrical systems. The Generator unit is used in place of the helicopter APU in ground test.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1128A
SA

FIGURE C-20. ELECTRICAL GENERATOR, DIESEL-POWERED, 60J20P



DESCRIPTION: The Analyzer Test Set, type 60B63-5A, provides a means of simulating the variations of electrical inputs to the generator control unit (GCU), and of monitoring the GCU's responses to the variations.

POWER REQUIREMENTS: Power supply, 0-28 VDC @ 4A.
Power supply, 115 VAC 1 ϕ 360-440 Hz.

CALIBRATION REQUIREMENTS:

FREQUENCY

Analyzer Test Set must be calibrated every 24 weeks, regardless of it's being used or not.

EQUIPMENT REQUIRED

- Behlman 3-200A AC Power Supply with EIP-3 Plug-In or equivalent.
- Fluke 8010A Digital Multimeter or equivalent.
- Hewlett Packard 5300A/5307A Counter or equivalent.
- Hewlett Packard 6002A DC Power Supply or equivalent.

FB1009_1
SA

FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 1 OF 7)

CALIBRATION REQUIREMENTS: (CONT)**EQUIPMENT REQUIRED (CONT)**

- Interstate F47 Function Generator or equivalent.
- Sikorsky TS 1060 Test Box.
- Decade Resistor 1100–3300 ohms.
- Oscilloscope – vertical.

ACCESSORIES REQUIRED

- Adapter, BNC plug to double banana jacks.
- Adapter Box, SKD 4850–3, or equivalent.
- Cable Assembly, 30–in., RG–58 / U; BNC plug terminations, 2 required.
- Lead, 24–in., single banana plug terminations (red), 4 required.
- Lead, 24–in., single banana plug terminations (black), 4 required.
- Test Hook Probe, SKC 4850–14 or equivalent.
- Tee Connector, BNC TGG or equivalent.

CALIBRATION PROCEDURES**Inspection / Preliminary**

1. Inspect ANALYZER for physical damage.
NO DAMAGE ALLOWED.
2. Set all ANALYZER switches to OFF or down positions.
3. Pull all ANALYZER GEN and PMG circuit breakers.
4. Check mechanical zero (or lowest reading) of all meters and adjust as required.
5. Set all TEST BOX switches to OFF.

CAUTION

ASSURE THAT THE LAB 200–A 400HZ BENCH POWER SUPPLY IS NOT BEING UTILIZED BEFORE PROCEEDING. OUTPUT VOLTAGE AND FREQUENCY WILL BE VARIED IN THIS PROCEDURE.

6. Connect test set–up shown in figure 1.
7. Set test equipment and ANALYZER power to ON and allow 30 minutes warm–up.

WARNING

HIGH VOLTAGE IS USED DURING THE PERFORMANCE OF THIS CALIBRATION. DEATH ON CONTACT MAY RESULT IF SAFETY PRECAUTIONS ARE NOT OBSERVED.

CAUTION

ASSURE THAT ALL ANALYZER CIRCUIT BREAKERS ARE PULLED AND THAT TEST SET A ϕ , B ϕ , AND C ϕ SWITCHES ARE SET TO OFF.

FREQUENCY METER

1. Adjust FUNCTION GENERATOR frequency for 400 Hz COUNTER indication – leave output set to minimum.
2. Provide a short across TEST BOX I terminals, set 28 VDC – ON and adjust POWER SUPPLY output for 28 VDC Digital Multimeter indication.
3. Reconnect Digital Multimeter, set for VAC, to 115 VAC test points.
4. Set TEST BOX 115 VAC switch to ON position.
5. Slowly increase FUNCTION GENERATOR output until Digital Multimeter indicates approximately 115 VAC.

FB1009_2
SA

FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 2 OF 7)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****FREQUENCY METER (CONT)****6. Set ANALYZER controls:**

a. GEN / PMG VOLTS selector to T1.

b. 400 GEN T1 breaker to IN.

7. Adjust FUNCTION GENERATOR frequency for ANALYZER FREQUENCY meter indications and note COUNTER indications are within tolerance of TABLE 1.**TABLE 1**

FREQUENCY METER	COUNTER Hz
360	356 to 364
400	396 to 404
440	436 to 444

8. Adjust FUNCTION GENERATOR frequency for COUNTER indication of 400 Hz.**9. Leave set-up for next test.****GEN / PMG VOLTS METER****1. Adjust FUNCTION GENERATOR output for ANALYZER GEN / PMG VOLTS meter 115 VAC indication and then note the Digital Multimeter indicates 113 to 117 VAC.**

2. Pull 400 ~ GEN T1 circuit breaker and depress T2.

3. Set GEN / PMG VOLTS selector to T2 and note 115 V meter indication.

4. Pull 400 ~ GEN T2 circuit breaker and depress T3.

5. Set GEN / PMG VOLTS selector to T3 and note 115 V meter indication.

6. Adjust FUNCTION GENERATOR for minimum output and leave set-up for the next test.

CT TEST CURRENT METER**1. Set ANALYZER controls:**

a. 400 ~ GEN T1 breaker to IN

b. 400 ~ GEN T2 breaker to OUT

c. 400 ~ GEN T3 breaker to OUT

d. TEST SELECT to A ϕ GEN CT

e. CT CUR READ to UP

f. GEN / PMG VOLTS to T1

2. Connect DECADE RESISTOR, set for 3300 ohms, across TEST BOX R terminals.

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SA

FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 3 OF 7)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****CT TEST CURRENT METER (CONT)**

3. Set TEST BOX switches
 - a. $A\phi$, $B\phi$, and $C\phi$ to OFF
 - b. 115 VAC to ON
 - c. 28 VDC to ON
4. Adjust FUNCTION GENERATOR output for 115 VAC indication on ANALYZER GEN / PMG VOLTS meter.
5. Set DECADE RESISTOR and note CT TEST CURRENT meter indications of TABLE 2.

TABLE 2

DECADE RESISTOR	CT TEST CURRENT
3300	0.9 to 1.1
2100	1.4 to 1.6
1500	1.9 to 2.1
1100	2.4 to 2.6

6. Adjust FUNCTION GENERATOR for minimum output and leave set-up for next test.

PMG VOLTS METER

1. Set ANALYZER controls:
 - a. 400 ~ GEN T1, T2, T3 breakers to OUT
 - b. PMG $A\phi$, $B\phi$, $C\phi$ breakers to OUT
 - c. GEN / PMG VOLTS selector to AB
2. Set TEST BOX controls:
 - a. 115 VAC to OFF
 - b. 28 VDC to OFF
 - c. $A\phi$ to ON
 - d. $B\phi$ to ∇
 - e. $C\phi$ to OFF
3. Remove DECADE RESISTOR from set-up.

CAUTION

ASSURE THAT LAB 400 Hz BENCH POWER IS NOT BEING UTILIZED AND THAT SWITCH SETTINGS ARE AS SPECIFIED BEFORE CONTINUING.

4. Remove Digital Multimeter from TEST BOX 115 VAC test points and reconnect to 18 VAC test points.
5. Adjust FUNCTION GENERATOR frequency for 1200 Hz COUNTER indication.
6. Slowly adjust FUNCTION GENERATOR output for approximate 18 VAC Digital Multimeter indication.

FB1009_4
SA**FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 4 OF 7)**

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****PMG VOLTS METER (CONT)**

7. Set ANALYZER PMG A ϕ , and B ϕ circuit breakers to IN and then adjust FUNCTION GENERATOR output for 18V GEN / PMG VOLTS meter indication.
8. Note Digital Multimeter indicates 17.6 to 18.4 VAC.
9. Adjust FUNCTION GENERATOR output for GEN / PMG VOLTS meter indications and note Digital Multimeter indications are within tolerances specified in TABLE 3.

TABLE 3

GEN/PMG VOLTS METER	DMM
21	20.6 to 21.4
24	23.6 to 24.4
27	26.6 to 27.4

10. Adjust FUNCTION GENERATOR output for 18V GEN / PMG VOLTS meter indication.
11. Set ANALYZER controls:
 - a. PMG A ϕ breaker to OUT
 - b. PMG B ϕ and C ϕ breaker to IN
 - c. GEN / PMG VOLTS to BC
12. Set TEST BOX switches:
 - a. A ϕ to OFF
 - b. C ϕ to ON
13. Note GEN / PMG VOLTS meter indicates 18 VAC.
14. Set ANALYZER controls:
 - a. PMG A ϕ breaker to IN
 - b. PMG B ϕ breaker to OUT
 - c. PMG C ϕ breaker to IN
 - d. GEN / PMG VOLTS to CA
15. Set TEST BOX switches:
 - a. A ϕ to ON
 - b. B ϕ to OFF
 - c. C ϕ to 
16. Note GEN / PMG VOLTS meter indicates 18 VAC.
17. Adjust FUNCTION GENERATOR for minimum output.

FB1009_5
SA

FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 5 OF 7)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****PMG VOLTS METER (CONT)**

18. Remove TEST BOX 115V power plug from bench source.

19. Set TEST BOX switches:

a. 115 VAC to OFF

b. $A\phi$, $B\phi$, AND $C\phi$ to OFF

NOTE

The lab 200-A 115V 400 Hz Power supply is not needed for the remainder of the procedure. Assure that the TEST BOX 115V power plug is removed from the bench supply plug.

FIELD AMPS / REG VOLTS METER

1. Connect Digital Multimeter, set for VDC, to TEST BOX 28 VDC test points.
2. Short TEST BOX I terminals.
3. Set ANALYZER READ REG BUS VOLTS-FLD CUR switch to up.
4. Set TEST BOX 28 VDC switch to ON and adjust DC POWER SUPPLY for 28 V FIELD AMPS / REG VOLTS meter indication.
5. Note Digital Multimeter indicates 27 to 29 VDC.
6. Adjust DC POWER SUPPLY for minimum output.
7. Connect Digital Multimeter DCA (10A) input across TEST BOX I terminals and set AMP to ON.
8. Set ANALYZER READ BUS VOLTS-FLD CUR switch to down.
9. Adjust DC POWER SUPPLY output for FIELD AMPS meter indications and note Digital Multimeter reads are within specified tolerances of TABLE 4.

TABLE 4

FIELD AMPS	DMM DCA
2.0	1.8 to 2.2
2.5	2.3 to 2.7
3.0	2.8 to 3.2
4.0	3.8 to 4.2

10. Adjust DC POWER SUPPLY for minimum output and leave set-up for next test.

TIMER TIME DELAY

1. Remove Digital Multimeter from TEST BOX I terminals and reconnect, set for VDC, to 28 VDC test points.
2. Provide a short across the TEST BOX I terminals and set AMP to OFF.
3. Adjust DC POWER SUPPLY output for 28 VDC Digital Multimeter indication.
4. Connect COUNTER A and B inputs to ANALYZER J2 (TRIG) and J4 (T2 CAL) bnc jacks respectively.
5. Set COUNTER for TIME INTERVAL measurement, separate inputs, and to start / stop trigger on a +20 VDC pulse.

FB1009_6
SA

FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 6 OF 7)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****TIMER TIME DELAY (CONT)**

6. Set ANALYZER TEST SELECT to 3 ϕ OVB.
7. Note as TRIP TIME START is toggled from down to up the COUNTER indicates 100 to 140 mSec. (Adjustment R1).
8. Make connections, set controls and then note COUNTER indications are with specified tolerances of TABLE 5 after TRIP TIME START switch is toggled.

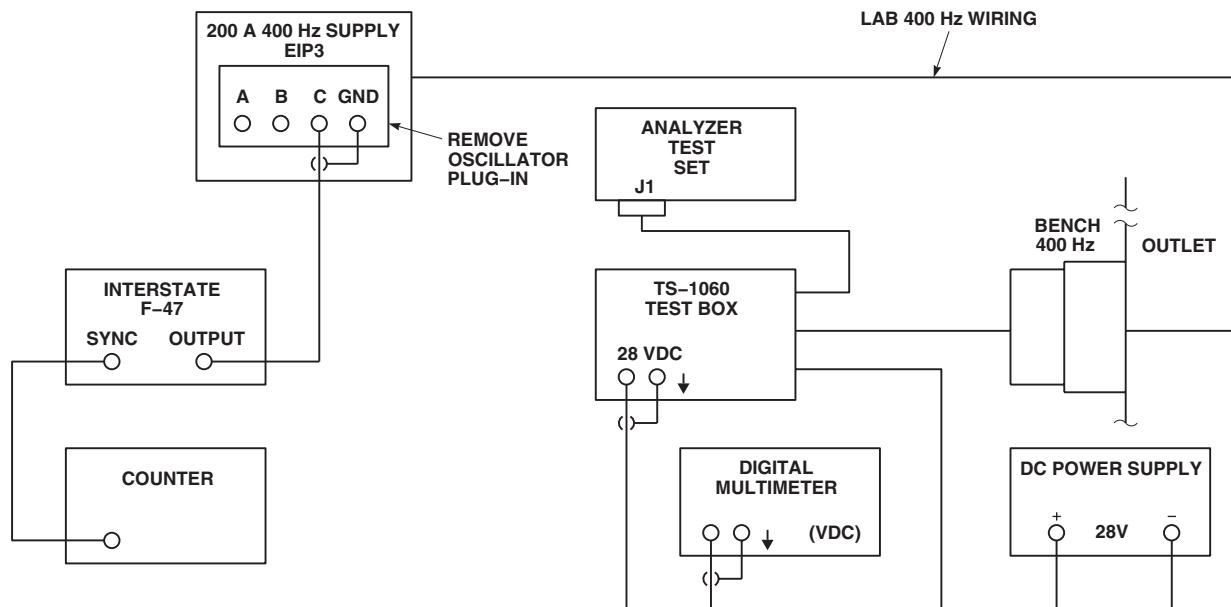
TABLE 5

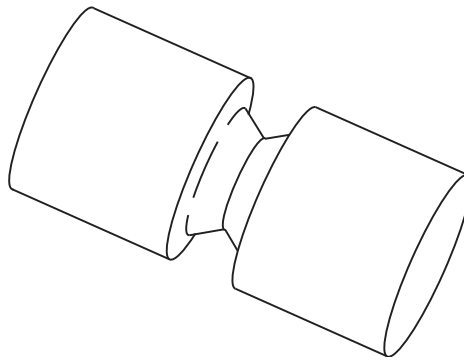
COUNTER B input	TEST SELECT	COUNTER T. I.	ADJUST
J4 (T2 CAL)	A ϕ UV	4.5 to 5.5 sec	R3
J4 (T2 CAL)	UND FREQ	3.0 to 4.0 sec	R7
J3 (T1 CAL)	A ϕ UV	3.0 to 4.0 sec	R5
J3 (T1 CAL)	UND FREQ	0.90 to 1.1 sec	R119

9. Leave set-up for next test.

UNDER FREQUENCY

1. Tee COUNTER A input and OSCILLOSCOPE input to J5 (UF CAL) bnc jack.
2. Set COUNTER to read frequency.
3. Set ANALYZER TEST SELECT to UND FREQ and TRIP TIME START to DOWN.
4. Note COUNTER indication of 1145 to 1165 Hz and 20 to 30 Vpp OSCILLOSCOPE display.
5. Set TRIP TIME START to UP, note 1085 to 1105 Hz COUNTER indication and 20 to 30 Vpp OSCILLOSCOPE display.
6. Unless further tests are to be made disconnect the set-up.

**FIGURE 1**FB1009_7
SA**FIGURE C-21. ELECTRICAL SYSTEM ANALYZER, 60B63-5A (SHEET 7 OF 7)**



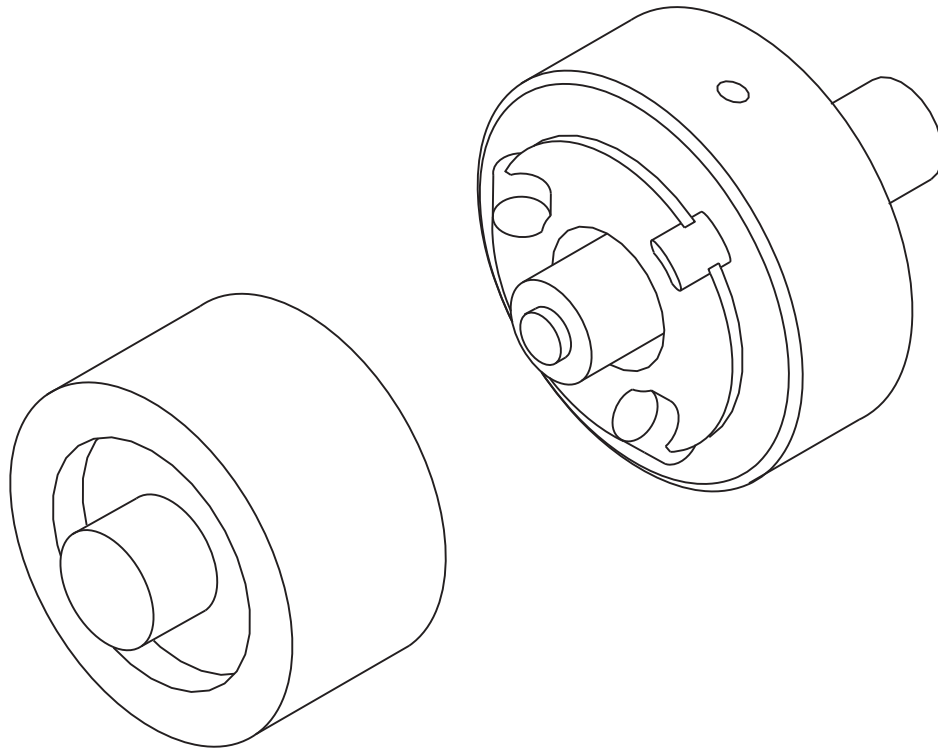
DESCRIPTION: The staking tool is used to stake bearings on the engine mount fittings.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:Not Applicable

FB1011A
SA

FIGURE C-23. ENGINE MOUNT STAKING TOOL, STD1850.451-005



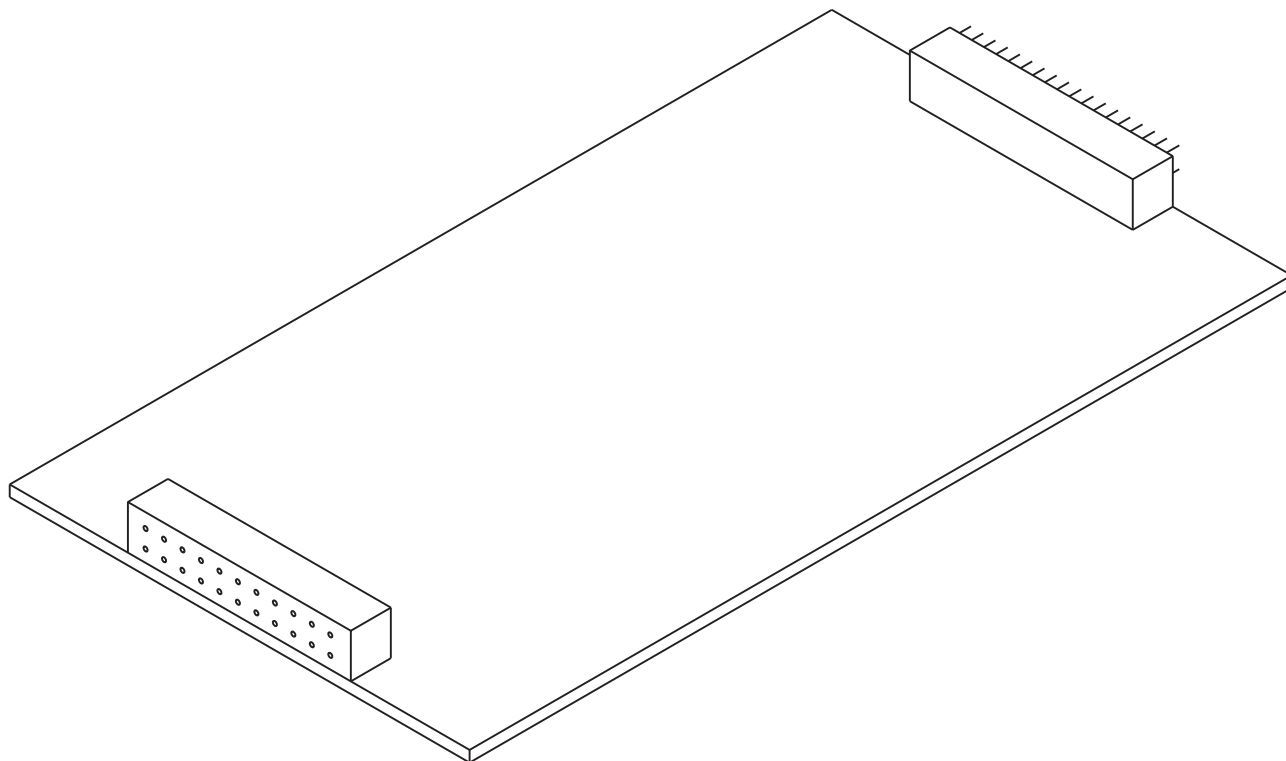
DESCRIPTION: This staking tool is used to roll stake bearings in engine mount struts.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FIGURE C-24. ENGINE STRUT STAKING TOOL, STD1850.458-024

FB1012
SA



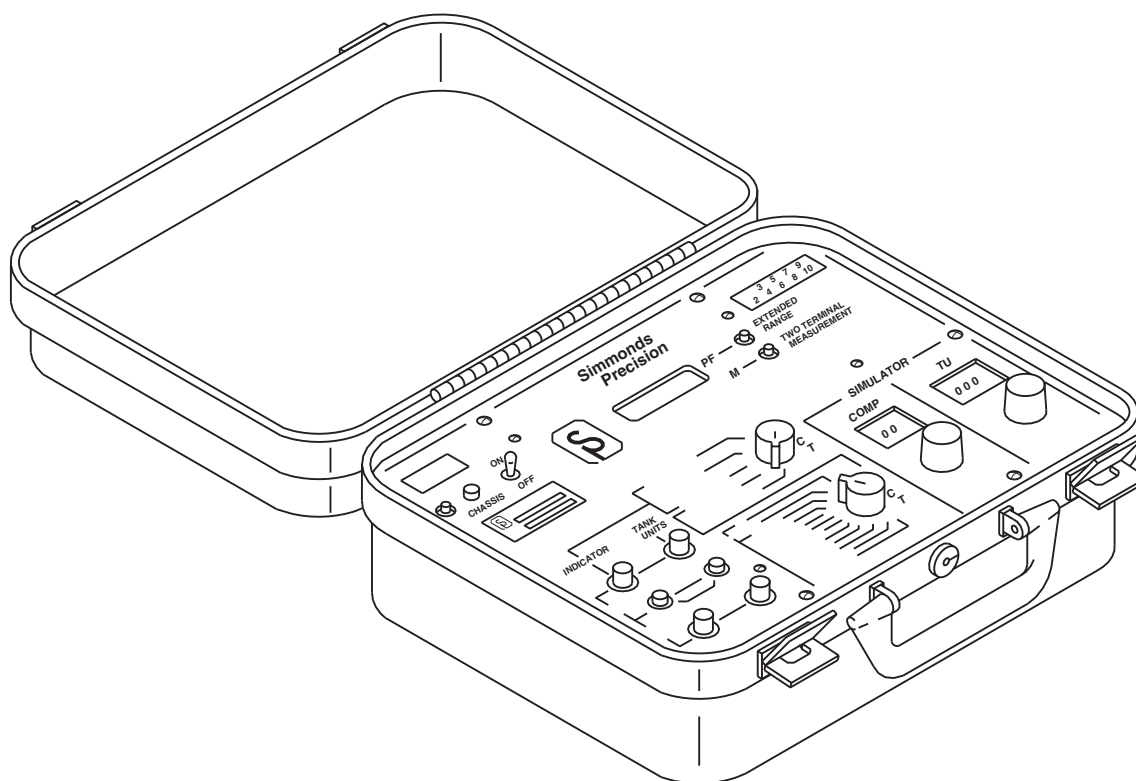
DESCRIPTION: The extender card provides an extension from vertical instrument display system (VIDS) components to line test sets during calibration or repair of VIDS.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1013A
SA

FIGURE C-25. EXTENDER CARD, 220-419916-000



DESCRIPTION: The fuel quantity test set is an instrument that permits complete functional checkout and calibration of an ac fuel gaging system, on or off the helicopter. The fuel quantity test set can accurately measure the capacitance of tank units, compensators, or entire systems. The fuel quantity test set can also simulate capacitance values for the operation of indicators as well as measure the insulation resistance of tank units and cabling.

POWER REQUIREMENTS: 18 1.5 Volt C-Cell Batteries

CALIBRATION REQUIREMENTS:

FREQUENCY

Fuel quantity test set must be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Simmonds capacitance transfer standard, 932005 (or equivalent).

NOTE

P / N 932005 consists of 14 fixed value capacitors, three terminal configurations, ranging from 10 pf to 10,000 pf with a temperature stability factor of ± 25 ppm per degree C, with consistent accuracy of $\pm .05$ pf at ambient temperature of 25°C.

- General Radio 1620A capacitance measuring system (or equivalent).
- High resistance decade – one hundred kilohms to one thousand megohms with $\pm 1\%$ tolerance.

FB1014_1C
SA

FIGURE C-26. FUEL QUANTITY TEST SET, PSD 60-2R (SHEET 1 OF 4)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURE

CAUTION:

THE PSD 60-2R CONTAINS ESD SENSITIVE DEVICES.
OPEN AND CALIBRATE AT ESD PROTECTED AREA ONLY.

1. Remove the panel from the case.
2. Turn power ON and allow a 5 minute warm-up before proceeding.

Calibration of Capacitance Bridge Section

1. FUNCT switch to MEASURE EXT. SELECT switch to TU. Adjust Capacitance Low Range Zero potentiometer to a zero reading on the display.

NOTE:

A true zero reading is when the "-" (minus) sign on the display is ON half the time and OFF the other half.

2. Adjust Capacitance Extended Range Zero potentiometer while the EXTENDED RANGE switch is depressed to a zero reading on the display.
3. Repeat Steps 1 and 2 to assure that zero is correct and no adjustment is required.
4. Connect 1000 pf (nominal) standard to J4 and J5 (TANK UNITS LO Z and HI Z). Adjust Capacitance Low Range Gain potentiometer to the standard's value within ± 1 LSD on the display (± 0.1 pf).
5. Connect 10,000 pf (nominal) standard to J4 and J5 (TANK UNITS LO Z and HI Z). Adjust Capacitance Extended Range Gain potentiometer to the standard's value within ± 1 LSD on the display (± 1 pf) while depressing the EXTENDED RANGE switch.
6. Repeat Steps 1 thru 5 until no adjustment is required.

Calibration of Megger Section

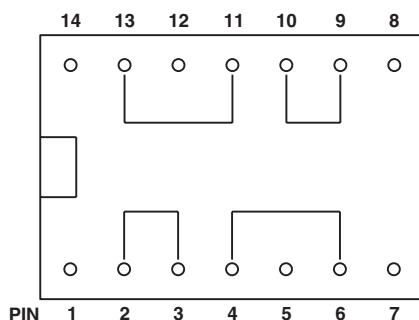
1. FUNCT switch to MEASURE EXT. SELECT switch to LO Z - HI Z.
2. Turn power OFF, remove U15 and short pins together as shown in View A then turn power ON.
3. With a 0.1 M (nominal) standard connected to TANK UNITS LO Z and HI Z (center conductors of J4 and J5) adjust Meg Low Range Gain potentiometer to the standard's value within $\pm 2 - 0$ LSD on the display ($\pm 2 - 0$ K ohms).
4. Turn power OFF and replace jumpers with U15, then turn power ON.
5. Replace the standard with a 10 M (nominal) standard and adjust Meg Low Range Zero potentiometer to the standard's value within ± 1 LSD on the display (± 0.1 M).
6. Replace the standard with a 100 M (nominal) standard and adjust Meg Mid Range Zero potentiometer to the standard's value within ± 1 LSD on the display (± 1 M).
7. Replace the standard with a 1000 M (nominal) standard and adjust Meg High Range Zero potentiometer to the standard's value within ± 1 LSD on the display (± 10 M).

FB1014_2A
SA

FIGURE C-26. FUEL QUANTITY TEST SET, PSD 60-2R (SHEET 2 OF 4)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURE (CONT)**
Calibration of Megger Section (cont)

8. Reassemble unit. To assure a correct calibration and test set functioning, perform the Calibration Check Procedure after the unit is reassembled.

**CALIBRATION JUMPERS****VIEW A****CALIBRATION CHECK PROCEDURE****Check Calibration of Capacitance Measuring Section**

1. Turn test set power ON and allow a five minute warm-up period.
2. Set SELECT switch (S3) to TU. Set FUNCT switch (S2) to MEASURE EXT.
3. Connect J4 and J5 panel connectors to capacitance source using suitable test leads (J5 to HI Z & J4 to LO Z).
4. Read capacitance directly on digital display (D3).

NOTE:

Tolerance for capacitance measuring section is $\pm 0.1\%$ of reading or 0.2 pf whichever is greater.

NOTE:

The same test can be performed by connecting J5 and J6 to external capacitance source and SELECT switch (S3) to COMP.

Check Operation of Capacitance Simulation Section

1. Turn test set power ON and allow five minute warm-up period.
2. Set SELECT switch (S3) to TU. Set FUNCT switch (S2) to MEASURE INT.
3. Connect test set J1 and J2 panel connectors to the General Radio 1620A capacitance measuring system "UNKNOWN" panel connectors (J1 to "L" and J2 to "H").
4. Set D1 capacitance decade to all zero's and adjust V1 vernier control for a reading of 20.0 pf on the digital display (D3). This sets the simulator to a calibrated value.
5. Set FUNCT switch (S2) to SIM TU ONLY.

NOTE:

With FUNCT switch in any simulate mode, the reading on the display is to be disregarded.

6. Read capacitance being simulated by the TU simulator on GR1620A bridge. The GR1620A should show the value of the TU SIMULATOR to be 20.0 ± 0.2 pf.

FB1014_3
SA**FIGURE C-26. FUEL QUANTITY TEST SET, PSD 60-2R (SHEET 3 OF 4)**

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION CHECK PROCEDURE (CONT)

Check Operation of Capacitance Simulation Section (CONT)

7. Set SELECT switch (S3) to COMP. Set FUNCT switch (S2) to MEASURE INT.
8. Connect test set J3 and J2 panel electrical connectors to the General Radio 1620A capacitance measuring system "UNKNOWN" panel electrical connectors (J3 to "L" and J2 to "H").
9. Using electrical jumper wire, connect the CHASSIS jack (J7) to the center conductor of J1.
10. Set D2 capacitance decade to all zero's and adjust V2 vernier control for a reading of 20.0 pf on the digital display (D3). This sets the simulator to a calibrated value.
11. Set FUNCT switch (S2) to SIM TU & COMP.

NOTE

With FUNCT switch in any simulate mode, the reading on the display is to be disregarded.

12. Read capacitance being simulated by the COMP simulator on GR1620A bridge. The GR1620A should show the value of the COMP SIMULATOR to be 20.0 ± 0.2 pf.

Check Accuracy of Megohmmeter Section

1. Turn test set power ON and allow a five minute warm-up period.
2. Set SELECT switch (S3) to LO Z – HI Z.
3. Set FUNCT Switch (S2) to MEASURE EXT.
4. Connect high resistance decade between the appropriate points of panel electrical connectors J4, J5 and / or J6 as indicated by the following list of SELECT switch positions for the megohmmeter functions:

LO Z – HI Z

COMP – HI Z

LO Z – COMP

LO Z – SHIELD

COMP – SHIELD

HI Z – SHIELD

SHIELD – GND

LO Z – GND

COMP – GND

HI Z – GND

NOTE

Shells of electrical connectors J4 and J6 are ground. Shell of J5 is shield.

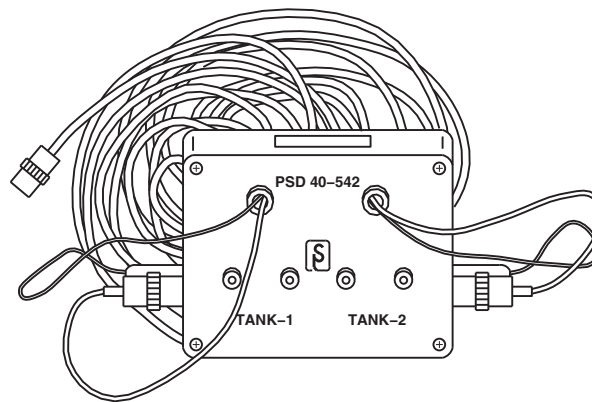
5. Read resistance values directly on digital display. Tolerance for megohmmeter section is $\pm 10\%$.

NOTE

The use of a commercial BNC to dual binding post adapter on J5 during this test may be helpful.

FB1014_4A
SA

FIGURE C-26. FUEL QUANTITY TEST SET, PSD 60-2R (SHEET 4 OF 4)



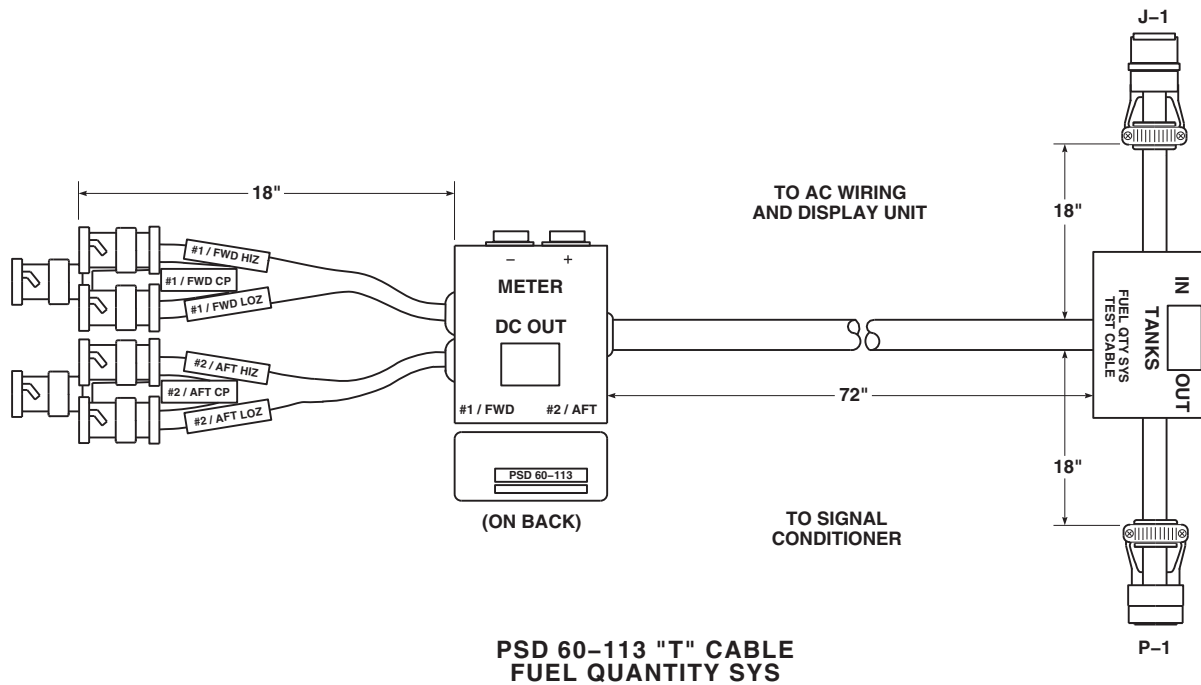
DESCRIPTION: The test harness is used to connect the fuel quantity test set to the fuel system for test purposes.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1015A
SA

FIGURE C-27. FUEL QUANTITY TEST HARNESS, PSD 40-542



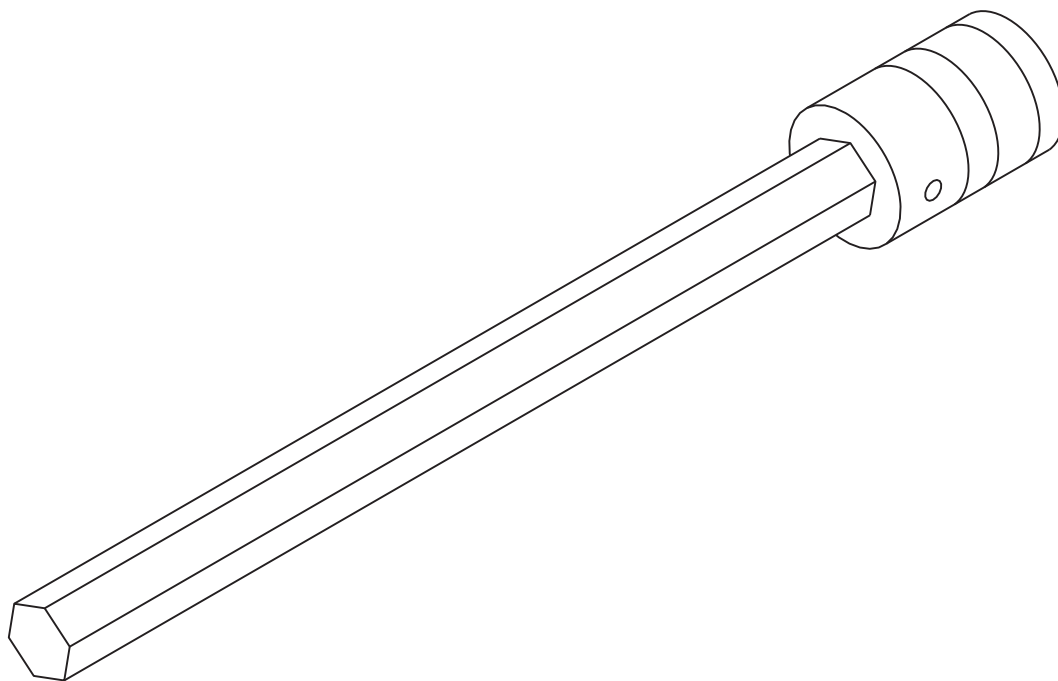
DESCRIPTION: The test cable is used to connect the fuel quantity tester set to the helicopter for test of fuel system.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1127A
SA

FIGURE C-28. FUEL QUANTITY CABLE ASSEMBLY, PSD 60-113



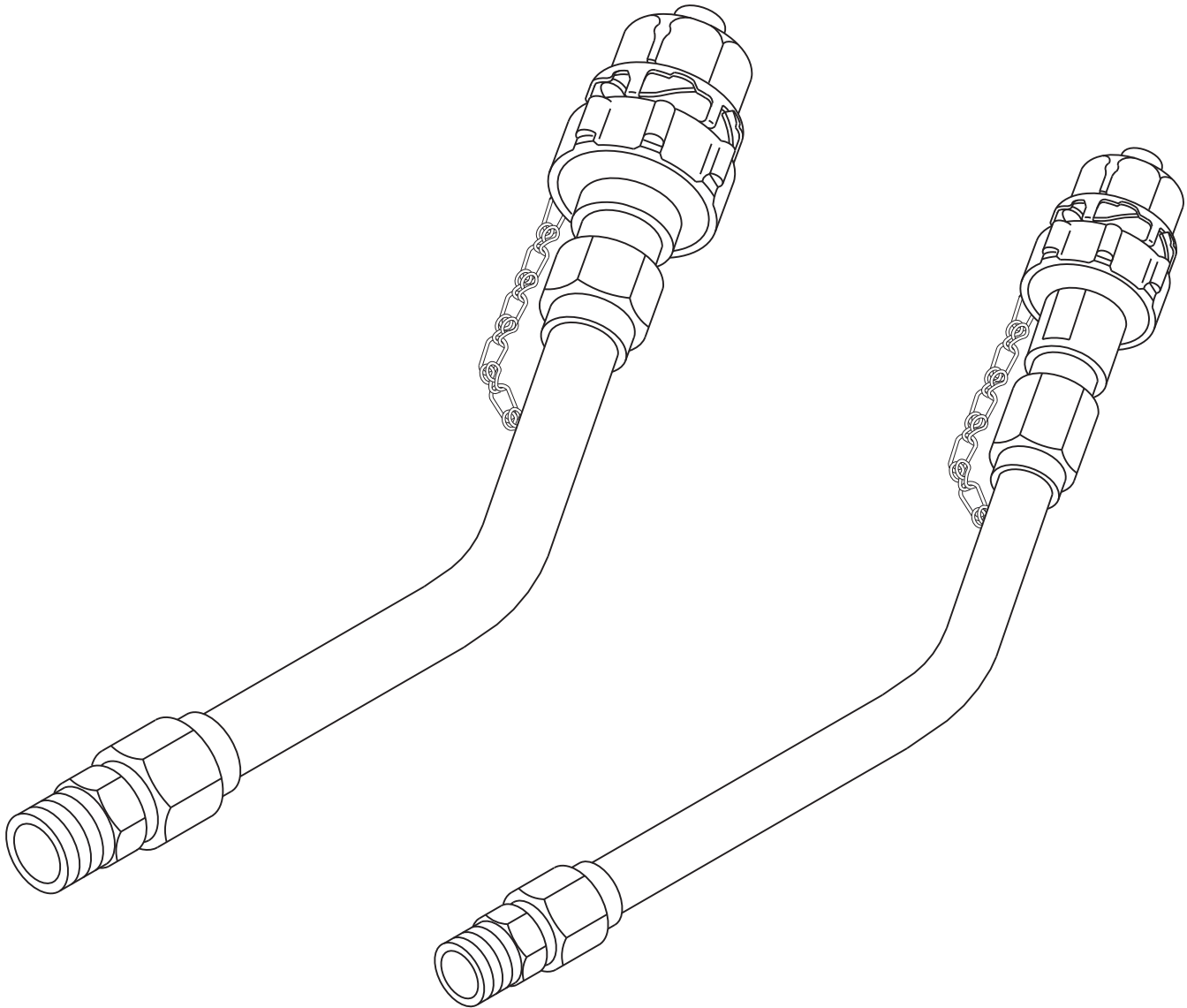
DESCRIPTION: The long hex head wrench is used on helicopter vibration absorbers.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FIGURE C-29. LONG HEX HEAD WRENCH, FA12L

FB1016
SA



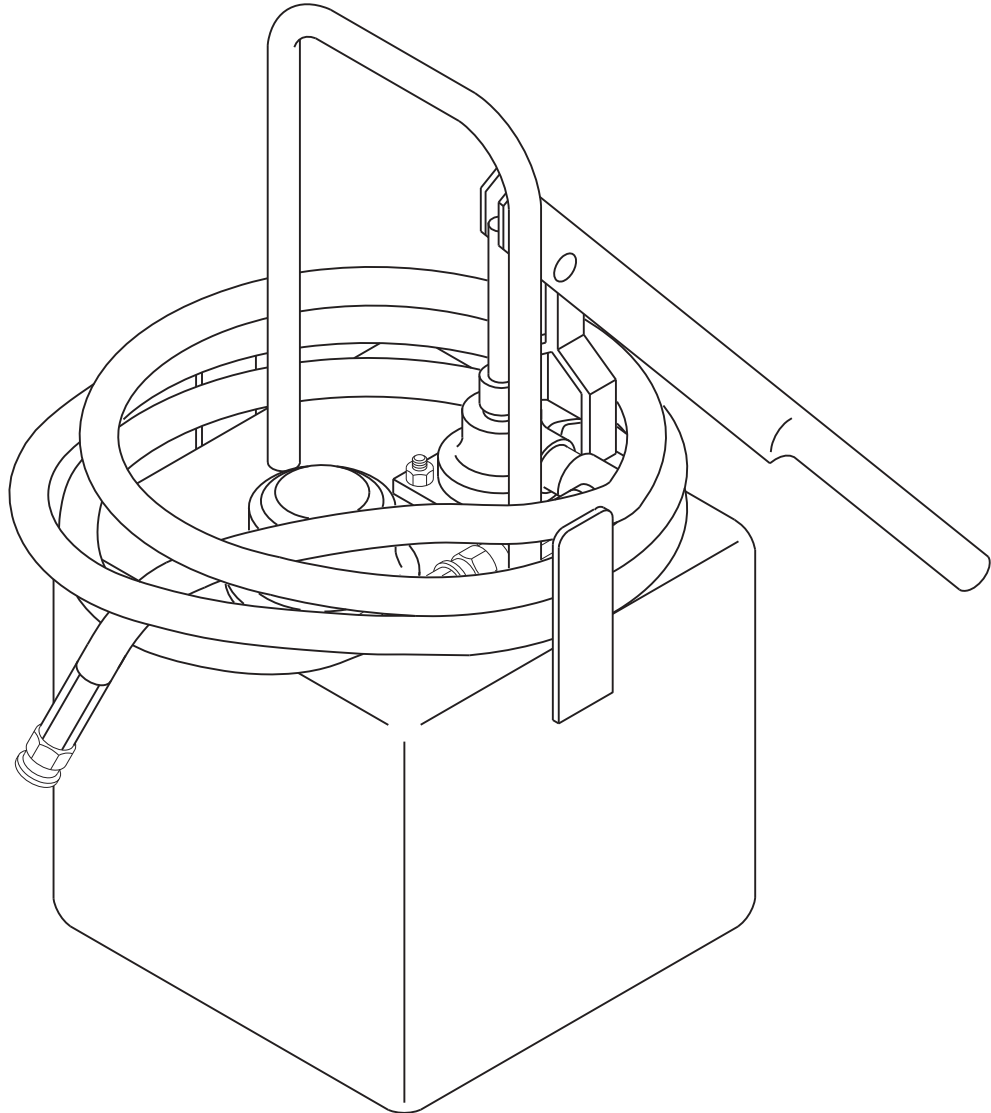
DESCRIPTION: The hydraulic cart adapter fittings are used to service the helicopter, and provide hydraulic power to move flight controls. The adapters are for hydraulic pressure and return to the power cart.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1017
SA

FIGURE C-30. HYDRAULIC CART ADAPTER FITTINGS, KHC 1004



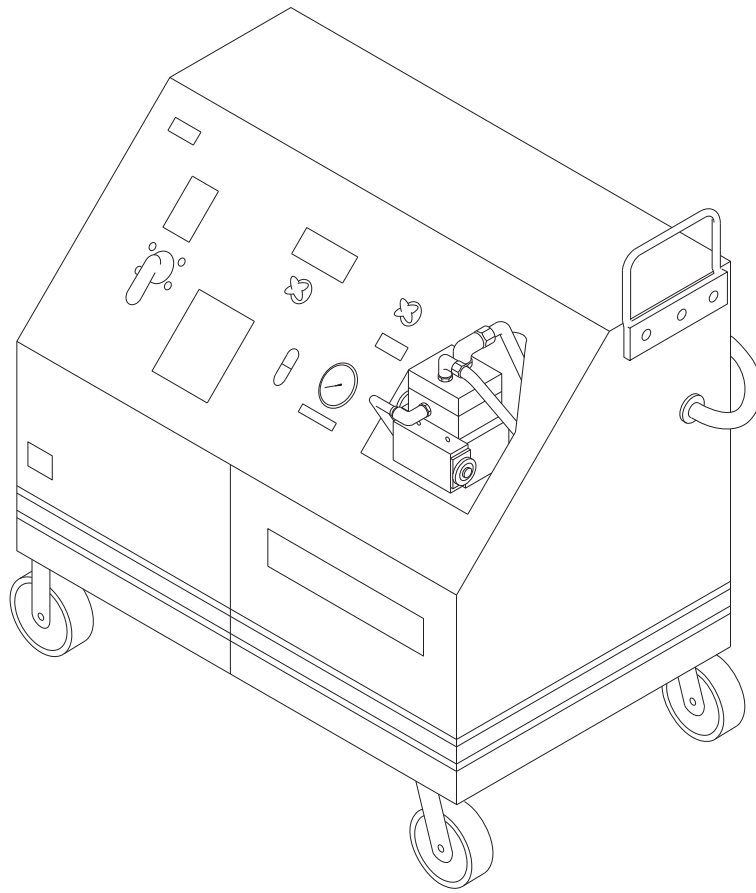
DESCRIPTION: The hydraulic filler / bleeder is a pressurized reservoir used to service hydraulic reservoir and bleed brake system on helicopter.

POWER REQUIREMENTS: External source of compressed air.

CALIBRATION REQUIREMENTS: Not Applicable

FB1018
SA

FIGURE C-31. HYDRAULIC FILLER/BLEEDER, 1102SP



DESCRIPTION: The hydraulic power unit is electrically powered to provide a source of clean, pressurized hydraulic fluid for performing required helicopter maintenance and servicing.

POWER REQUIREMENTS: 440 volts, 60 Hz

CALIBRATION REQUIREMENTS:

FREQUENCY

Hydraulic power unit must be calibrated every 52 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Hydraulic Hand Pump, model 10 or equivalent.
- Portable Hydraulic Circuit Tester, Model PHS-60-6 or equivalent flow and temperature measuring equipment.
- Standard Oil Service Pressure Gages, 0-2000 and 0-5000 psi.
- Hydraulic Pressure and Return Lines.

FB1129_1B
SA

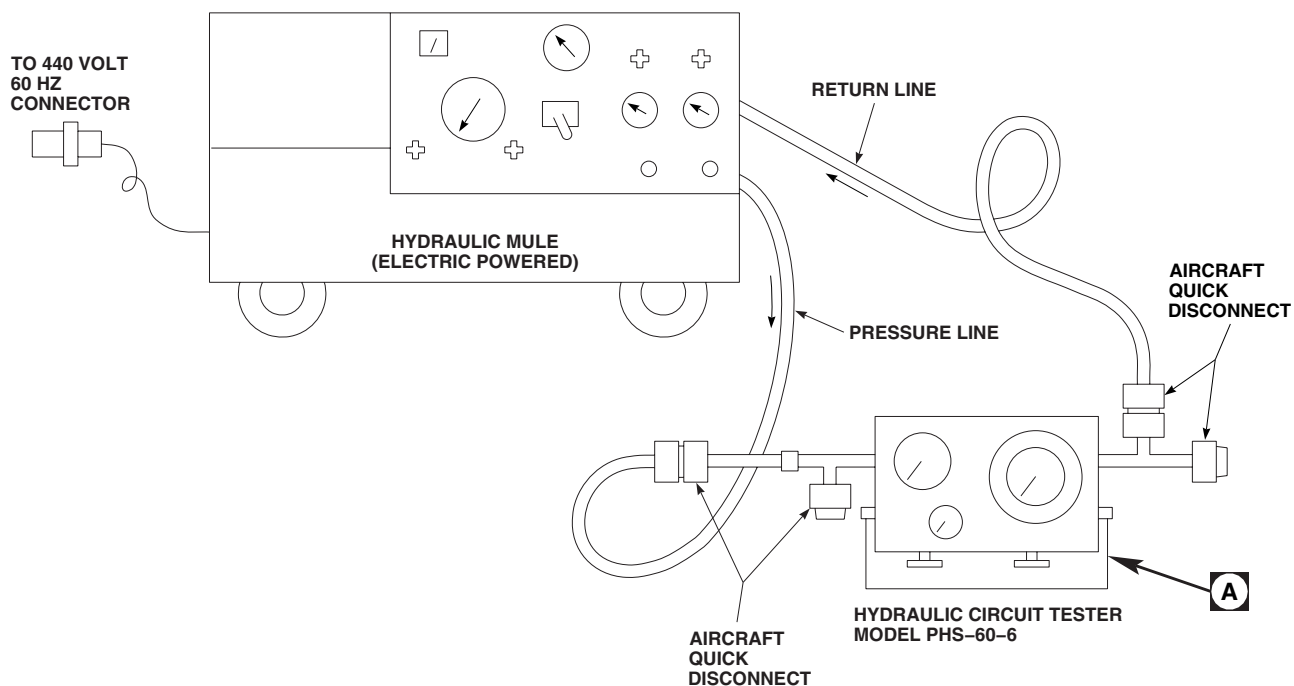
FIGURE C-32. HYDRAULIC POWER UNIT, 05-7040-1200 (SHEET 1 OF 4)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES****Inspection**

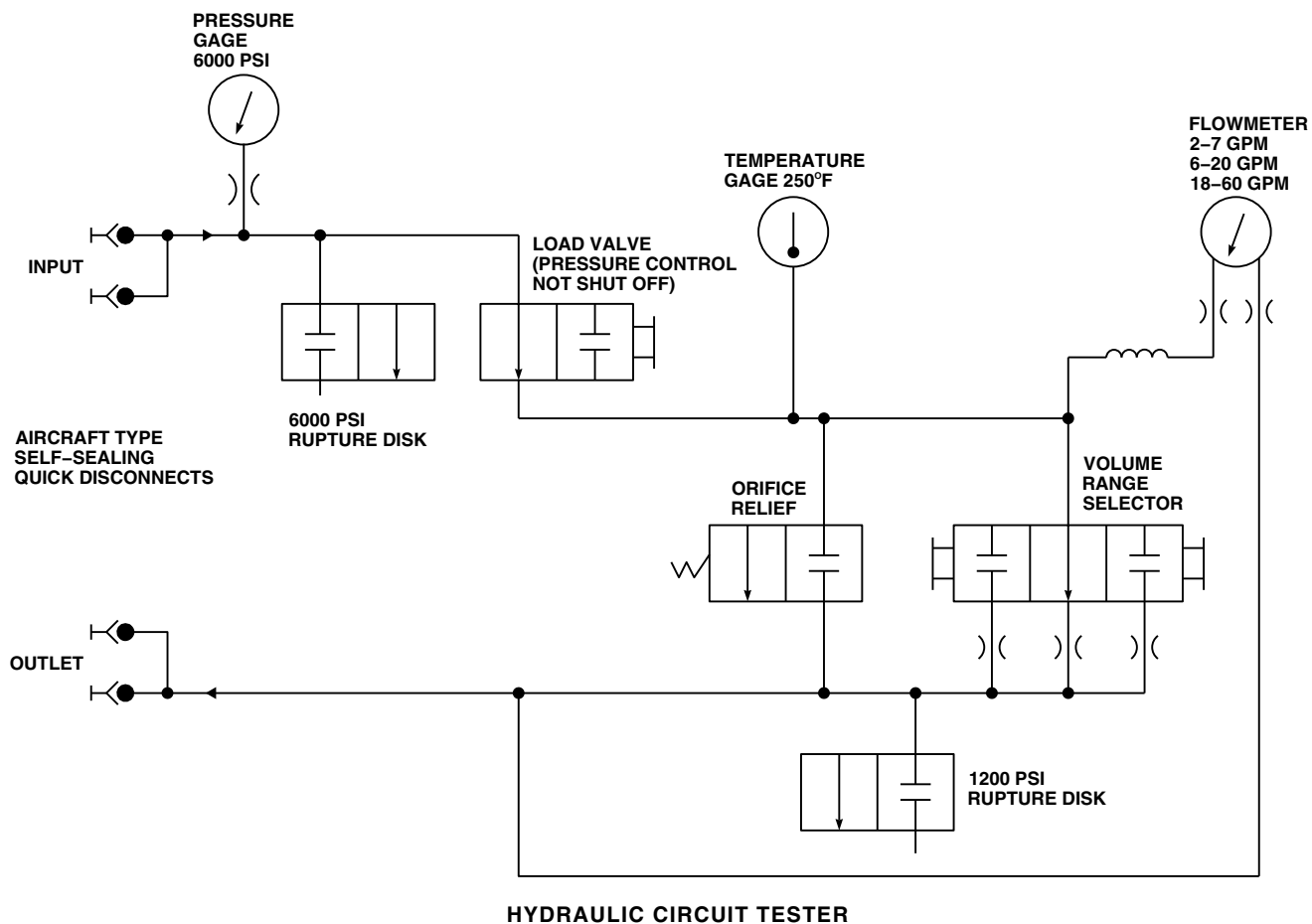
1. Check hydraulic fluid reservoir. Make sure reservoir is at least 1/2 full before starting calibration.
2. Check hoses for damage, cracks, loose or missing connectors, and dirt.
3. Check all gages for broken glass, bent pointers, and dirt.
4. Check drip pan and floor for signs of hydraulic fluid leaks. Repair any leaks.

Calibration

1. Connect pressure and return lines from hydraulic power unit to PHS-60-6, portable hydraulic circuit tester. Return line is larger than pressure line.
2. Place volume range selector on, PHS-60-6 circuit tester in #20 position.
3. Make sure power switch on hydraulic power unit is off. Connect power cord to 440 volt power receptacle.
4. Make sure fittings are capped and output valves are closed. Open by-pass valve on hydraulic power unit. Open load valve of PHS-60-6 circuit tester. Place selector valve on hydraulic unit to STAND RESERVOIR position.
5. Turn on power of hydraulic unit. With pump operating, check for leaks. If leaking, stop pump and repair. Restart pump.
6. Check hydraulic unit and circuit tester gages for high pressure readings. If readings are high, replace filters.
7. Operate controls on hydraulic power unit through full range of flow and full range of pressure. Do not exceed pressure or flow range of circuit tester, PHS-60-6 gages.

FB1129_2A
SA**FIGURE C-32. HYDRAULIC POWER UNIT, 05-7040-1200 (SHEET 2 OF 4)**

A

**CALIBRATION REQUIREMENTS (CONT)****CALIBRATION PROCEDURES (CONT)****Calibration (CONT)**

8. Increase pressure by closing load valve on circuit tester, PHS-60-6. Open by-pass valve and turn off hydraulic power unit. Repair as required.
9. Calibration of pressure gages on hydraulic power unit is checked using external hydraulic hand pump and standard pressure gage. Make sure by-pass valve is open and hydraulic power unit is turned off.
10. Connect hydraulic hand pump and pressure gage to gage port on hydraulic power unit with a high pressure line. Close gage shut off valve and power unit to isolate gage.
11. Increase pressure using hand pump on gage being tested. Make sure pressure on standard gage is not exceeded. Release pressure on hand pump system. Using hand pump increase pressure once more to full scale on gage being tested.
12. Observe pointer of gage being tested, any deviation from smooth operation may indicate worn gears or sector. Remove or repair gage as required.

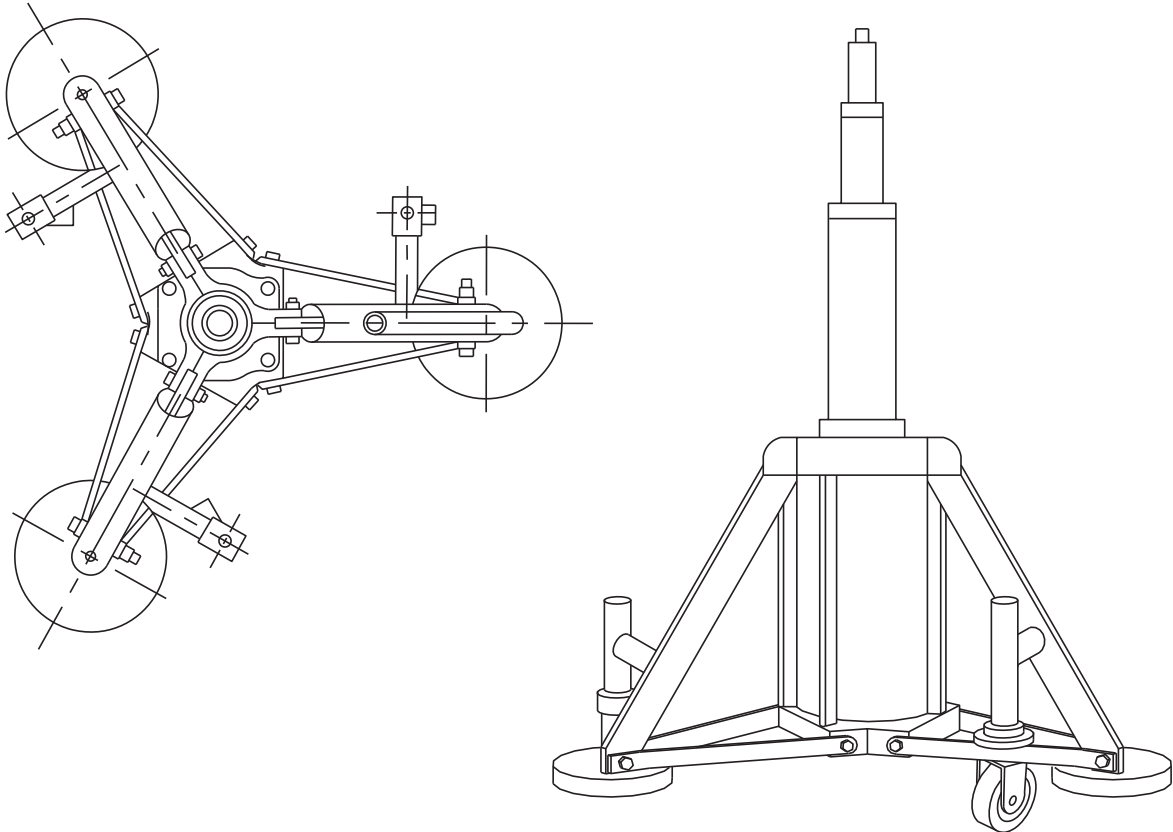
FB1129_3A
SA**FIGURE C-32. HYDRAULIC POWER UNIT, 05-7040-1200 (SHEET 3 OF 4)**

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES (CONT)****Calibration (CONT)**

13. Hold pressure at full scale on gage being tested for one minute. Any detectable movement of pointer down scale is a sign of system leak. Check system and repair leaks as required. Release pressure.
14. Increase pressure in system until standard pressure gage reads exactly 50% of full scale on gage being tested. Lightly tap test gage and observe gage reading. Adjust test gage to read exact pressure as on standard gage by adjusting screw or resetting pointer on gage. After adjustment, lightly tap gage and check reading. Reading must be same as standard pressure gage.
15. Release system pressure down to minimum value.
16. Increase pressure in system until standard pressure gage reads 20%, 40%, 60%, and 80% of test gage. Check test gage at each point. Tap test gage and check readings at each point.
17. Adjust test gage, if necessary, to bring the center 2/3 of its span within one scale division on gage.
18. Repeat steps 9 through 17 for any other pressure gages on hydraulic power unit, not used for filter condition monitoring.
19. Remove hydraulic hand pump, pressure hose, and standard pressure gage.
20. Calibration of flow gage on hydraulic power unit is checked using standard flowmeter in circuit tester, PHS-60-6, at an oil temperature of $100^{\circ}\text{F} \pm 20^{\circ}\text{F}$.
21. Make sure by-pass valve is open. Turn on hydraulic power unit. Open load valve on circuit tester to allow flow through system loop. Make sure volume range selector is on #20 position on circuit tester.
22. Adjust flow on hydraulic power unit to full flow. If under 20 GPM, also check flow indication on circuit tester to insure flow is not running over 20 GPM. If full flow can't be reached, open bypass valve and turn off power unit. Repair as required.
23. Check oil flow through system loop, while observing oil temperature on gage of circuit tester, PHS-60-6. Temperature shall read $100^{\circ}\text{F} \pm 20^{\circ}\text{F}$. Adjust temperature by adding pressure to raise temperature and release pressure to lower temperature.
24. Lower pressure and flow to minimum and place circuit tester range selector to #7.
25. When oil temperature is reasonably stable, adjust flow and pressure to achieve 20% of full flow on power unit gage. Check flow reading on power unit flow meter and compare with circuit tester reading.
26. Make sure oil temperature is $100^{\circ}\text{F} \pm 20^{\circ}\text{F}$. If necessary, adjust to maintain oil temperature. Bring oil flow and pressure to a minimum, place range selector on circuit tester to #20 position.
27. Check flow readings at 40%, 60%, and 80% on power unit gage with readings on circuit tester, PHS-60-6. Readings shall be within $\pm 8\%$ indicated, ± 1 division of circuit tester gage.
28. Adjust flowmeter, if necessary, and recheck readings. Final adjustment should leave a $\pm 8\%$ difference, ± 1 division on flow rate throughout the 25% to 75% scale range.
29. Install limited use calibration sticker to hydraulic power unit.

FB1129_4A
SA**FIGURE C-32. HYDRAULIC POWER UNIT, 05-7040-1200 (SHEET 4 OF 4)**

VERT. CAP.	MIN. CLOSED HEIGHT	HYD. LIFT	SCREW EXT.	MAX. EXTENDED HEIGHT	NET WEIGHT	CRATED WEIGHT
TONS	INCH	INCH	INCH	INCH	LBS	LBS
12	14	23	3	40	100	125



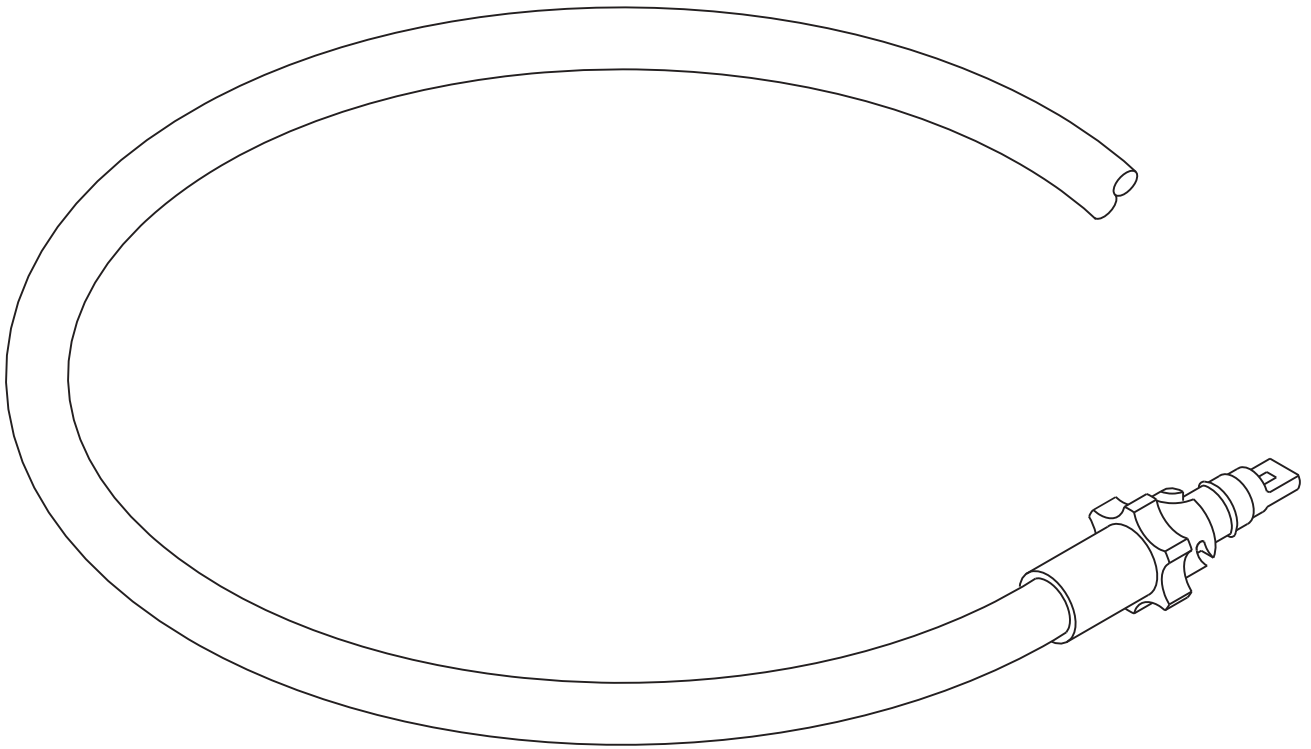
DESCRIPTION: The hydraulic jack has a load capacity of 12 tons, and a hydraulic lift height of 23 inches to a maximum of 40 inches.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1130
SA

FIGURE C-33. HYDRAULIC TRIPOD JACK, 12-3-14



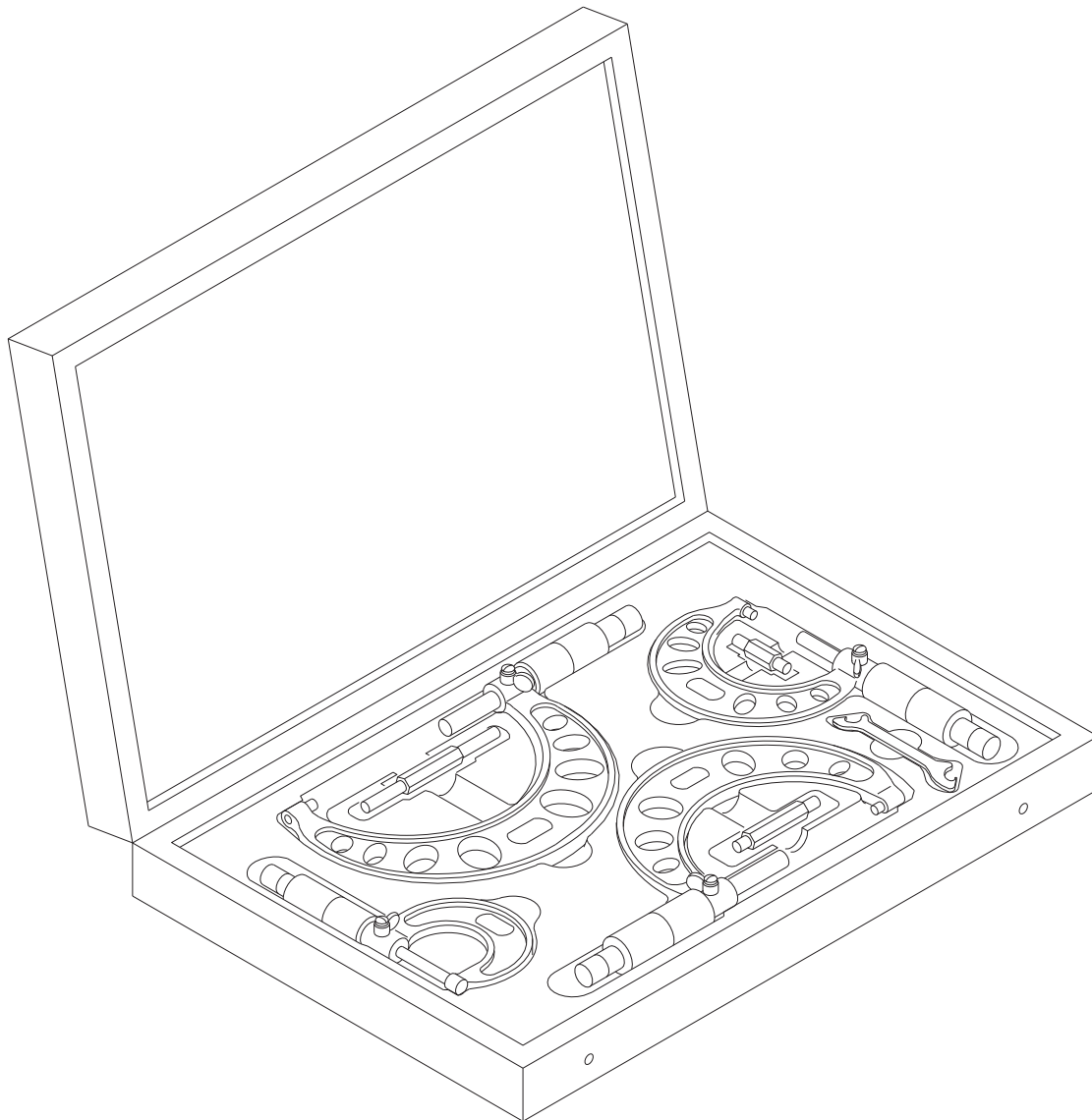
DESCRIPTION: The drain line is used to drain oil from tail gear box and intermediate gear box into container to prevent splashing or spilling.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1019
SA

FIGURE C-34. INTERMEDIATE/TAIL GEAR BOX DRAIN LINE, H538A-3



DESCRIPTION: The micrometers are used for obtaining precise measurements of an object.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Micrometers must be calibrated every 12 weeks regardless of being used or not.

NOTE

If a micrometer is dropped or knocked against a hard object, it must be recalibrated before it is used again.

EQUIPMENT REQUIRED

- Gage Block Set, Hoke-style, 0.05–4.0–inch $\pm$ 0.000010–inch (or equivalent).

FB1021_1C
SA

FIGURE C-35. MICROMETER SET, 0 - 4-INCH, PMF114 (SHEET 1 OF 2)

- Gage Block Set, Hoke-style, 5.0–20.0–in., ± 0.000010 –in. (or equivalent).
- Measuring Rods, 1.0–11.0–in., ± 0.000010 –in. Browne & Sharp #655 (or equivalent).
- Measuring Rods, 12.0–16.0–in., ± 0.000010 –in. Starrett #234 (or equivalent).

CLEANING MATERIALS

- Isopropyl Alcohol, Item 72, Appendix D
- Machinery Towel, Item 211, Appendix D
- Soap and Water

CALIBRATION PROCEDURES**Cleaning / Inspection**

1. Clean spindle and anvil using Isopropyl Alcohol, Item 72, Appendix D. Wipe dry using Machinery Towel, Item 211, Appendix D.
2. Check that ratchet screw and thimble move freely.
3. Check spindle and anvil surfaces for damage. **NONE ALLOWED.**

Calibration

1. Loosen ratchet screw and rotate thimble until zero is indicated on scale.

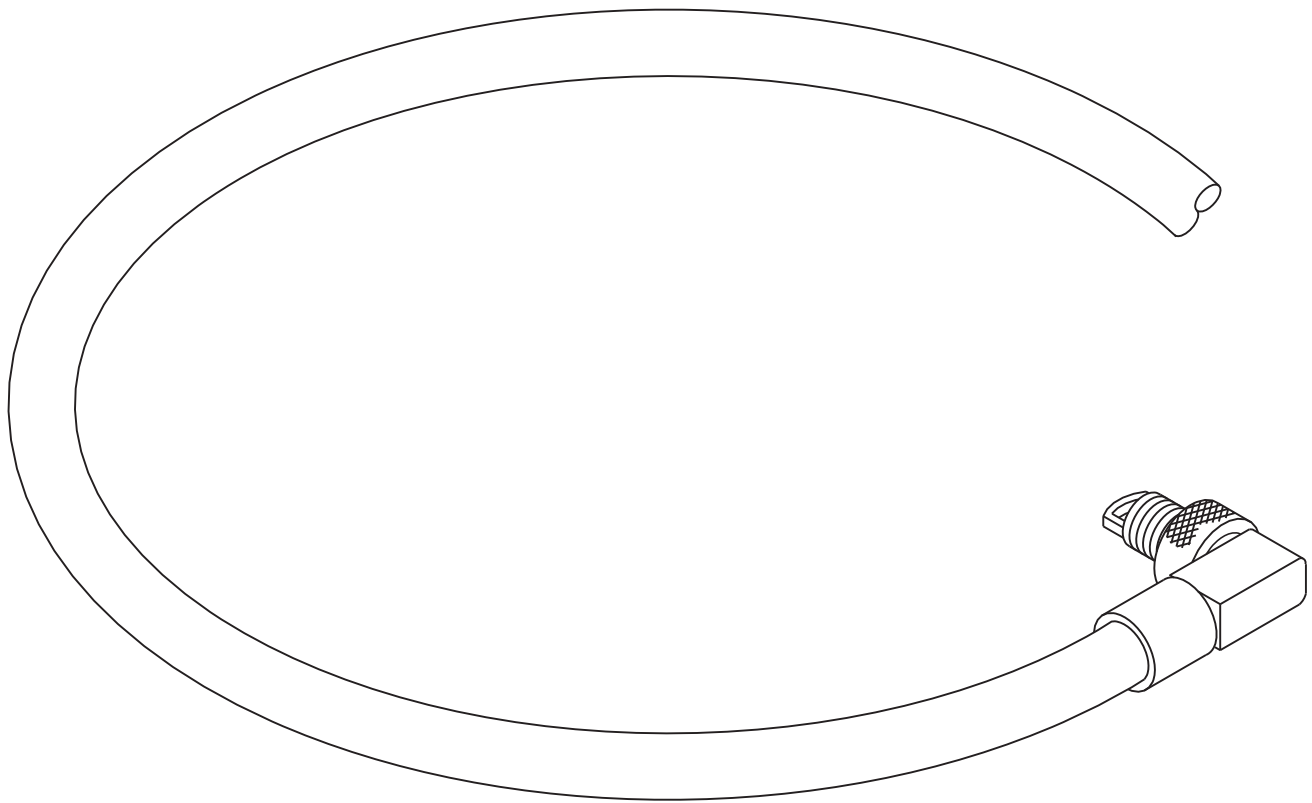
NOTE

Since the actual micrometer adjustment of all test instruments is limited to one in., an additional gage block must be used when calibrating test instruments larger than one in. For example, an additional three-in. block must be used when calibrating a four-in. test instrument.

2. Calibrate the test instruments using the 0.0625, 0.125, 0.25, 0.5, and 1.0–in. gaging blocks. Measurements must be within tolerance given in Table 1. If tolerances cannot be achieved, adjust instrument and recalibrate.

TABLE 1	
INSTRUMENT RANGE	ACCEPTABLE TOLERANCE
0–1–in.	0.0002–in.
1–2–in.	0.0002–in.
2–3–in.	0.0003–in.
3–4–in.	0.0003–in.
4–5–in.	0.0003–in.
5–6–in.	0.0003–in.
6–7–in.	0.0004–in.
7–8–in.	0.0004–in.
8–9–in.	0.0004–in.
9–10–in.	0.0005–in.
10–11–in.	0.0005–in.
11–12–in.	0.0005–in.
12–18–in.	0.0008–in.
18–20–in.	0.001–in.

FK2213_2C
SA**FIGURE C-35. MICROMETER SET, 0 - 4-INCH, PMF114 (SHEET 2 OF 2)**



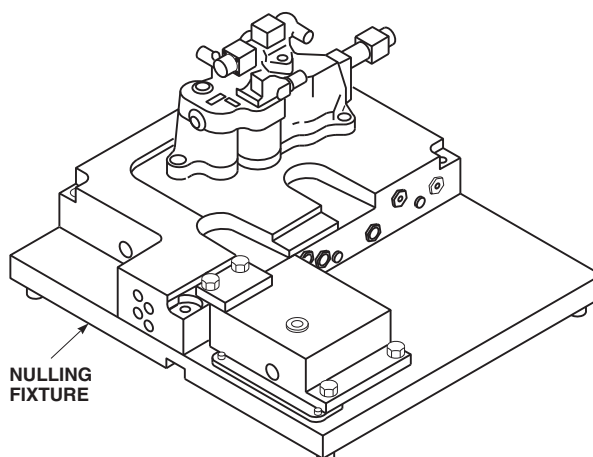
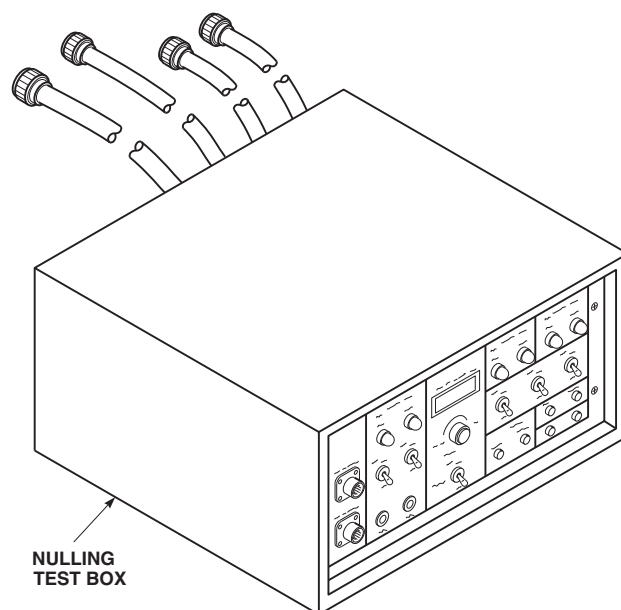
DESCRIPTION: The drain line is used to drain oil from main transmission into containers preventing splashing or spilling oil.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1022
SA

FIGURE C-36. MAIN GEAR BOX DRAIN LINE, D-730-L



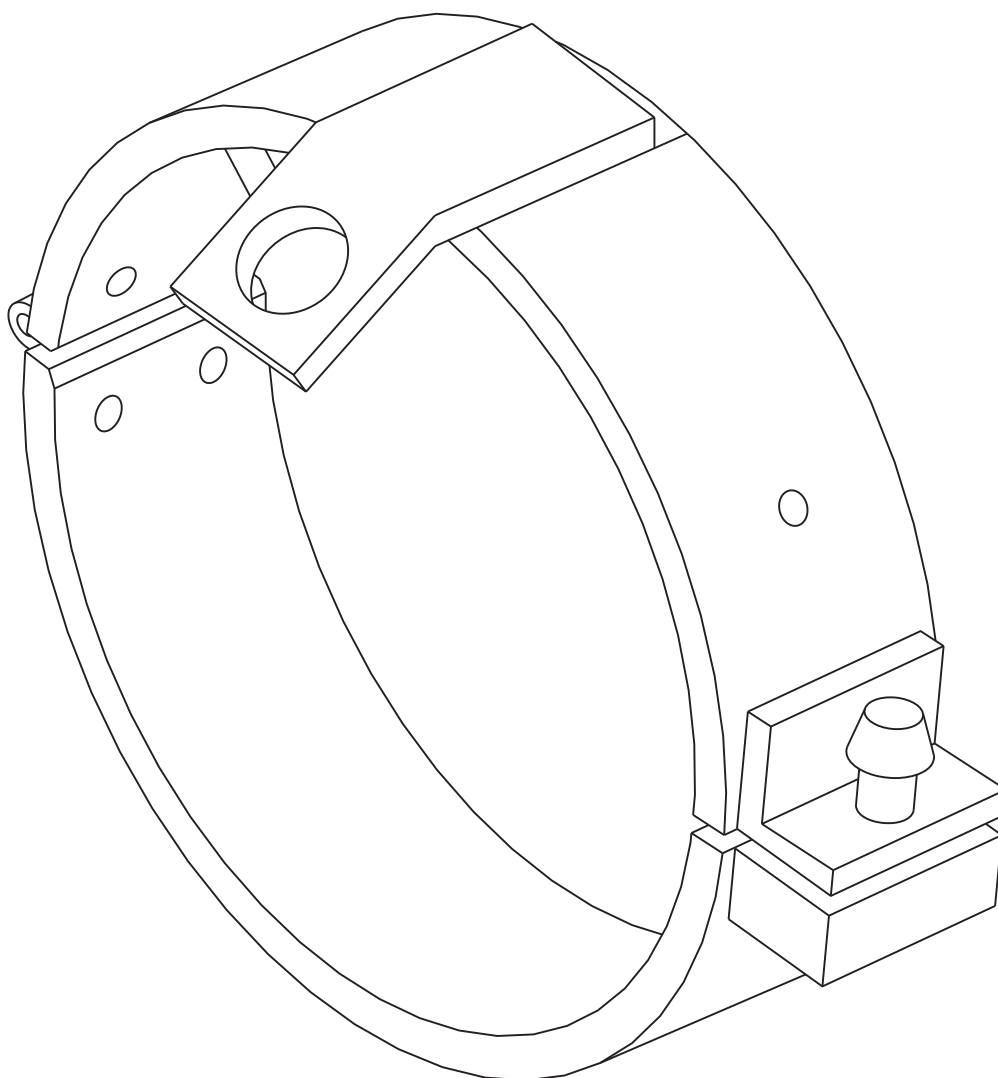
DESCRIPTION: The nulling fixture is used to operationally test pilot assist module, roll trim, pitch trim and yaw boost servos after repair.

POWER REQUIREMENTS: The nulling fixture operates on 115 volts AC, single phase, 60 Hz, 2 amp and 28 volt DC, 5 amps. Also needed is hydraulic power at a rate of 6 GPM, at 3000 psi.

CALIBRATION REQUIREMENTS: Not Applicable

FB1131
SA

FIGURE C-37. NULLING FIXTURE, 70700-20675-041



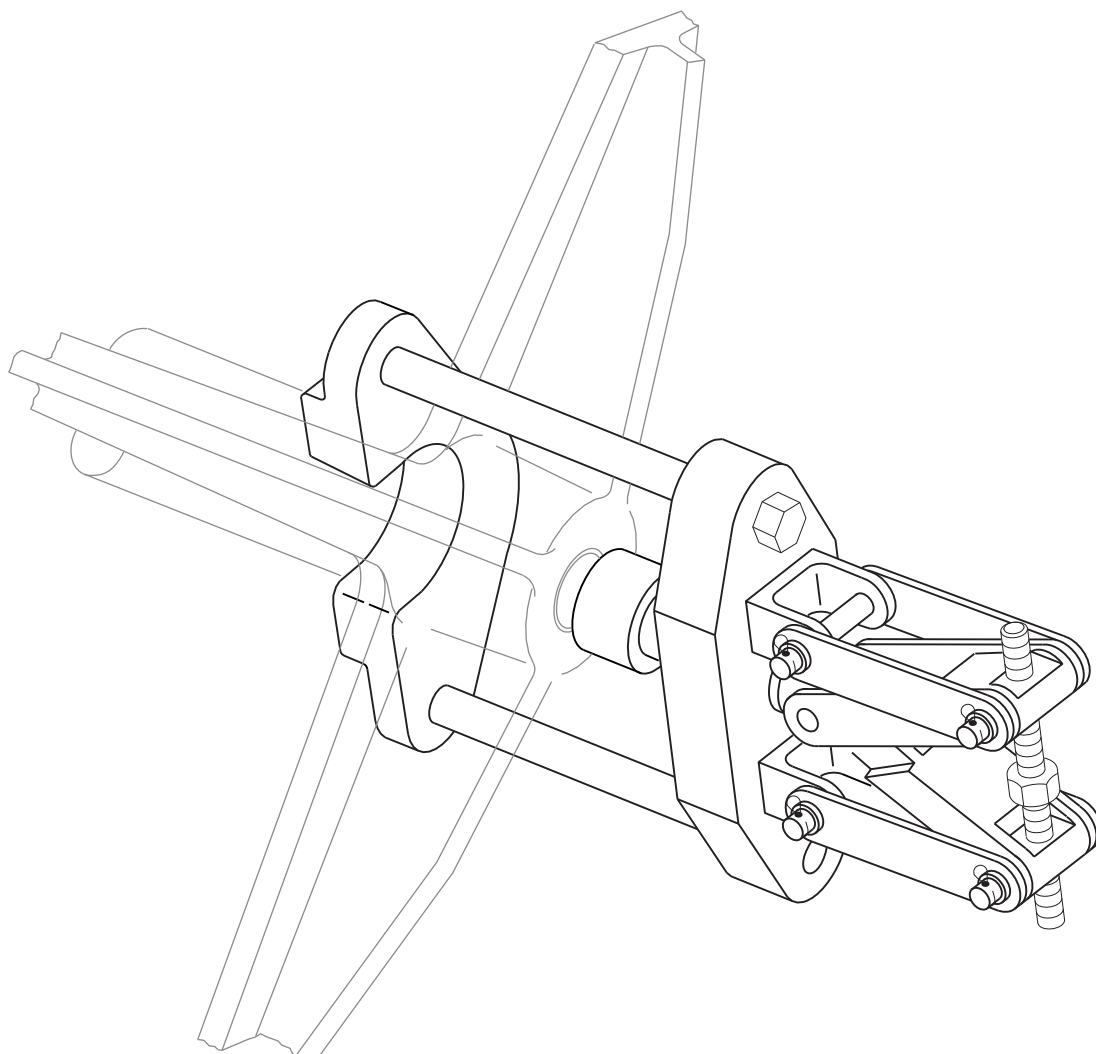
DESCRIPTION: The photocell bracket is used in balancing engine output shafts. The bracket is attached to forward support tube assembly, to provide a location to install a photocell sensor to measure shaft rotational speed.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2070
SA

FIGURE C-38. PHOTOCCELL BRACKET, 70700-20114-044



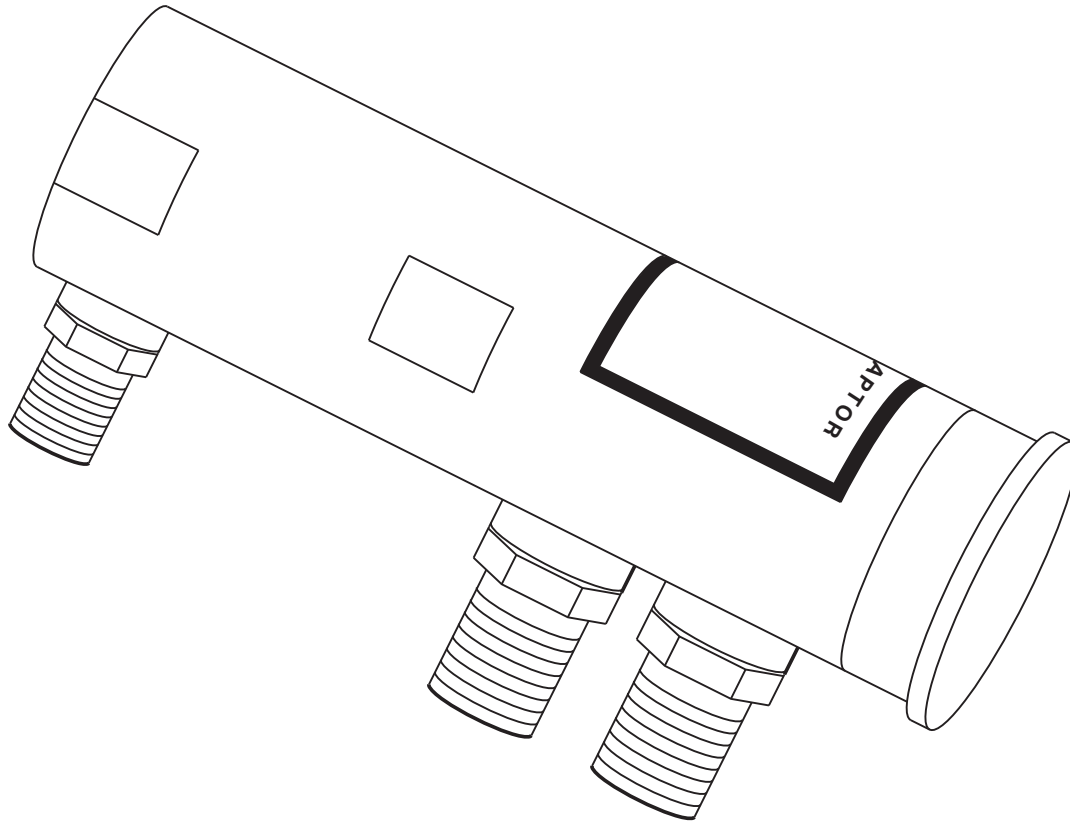
DESCRIPTION: The pitch beam puller is a mechanical puller, used to remove pitch beam from pitch change shaft on tail gear box.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1132
SA

FIGURE C-39. PITCH BEAM PULLER, 70700-77389-041



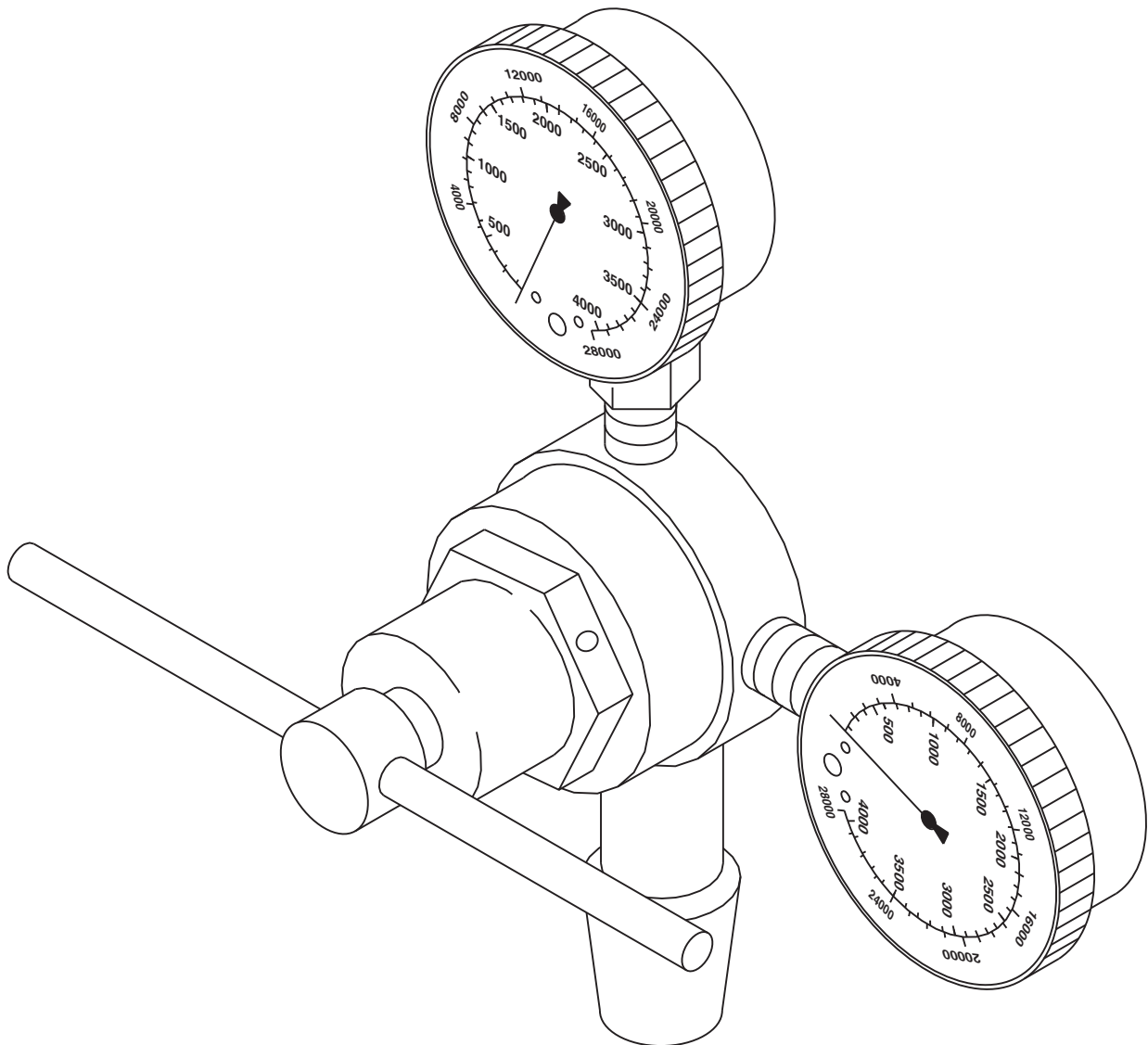
DESCRIPTION: The pitot pressure adapter consists of a pressure sleeve that conforms to the contour of pitot / static heads, with replaceable ring seals to provide air tightness when pressure / temperature test set is installed.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1024
SA

FIGURE C-40. PITOT PRESSURE ADAPTER, 884DB



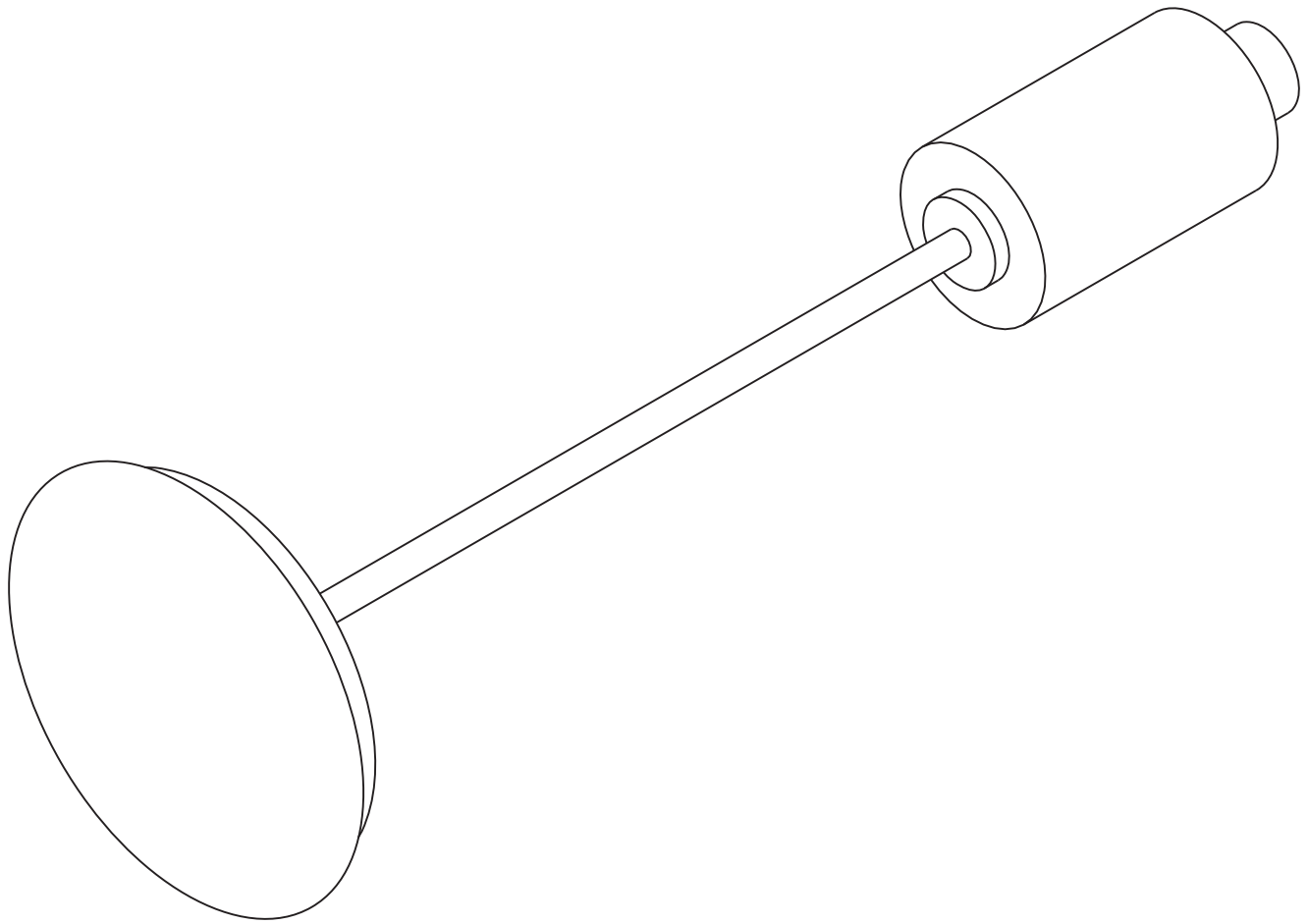
DESCRIPTION: The pressure regulator is used to read servicing pressures when servicing helicopter shock struts, tires, and hydraulic reservoirs.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Annually.

FB1025A
SA

FIGURE C-41. PRESSURE REGULATOR, 14-6804-6000



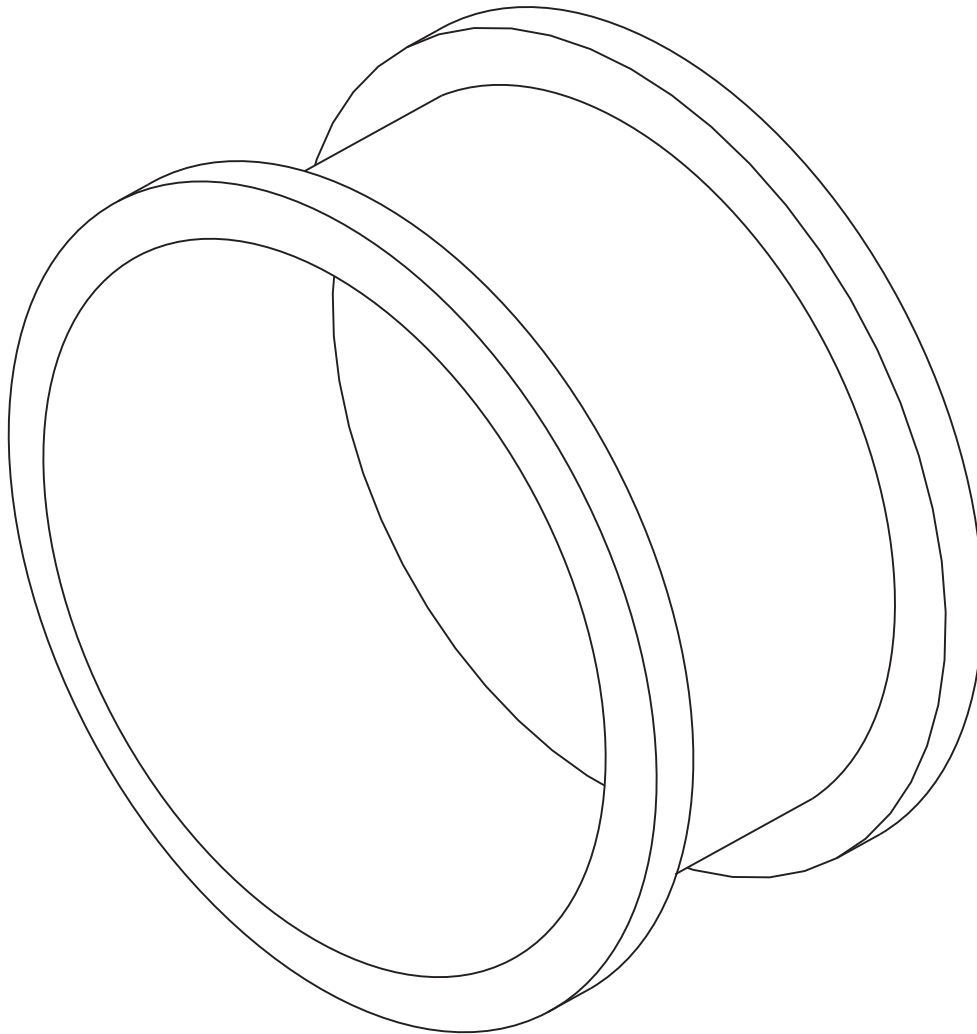
DESCRIPTION: The APU combustor puller is designed to be used with combustor puller adapter, ST9314 to prevent damage to combustor section during separation of APU combustor.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1026
SA

FIGURE C-42. APU COMBUSTOR PULLER, ST 91125



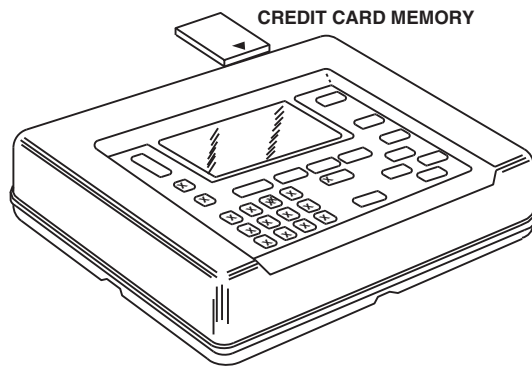
DESCRIPTION: The APU combustor adapter is designed to be used with APU combustor puller, ST 91125 to prevent damage to combustor section during separation of APU combustor.

POWER REQUIREMENTS: Not Applicable

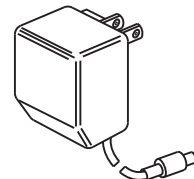
CALIBRATION REQUIREMENTS: Not Applicable

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SA

FIGURE C-43. APU COMBUSTOR ADAPTER, ST 93014

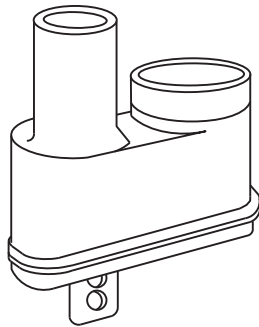


CREDIT CARD MEMORY

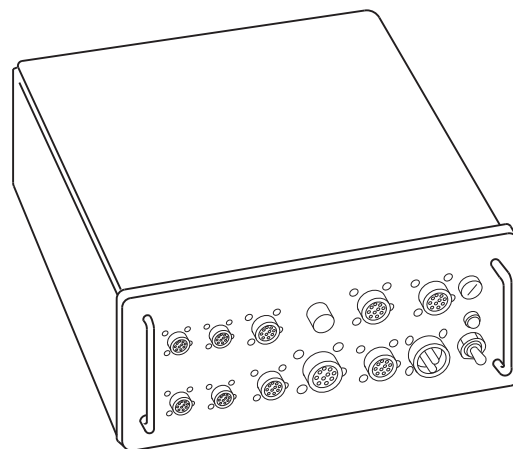


BATTERY CHARGER

CONTROL AND DISPLAY UNIT (CADU)



UNIVERSAL TRACKING DEVICE (UTD)



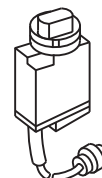
DATA ACQUISITION UNIT (DAU)



ACCELEROMETER



MAGNETIC RPM SENSOR



OPTICAL RPM SENSOR

DESCRIPTION: The RADS-AT is a portable test set, weighing less than 25 pounds, which contains sophisticated measurement technology that operates with a minimum of in-flight control, using fast parallel data acquisition techniques to reduce flight test time. The RADS-AT utilizes a diagnostic programming language in an OS-9 environment to provide a high level, easy to use method of performing rotor vibration analysis. The automatic

FB1133_1
SA

FIGURE C-44. RADS-AT VIB ANALYZER, 29313100 (SHEET 1 OF 2)

DESCRIPTION: (CONT)

blade tracking is a universal tracking device which can be used in daylight or at night, it contains a data acquisition unit (DAU) which processes the tracker and vibration signals, a handheld control and display unit (CADU), to control data acquisition, display measurements and analysis results, print reports, and transfer data to or receive data from an off-line computer.

POWER REQUIREMENTS: 17-36 VDC (DAU), 12 VDC (CADU)

CALIBRATION REQUIREMENTS:**FREQUENCY**

The RADS-AT vib analyzer must be calibrated every 24 weeks regardless of being used or not.

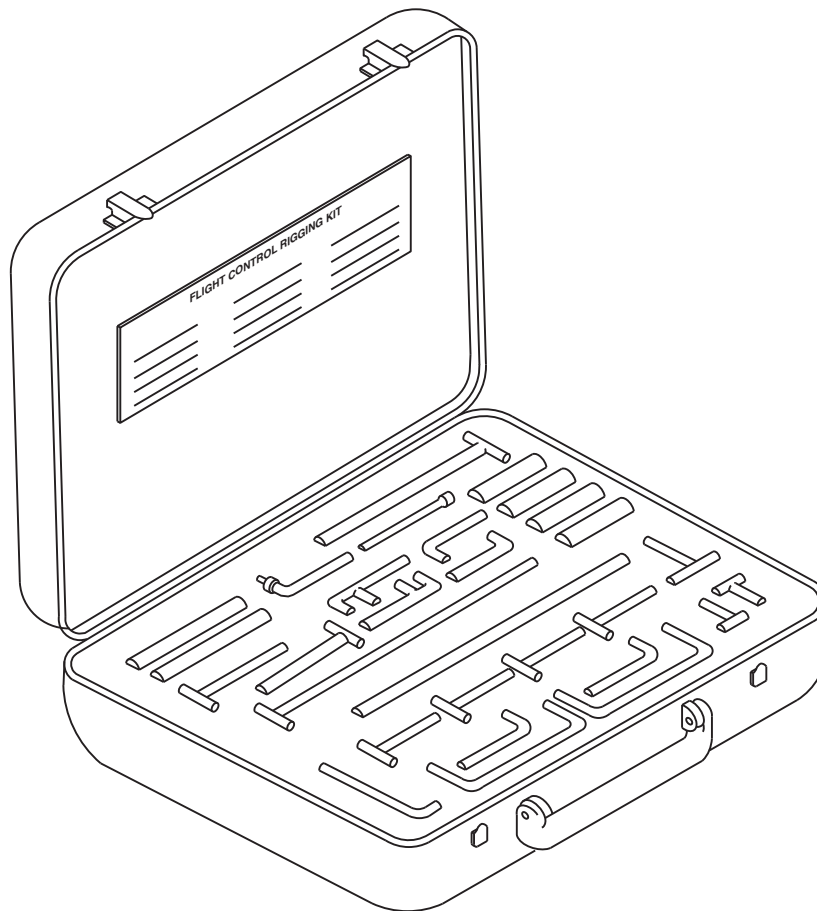
CALIBRATION PROCEDURES

Refer to manufacturer's maintenance manual for calibration procedures.

FB1133_2
SA

FIGURE C-44. RADS-AT VIB ANALYZER, 29313100 (SHEET 2 OF 2)

Change 10**C-75**



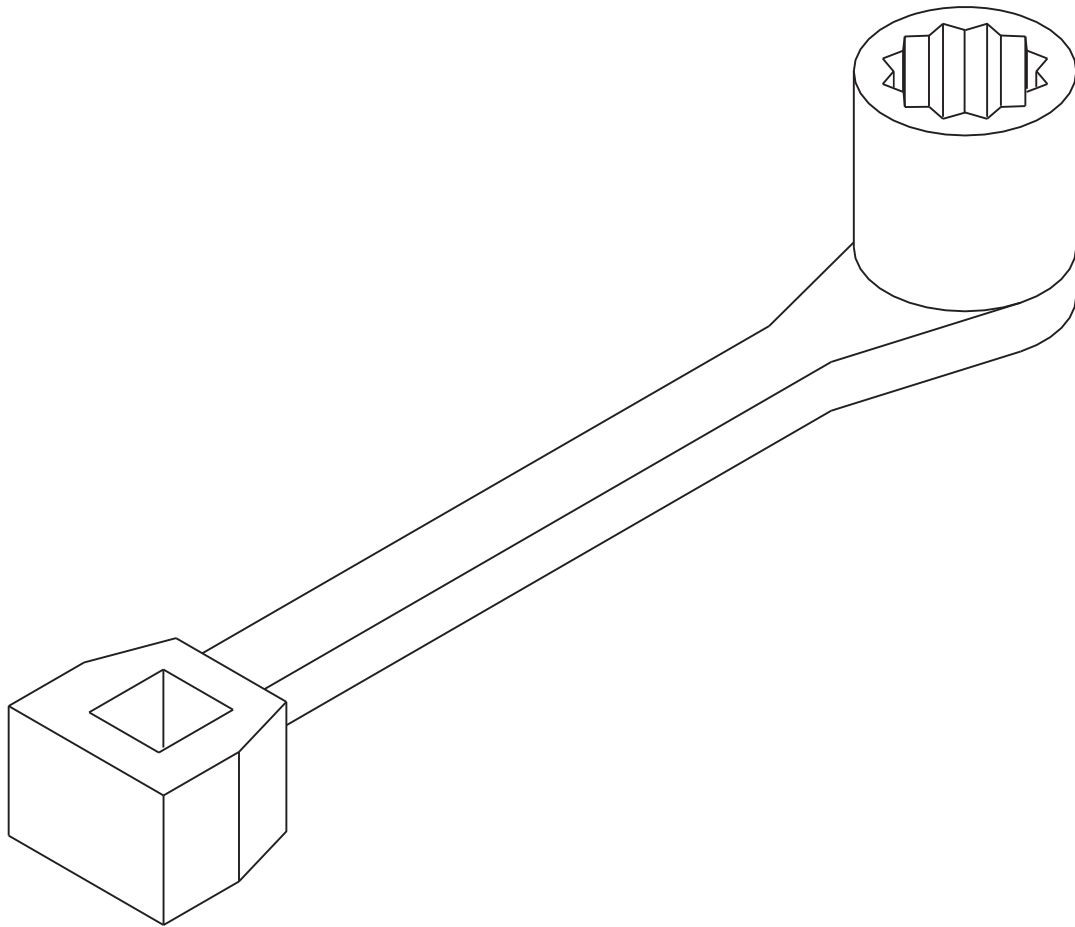
DESCRIPTION: The rigging set is used to hold the flight control system at a predetermined position to ensure proper adjustment of controls. The rigging set contains rig pins with red flags attached to ensure removal of all pins and blocks from flight control system.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1027
SA

FIGURE C-45. FLIGHT CONTROL RIGGING SET, 70700-20389-042



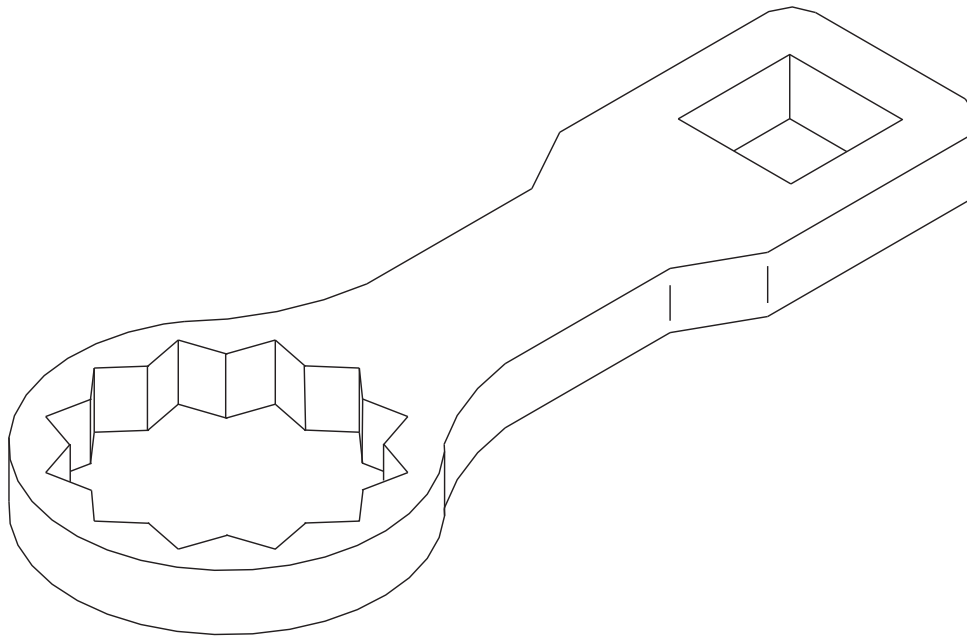
DESCRIPTION: The rotor brake wrench is used to check required torque on mounting nuts, rotor brake to main module, without disassembling a major portion of the rotor brake assembly.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FIGURE C-46. ROTOR BRAKE WRENCH, 70700-77117-041

FB1028
SA



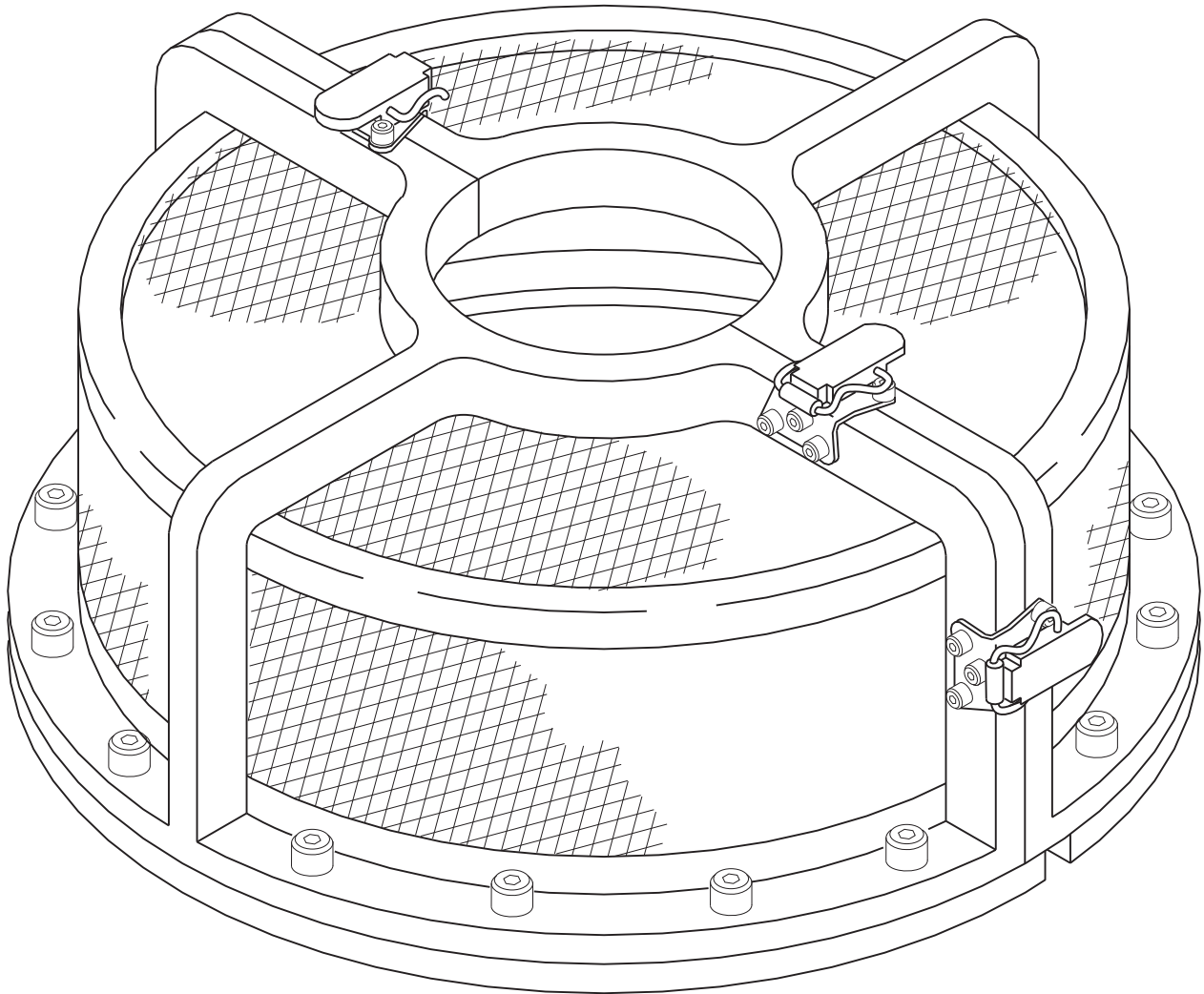
DESCRIPTION: The main rotor head wrench is a special wrench for torquing the rotor head bolts to 1000 inch-pounds of torque, which exceeds the recommended torque limits of standard crowfoot wrench use.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1029A
SA

FIGURE C-47. MAIN ROTOR HEAD WRENCH, 70700-20210-101



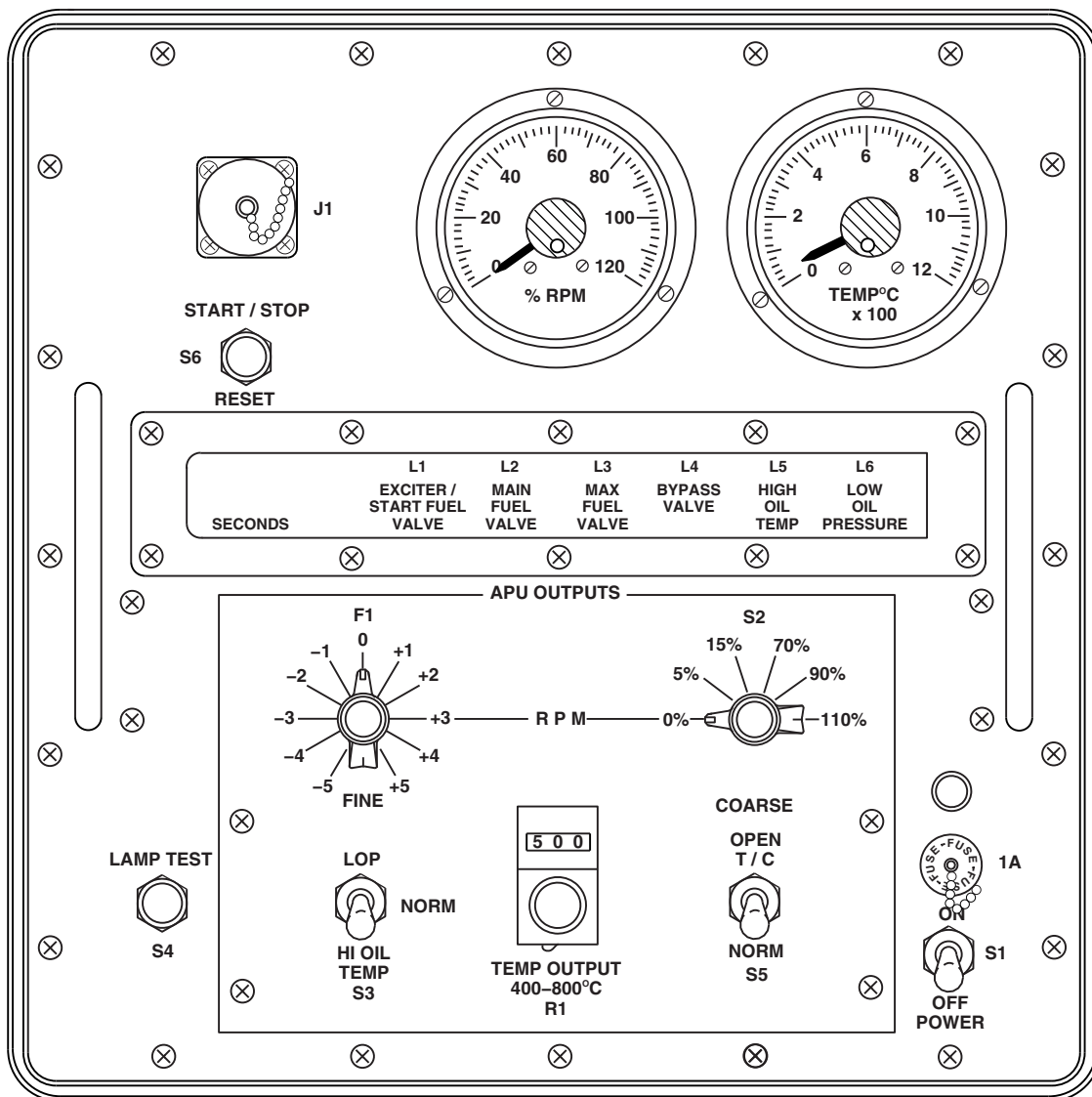
DESCRIPTION: The engine inlet screen is used when the engine inlet is removed to expose output shaft when performing shaft balance. The screen assembly is clamped to front of engine to prevent ingestion of foreign objects into engine during run up and balancing of engine output shaft.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1030
SA

FIGURE C-48. ENGINE INLET SCREEN, 70700-20222-044



DESCRIPTION: The APU simulator test set temp / speed, simulates the auxiliary power unit (APU) when statically testing the electronic sequencing unit (ESU) of the helicopter. The test set provides selectable frequency, temperature output signals, contacts to simulate low oil pressure, high oil temperature and open thermo-couple conditions. Six lamps (L1-L6) monitor the closing of six switches in the ESU. The test set also monitors APU temperature and RPM during dynamic testing of the APU.

POWER REQUIREMENTS: 28 volts DC, 1 amp

CALIBRATION REQUIREMENTS:

FREQUENCY

APU simulator test set must be calibrated every 24 weeks regardless of being used or not.

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SA

FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 1 OF 6)

CALIBRATION REQUIREMENTS: (CONT)**EQUIPMENT REQUIRED**

- Frequency Generator, Wavetek 130, 0 to 6K Hz (para 5–8), $\pm 0.05\%$ of setting for 10 minutes
- Frequency Counter, General Radio Co. 1192, dc to 32 megaHz, ± 1 count
- Oscilloscope, Hewlett–Packard Co. 1 30C, sweep: 0.1 ms / cm $\pm 3\%$, vertical: 1v / cm $\pm 3\%$
- Digital Multimeter, Data Technology Corp. 370, 100 mvdc, $\pm 0.003\%$ rdng $\pm 0.004\%$ fsc, 100.000 ohms, $\pm 0.006\%$ rdng $\pm 0.004\%$ fsc
- Ice–Bath Reference Junction, Howell Instruments BH14106, $\pm 0.25^\circ\text{C}$
- Voltage Reference, General Resistance, 0 to 50 mvdc, $\pm 0.0025\%$ of setting +5uv

CALIBRATION PROCEDURE**Inspect**

1. Inspect test set for physical damage, dirt, loose, and broken knobs. Clean and repair as required.

CALIBRATION**Initial Switch Positions**

1. Turn power switch to off, FINE RPM switch to 0, COARSE RPM switch to 0%, OPEN T / C–NORM switch to NORM, and LOP–HI OIL TEMP switch to NORM.

Functional Checks

1. Continuity Tests. Connect a VOM between pins N and K of receptacle J1. VOM shall show continuity when switch S3 is in NOR or HI OIL TEMP positions and shall show an open circuit when switch is in LOP position.
2. Connect VOM between pins N and M of J1. VOM shall show continuity when S3 is in NORM or LOP positions and shall show open circuit when S3 is in HI OIL TEMP position.
3. Use VOM and test cable schematics in Section VI to check cable continuity.
4. Using a 28 VDC power supply rated 1.5 amps or more, apply 28 VDC to terminals shown in Table 1. Corresponding lamps shall light.

Table 1. Lamp Test Table

RECEPTACLE J1 PIN		LAMP	S3 POSITION
+ 28 VDC	28 VDC RTN		
S	T	L1	NORM
P	T	L2	NORM
R	T	L3	NORM
V	W	L4	NORM
b&c&N	T	L5	HI OIL TEMP
b&c&N	T	L6	LOP

5. Power Up Tests. Connect 28 VDC power supply to J1, N(+) and T(–). Also jumper +28 VDC to pins b and c. Connect a jumper (see note) between J1–B and J1–a.

NOTE:

Jumper should be 1–inch or less in length.

6. Turn POWER switch on. Power lamp shall light.
7. Press LAMP TEST switch. Lamps L1–L6 shall light.
8. Momentarily remove 28 VDC from J1, pin b. L5 shall light.
9. Momentarily remove 28 VDC from J1, pin c. L6 shall light.

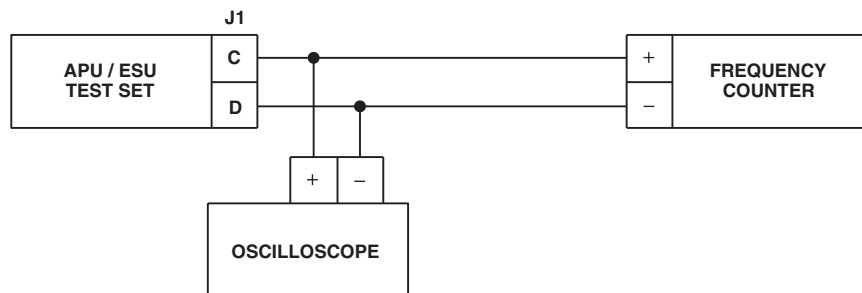
FB1134_2B
SA**FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 2 OF 6)**

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION (CONT)

Accuracy Checks

1. **Timer Accuracy.** Observe the second hand of a watch and press Start / Stop-Reset switch to start timer. Press switch again to stop timer after 95 seconds have elapsed on watch. Timer shall read 95 ± 3 seconds. Press switch again to reset timer. All timer display segments shall be operative.
2. Remove jumper from J1-B to J1-a and connect it from J1-B to J1-X. See note, step 4, power up test.
3. Repeat step 1.
4. **Frequency Output Accuracy.** Make frequency output test setup as in view 1.



View 1. Frequency Output Test Setup

5. Check output frequencies of Table 2. (4666 Hz = 100% RPM) against the switch positions. Output voltage at 70% rpm shall be at least 8 volts peak-to-peak.
6. Remove jumper from J1-B to J1-X and connect it from J1-B to J1-a. See note, step 4, power up test.
7. Check output frequencies of Table 2 (4800 Hz = 100% RPM) against the switch positions. Output voltage at 70% rpm shall be at least 8 volts peak-to-peak.
8. **Frequency Input Accuracy.** Jumper J1-Z to J1-Y. See note, step 4, power up test.
9. Disconnect frequency output test setup from test set and make frequency input test setup as in View 2.
10. Check %RPM meter accuracy (4800 Hz = 100% RPM) per Table 3. Lightly tap meter before taking a reading to minimize pointer friction.

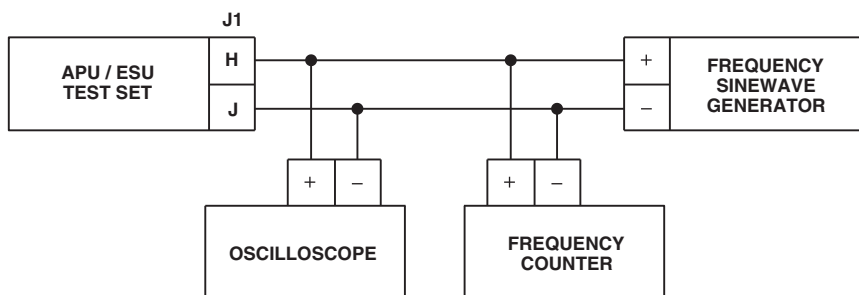
Table 2. Frequency Output Test Table

COARSE SWITCH S2	FINE SWITCH F1	OUTPUT FREQUENCY 4666 Hz = 100% RPM (Hz $\pm$ TOL)	OUTPUT FREQUENCY 4800 Hz = 100% RPM (Hz $\pm$ TOL)
0%	any position	0 $\pm$ 0.00	0 $\pm$ 0.00
5%	0	233.3 $\pm$ 0.47	240 $\pm$ 0.48
5%	+5	466.6 $\pm$ 0.93	480 $\pm$ 0.96
5%	-4	46.66 $\pm$ 0.09	48 $\pm$ 0.096
5%	-5	46.66 $\pm$ 0.09	48 $\pm$ 0.096
15%	0	699.9 $\pm$ 1.40	720 $\pm$ 1.44
70%	0	3266.2 $\pm$ 6.53	3360 $\pm$ 6.72
90%	0	4199.4 $\pm$ 8.40	4320 $\pm$ 8.64
110%	0	5132.6 $\pm$ 10.27	5280 $\pm$ 10.56
110%	-5	4899.3 $\pm$ 9.80	5040 $\pm$ 10.08
110%	-4	4945.96 $\pm$ 9.89	5088 $\pm$ 10.18
110%	-3	4992.62 $\pm$ 9.99	5136 $\pm$ 10.27
110%	-2	5039.28 $\pm$ 10.08	5184 $\pm$ 10.37
110%	-1	5085.94 $\pm$ 10.17	5232 $\pm$ 10.46
110%	+1	5179.26 $\pm$ 10.36	5328 $\pm$ 10.66
110%	+2	5225.92 $\pm$ 10.45	5376 $\pm$ 10.75
110%	+3	5272.58 $\pm$ 10.55	5424 $\pm$ 10.85
110%	+4	5319.24 $\pm$ 10.64	5472 $\pm$ 10.94
110%	+5	5365.9 $\pm$ 10.73	5520 $\pm$ 11.04

FB1134_3A
SA

FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 3 OF 6)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION (CONT)
Accuracy Checks (cont)

View 2. Frequency Input Test Setup

11. Remove jumper from J1–Z to J1–Y.
12. Check % RPM meter accuracy (4666 Hz = 100% RPM) per Table 3. Lightly tap meter before taking a reading to minimize pointer friction.

Table 3. Frequency Input Test Table

FREQUENCY 4800 Hz = 100% RPM (3V P-P MIN)	FREQUENCY 4666 Hz = 100% RPM (3V P-P MIN)	% RPM METER (% RPM $\pm$ 2% RPM)
0	0	0
480	466.6	10
960	933.2	20
1440	1399.8	30
1920	1866.4	40
2400	2333.0	50
2880	2799.6	60
3360	3266.2	70
3840	3732.8	80
4320	4199.4	90
4800	4666.0	100
5280	5132.6	110

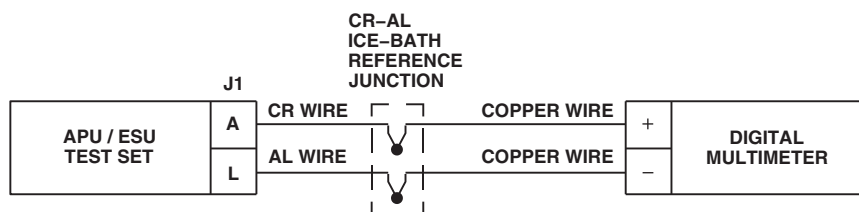
13. Temperature Output Accuracy. Disconnect frequency input test setup from test set and make temperature output test setup as in View 3.
14. Use Table 4 to check the temperature output voltages against the TEMP OUTPUT dial settings.

FB1134_4
SA

FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 4 OF 6)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION (CONT)
Accuracy Checks (cont)

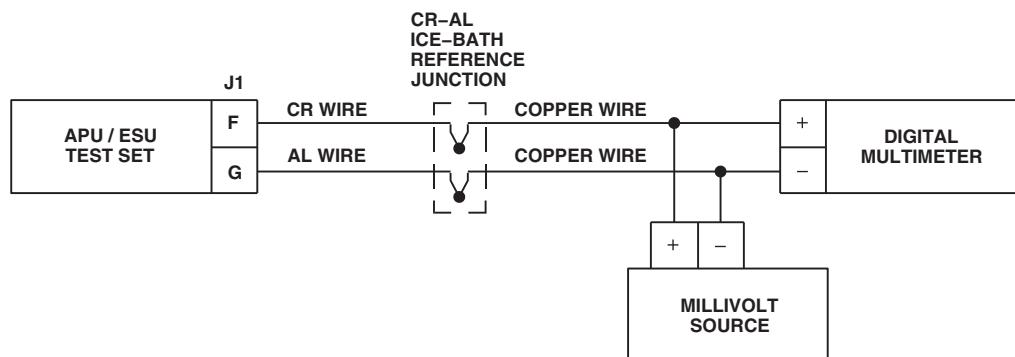


View 3. Temperature Output Test Setup

Table 4. Temperature Output Test Table

TEMP OUTPUT (°C)	TEMPERATURE OUTPUT (MVDC ± 0.21)
400	16.40
500	20.65
600	24.91
700	29.14
800	33.30

15. Momentarily position switch S5 to OPEN T / C position. Digital multimeter shall display 0 mvdc (open circuit).
16. Temperature Input Accuracy. Disconnect temperature output test setup from test set and make temperature input test setup as in View 4.



View 4. Temperature Input Test Setup

FB1134_5
SA

FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 5 OF 6)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION (CONT)****Accuracy Checks (cont)**

17. Check TEMP °C x 100 meter accuracy per Table 5. Lightly tap meter before taking a reading to minimize pointer friction.

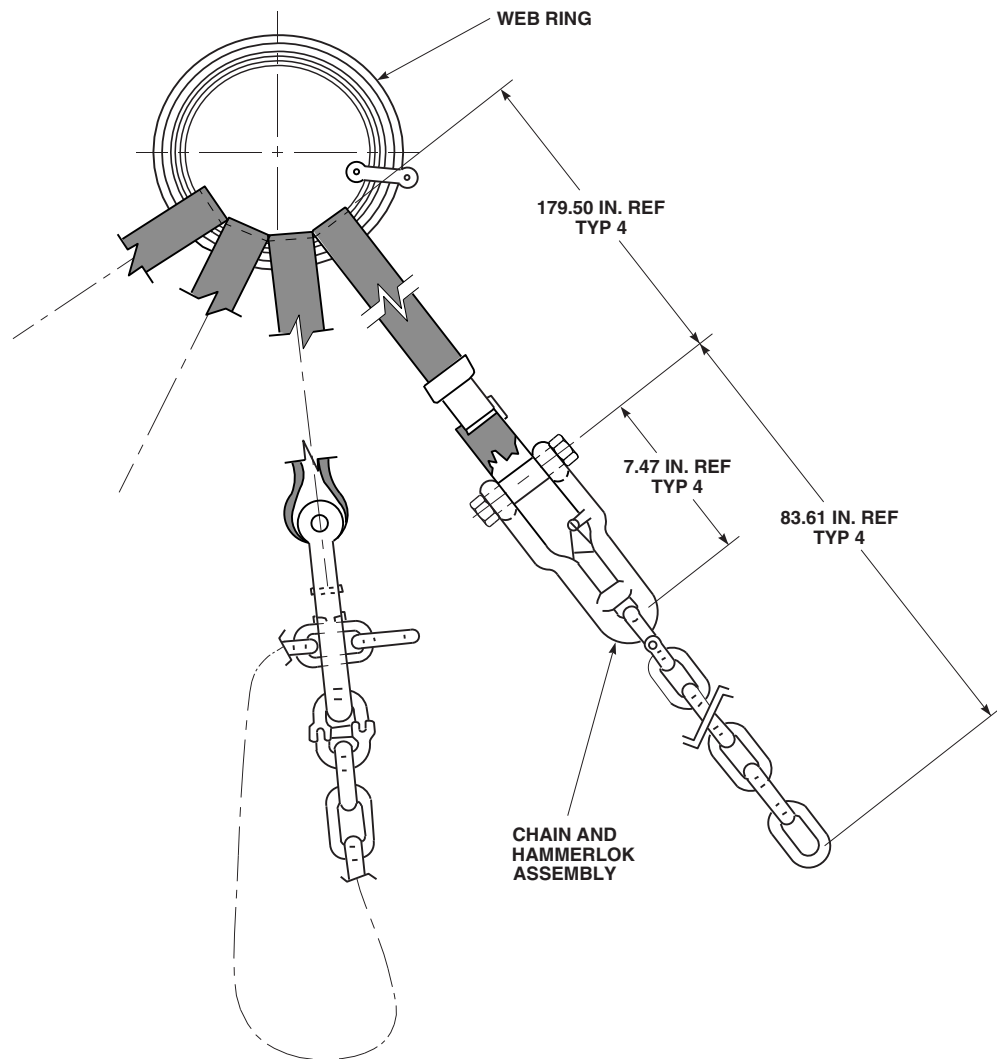
Table 5. Temperature Input Test Table

MILLIVOLT OUPUT (MVDC)	TEMP °C METER (°C ± 20 °C)
0.00	0
4.10	100
8.13	200
12.21	300
16.40	400
20.65	500
24.91	600
29.14	700
33.30	800
37.36	900
41.31	1000
45.16	1100
48.89	1200

18. Set all POWER switches to OFF and disconnect the test set-up.

19. Clean and secure test equipment per prescribed lab practice and affix the appropriate calibration label.

FB1134_6
SA**FIGURE C-49. SIMULATOR TEMP/SPEED APU TEST SET, H296B-1 (SHEET 6 OF 6)**



DESCRIPTION: The cargo sling assembly is used to transport an external load of cargo or equipment. The cargo sling assembly has a load capacity of 15000 pounds.

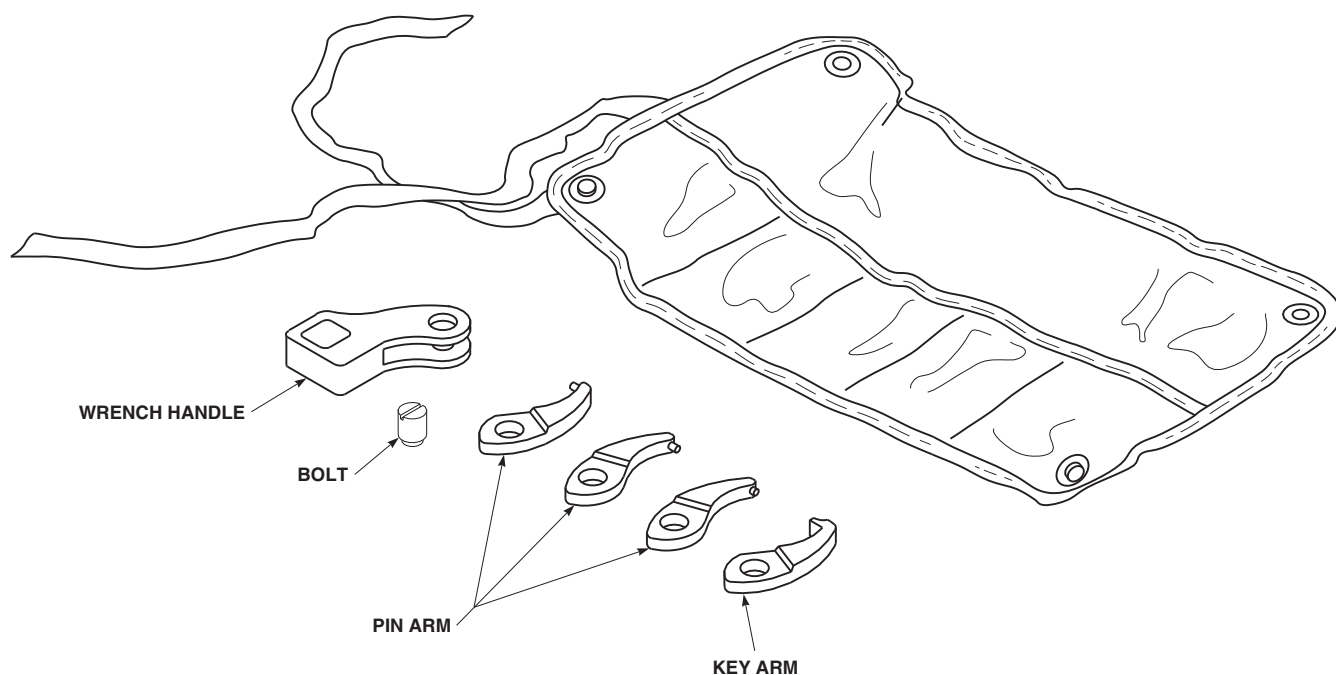
POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1135A
SA

FIGURE C-50. CARGO SLING ASSEMBLY, 3900061 OR XB-4-20-16

AN PART NO.	ITEM	PIN DIMENSION	
		LENGTH ± .005	DIA ± .003
AN8514-1	COMPLETE SET (AN8514-2 TO AN8514-7 INCL)		
AN8514-2	WRENCH HANDLE		
AN8514-3	BOLT		
AN8514-4	.09 PIN ARM	.060	.089
AN8514-5	.12 PIN ARM	.120	.121
AN8514-6	.19 PIN ARM	.190	.183
AN8514-7	KEY ARM		



DESCRIPTION: The spanner wrench set is used on main gear box and main rotor head to remove bearing seal retaining nuts.

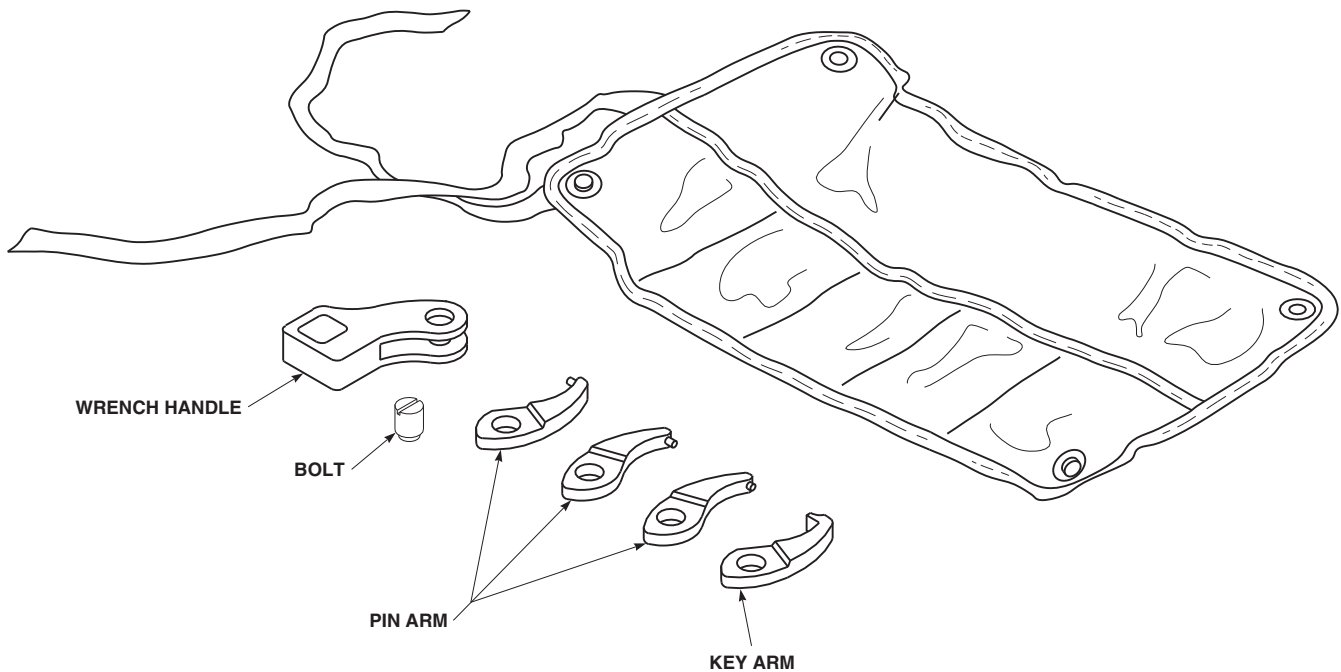
POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1031
SA

FIGURE C-51. SPANNER WRENCH SET, AN8514-1

AN PART NO.	ITEM	PIN DIMENSION	
		LENGTH ±.005	DIA ±.003
AN8515-1	COMPLETE SET (AN8515-2 TO AN8515-7 INCL)		
AN8515-2	WRENCH HANDLE		
AN8515-3	BOLT		
AN8515-4	.16 PIN ARM	.160	.152
AN8515-5	.25 PIN ARM	.190	.246
AN8515-6	.31 PIN ARM	.250	.308
AN8515-7	KEY ARM		



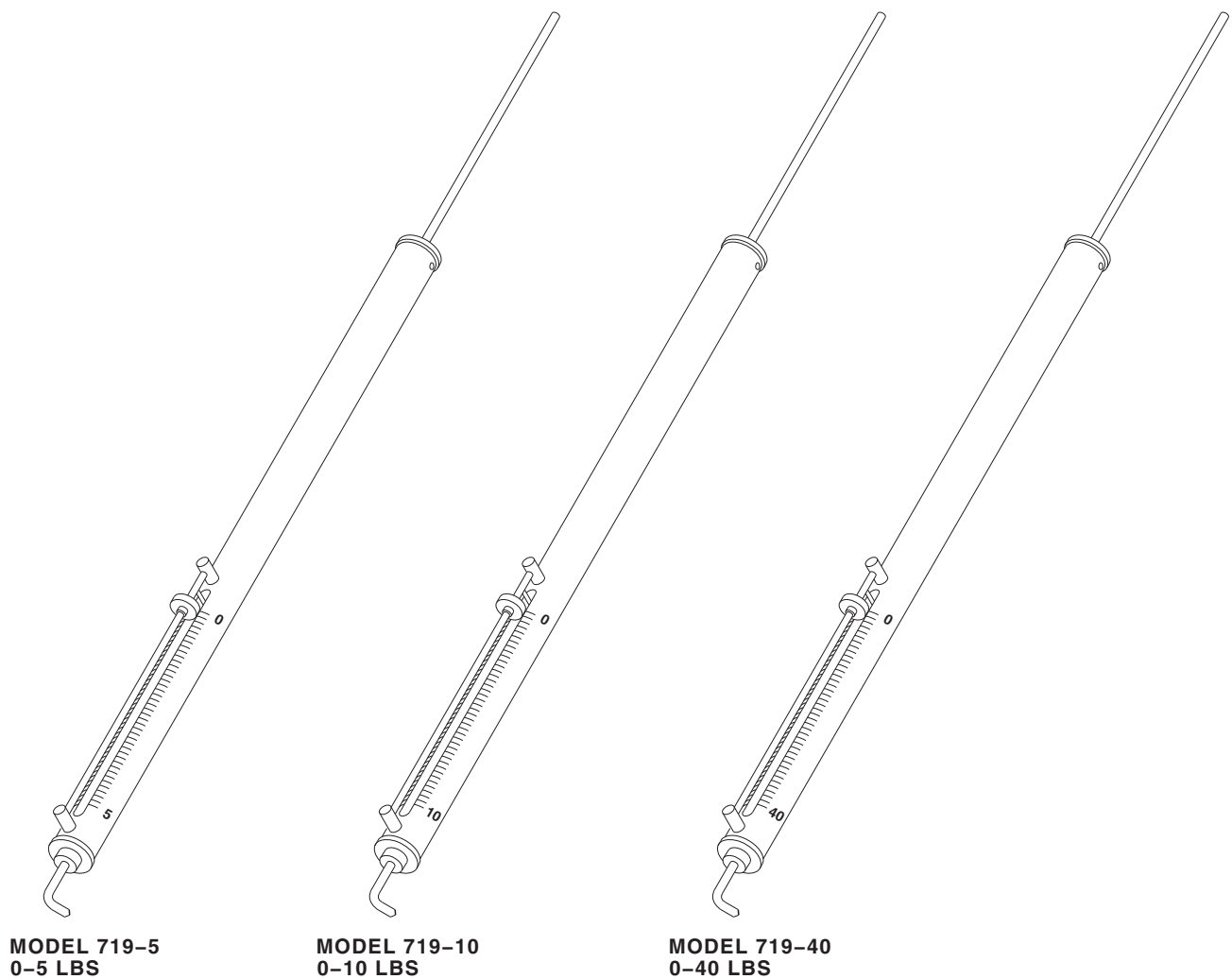
DESCRIPTION: The spanner wrench set is used on main gear box and main rotor head to remove bearing seal retaining nuts.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1032
SA

FIGURE C-52. SPANNER WRENCH SET, AN8515-1



DESCRIPTION: The spring testers are portable gages in a variety of sizes, the three described have capacity of 0-5 lbs in graduations of 1 oz., 0-10 lbs in graduations of 2 oz., and 0-40 lbs in graduations of 8 oz. The spring testers are adapted to a variety of uses.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Spring testers must be calibrated every 27 weeks regardless of being used or not.

FB1035_1
SA

FIGURE C-53. SPRING TESTERS, 719-5, 719-10, 719-40 (SHEET 1 OF 2)

CALIBRATION REQUIREMENTS: (CONT)

EQUIPMENT REQUIRED

- Dead Weights $\pm .01$ LBS.

INSPECTION

Inspect spring tester for any obvious damage, dents, nicks, or missing parts. Check push rod for binding; scale graduations for clear readability.

CALIBRATION

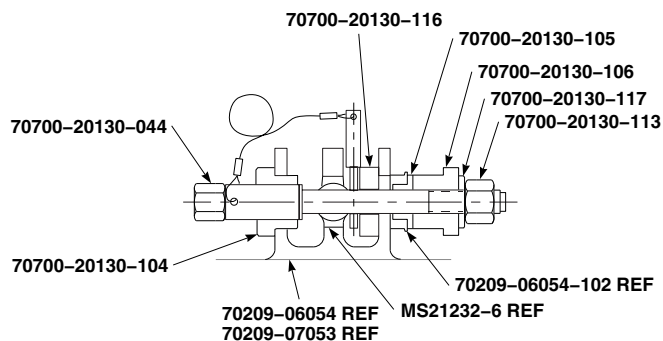
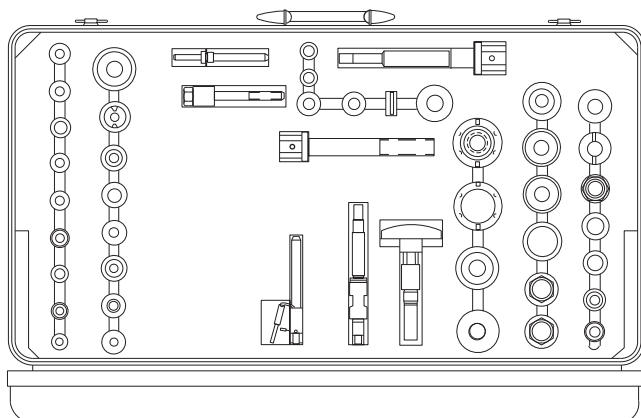


Do not load spring tester past its operating range.

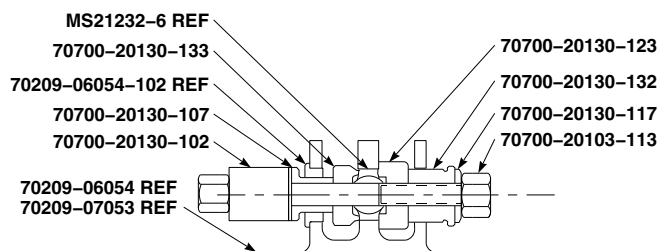
1. Operate spring tester through entire scale range, checking for any binding.
2. Attach dead weights to spring tester and check scale range at five points throughout scale. Scale shall not deviate more than one minor graduation from the dead weight nominal value. Adjust as required.
3. Adjust spring tester by use of knurled end cap on spring tester. Recheck calibration.

FB1035_2
SA

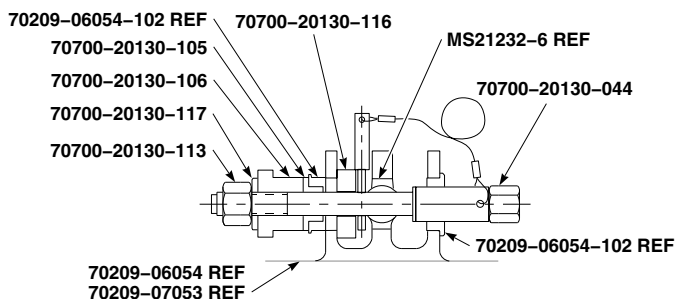
FIGURE C-53. SPRING TESTERS, 719-5, 719-10, 719-40 (SHEET 2 OF 2)



**APPLICATION DIAGRAM REF
DUAL BUSHING INSTALLATION 70209-06054-102**



**APPLICATION DIAGRAM REF
(BEARING STAKING MS21232-6
ONE BUSHING REMOVED)**



**APPLICATION DIAGRAM REF
(SINGLE BUSHING INSTALLATION 70209-06054-102)**

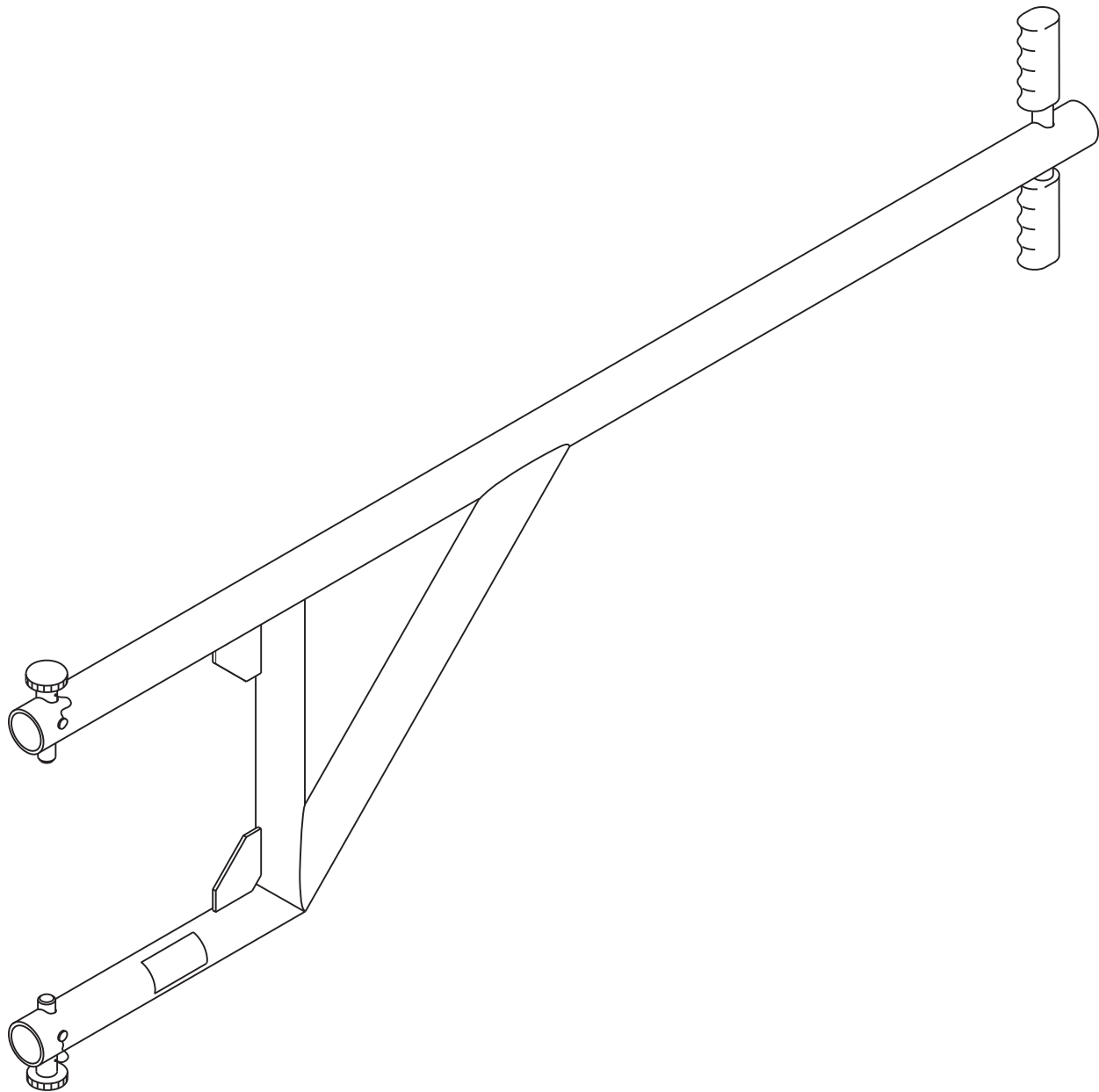
DESCRIPTION: The staking tool kit is used to remove and install bushings in stabilator fittings. The kit contains cutting tools for removing bearings and roll staking tools for installing bearings.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1137
SA

FIGURE C-54. STAKING TOOL KIT, STABILATOR FITTINGS, 70700-20129-041



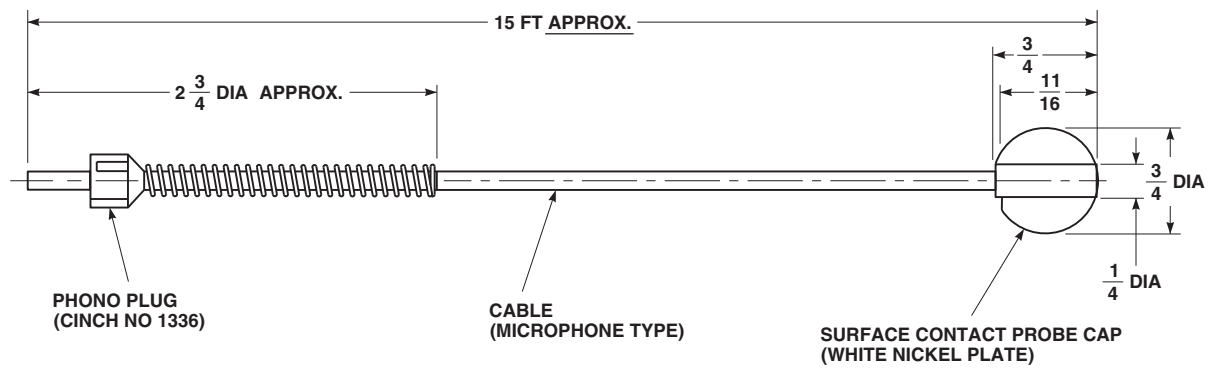
DESCRIPTION: The steering bar is used by the ground handling crew as a means of maneuvering the helicopter. The steering bar is constructed of welded aluminum tubing with tee handle.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1138
SA

FIGURE C-55. STEERING BAR, 70700-20374-041



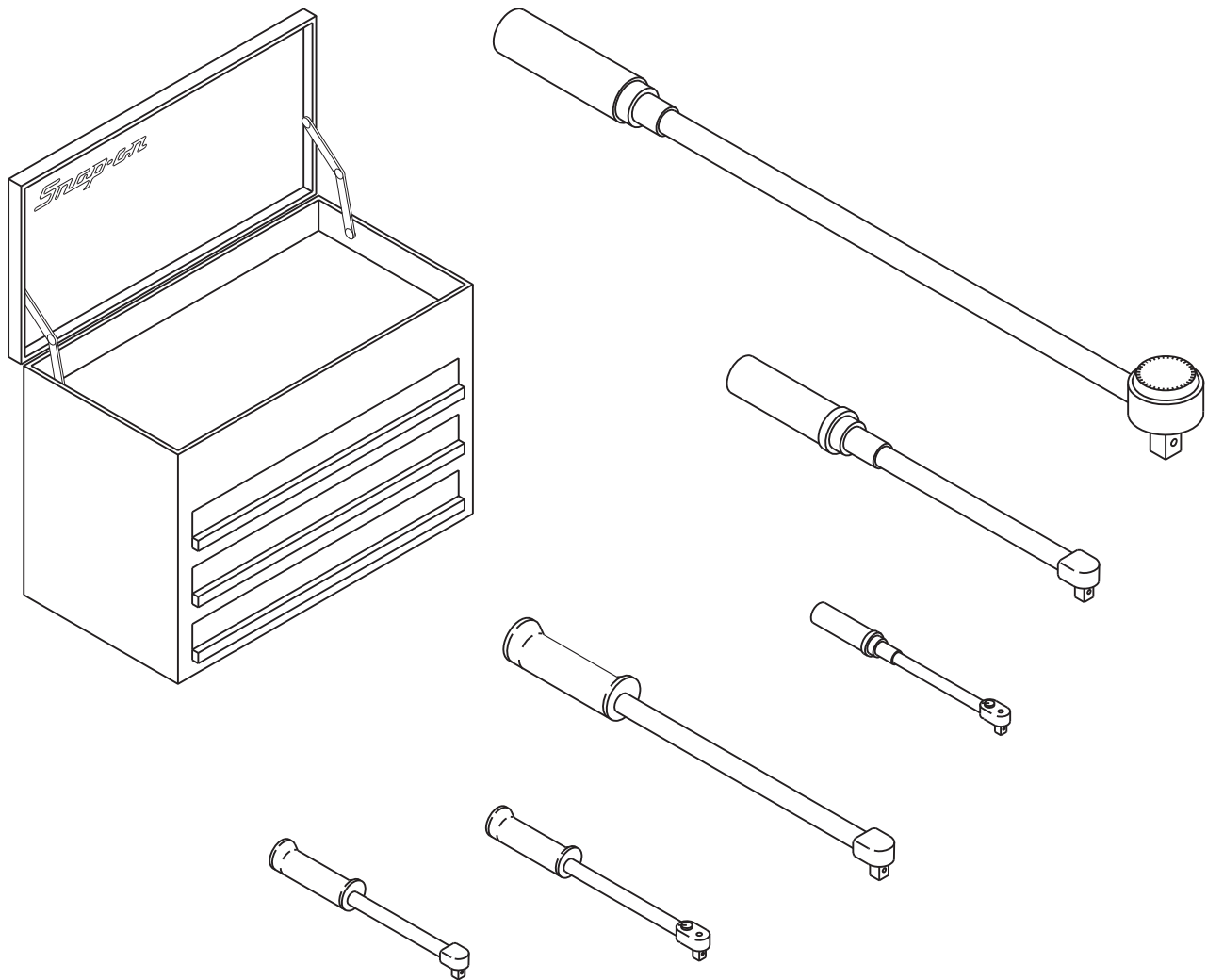
DESCRIPTION: The surface probe is used to determine if engine anti-ice and windshield anti-ice are operative within designed temperature limits. The engine inlet cycles between 100°F and 130°F minimum to 196°F maximum in operation. The windshield cycles in low mode 84°F to 110°F, high mode 90°F to 125°F.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1139
SA

FIGURE C-56. SURFACE PROBE, 00790



DESCRIPTION: The torque wrench kit will support all torque requirements to maintain the helicopter.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Torque wrenches will be calibrated every 12 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Model 10 Torque Wrench Tester, Chatillon, or similar equipment, provided it has required measuring capabilities.

FB1036_1
SA

FIGURE C-57. TORQUE WRENCH (SHEET 1 OF 2)

CALIBRATION REQUIREMENTS: (CONT)**CLEANING MATERIALS**

- Isopropyl Alcohol, Item 72, Appendix D
- Machinery Towel, Item 211, Appendix D

CALIBRATION PROCEDURES**Cleaning / Inspection**

1. Clean torque wrench using isopropyl alcohol, Item 72, Appendix D. Wipe dry with machinery towel, Item 211, Appendix D.
2. Inspect torque wrench for any defects or damage. NONE ALLOWED.

Calibration

1. Set up torque wrench tester to zero.
2. Select five equally spaced calibration points on the torque wrench including zero and full scale. Record these values on a calibration work sheet.

NOTE

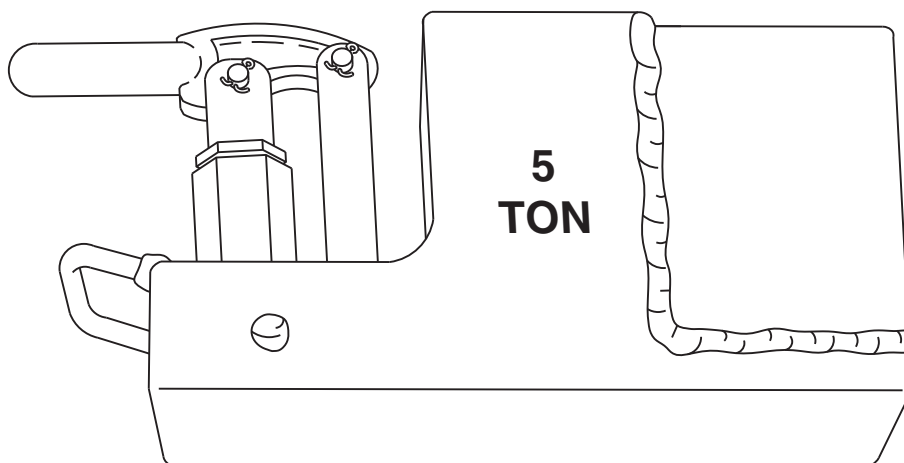
Acceptable tolerance is $\pm 3\%$ of calibration setting on torque wrench as shown in calibration work sheet example.

CALIBRATION DATA SHEET			
TORQUE WRENCH/DRIVER			
RANGE: 50-in. lbs			
ft lbs			
CALIBRATION SETTING	ACTUAL READING	ACCEPTANCE TOLERANCE	ADJUSTED VALUE
10		9.7–10.3	
20		19.4–20.6	
30		29.1–30.9	
40		38.8–41.2	
50		48.5–51.5	

EXAMPLE CALIBRATION WORK SHEET

3. Attach torque wrench to appropriate input on torque tester.
4. Pull torque wrench in a clockwise direction until torque wrench indicates first pre-selected calibration point. Record actual readings on calibration work sheet.
5. Remove torque wrench from torque tester and verify torque wrench zero. Adjust torque wrench to zero if required and note adjustment on calibration work sheet.
6. If torque wrench has been adjusted to zero, repeat steps 3 through 6.
7. Check remaining torque wrench calibration points and record on calibration work sheet.
8. Install calibration label on torque wrench after calibration acceptance, identified by acceptance stamp, date of calibration, and scheduled calibration due date.

 FK2214_2A
SA
FIGURE C-57. TORQUE WRENCH (SHEET 2 OF 2)



REGENT MODEL NUMBER	VERTICAL CAPACITY (TONS)	MINIMUM CLOSED HEIGHT (INCHES)	HYDRAULIC LIFT (INCHES)	SCREW EXTENSION (INCHES)	MAXIMUM EXTENDED HEIGHT (INCHES)	NET WEIGHT (LBS)	CRATED WEIGHT (LBS)	CUBIC FEET CRATED
1926-010	5	5.5	9	3	17.5	26	37	1.0

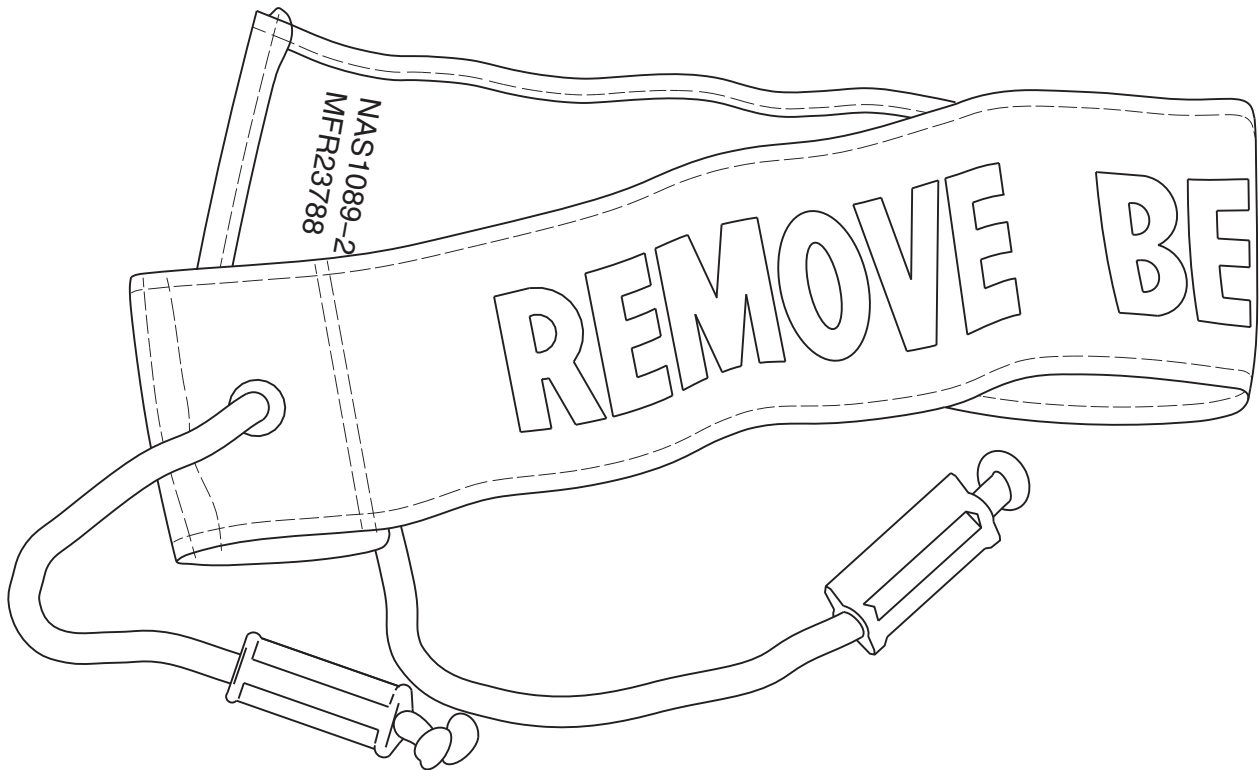
DESCRIPTION: The tail wheel jack is a hand carried jack, used for removal and replacement of tail landing gear and tail wheel assemblies.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1037
SA

FIGURE C-58. TAIL WHEEL JACK, 1926-10



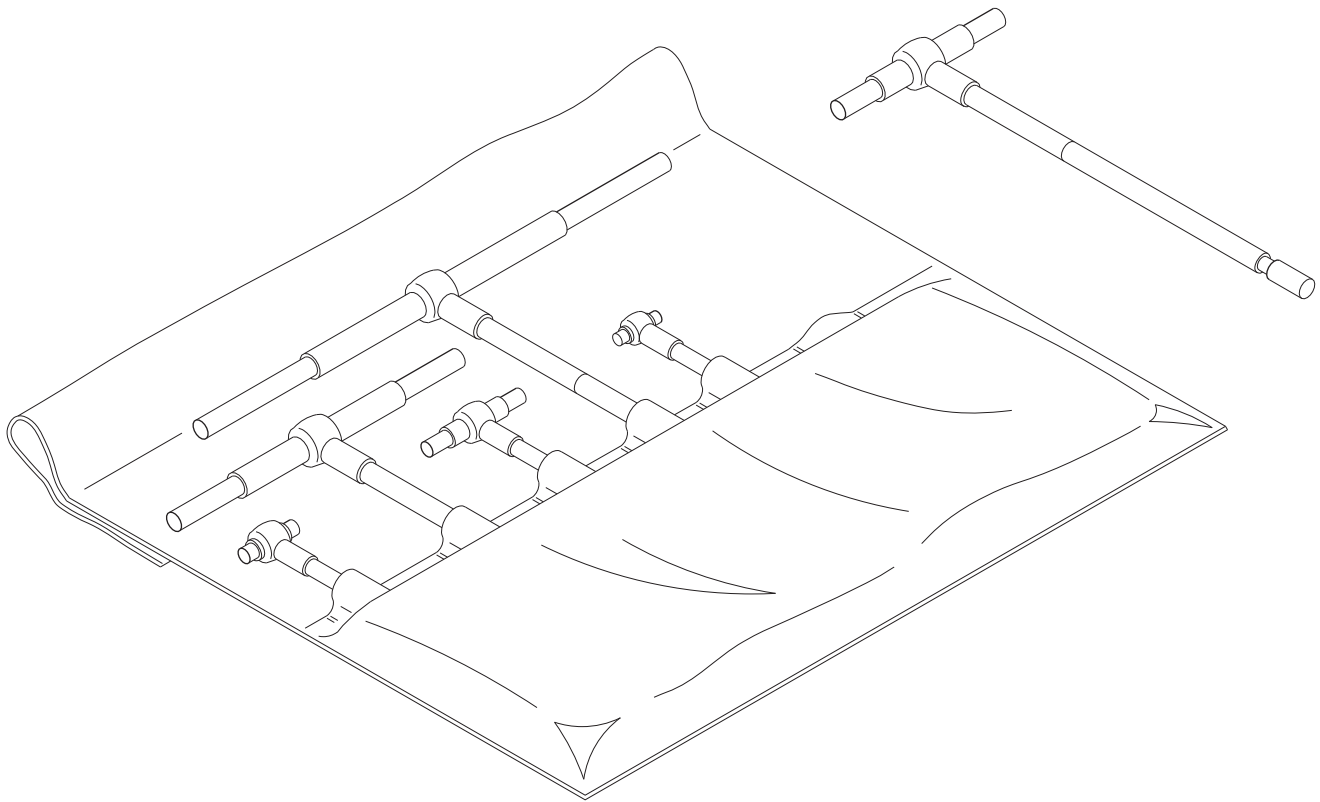
DESCRIPTION: The lockout block is composed of two phenolic blocks connected by a 16-inch rope knotted at each end, with a red warning flag to remove before flight. The lockout block is used to retain tail rotor spring pressure when rigging tail rotor system.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1038
SA

FIGURE C-59. TAIL ROTOR LOCKOUT BLOCKS, 70700-20463-041



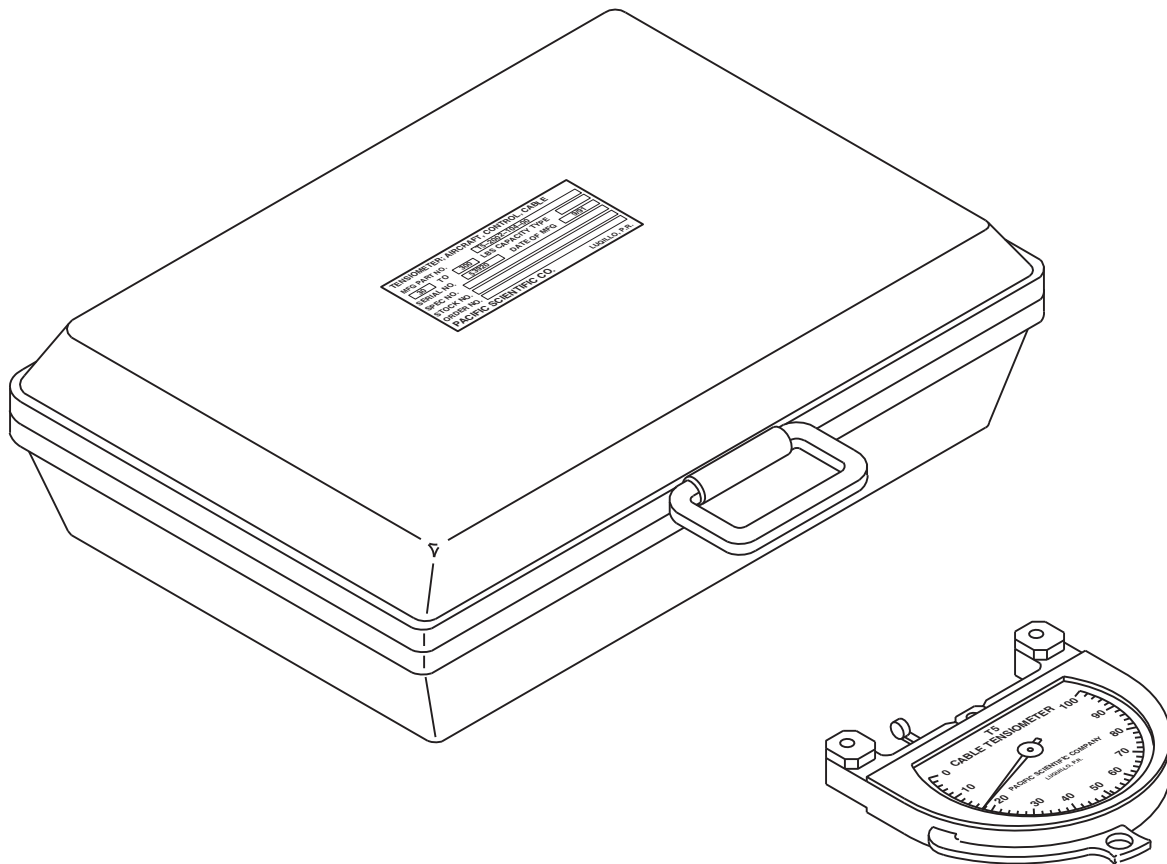
DESCRIPTION: The telescoping gage set is used for accurate measurements of inside diameters on flight controls, main, and tail rotor head components. The gage set has a measuring range of 5/16 inch to 6 inches and a measurable depth of up to 5 inches. Final hole size is obtained by measuring the contacts of the gage using micrometer or vernier calipers.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1039
SA

FIGURE C-60. TELESCOPING GAGE SET, PMF119



DESCRIPTION: The tensiometer is a small compact unit designed to determine tension on flight control and tail rotor cables when rigging helicopter, or when cables are disconnected for maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Tensiometer shall be calibrated every 6 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Morehouse Dead Weight Calibrating Machine or similar equipment, provided it has required measuring capability.
- 3/32 inch Test Cable
- 3/16 inch Test Cable
- 7/16 inch Test Cable

FB1040_1
SA

FIGURE C-61. TENSIONER, T5-2002-104-00 (SHEET 1 OF 2)

CALIBRATION REQUIREMENTS: (cont)

CLEANING MATERIALS

- Isopropyl Alcohol, Item 72, Appendix D
- Machinery Towel, Item 211, Appendix D

CALIBRATION PROCEDURES

Cleaning / Inspection

1. Clean tensiometer with isopropyl alcohol, Item 72, Appendix D. Wipe dry with machinery towel, Item 211, Appendix D.
2. Inspect tensiometer for obvious damage, nicks, broken glass and full indicator range without skipping or binding.

Calibration

1. Make sure temperature in area of calibration performance is within 68°F to 73°F. Calibration equipment should stabilize for 24 hour period.

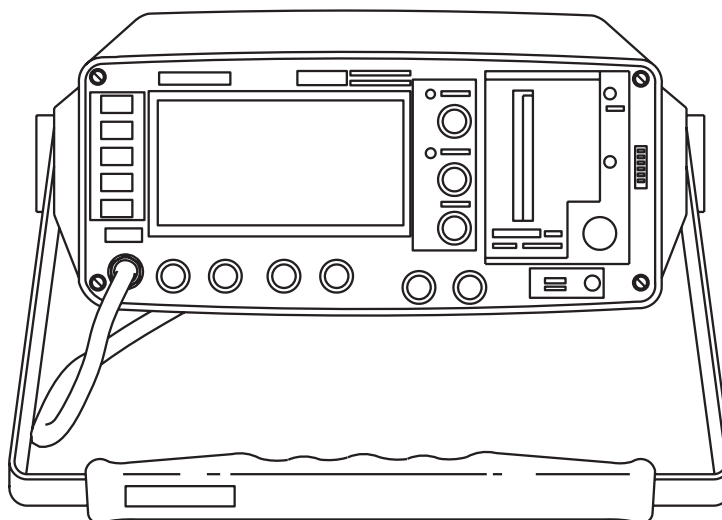
NOTE

Do not exceed tensiometer range when calibrating. Only one size cable is required for calibration of tensiometer.

2. Select test cable to match riser available with tensiometer and install test cable in dead weight calibrator.
3. Check tensiometer with dead weight calibrator set at four points. 50 lbs, 100 lbs, 150 lbs and 200 lbs.
4. Attach tensiometer to test cable and check cable tension, refer to tensiometer calibration table enclosed with tensiometer. Compare values specified at 50 lbs, 100 lbs, 150 lbs, and 200 lbs for size test cable used. Values specified shall agree with actual tensiometer indication within ± 5 counts on tensiometer.
5. If values are not as specified, adjust tensiometer as required and repeat steps 3 and 4.

FK2215_2B
SA

FIGURE C-61. TENSIO METER, T5-2002-104-00 (SHEET 2 OF 2)



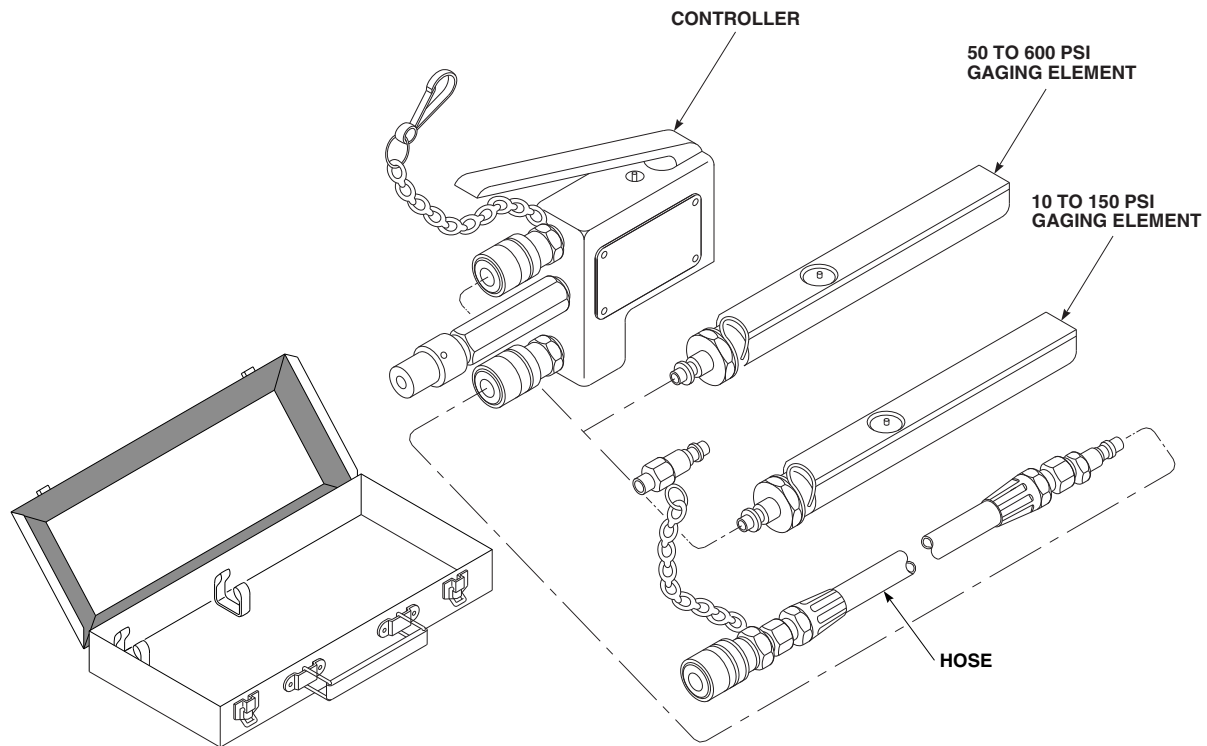
DESCRIPTION: The reflectometer is designed to check faults in metallic cables such as opens, shorts, crimps, and other defects in metallic cables to lengths of 50,000 feet. Using a process similar to radar, electrical pulses transmitted down cables under test, reflect pulses back to reflectometer, allowing user to determine nature of cable fault.

POWER REQUIREMENTS: Internal battery 12-volt, external ac power 115 / 230 volts.

CALIBRATION REQUIREMENTS: Not Applicable

FB1140
SA

FIGURE C-62. TIME DOMAIN REFLECTOMETER CABLE TESTER, 1503C



DESCRIPTION: The inflator kit is a remote-controlled, direct-reading pneumatic inflator used on helicopter tires, shock struts, and accumulators having pneumatic pressure requirements of 10 to 600 psi.

POWER REQUIREMENTS: 500 psi Nitrogen Gas Tank

CALIBRATION REQUIREMENTS:

FREQUENCY

Inflator kit shall be calibrated every 6 weeks regardless of being used or not.

FB1041_1
SA

FIGURE C-63. TIRE INFLATOR, M85352/1 (SHEET 1 OF 4)

CALIBRATION REQUIREMENTS: (CONT)**EQUIPMENT REQUIRED**

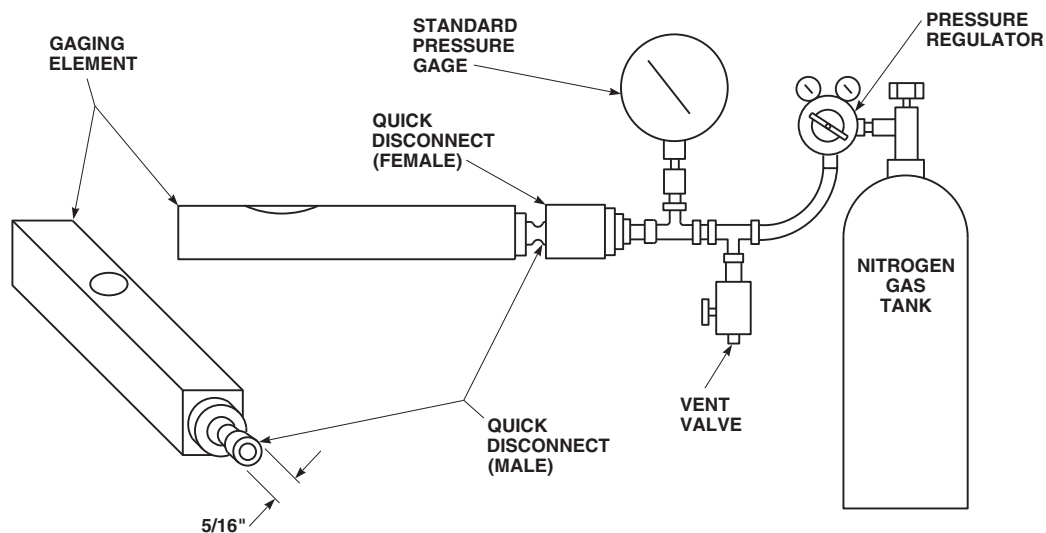
- Nitrogen Gas Tank, minimum pressure 500 psi.
- Pressure Regulator, adjustable to 200 psi.
- Inline Pressure Gage, 0–200 psi range.
- Quick Disconnects, male and female.
- Pneumatic Hose, Adapters, Vent valve, and Plugs as required.

CALIBRATION PROCEDURES**INSPECTION**

- Check gaging elements for obvious damage, broken viewing window, inoperative quick disconnects.
- Check that pressure relief hole on adjusting cylinder is not blocked or restricted.
- Check hoses for damage to quick disconnects, cuts or damage to hose.
- Clean and repair as required.

CALIBRATION

1. Set up test equipment as shown in figure below for calibration of gaging elements.

**CALIBRATION SETUP FOR GAGING ELEMENT****NOTE:**

Make sure all pressure is set to zero and make sure all test equipment controls are set to proper position before making any connections or applying pressure to items being calibrated.

FB1041_2
SA

FIGURE C-63. TIRE INFLATOR, M85352/1 (SHEET 2 OF 4)

CALIBRATION PROCEDURES: (CONT)

CALIBRATION: (CONT)

2. Connect high range 50 to 600 psi gaging element to quick disconnect after pressure gage on hose.
3. Make sure pressure regulator is closed. Open valve on nitrogen tank. Tank pressure on regulator gage shall read above 500 psi. If below 500 psi, replace tank.
4. Make sure vent valve is closed. Slowly open regulator while observing pressure on gaging element, pressure shall not exceed 200 psi on gaging element or standard pressure gage. Open vent valve to release pressure.
5. Open pressure regulator slowly, to bring pressure reading on standard pressure gage to first actual reading listed in Table 1. If system is over-pressurized, open vent valve to lower pressure below value in Table 1. Close vent valve, and open regulator slowly to bring pressure reading on gage to first actual reading listed in Table 1.

TABLE 1			
LOW RANGE GAGING ELEMENT (10 TO 150 PSI)		HIGH RANGE GAGING ELEMENT (50 TO 600 PSI)	
ACTUAL	TOLERANCE	ACTUAL	TOLERANCE
10 PSI	8 TO 12 PSI	60 PSI	54 TO 66 PSI
30 PSI	28 TO 32 PSI	100 PSI	94 TO 106 PSI
50 PSI	48 TO 52 PSI	120 PSI	114 TO 126 PSI
70 PSI	68 TO 72 PSI	140 PSI	134 TO 146 PSI
100 PSI	96 TO 104 PSI	160 PSI	154 TO 166 PSI
130 PSI	126 TO 134 PSI	180 PSI	174 TO 186 PSI
150 PSI	146 TO 154 PSI	200 PSI	194 TO 206 PSI

6. Check pressure on standard pressure gage and on gaging element. Tap both gages with finger to reduce any mechanical friction in gages.
7. With desired pressure readings on standard gage and gaging element, record readings. Repeat procedure for remaining values in Table 1.
8. When check is complete, close regulator and tank valves. Open vent valve, and remove gaging element.
9. Install low range gaging element 10 to 150 psi onto test equipment and repeat steps 1 through 8 using values in Table 1.

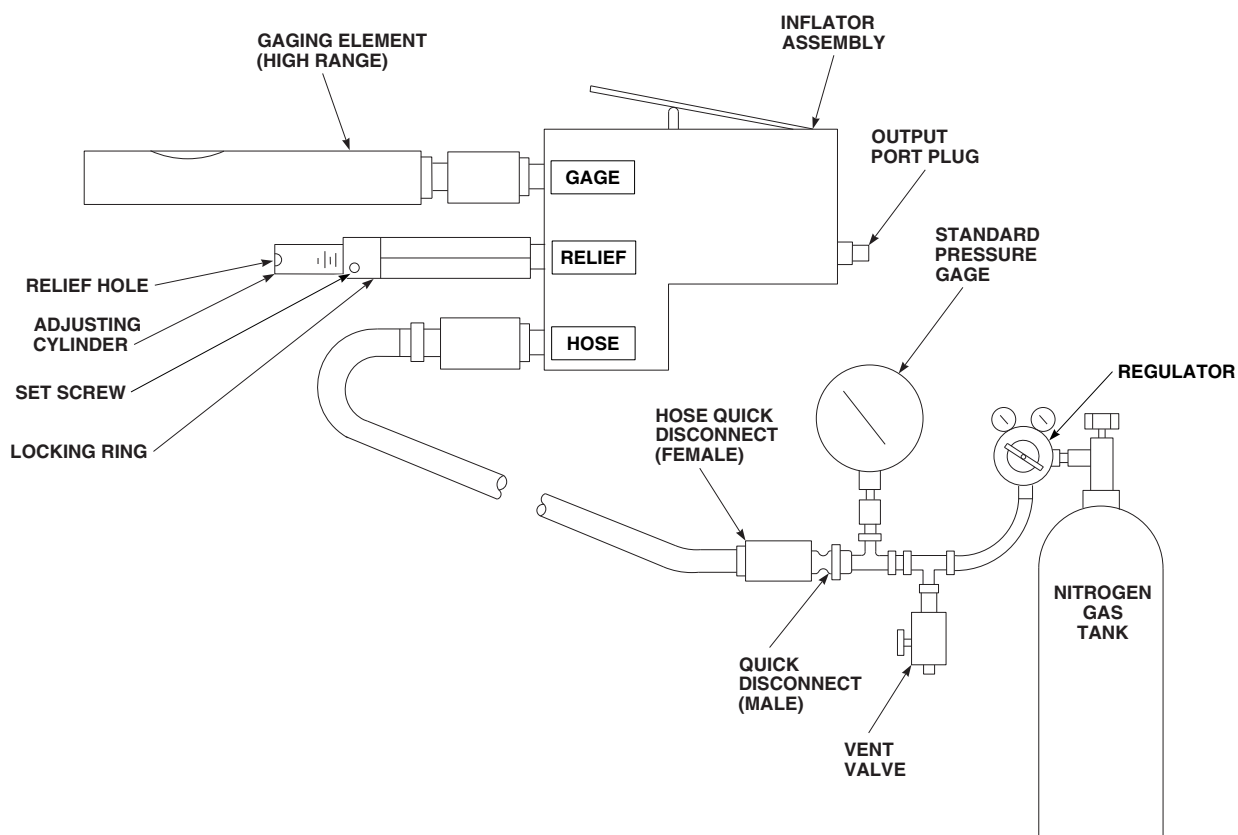
FB1041_3
SA

FIGURE C-63. TIRE INFLATOR, M85352/1 (SHEET 3 OF 4)

CALIBRATION PROCEDURES: (CONT)

ADJUSTMENT AND CALIBRATION OF RELIEF VALVE

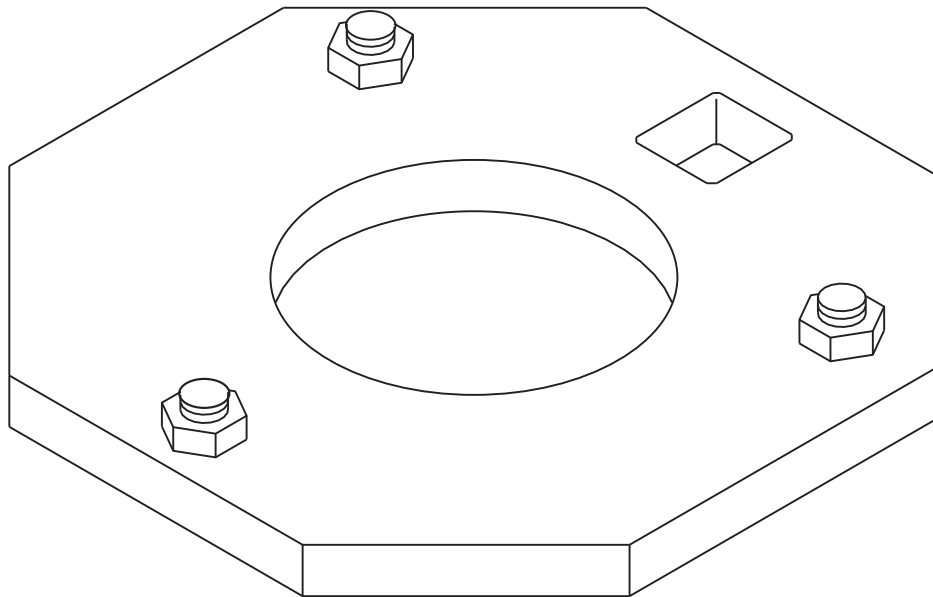
1. Set up test equipment as shown in figure below for adjustment and calibration of inflator assembly relief valve.



2. Remove adapter from output side of inflator assembly and install plug. Install high range gaging element, 50 to 600 psi, to inflator assembly.
3. Close vent valve and open valve on nitrogen tank. Open pressure regulator until 200 psi is read on standard pressure gage and gaging element.
4. Using proper hex head wrench, loosen allen head set screw and relief valve locking ring. Screw relief valve adjusting cylinder in or out, until valve relief hole begins to leak pressure, tighten set screw on locking ring.
5. Open vent valve and release pressure. Close vent valve and retest valve for pressure leakage at the same valve as in step 4.
6. Open vent valve to release pressure, remove all test equipment.
7. Install limited use calibration sticker, stating calibration limited to 200 psi maximum pressure.

FB1041_4A
SA

FIGURE C-63. TIRE INFLATOR, M85352/1 (SHEET 4 OF 4)



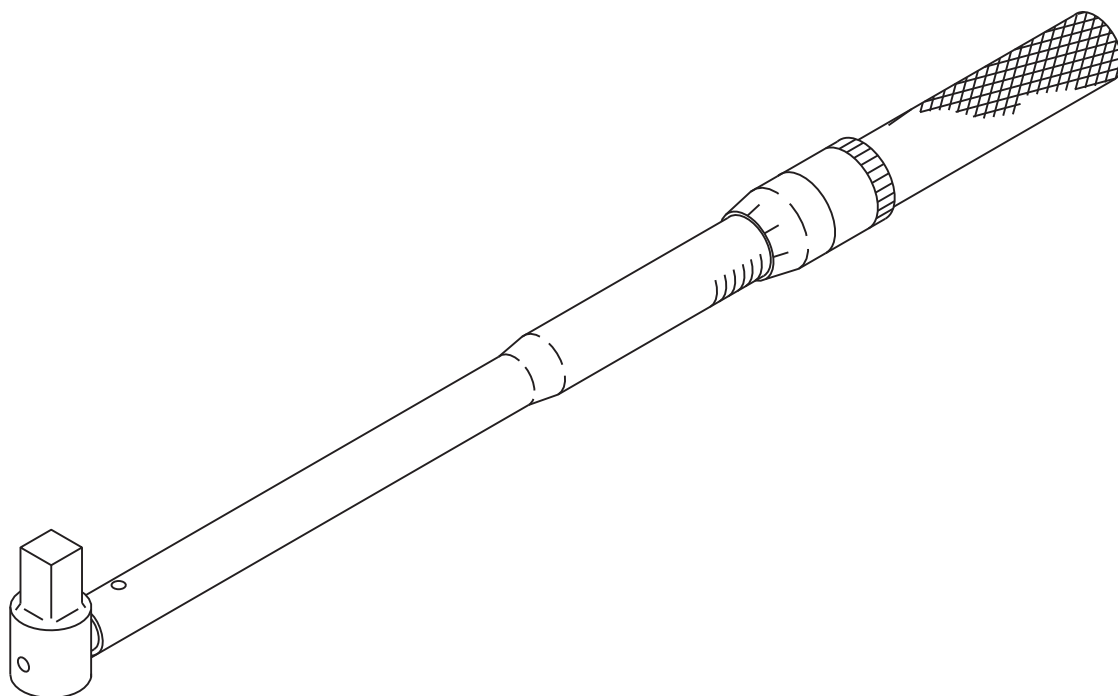
DESCRIPTION: The torque reactor is a machined 0.500 inch steel plate, 6.12 inches square. A 2.670-inch diameter hole thru center of reactor provides access for 70700-20687-101 wrench to reach nut. There are three equally spaced holes that mate with hole pattern on flange. The torque reactor is used to remove and install the nut that retains the flange against shoulder of input or output shafts.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1042
SA

FIGURE C-64. TORQUE REACTOR, 70700-20688-041



DESCRIPTION: The torque wrench is used to torque pylon attach bolts. The torque wrench is a click type wrench with a torque range of 700 to 1600 inch-pounds in 10-pound graduations.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Torque wrenches will be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Model 10 torque wrench tester, Chatillon, or similar equipment, provided it has required measuring capabilities.

CLEANING MATERIALS

- Isopropyl Alcohol, Item 72, Appendix D
- Machinery Towel, Item 211, Appendix D

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SA

FIGURE C-65. TORQUE WRENCH, PYLON FOLD, 28420 (SHEET 1 OF 2)

CALIBRATION REQUIREMENTS: (cont)**CALIBRATION PROCEDRES**
Cleaning / Inspection

1. Clean torque wrench using isopropyl alcohol, Item 72, Appendix D. Wipe dry with machinery towel, Item 211, Appendix D.
2. Inspect torque wrench for any defects or damage. NONE ALLOWED.

Calibration

1. Set up torque wrench tester to zero.
2. Select five equally spaced calibration points on the torque wrench including zero and full scale. Record these values on a calibration work sheet.

NOTE

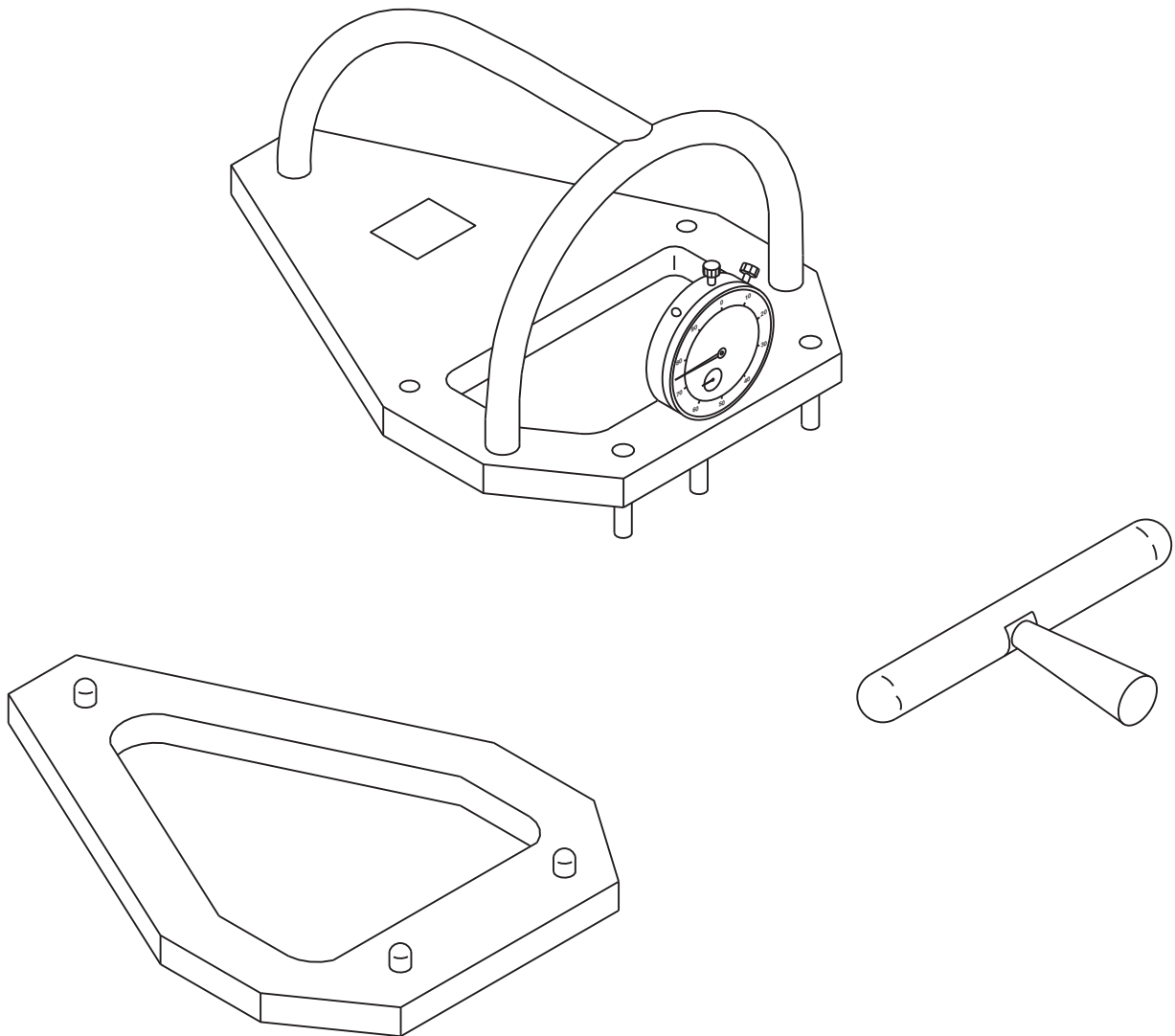
Acceptable tolerance is $\pm 3\%$ of calibration setting on torque wrench as shown in calibration work sheet example.

CALIBRATION DATA SHEET TORQUE WRENCH/DRIVER			
RANGE: 50–in. lbs ☐			
ft lbs ☐			
CALIBRATION SETTING	ACTUAL READING	ACCEPTANCE TOLERANCE	ADJUSTED VALUE
10		9.7–10.3	
20		19.4–20.6	
30		29.1–30.9	
40		38.8–41.2	
50		48.5–51.5	

EXAMPLE CALIBRATION WORK SHEET

3. Attach torque wrench to appropriate input on torque tester.
4. Pull torque wrench in a clockwise direction until torque wrench indicates first pre–selected calibration point. Record actual readings on calibration work sheet.
5. Remove torque wrench from torque tester and verify torque wrench zero. Adjust torque wrench to zero if required and note adjustment on calibration work sheet.
6. If torque wrench has been adjusted to zero, repeat steps 3 through 5.
7. Check remaining torque wrench calibration points and record on calibration work sheet.
8. Install calibration label on torque wrench after calibration acceptance, identified by acceptance stamp, date of calibration, and scheduled calibration due date.

 FB1043_2C
 SA
FIGURE C-65. TORQUE WRENCH, PYLON FOLD, 28420 (SHEET 2 OF 2)



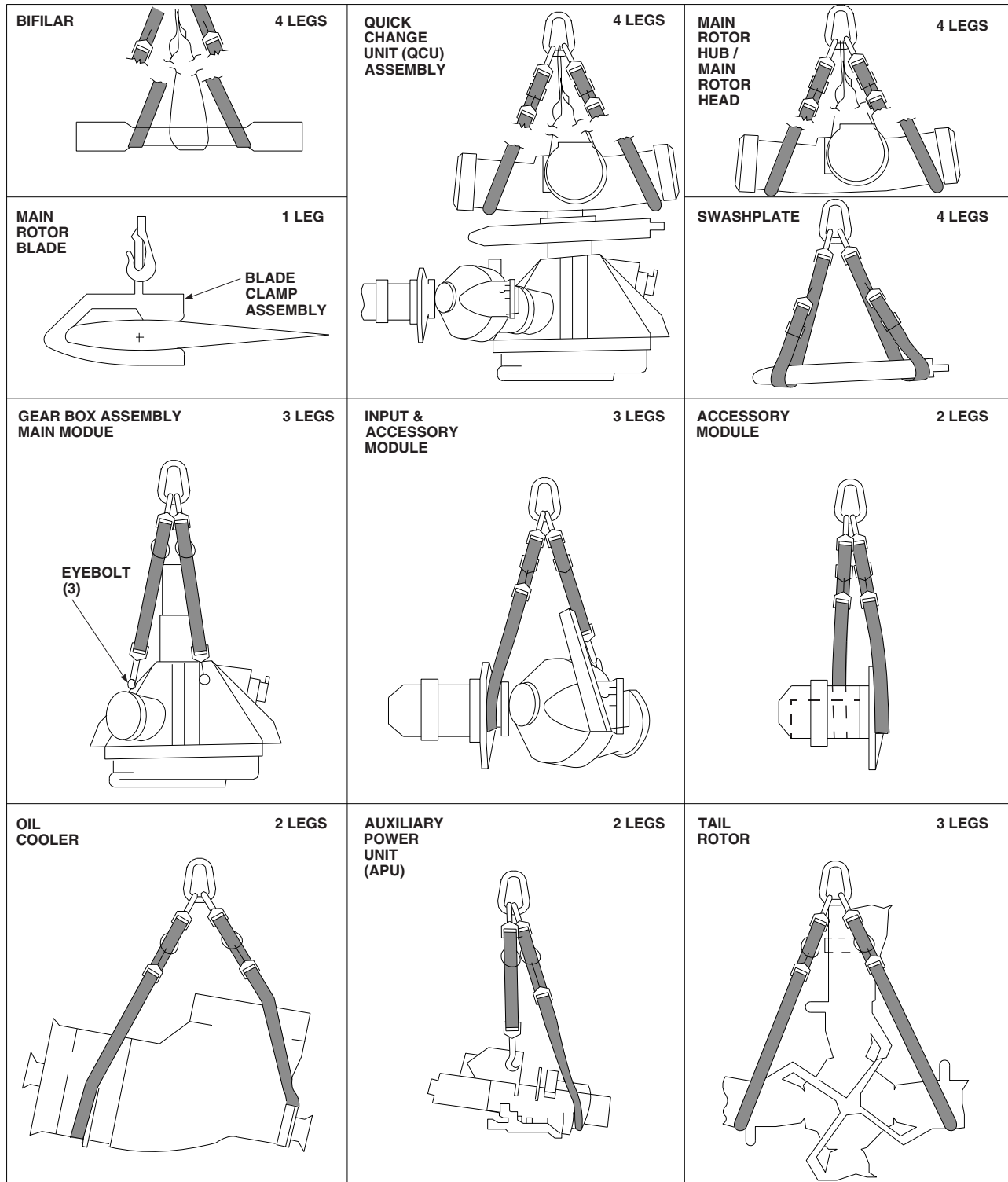
DESCRIPTION: The trim tab adjustment set is required to adjust the trailing edge trim tab when setting main rotor blade pretrack.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1044
SA

FIGURE C-66. TRIM TAB ADJUSTMENT SET, 70700-20308-041



FB1141_1A
SA

FIGURE C-67. UNIVERSAL HOISTING SLING ASSEMBLY, 70700-20320-045 (SHEET 1 OF 2)

DESCRIPTION: The universal hoisting sling assembly is used to remove and install components on helicopters. The universal hoisting sling assembly is adaptable to items being lifted by removing one or more straps (legs) to adjust for the different components.

POWER REQUIREMENTS: Not Applicable

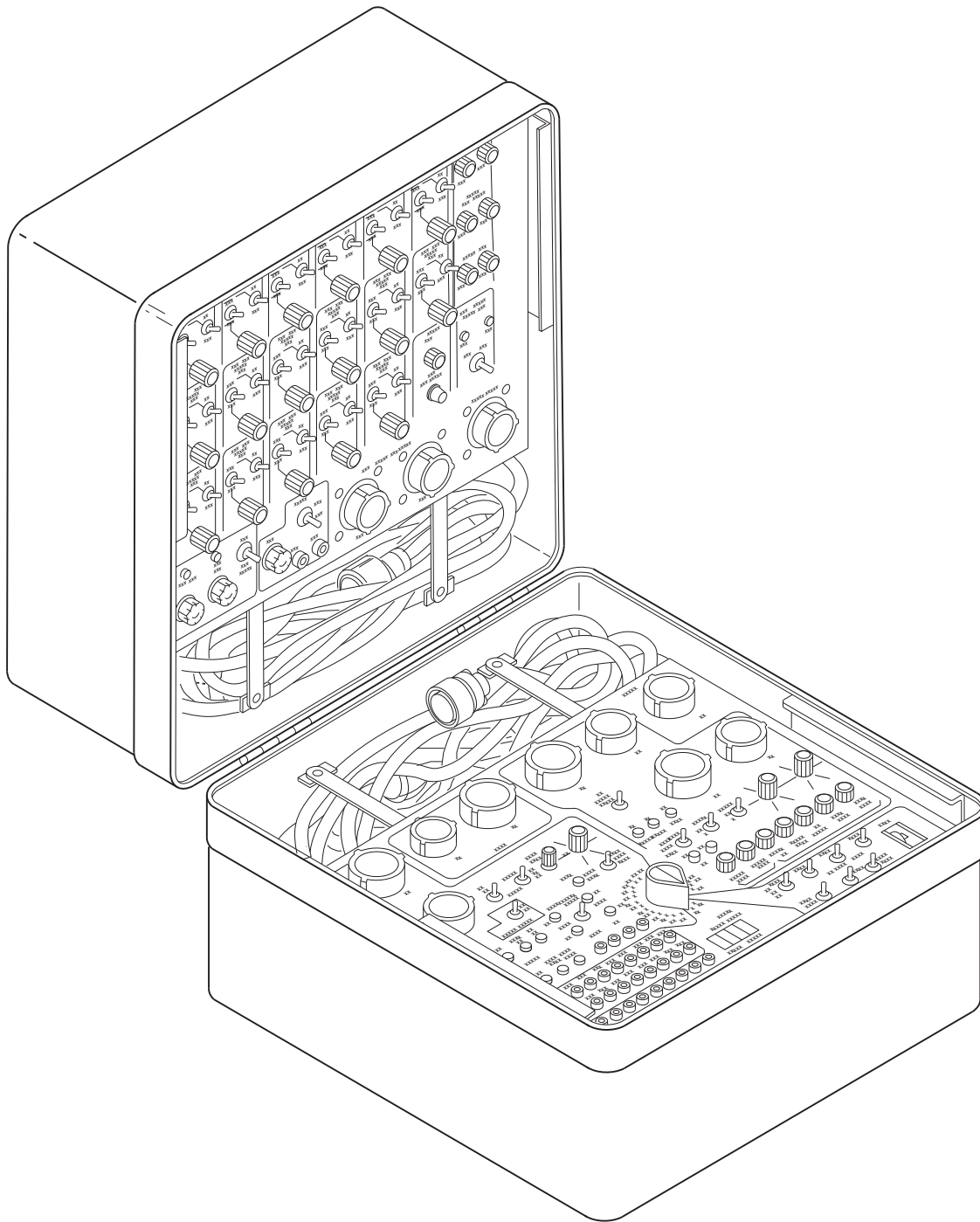
CALIBRATION REQUIREMENTS: Not Applicable

FB1141_2A
SA

FIGURE C-67. UNIVERSAL HOISTING SLING ASSEMBLY, 70700-20320-045 (SHEET 2 OF 2)

Change 10

C-111



DESCRIPTION: The VIDS bench test set is a two part instrument comprised of a simulator and a unit tester. The simulator is primarily a system tester. The simulator provides the power, switches, variable resistors and indicators to operate and test the VIDS. The switches and variable resistors mounted to the front panel allow for operator selection and control of the simulated sensor signal outputs normally provided by helicopter engine sensors. The outputs of the VIDS that normally provide warning information to the aircraft annunciator panel are displayed on the simulator.

An interconnect cable to the unit tester routes power and LAMP TEST parameters to the unit tester. By the same interconnect cable, AC excitation signals are routed to the simulator.

FK2216_1
SA

FIGURE C-68. VIDS BENCH TEST SET, 100-601165-000 (SHEET 1 OF 2)

DESCRIPTION: (CONT)

The unit tester provides power, inputs, switches, controls, indicators, and a digital monitor to operate and test each unit of the VIDS independently. A test point field on the unit tester front panel permits measurement of power supply and regulated voltages.

POWER REQUIREMENTS: 115 VAC, 400 Hz ± 5 Hz
28 VDC, ± 0.5 VDC

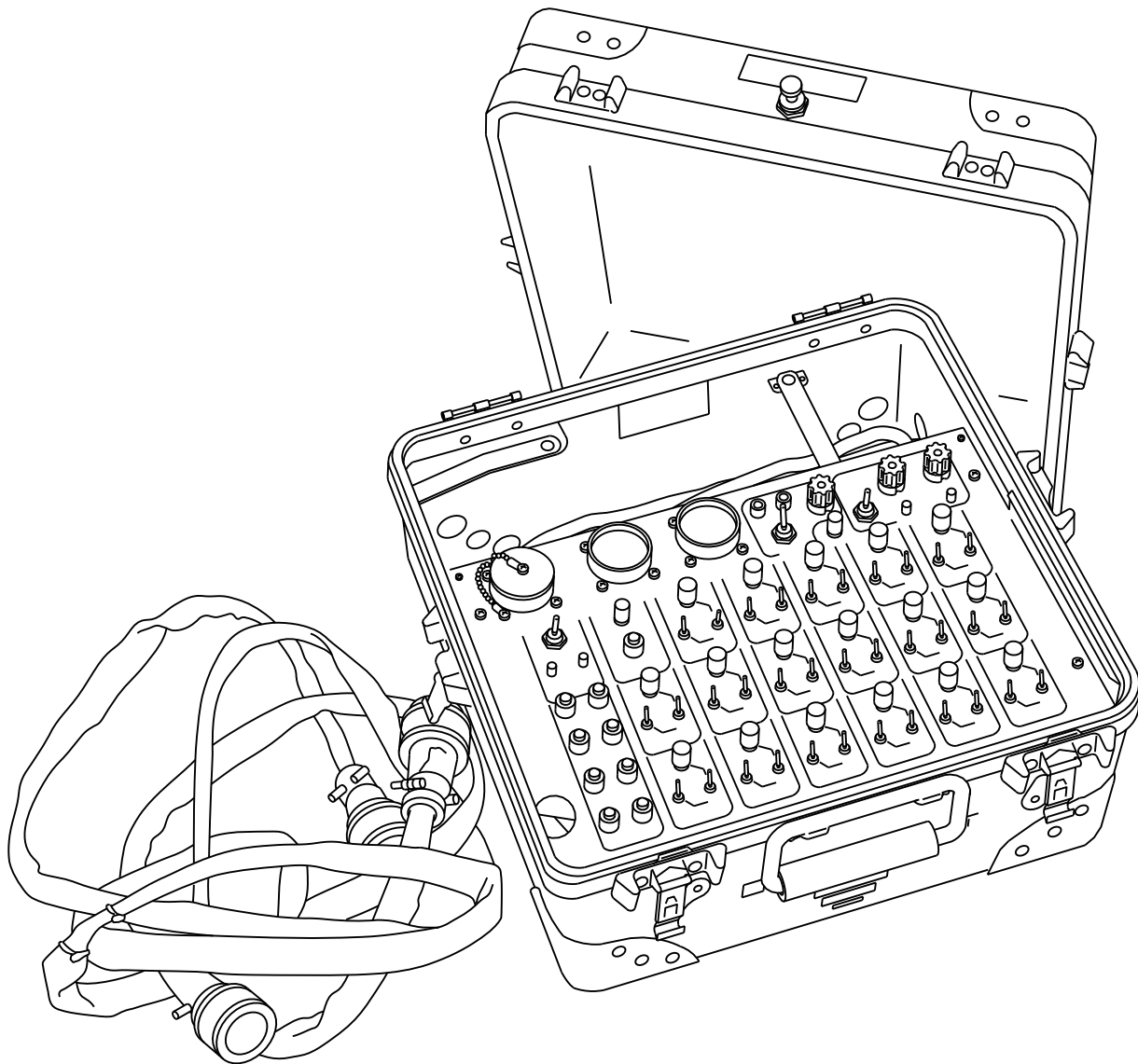
CALIBRATION REQUIREMENTS:**FREQUENCY**

The VIDS bench test set must be calibrated every 24 weeks regardless of being used or not, using same calibration procedure as in Figure 69, VIDS test set.

FB1136_2
SA

FIGURE C-68. VIDS BENCH TEST SET, 100-601165-000 (SHEET 2 OF 2)

Change 10**C-113**



DESCRIPTION: The VIDS line test set provides the necessary power, simulated signals, and signal control in order to operate and test the Vertical Instrument Display System (VIDS). It also provides the VIDS with simulated signals for all indicators.

POWER REQUIREMENTS: 115 Volts AC ± 1.0 , Single phase 400 ± 5.0 Hz
28 Volt DC ± 0.5

FB1046_1
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 1 OF 12)

CALIBRATION REQUIREMENTS:**FREQUENCY**

The VIDS line test set must be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

- Hewlett Packard 5300 Series Counter, frequency .001%
- Fluke 8500 or 8800 Series Digital Multimeter, DCV 0.01%, ACV 1%, OHMS 0.1%
- North Atlantic 225 Phase Angle Voltmeter (PAV), phase sensitive nulls @ 400 Hz, voltage 1%
- Gertsch RT7 Ratotrans, 400 Hz ratio 0.001%
- Ectron 1100 Thermocouple Simulator / Calibrator, 40 mV nominal 0.05% CH / AL thermocouple
- Sikorsky TS-1028 400 Hz Filter or CMS T-750477 Oil Pressure Excitation Test Box, Simulate sensor
- Thermocouple lead wire Ch to Cu, A1 to Cu, approximate 4' length
- Variac 400 Hz, adjust voltage
- Sikorsky TS-1058 VIDS Test Set, provide connections
- Sikorsky TCDC-005 Test Cable, provide bench 28 VDC
- Sikorsky TC400-013 Test Cable, provide bench 115V, 400 Hz
- Oscilloscope Monitor, waveforms only
- DC Power Supply adjustable 5 VDC with 25 mA current limit, lamp test
- Cables and adaptors as required, make connections

CALIBRATION PROCEDURES**INSPECT**

Check test set for damaged and missing knobs, switches, dirt or any physical damage.

CALIBRATION

Connect the VIDS TEST SET power cables to bench 28 VDC and 115V 400 Hz power using TCDC-005 Test Cable and TC400-013 Test Cable respectively.

Set VIDS TEST SET INPUT POWER to ON position and then note 115 VAC and 28 VDC indicator lamps illuminate.

Connect TS-1058 VIDS TEST SET to TEST SET No. 1 connector with VIDS TEST SET to SDC cable.

Press VIDS TEST SET LAMP TEST and then note the nine (9) WARNING lamps and two (2) SUPPLY OVERLOAD lamps illuminate.

Connect the powered equipment to bench power, set POWER to ON position and then allow at least a thrity (30) minute warmup.

ROTOR SPEED

1. Connect COUNTER and OSCILLOSCOPE inputs to VIDS TEST SET SIGNAL bnc connector.
2. Set VIDS TEST SET ROTOR SPEED controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Set VIDS TEST SET SIGNAL switch to 1 position.
4. Set VIDS TEST SET ROTOR SPEED controls and note outputs are within specified tolerance of Table 1.

FB1046_2
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 2 OF 12)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURE (CONT)
ROTOR SPEED (CONT)

TABLE 1

NOR/CAL	HI/LO	VARIABLE	COUNTER (Hz)
CAL	HI	full CW	11,245.1 to 11,247.1
CAL	LO	full CW	0 (no output)
NOR	LO	full CW	12,200 to 18,000
NOR	LO	full CCW	0 (no output)

5. Leave set-up for next test.

ENGINE % RPM #1 and #2

1. Set VIDS TEST SET SIGNAL switch to 2 position.
2. Set VIDS TEST SET ENGINE % RPM #1 controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Set VIDS TEST SET ENGINE % RPM #1 controls and note outputs are within specified tolerance of Table 2.

TABLE 2

NOR/CAL	HI/LO	VARIABLE	COUNTER (Hz)
CAL	HI	full CW	1,332.5 to 1,334.5
CAL	LO	full CW	0 (no output)
NOR	LO	full CW	1,830 to 2,030
NOR	LO	full CCW	0 (no output)

4. Set VIDS TEST SET SIGNAL to position 3.
5. Repeat from step (2) for ENGINE % RPM #2.
6. Leave set-up for next test.

GAS GEN SPEED #1 and #2

1. Set VIDS TEST SET SIGNAL switch to 4 position.
2. Set VIDS TEST SET GAS GEN SPEED #1 controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Set VIDS TEST SET GAS GEN SPEED #1 controls and note outputs are within specified tolerance of Table 3.

FB1046_3
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 3 OF 12)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURE (CONT)**
GAS GEN SPEED #1 and #2 (CONT)**TABLE 3**

NOR/CAL	HI/LO	VARIABLE	COUNTER (Hz)
CAL	HI	full CW	2,134.3 to 2,136.3
CAL	LO	full CW	0 (no output)
NOR	LO	full CW	2,400 to 2,480
NOR	LO	full CCW	0 (no output)

4. Move cable from VIDS TEST SET No. 1 and then reconnect to #2.

5. Repeat from Step (2) for GAS GEN SPEED #2.

6. Remove the COUNTER and OSCILLOSCOPE from the set-up.

TORQUE #1 and #2

1. Set VIDS TEST SET SIGNAL switch to 5 position.

2. Connect DIGITAL MULTIMETER input to VIDS TEST SET SIGNAL bnc.

3. Set VIDS TEST SET TORQUE #1 controls:

a. NOR / CAL to CAL

b. HI / LO to HI

c. variable to full CW

4. Set VIDS TEST SET TORQUE #1 controls and note outputs are within specified tolerance of Table 4.

TABLE 4

NOR/CAL	HI/LO	VARIABLE	DIGITAL MULTIMETER (DCV)
CAL	HI	full CW	3.6963 to 3.7037
CAL	LO	full CW	-1 to +1 mV
NOR	LO	full CW	5.4150 to 5.8150
NOR	LO	full CCW	-3 to +3 mV

5. Set VIDS TEST SET SIGNAL to 6 position.

6. Repeat from Step (3) for TORQUE #2.

7. Leave set-up for the next test.

FUEL QUANTITY #1 and #2

1. Move VIDS TEST SET cable from No. 2 to No. 1 connector.

2. Set VIDS TEST SET SIGNAL switch to 7 position.

FB1046_4
SA**FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 4 OF 12)**

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURE (CONT)**
FUEL QUANTITY #1 and #2 (CONT)**3. Set VIDS TEST SET FUEL QUANTITY #1 controls:**

- a. NOR / CAL to CAL
- b. HI / LO to HI
- c. variable to full CW

4. Set VIDS TEST SET FUEL QUANTITY #1 controls and note outputs are within specified tolerance of Table 5.**TABLE 5**

NOR/CAL	HI/LO	VARIABLE	DIGITAL MULTIMETER (DCV)
CAL	HI	full CW	6.120 to 6.130
CAL	LO	full CW	-1 to +1 mV
NOR	LO	full CW	7.630 to 8.030
NOR	LO	full CCW	-10 to +10 mV

5. Move VIDS TEST SET cable from No. 1 to No. 2 connector.**6. Repeat from Step (3) for FUEL QUANTITY #2.****7. Leave set-up for the next test.****ENGINE OIL TEMP #1 and #2****1. Move VIDS TEST SET cable from No. 2 to No. 1 connector.****2. Set VIDS TEST SET SIGNAL switch to 8 position.****3. Set VIDS TEST SET ENGINE OIL TEMP #1 controls:**

- a. NOR / CAL to CAL
- b. HI / LO to HI
- c. variable to full CW

4. Set VIDS TEST SET ENGINE OIL TEMP #1 controls and note outputs are within specified tolerance of Table 6.**TABLE 6**

NOR/CAL	HI/LO	VARIABLE	DIGITAL MULTIMETER (OHMS)*
CAL	HI	full CW	150.2 to 151.0
CAL	LO	full CW	75.3 to 75.9
NOR	LO	full CW	170.0 to 190.0
NOR	LO	full CCW	73.6 to 77.6

* Compensate for lead resistance.

5. Move VIDS TEST SET cable from No. 1 to No. 2 connector.**6. Repeat from Step (3) for ENGINE OIL TEMP #2.****7. Leave set-up for the next test.**FB1046_5
SA**FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 5 OF 12)**

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURE (CONT)**
MAIN TRANSMISSION OIL TEMP

1. Set VIDS TEST SET SIGNAL switch to 9 position.
2. Set VIDS TEST SET MAIN TRANSMISSION OIL TEMP controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Set VIDS TEST SET MAIN TRANSMISSION OIL TEMP controls and note outputs are within specified tolerance of Table 7.

TABLE 7

NOR/CAL	HI/LO	VARIABLE	DIGITAL MULTIMETER (OHMS)*
CAL	HI	full CW	234.16 to 234.96
CAL	LO	full CW	86.70 to 87.30
NOR	LO	full CW	270.0 to 290.0
NOR	LO	full CCW	84.0 to 90.0

* Compensate for lead resistance.

ENGINE OIL PRESS #1 and #2

1. Set VIDS TEST SET controls:
 - a. SIGNAL to OFF
 - b. ENG / XSMN to ENG
2. Set VIDS TEST SET ENGINE OIL PRESS #1 controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Connect the test set-up shown in View A.
4. Adjust the VARIAC for approximately 10 VAC Digital Multimeter indication.
5. Set PAV to INPHASE and AUTO ranging modes.
6. Adjust the RATIOTRANS as required for PAV INPHASE null indication and note setting is within 0.48467 and 0.48667.
7. Set PAV to QUAD and then note reading is less than 42 mV.
8. Set VIDS TEST SET ENGINE OIL PRESS #1 HI / LO to LO.
9. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.59751 and 0.59951.
10. Set PAV to QUAD and then note reading is within -148 and -178 mV.
11. Set VIDS TEST SET ENGINE OIL PRESS #1 CAL / NOR to NOR.

FB1046_6
SA**FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 6 OF 12)**

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURE (CONT) ENGINE OIL PRESS #1 and #2 (CONT)

12. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.39900 and 0.40100.
13. Set VIDS TEST SET ENGINE OIL PRESS #1 variable to full CCW.
14. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.59751 and 0.59951.
15. Move VIDS TEST SET cable from No. 1 to No. 2 connector.
16. Repeat from Step (4) for ENGINE OIL PRESSURE #2.
17. Leave set-up for the next test.

TRANSMISSION OIL PRESSURE

1. Set VIDS TEST SET ENG / XSMN switch to XSMN position.
2. Set VIDS TEST SET TRANSMISSION OIL PRESSURE controls:
 - a. NOR / CAL to CAL
 - b. HI / LO to HI
 - c. variable to full CW
3. Set PAV to INPHASE and AUTO ranging modes.
4. Adjust the RATIOTRANS as required for PAV INPHASE null indication and note setting is within 0.46015 and 0.46215.
5. Set PAV to QUAD and then note reading is less than 42 mV.
6. Set VIDS TEST SET TRANSMISSION OIL PRESSURE HI / LO to LO.
7. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.59751 and 0.59951.
8. Set PAV to QUAD and then note reading is within -148 and -178 mV.
9. Set VIDS TEST SET TRANSMISSION OIL PRESSURE CAL / NOR to NOR.
10. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.39900 and 0.40100.
11. Set VIDS TEST SET TRANSMISSION OIL PRESSURE variable to full CCW.
12. Set PAV to INPHASE and adjust the RATIOTRANS as required for PAV null indication and note setting is within 0.59751 and 0.59951.

WARNING LAMPS

1. Connect VIDS TEST SET to connector No. 1.
2. Set VIDS TEST SET LAMPS switch to ON and note listed lamps illuminate:
 - a. LOW OIL PRESS #1
 - b. HI OIL TEMP #1
 - c. LO ROTOR SPEED
 - d. ENG OUT No. 1
 - e. HI TGT #1

FB1046_7A
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 7 OF 12)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURE (CONT)**
WARNING LAMPS (CONT)

3. Move VIDS TEST SET cable from No. 1 to No. 2 connector.
4. Repeat Step (2) for #2 lamps.

TURBINE GAS TEMP #1 AND #2

1. Set VIDS TEST SET INTEGRAL LIGHTING switch to INT position.
2. Connect OSCILLOSCOPE and Digital Multimeter input across VIDS TEST SET ILL binding posts.
3. Hold VIDS TEST SET ILL toggle switch up and then note OSCILLOSCOPE display of a symmetrical and noise free sine wave of 11.3 to 17.0 Vpp amplitude.
4. Release the VIDS TEST SET ILL switch.
5. Connect test set-up shown in View B.
6. Set VIDS TEST SET TURBINE GAS TEMP #1 controls:
 - a) NOR / CAL to CAL
 - b) HI / LO to HI
 - c) variable to full CW
7. Set VIDS TEST SET TURBINE GAS TEMP #1 controls, measure output and note value is within specified tolerance of Table 8.

TABLE 8

NOR/CAL	HI/LO	VARIABLE	DIGITAL MULTIMETER (mVDC)
CAL	HI	full CW	33.69 to 33.89
CAL	LO	full CW	-0.05 to +0.05
NOR	LO	full CW	41.3 to 51.3
NOR	LO	full CCW	-0.15 to +0.15

8. Move VIDS TEST SET cable from No. 1 to No. 2 connector.
9. Repeat Step 6 for #2.

VOLTAGE MEASUREMENTS

1. On initial checks or for troubleshooting perform the measurements of APPENDIX A.

CONTINUITY MEASUREMENTS

1. On initial checks or for troubleshooting perform the measurements of APPENDIX B.

OVERLOAD LAMPS

1. On initial checks or for troubleshooting perform the measurements of APPENDIX C.

Set all POWER to OFF.

Disconnect the set-up.

FB1046_8
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 8 OF 12)

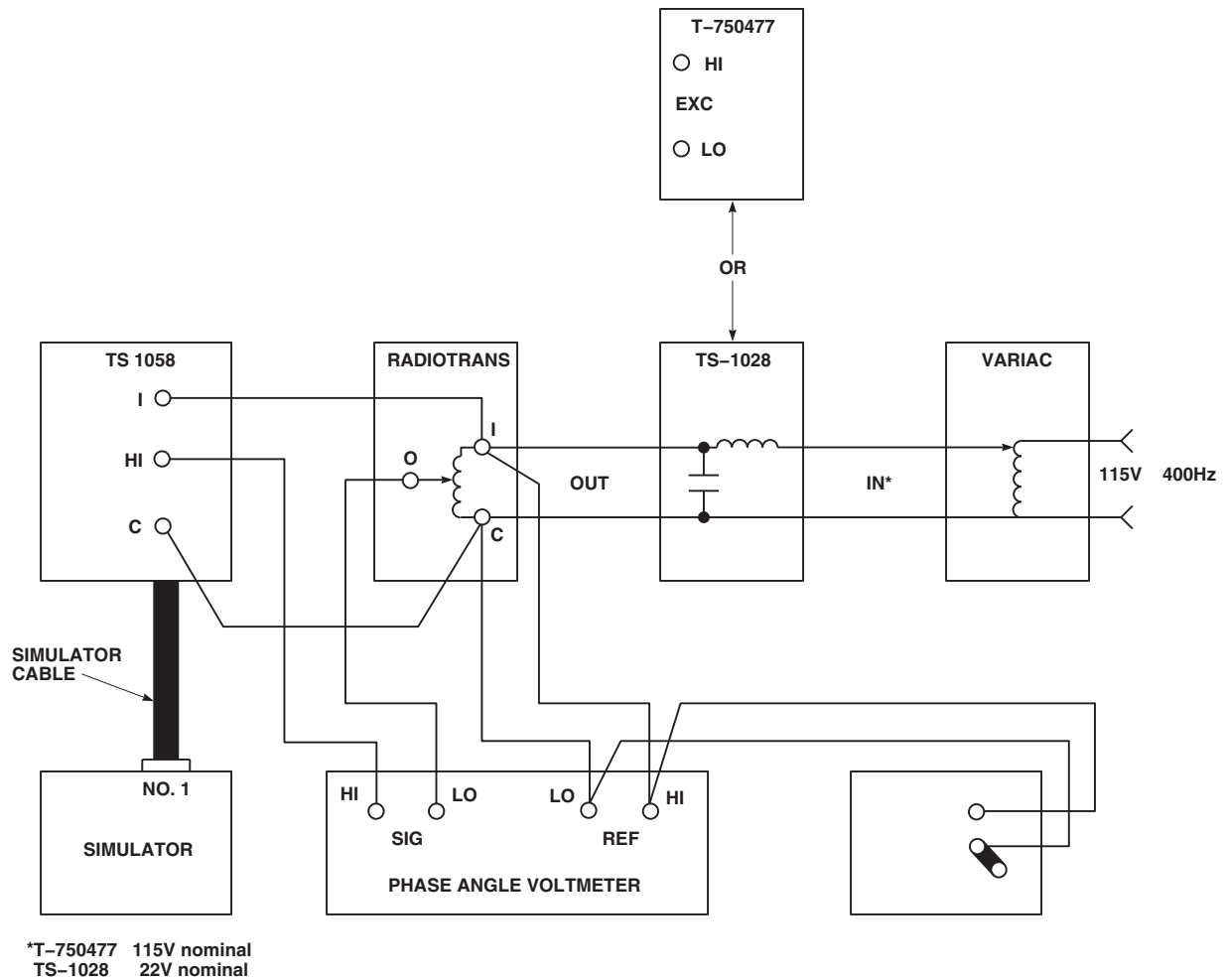
CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURE (CONT)
OVERLOAD LAMPS (CONT)

Secure all test equipment.

Apply "CALIBRATED" status label.

If VIDS TEST SET failed calibration, return to manufacturer.

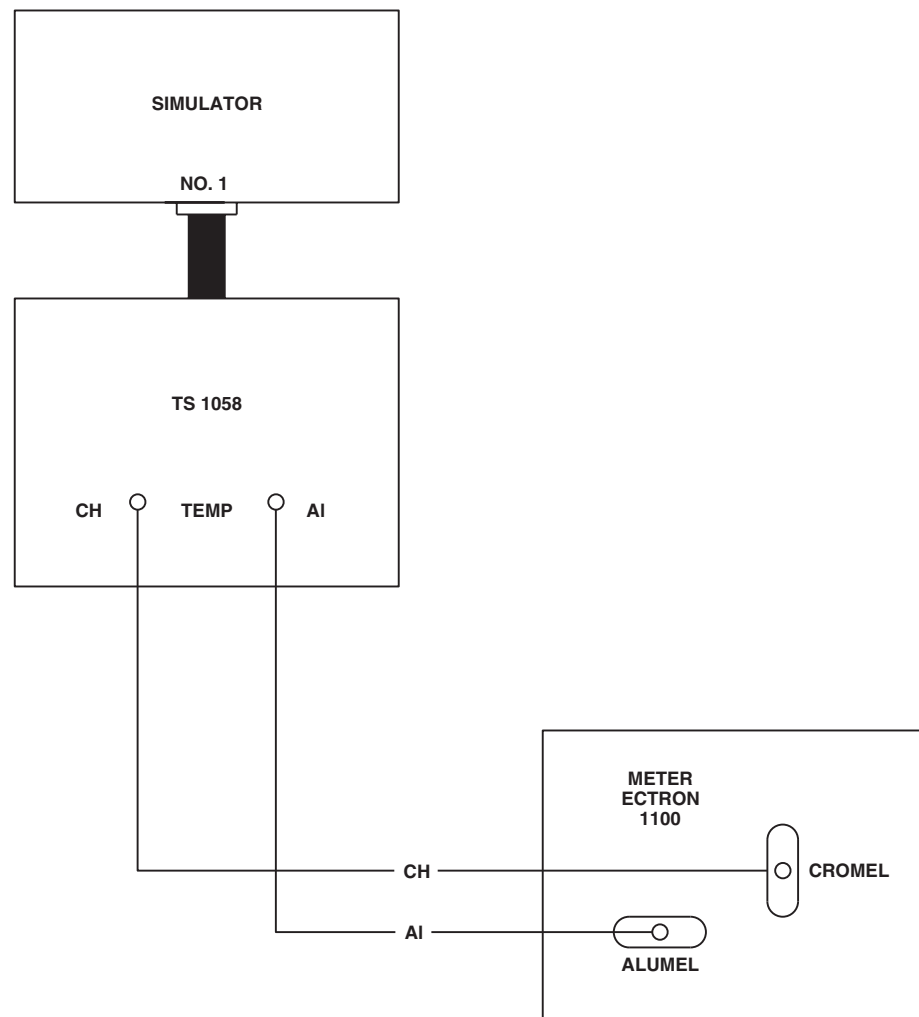


VIEW A
ENGINE OIL PRESSURE SET-UP

FB1046_9
 SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 9 OF 12)

CALIBRATION REQUIREMENTS: (CONT)
CALIBRATION PROCEDURE (CONT)



VIEW B
TURBINE GAS TEMPERATURE SET-UP

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 10 OF 12)

FB1046_10
 SA

CALIBRATION REQUIREMENTS: (CONT)
CALIBRATION PROCEDURE (CONT)

CONNECTIONS*		
HI	LO	MEASURE
P3-V	No. 1-EE	26 to 30 VDC (LAMP TEST depressed)
No. 1-AA	No. 1-BB	26 to 30 VDC
No. 2-AA	No. 2-BB	26 to 30 VDC
P3-M	P3-L	26 to 30 VDC
P3-M	P3-K	26 to 30 VDC
No. 1-CC	No. 1-DD	110 to 120 VAC
No. 2-CC	No. 2-DD	110 to 120 VAC
P3-P	P3-N	110 to 120 VAC**
No. 1-FF	No. 1-HH	4 to 6 VAC
No. 1-GG	No. 1-HH	4 to 6 VAC
No. 2-FF	No. 2-GG	4 to 6 VAC
No. 2-FF	No. 2-HH	4 to 6 VAC
P3-R	P3-S	4 to 6 VAC

Set INTEGRAL LIGHTING to INT.

* P3 is INTERCONNECT

** POWER-ON (0 VAC when POWER-SIM only)

APPENDIX A
VOLTAGE MEASUREMENTS

FB1046_11
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 11 OF 12)

CALIBRATION REQUIREMENTS: (CONT)
CALIBRATION PROCEDURE (CONT)

CONNECTIONS*		
HI	LO	CONDITIONS FOR CONTINUITY
No. 1-EE	P3-U	LAMP TEST depressed
No. 1-p	No. 1-EE	SIMULATOR-OFF ROTOR OVERSPEED RESET depressed
No. 2-p	No. 2-EE	SIMULATOR-OFF ROTOR OVERSPEED RESET depressed
P3-c	No. 2-j	- - - - -
P3-d	No. 2-W	- - - - -
P3-g	No. 2-H	- - - - -
P3-h	No. 2-G	- - - - -
P3-i	No. 1-G	- - - - -
P3-j	No. 1-H	- - - - -
P3-k	No. 1-i	- - - - -
P3-m	No. 1-W	- - - - -
P3-n	No. 2-j	- - - - -
P3-v	No. 2-K	- - - - -
P3-w	No. 2-J	- - - - -

* P3 is INTERCONNECT

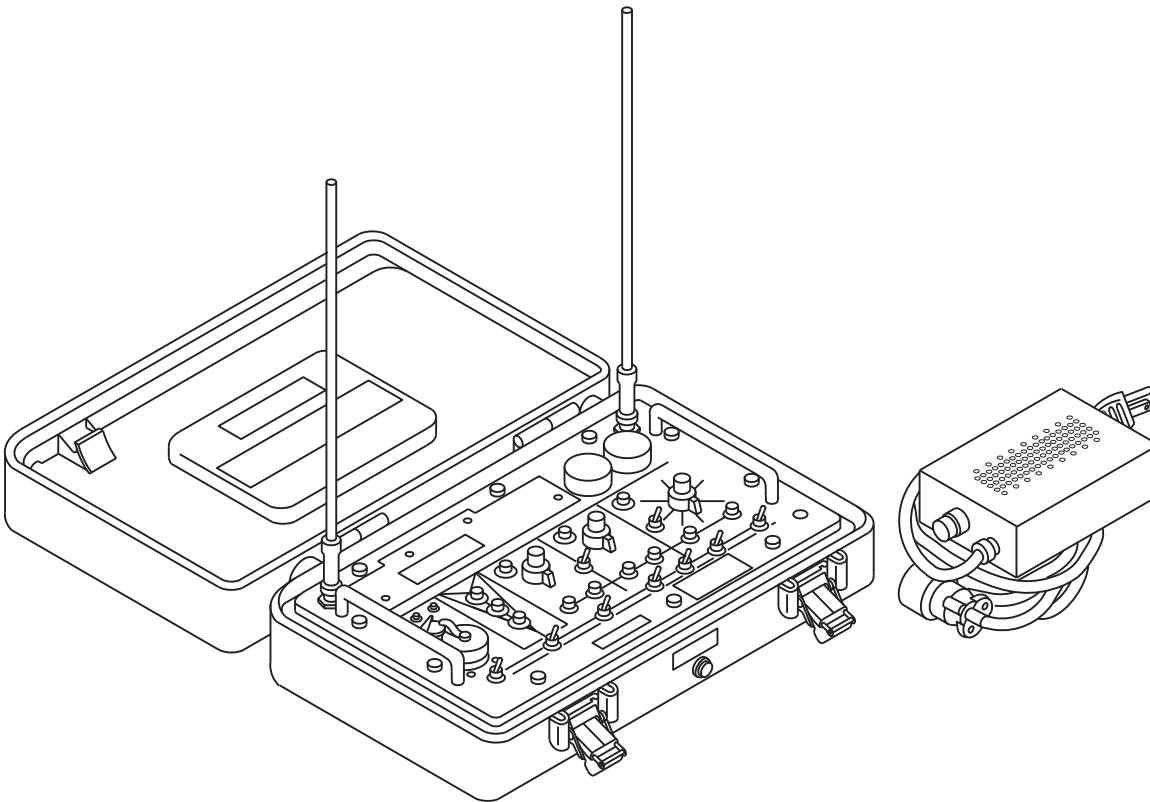
APPENDIX B
CONTINUITY MEASUREMENTS

APPENDIX C
OVERLOAD LAMPS

1. Connect POWER SUPPLY, set for 5 VDC with 25 mA current limit, across No. 1 pins U and EE (Io).
2. Note LAMP SUPPLY OVERLOAD No. 1 illuminates.
3. Press OVERLOAD switch to the left and note lamp extinguishes.
4. Repeat from (1) for No. 2 pins and pressing the switch to the right to extinguish the lamp.

FB1046_12
SA

FIGURE C-69. VIDS LINE TEST SET, 100-601164-000 (SHEET 12 OF 12)



DESCRIPTION: The VOR / ILS / MB / RAMP TEST SET, 972Q-4 provides outputs that allow checking of the VOR, ILS, and MB (marker beacon) avionic systems while they are installed in the helicopter.

POWER REQUIREMENTS: 115 volt AC or 230 volt AC, and 8 volt DC rechargeable battery.

CALIBRATION REQUIREMENTS:

FREQUENCY

VOR / ILS / MB ramp test set must be calibrated every 48 weeks regardless of being used or not.

FB1047_1A
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 1 OF 11)

CALIBRATION REQUIREMENTS: (CONT)**EQUIPMENT REQUIRED**

- Hewlett Packard 8902A Measuring Receiver no direct substitute
- Hewlett Packard 11722A or 11792A Sensor used with item 1
- Aircraft Radio Type H16 Standard Course Checker VOR bearing 0.1 deg
- Sikorsky TS-2000 VOR Test Box
- Dranetz 305 / 3007 Phasemeter 30 Hz phase 0.1 deg
- Sikorsky TS-1034 Antenna Coupler couple antenna RF to item 1
- Fluke 8000A Dmm VDC 0.1% VAC 0.5%
- Sikorsky TS-6442 Power Supply 15 vdc for item 4
- Khron Hite 6900 Distortion Analyzer audio frequency distortion

CALIBRATION PROCEDURES**INSPECTION**

- Inspect Test Set / Battery Charger for physical damage, (switches, antenna's, etc.), and repair as required.

CALIBRATION

1. Set all TEST SET POWER switches to ON (battery operation).

NOTE:

GREEN BATTERY lamp shall be illuminated for V+ AND V- positions of the CHECK switch.

For TEST SETS with lighted GLIDE SLOPE, LOC and VOR VAR buttons only, depress and note flashing indicator. Depress to set OFF after test.

2. Set all TEST SET POWER switches to OFF.

- Connect test set-ups as shown in Views A or B and View C.

NOTE:

View C set-up is used for VOR module only.

- Set the TEST SET ATTEN to 0 db position.

NOTE:

TEST SET operation from either internal batteries or BATTERY CHARGER is acceptable. Re-charge batteries as required after use.

MKR BCN MODULE

1. Connect TEST SET J1 (MKR BCN) output to SENSOR with a short cable.
2. Connect RECEIVER MODULATION OUTPUT to DISTORTION ANALYZER input.
3. Set PWR and MKR BCN power switch in the ON position.
4. Press on RECEIVER:
 - a. (BLUE) , (AUTOMATIC OPERATION)
 - b. (FREQ) , (7), (5), (MHz)
5. Press RECEIVER (RF POWER) and LOG / LIN for dBm scaling.
6. Note reading is -20dBm minimum.
7. Depress RECEIVER (FREQ) button.

FB1047_2
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 2 OF 11)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES: (CONT)

CALIBRATION (CONT)

MKR BCN MODULE (CONT)

8. Note RECEIVER 74.99625 to 75.00375 MHz indication.
9. Depress and hold TEST SET MKR BCN WHITE button.
10. Press RECEIVER (AM) and (PEAK $\pm$ / 2) buttons and note 90 to 100% indication.
11. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 2940 to 3060 Hz indication.
12. Note DISTORTION ANALYZER distortion indication of 15% maximum.
13. Depress and hold TEST SET MKR BCN BLUE button.
14. Press RECEIVER (AM) button and note 90 to 100% indication.
15. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 392 to 408 Hz indication.
16. Note DISTORTION ANALYZER distortion indication of 15% maximum.
17. Depress and hold TEST SET MKR BCN YELLOW button.
18. Press RECEIVER (AM) button and note 90 to 100% indication.
19. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 1274 to 1326 Hz indication.
20. Note DISTORTION ANALYZER distortion indication of 15% maximum.
21. Set all TEST SET MKR BCN switches to OFF and disconnect SENSOR.

LOC MODULE

1. Connect TEST SET J3 (GS/VOR/LOC) output to SENSOR with a short cable.
2. Connect RECEIVER MODULATION OUTPUT to DISTORTION ANALYZER input.
3. Set PWR and LOC power switch in ON position and FREQ SW to 108.1.
4. Press on RECEIVER:
 - a. (BLUE) , (AUTOMATIC OPERATION)
 - b. (FREQ), (1), (0), (8), (.), (1), (MHz)
5. Press RECEIVER (RF POWER) and LOG/LIN for dBm scaling.
6. Note reading is -10 dBm minimum.
7. Depress RECEIVER (FREQ) button.
8. Note RECEIVER 108.0973 to 108.1027 MHz indication.
9. Set LOC FREQ SW to 108.15 position.
10. Press on RECEIVER if required to tune:
 - a. (FREQ), (1), (0), (8), (.), (1), (5), (MHz)
11. Note RECEIVER 108.1473 to 108.1527 MHz indication.

FB1047_3
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 3 OF 11)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****CALIBRATION (CONT)****LOC MODULE (CONT)**

12. Set LOC to OC and hold 150 Hz DELETE ON.
13. Press RECEIVER (AM) and (PEAK $\pm$ / 2) buttons and note approximate 20% indication. Record value.
14. Press the RECEIVER (SHIFT), (AUDIO FREQ) and note 89.1 to 90.9 Hz indication.
15. Note DISTORTION ANALYZER indication of 5% maximum.
16. Release 150 Hz DELETE and hold 90 Hz DELETE to ON.
17. Press RECEIVER (AM) button and note approximate 20% indication. Record value.
18. Subtract Step (13) from (17) and note computed value is 0 within 0.5.
19. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 148.5 to 151.5 Hz indication.
20. Note DISTORTION ANALYZER indication of 5% maximum.
21. Press RECEIVER (AM) button.
22. Set LOC COURSE switch, hold DELETE switch ON and note % AM indication differences are within specified tolerances of TABLE 1.

TABLE 1			
STEP	COURSE	DELETE	%AM
1	LEFT	90 Hz	28 Nominal
2	LEFT	150 Hz	12 Nominal
Step 1 minus Step 2 = 13.5 to 17.5			
3	RIGHT	150 Hz	28 Nominal
4	RIGHT	90 Hz	12 Nominal
Step 3 minus Step 4 = 13.5 to 17.5			

23. Press RECEIVER 300 Hz HP AND 3 KHz LP FILTERS to ON (all other filters OFF).
24. Hold 90 Hz and 150 Hz DELETE switches ON.
25. Set 1020 Hz to ON and note 5 to 20% indication.
26. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 995 to 1045 Hz indication (DELETE switches held ON).
27. Note DISTORTION ANALYZER indication of 10% maximum (DELETE switches held ON).
28. Set all TEST SET LOC switches to OFF and disconnect SENSOR.

FB1047_4
SA**FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 4 OF 11)**

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES: (CONT)

CALIBRATION (CONT)

GS MODULE

1. Connect TEST SET J3 (GS / VOR / LOC) output to the SENSOR with a short cable.
2. Connect RECEIVER MODULATION OUTPUT to DISTORTION ANALYZER input.
3. Set PWR and GLIDESLOPE power switch in the ON position and FREQ. SW to 334.70.
4. Press on RECEIVER:
 - a. (BLUE), (AUTOMATIC OPERATION)
 - b. (FREQ), (3), (3), (4), (.), (7), (MHz)
5. Press RECEIVER (RF POWER) and LOG / LIN for dBm scaling.
6. Note reading is -20 dBm minimum.
7. Depress RECEIVER (FREQ) button.
8. Note RECEIVER 334.6916 TO 334.7084 MHz indication.
9. Set GLIDESLOPE FREQ SW to 334.55 position.
10. Press on RECEIVER if required to tune:
 - a. (FREQ), (3), (3), (4), (.), (5), (5), (MHz)
11. Note RECEIVER 334.5416 TO 334.5584 MHz indication.
12. Press RECEIVER (RF POWER) and LOG / LIN for dBm scaling.
13. Note reading is -20 dBm minimum.
14. Set GLIDESLOPE to OC and hold 150 Hz DELETE ON.
15. Press RECEIVER (AM) and (PEAK $\pm$ / 2) buttons and note approximate 40% indication. Record value.
16. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 89.1 to 90.9 Hz indication.
17. Note DISTORTION ANALYZER indication of 5% maximum.
18. Release 150 Hz DELETE and hold 90 Hz DELETE to ON.
19. Press RECEIVER (AM) button and note approximate 40% indication. Record value.
20. Subtract Step (15) from (19) and note computed value is 0 within 1.
21. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 148.5 to 151.5 Hz indication.
22. Press RECEIVER (AM) button.

FB1047_5A
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 5 OF 11)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****CALIBRATION (CONT)****GS MODULE (CONT)**

23. Set GLIDESLOPES COURSE switch, hold DELETE switch ON and note % AM indication differences within specified tolerances of TABLE 2.

TABLE 2			
STEP	COURSE	DELETE	%AM
1	UP	90 Hz	49 Nominal
2	UP	150 Hz	31 Nominal
Step 1 minus Step 2 = 15 to 20			
3	DOWN	150 Hz	49 Nominal
4	DOWN	90 Hz	31 Nominal
Step 3 minus Step 4 = 15 to 20			

24. Set all TEST SET GLIDESLOPE switches to OFF and disconnect SENSOR.

VOR MODULE

1. Connect TEST SET J3 (GS / VOR / LOC) output to SENSOR with a short cable.
2. Connect RECEIVER MODULATION OUTPUT to DISTORTION ANALYZER input.
3. Set PWR and VOR power switch in the ON position and FREQ. SW to 108.00.
4. Press on RECEIVER:
 - a. (BLUE), (AUTOMATIC OPERATION)
 - b. (FREQ), (1), (0), (8), (MHz)
5. Press RECEIVER (RF POWER) and LOG / LIN for dBm scaling.
6. Note reading is -10 dBm minimum.
7. Depress RECEIVER (FREQ) button.
8. Note RECEIVER 107.9973 to 108.0027 MHz indication.
9. Set VOR FREQ SW to 108.05 position.
10. Press on RECEIVER if required to tune:
 - a. (FREQ), (1), (0), (8), (.), (0), (5), (MHz)
11. Note RECEIVER 108.0473 to 108.0527 MHz indication.

FB1047_6A
SA**FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 6 OF 11)**

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES: (CONT)

CALIBRATION (CONT)

VOR MODULE (CONT)

12. Press RECEIVER (AM), (PEAK $\pm$ / 2) buttons and 3 KHz LP FILTER to ON.
13. Note indication of 27.0 to 33.0%.
14. Press RECEIVER (SHIFT), (AUDIO FREQ) and note 29.7 to 30.3 Hz indication.
15. Depress and hold DELETE 30 Hz REF and note DISTORTION ANALYZER reading of 7% maximum.
16. Press RECEIVER AM button, 15 KHz LP AND 300 Hz HP FILTERS to ON (all other filters OFF).
17. Note indication of 27.0 to 33.0%.
18. Depress and hold DELETE 30 Hz REF and 30 Hz VAR and then note DISTORTION ANALYZER reading of 3% maximum.
19. Press RECEIVER (SHIFT), (AUDIO FREQ).
20. Press and hold DELETE 30 Hz REF and 30 Hz VAR and note 9860 to 10060 Hz indication of RECEIVER.
21. Press the RECEIVER 300 Hz HP and 3 KHz LP FILTERS to ON (all other filters OFF).
22. Set 1020 Hz to ON and note 970 to 1070 Hz indication.
23. Press RECEIVER (AM) button and note 5 to 15% indication.
24. Set 1020 Hz switch to OFF.
25. Press RECEIVER AM button, and all FILTERS OFF.
26. Leave present set-up for next test.

VOR MODULE BEARING

1. Add VIEW C set-up if not already done.
2. Set STANDARD COURSE CHECKER controls:
 - a. CAL / DEV to center position
 - b. FUNCTION to SET INPUT (RED LINE)
 - c. READOUT to NORM
 - d. COURSE to ZERO
3. Preset STANDARD COURSE CHECKER INPUT pot approximately 3/4 full CW and adjust VOR TEST BOX GAIN and / or STANDARD COURSE CHECKER INPUT pot for "RED LINE" meter reading on COURSE CHECKER.
4. Set STANDARD COURSE CHECKER FUNCTION to NOTE METER READING and note the reading.
5. Set STANDARD COURSE CHECKER FUNCTION to EQUALIZE 9960 and adjust 9960 pot for indication of Step 4.
6. Set FUNCTION to ADJUST PHASE-BAL and adjust PHASE pot to bring meter indication one-half way from present indication toward zero.
7. Set COURSE to 180 and adjust BALANCE pot to bring meter indication one-half way from present indication toward zero.

FB1047_7
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 7 OF 11)

CALIBRATION REQUIREMENTS: (CONT)**CALIBRATION PROCEDURES: (CONT)****CALIBRATION (CONT)****VOR MODULE BEARING (CONT)**

8. Set COURSE to 0 and repeat from STEP 6 until adjustment is not required.
9. Set STANDARD COURSE CHECKER READOUT to EXT and adjust VOR TEST BOX PHASE control for TEST BOX "CAL PHASE" label reading on the PHASEMETER.
10. Set STANDARD COURSE CHECKER FUNCTION to READ COURSE ERROR.
11. Set VOR BEARING "to" switch and note the PHASEMETER indications are within TABLE 3 tolerances.

TABLE 3

VOR BEARING	PHASEMETER
000	359.3 to 000.7
045	044.0 to 046.0
090	089.0 to 091.0
135	134.0 to 136.0
180	179.0 to 181.0
225	224.0 to 226.0
270	269.0 to 271.0
315	314.0 to 316.0

12. Set STANDARD COURSE CHECKER controls:
 - a. CAL / DEV to CAL
 - b. FUNCTION to EQUALIZE 9960
 - c. 9960 pot to full CW
13. Set VOR BEARING to 0 TO.
14. Adjust VOR TEST BOX GAIN control for exact 1 VAC reading on Dmm.
15. Set STANDARD COURSE CHECKER CAL / DEV to DEV.
16. Record Dmm mVAC indication.
17. Compute Step 16 times TEST BOX label VAC value and note result is 900 to 1020.

NOTE:

Result is p-p deviation in Hz of 9960 Hz carrier.

18. Set all TEST SET VOR switches to OFF and disconnect SENSOR.

FB1047_8

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 8 OF 11)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES: (CONT)

CALIBRATION (CONT)

RF OUTPUT LEVEL

1. Connect TEST SET J3 (GS / VOR / LOC) output to SENSOR with a short cable.
2. Set PWR and LOC power switch in the ON position and FREQ switch to 108.1.
3. Press on RECEIVER:
 - a. (BLUE), (AUTOMATIC OPERATION)
 - b. (FREQ), (1), (0), (8), (.), (1), (MHz)
4. Press RECEIVER (RF POWER) and LOG / LIN for dBm scaling.
5. Note reading is -10 dBm minimum.
6. Press on RECEIVER:
 - a. (SHIFT), (TUNED RF LEVEL)
 - b. (RATIO)
7. RECEIVER display should be 0.000 REL nominal.

NOTE:

Allow time for RECEIVER to display result. If RECAL indication is displayed press CALIBRATE button.

8. Step TEST SET ATTN (0-10 dB /) from 0 to 10 and note display agrees with setting within 0.2 dB / step.
9. Set TEST SET ATTN to 0.
10. Step TEST SET ATTN (0-100 dB /) from 0 to 100 and note display agrees with setting within 2 dB / step.

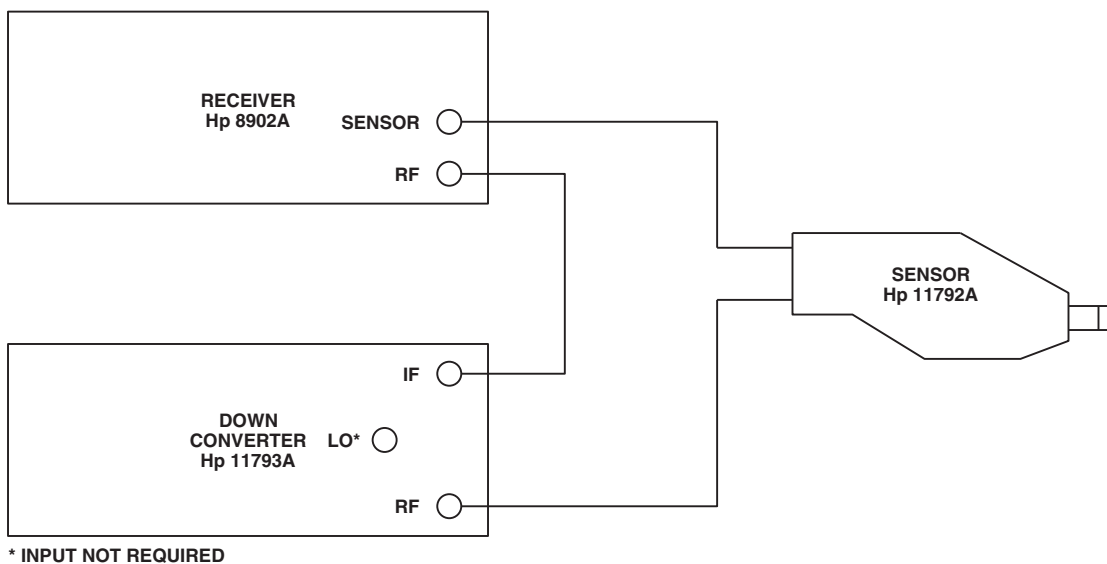
NOTE:

Allow time for RECEIVER to display result at each step. If RECAL indication is displayed press CALIBRATE button.

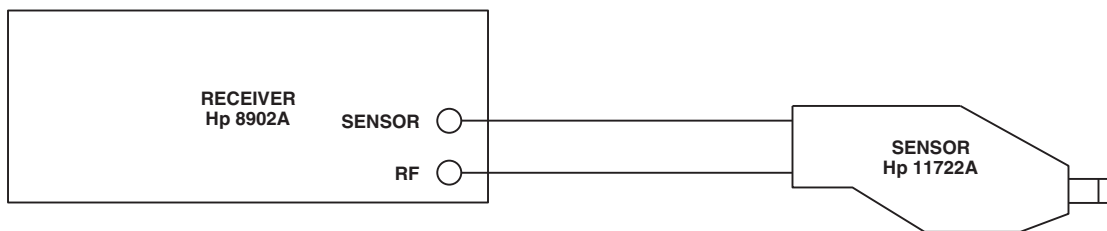
11. Set all TEST SET LOC switches to OFF and disconnect the SENSOR.
12. Set all power to OFF.
13. Secure all test equipment.
14. Apply applicable calibration label.

FB1047_9
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 9 OF 11)



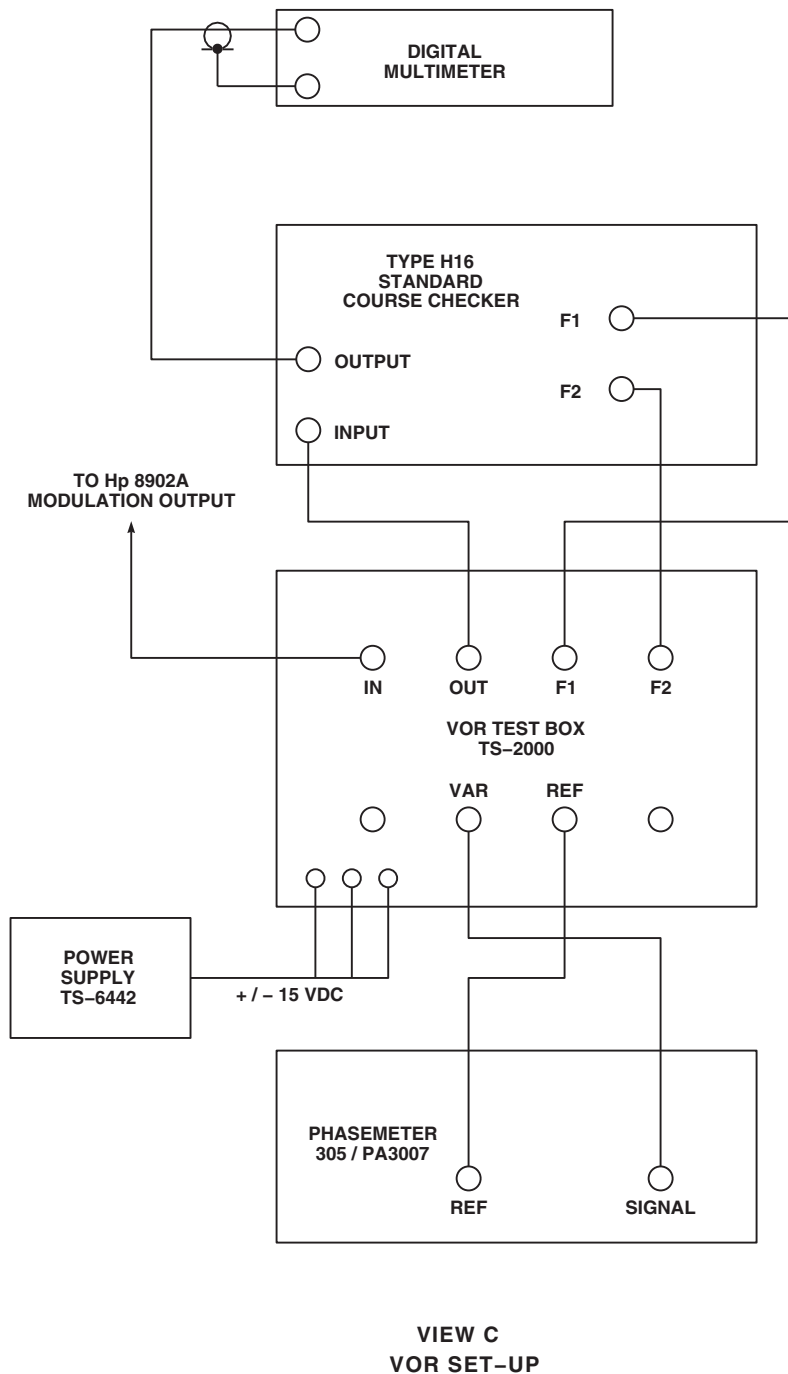
VIEW A
RECEIVER SET-UP



VIEW B
RECEIVER SET-UP

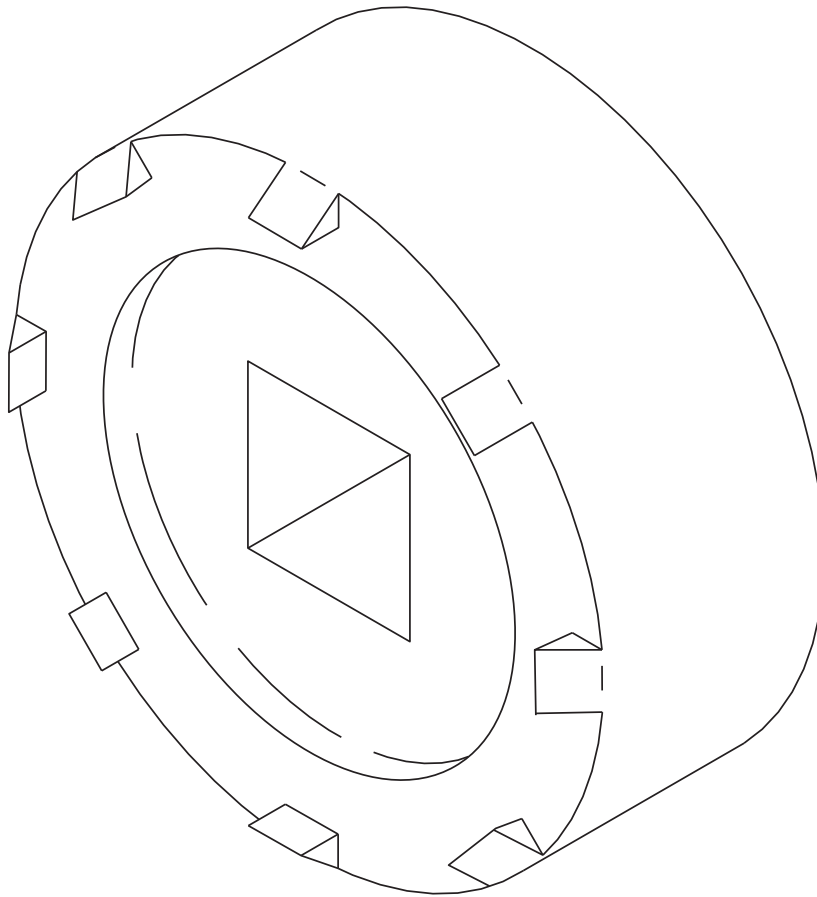
FB1047_10
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 10 OF 11)



FB1047_11
SA

FIGURE C-70. VOR/ILS/MB RAMP TEST SET, 972Q-4 (SHEET 11 OF 11)



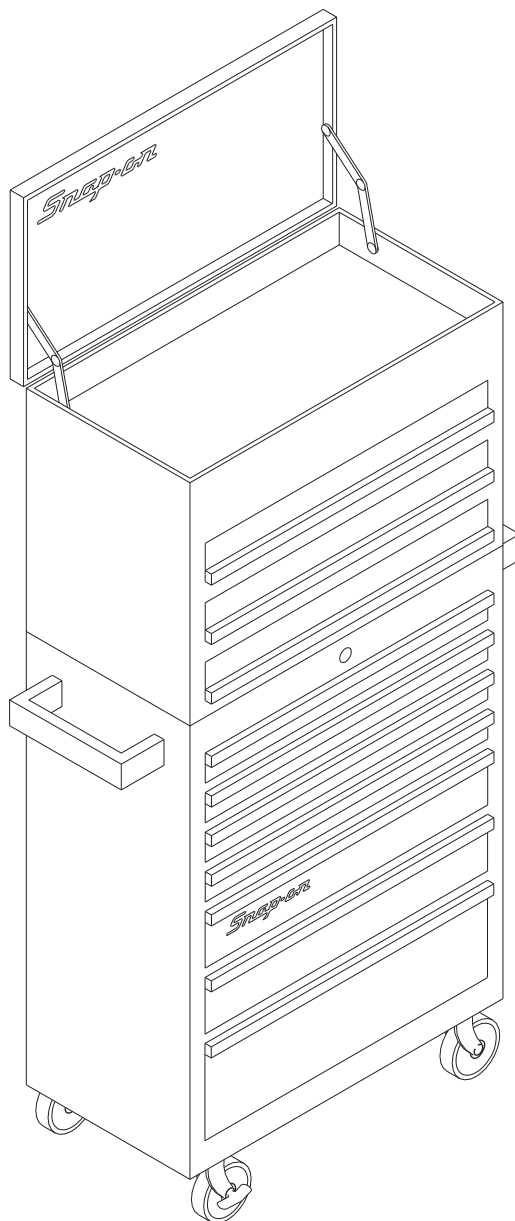
DESCRIPTION: The tail drive flange wrench is used with torque reactor, 70700-20688-102, to remove and install nut that retains the input or output flanges on tail drive shafts, tail gear box input, and rotor brake output flange.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB1048B
SA

FIGURE C-71. TAIL DRIVE FLANGE WRENCH, 70700-20687-102



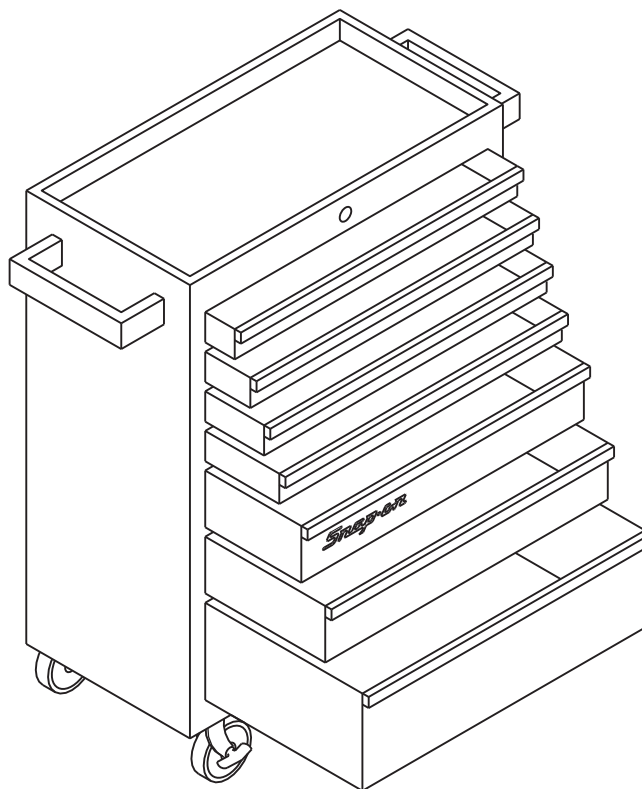
DESCRIPTION: The aircraft mechanic's toolkit contains tools required to support all requirements of the helicopter.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2072A
SA

FIGURE C-72. AIRCRAFT MECHANIC'S TOOLKIT



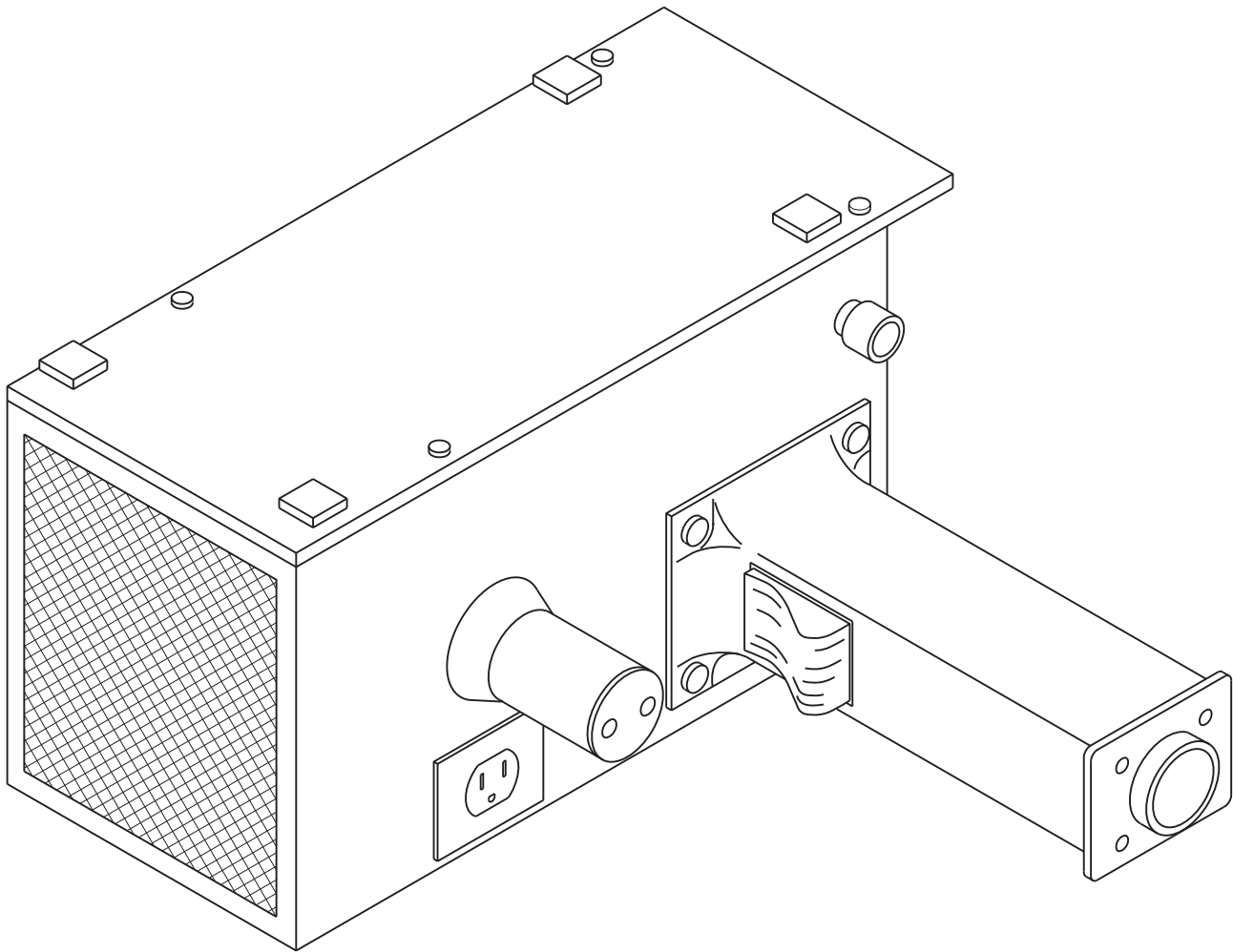
DESCRIPTION: The airframe repairer's toolkit contains tools required to support all requirements to maintain and repair helicopter airframe.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2073A
SA

FIGURE C-73. AIRFRAME REPAIRER'S TOOLKIT, SIK-AF-92



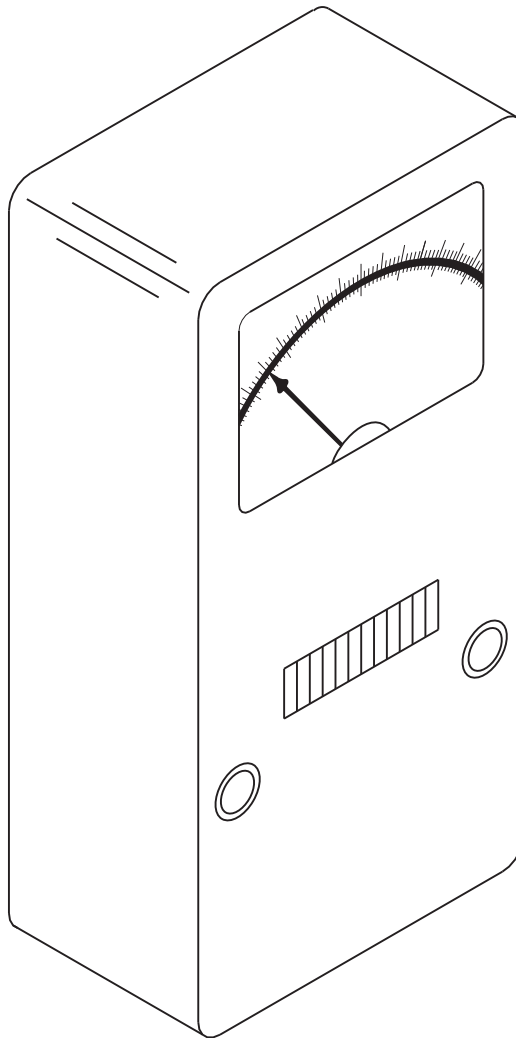
DESCRIPTION: The simulator is a handheld, portable source of pulsed radar frequency energy. Powered by two self-contained rechargeable batteries, a trigger switch on handgrip energizes the simulator.

POWER REQUIREMENTS: Two rechargeable, size D type batteries.

CALIBRATION REQUIREMENTS: Not Applicable

FB2074
SA

FIGURE C-74. SIMULATOR, SM-C-877251



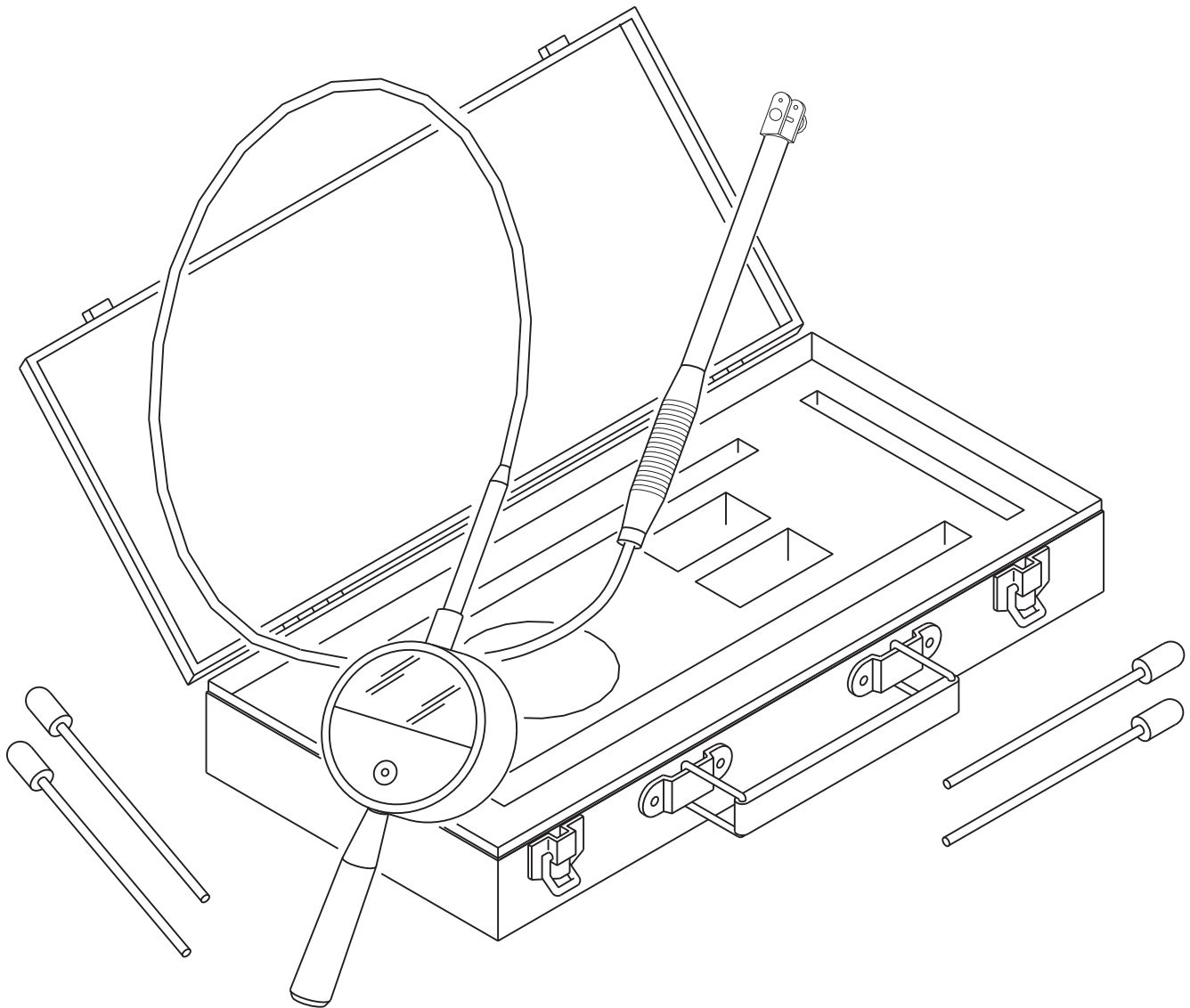
DESCRIPTION: The temperature meter is designed to provide accurate temperature measurements over the range of -50°F to $+70^{\circ}\text{F}$ (-45°C to $+21^{\circ}\text{C}$). A total of three separate thermistor probes can be attached to allow readings from three separate locations up to 150 feet away.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2075
SA

FIGURE C-75. TEMPERATURE METER, 385-3L



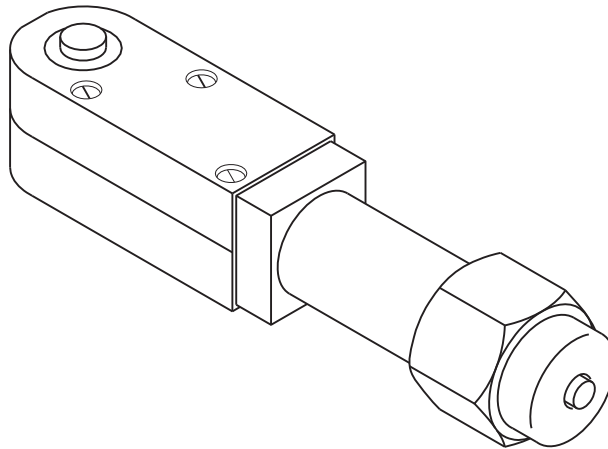
DESCRIPTION: The pyrometer is a handheld, dial type, temperature sensing device. This tool is used to accurately determine main rotor blade temperature on or off the helicopter. It may also be used to measure temperature anywhere on the helicopter where an accurate surface temperature determination is required. The pyrometer must be used with the type 4020 flexible extension arm, and type 4040 thermocouple.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2076
SA

FIGURE C-76. PYROMETER, 4000A



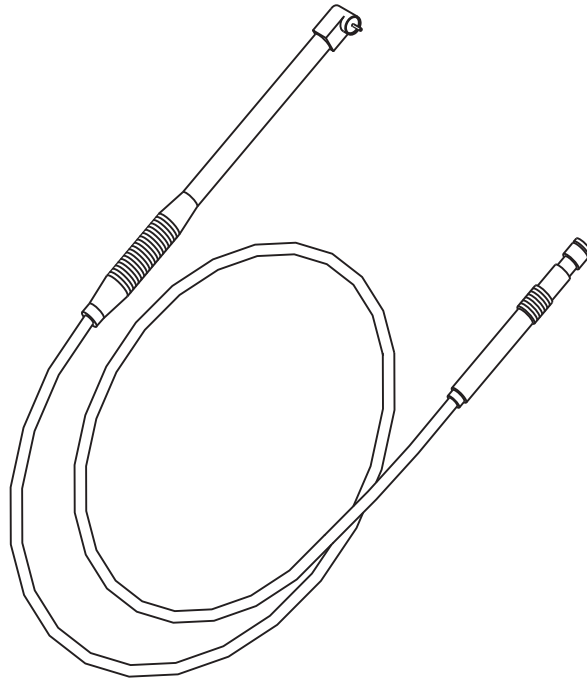
DESCRIPTION: The thermocouple is an accessory for the type 4000A pyrometer. It has a fine wire element in a retractory semi-swivel mounting for automatic self-alignment. A shield is provided to protect the fine wire tip. Use to maximum temperature 1200°F.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2077
SA

FIGURE C-77. THERMOCOUPLE, TYPE 4040



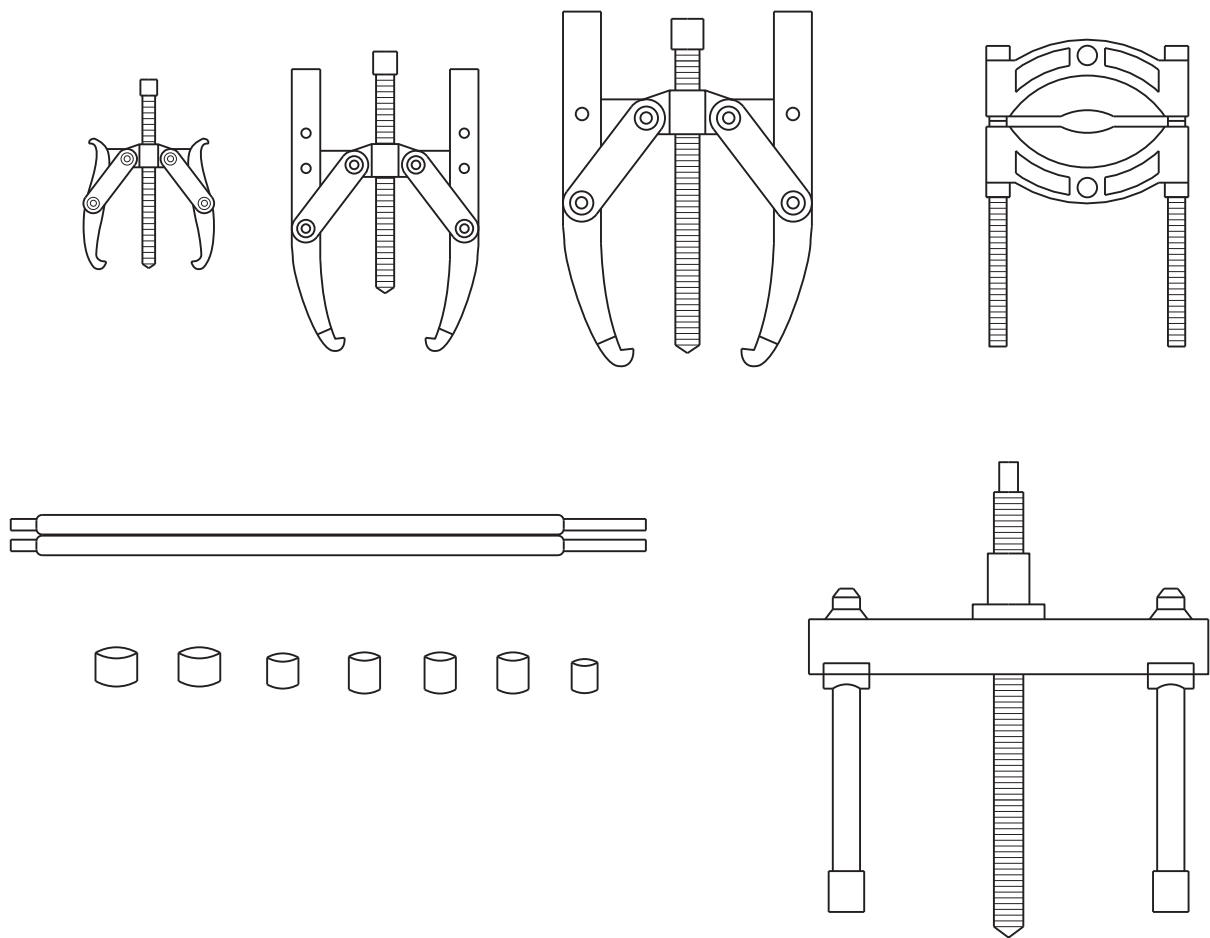
DESCRIPTION: The flexible extension arm is an accessory for type 4000A pyrometer. It has a flexible section, 36 inches long and rigid section 10 inches long. The flexible arm permits placing the thermocouple in hand to reach places and still hold the pyrometer in a convenient reading position.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2078
SA

FIGURE C-78. FLEXIBLE EXTENSION ARM, TYPE 4020



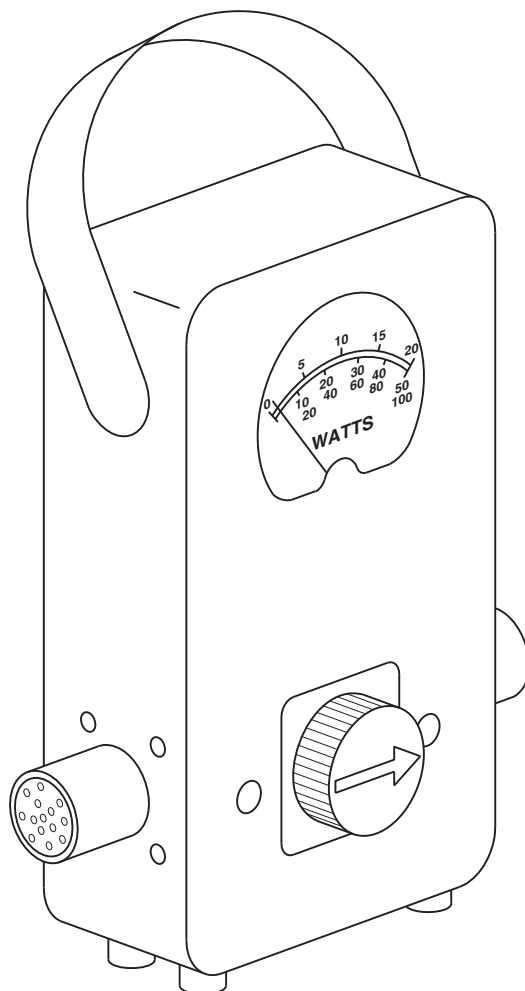
DESCRIPTION: The puller kit is used to remove and replace gears, bearings, wheels and any other press fit parts on the helicopter. Pullers are versatile in use, based on tonnage capacity required.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2091
SA

FIGURE C-79. PULLER SET, MECHANICAL, 1677



DESCRIPTION: The wattmeter is a portable insertion-type instrument. It accurately measures forward or reflected power in coaxial transmission lines under any load condition.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS:

FREQUENCY

Wattmeter must be calibrated every 12 weeks regardless of being used or not.

FB2101_1
SA

FIGURE C-80. WATTMETER, 43 (SHEET 1 OF 4)

CALIBRATION REQUIREMENTS: (CONT)**EQUIPMENT REQUIRED**

- Bird 43 RF Wattmeter
- Bird 100H Element
50B Element
50C Element
50D Element
50E Element
- Bird 8135 Termination
- Fluke 853A Multimeter
0–30 A DC $\pm 0.5\%$
- Narda 766 Attenuators 3 & 10 dB 20W
- Amplifier Research 50W 1000 RF Power Amplifier
10–1000 MHz 50W CW
- Sikorsky TS 7024 Dummy Element
- Weinschel GS–1 Connector Gage Set
- Current Source 0–30 A DC
- Hewlett Packard 8640b Signal Generator
1 mW 10–1000 Mhz CW

Calibrated $\pm 3\%$
10 – 50W
10 – 1000 MHz

NOTE

Calibration limited to 10–500 MHz and 1000 MHz at 50 watts maximum.

**CALIBRATION PROCEDURE
INSPECTION**

Inspect WATTMETER for physical damage / cleanliness and repair / clean as required.

CALIBRATION

1. Adjust WATTMETER mechanical zero if required.
2. Verify WATTMETER type N connectors are within dimensional tolerances, see View A.
3. Perform calibration section 4, 5, or 6 as applicable.
 - a. Section 4: Bird 43 without element.
 - b. Section 5: Bird 43 with element.
 - c. Section 6: Wattmeter

WATTMETER WITHOUT ELEMENT

1. Connect test set–up shown in View B.
2. Adjust current source for WATTMETER readings of Table 1 and note input current is within tolerance.

TABLE 1

WATTMETER	CURRENT (μ A)
100	29.4 to 30.6
80	26.2 to 27.4
60	21.6 to 22.8
40	16.0 to 17.2
20	9.23 to 10.4

FB2101_2
SA

FIGURE C-80. WATTMETER, 43 (SHEET 2 OF 4)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES (CONT)

WATTMETER WITH ELEMENT

1. Perform Section 4 with element removed.
2. Perform Section 5 with element installed.

WATTMETER

1. Determine mid-frequency* range of WATTMETER as in note and connect test set-up as shown in View C using applicable equipment.

NOTE

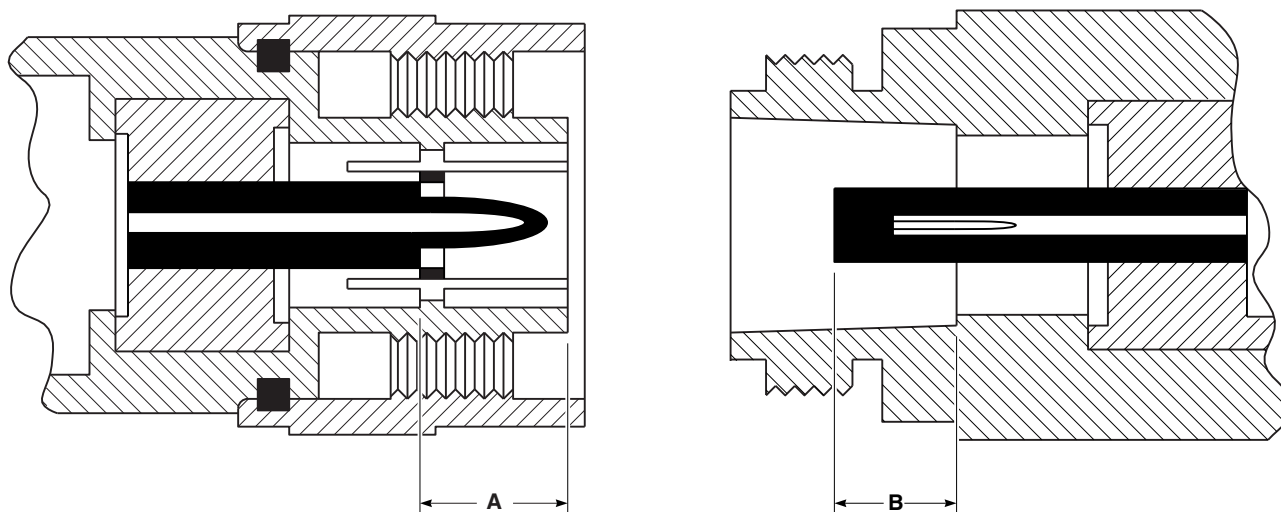
*500 MHz maximum; If frequency range of WATTMETER extends above 1000 MHz but not below, calibrate at 1000 MHz only.

2. Check scale linearity at cardinal points (50W maximum) to $\pm 5\%$ full scale using corrections for WATTMETER and standard wattmeter as required.
3. Check frequency response at low (10 MHz min) mid and high (500 MHz max) ends of frequency range at full scale power (50W Max). Tolerance is $\pm 5\%$ using corrections for WATTMETER and standard wattmeter as required.

NOTE

Check at 3 frequencies 500 MHz and below even if response extends above 500 MHz. If minimum frequency of WATTMETER is 1000 MHz, omit this test.

4. Apply standard calibration label stamped "LIMITED USE".



A INCHES	B INCHES
0.233 ± 0.009	0.197 ± 0.010

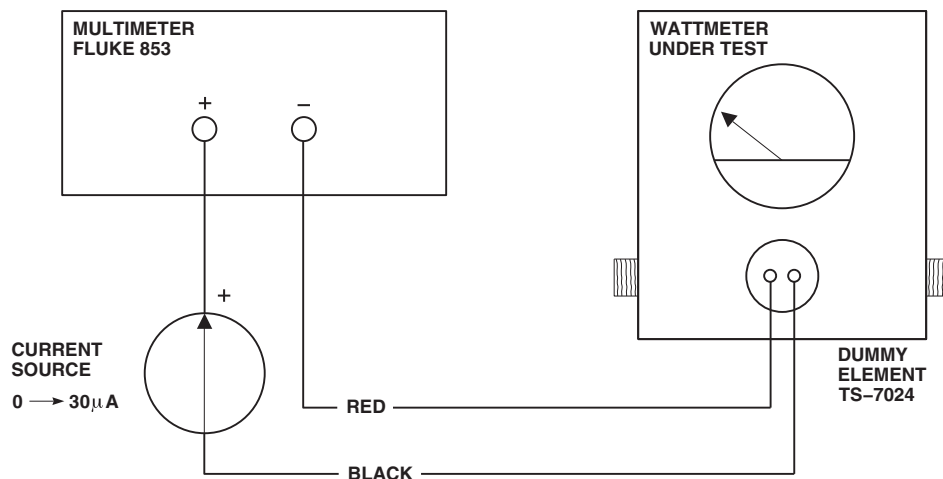
VIEW A
TYPE N CONNECTORS

FB2101_3
SA

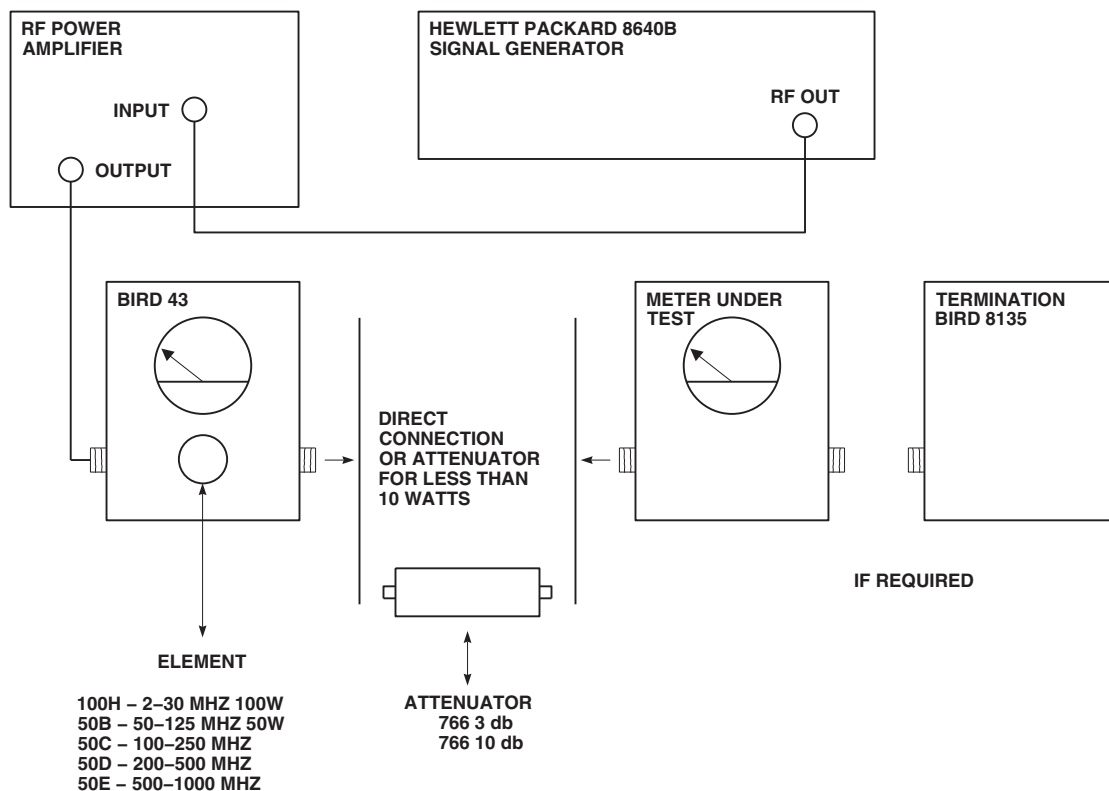
FIGURE C-80. WATTMETER, 43 (SHEET 3 OF 4)

CALIBRATION REQUIREMENTS: (CONT)

CALIBRATION PROCEDURES (CONT)



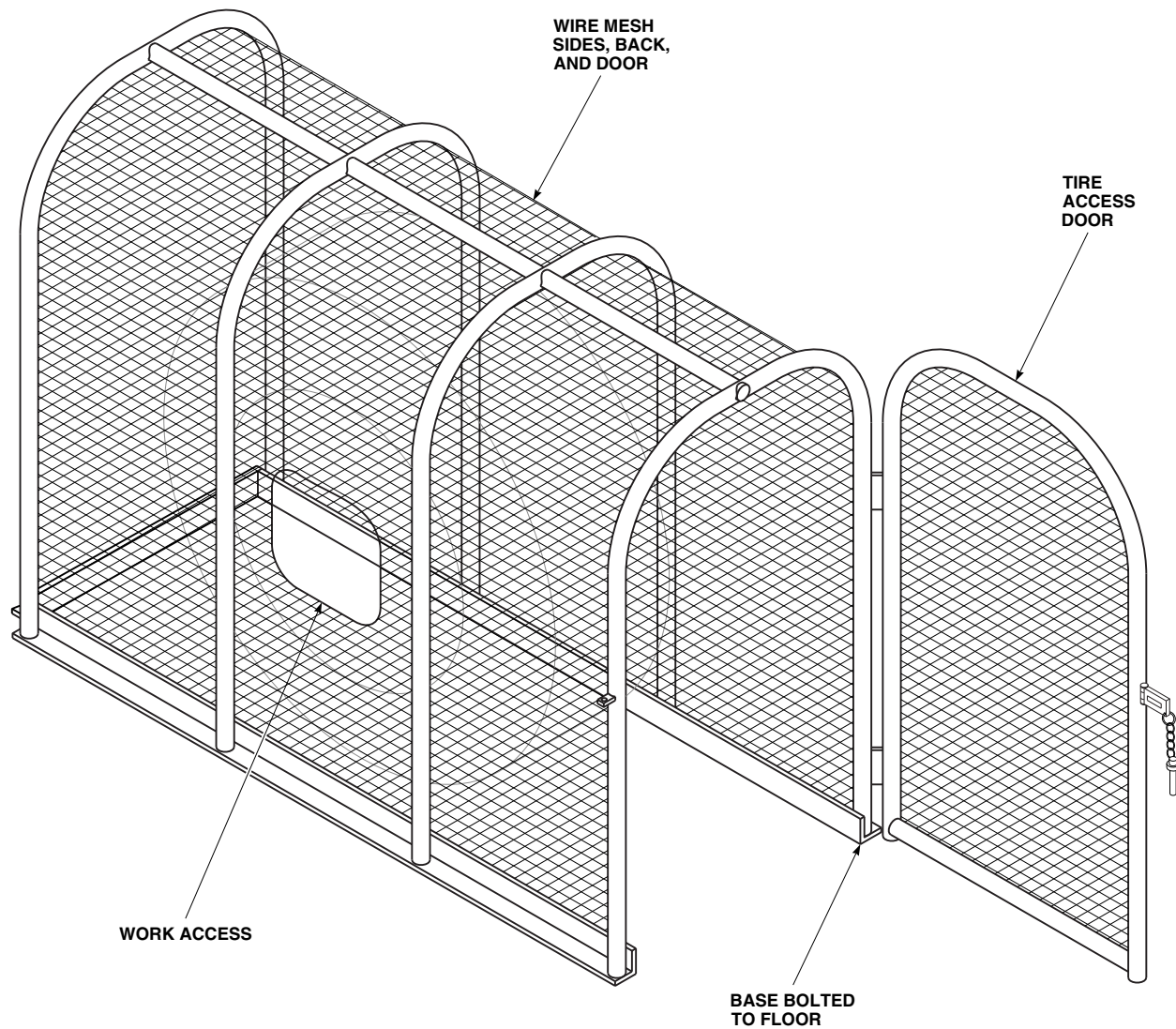
VIEW B



VIEW C

FIGURE C-80. WATTMETER, 43 (SHEET 4 OF 4)

FB2101_4
SA



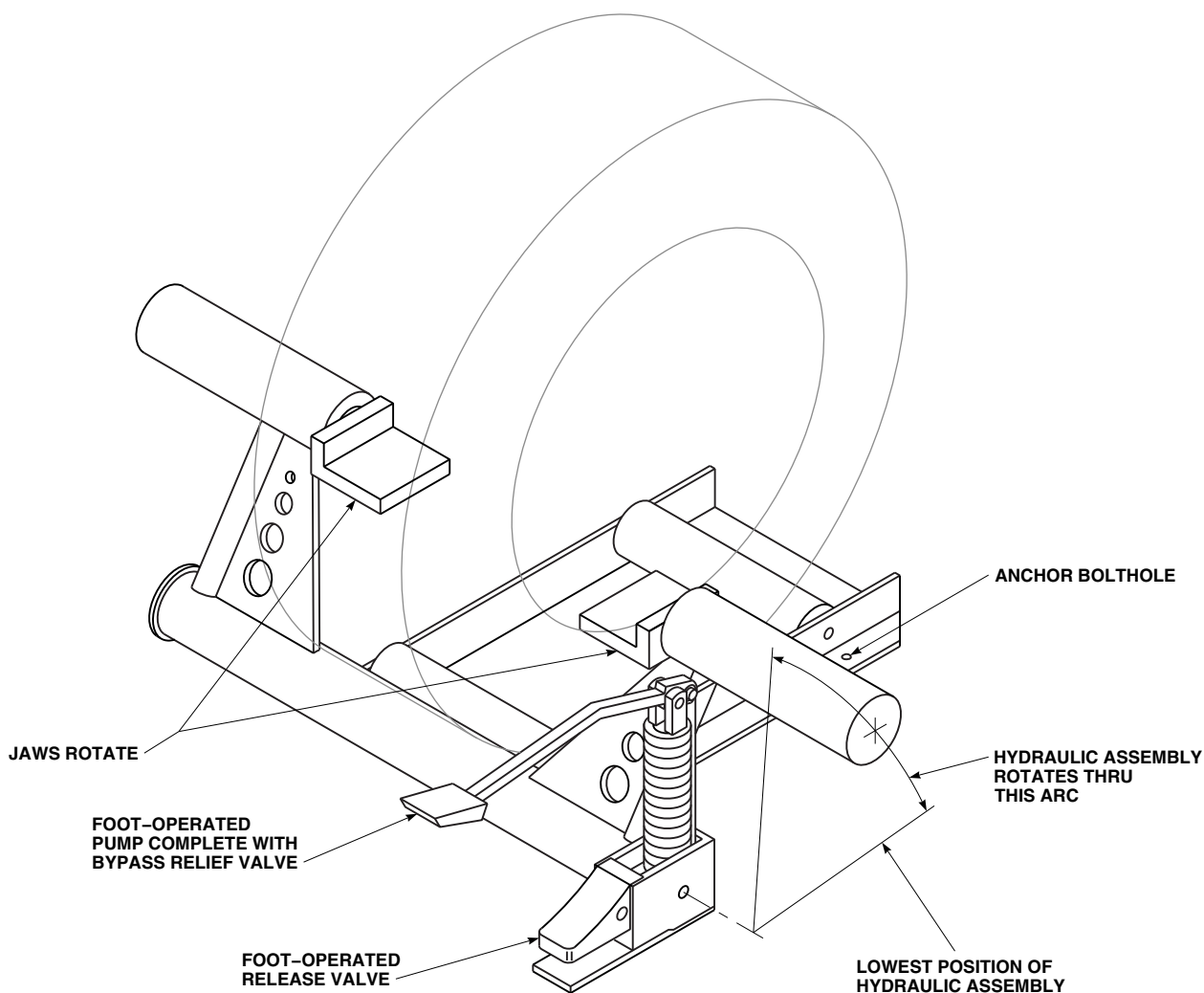
DESCRIPTION: The aircraft tire inflation cage is a metal tubular structure, covered with wire mesh. It has a locking door and a work access to inflate tires. The aircraft tire inflation cage is used to prevent injury to personnel. The aircraft tire inflation cage must be anchored to the floor.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FK1857A
SA

FIGURE C-81. AIRCRAFT TIRE INFLATION CAGE, 7201-010



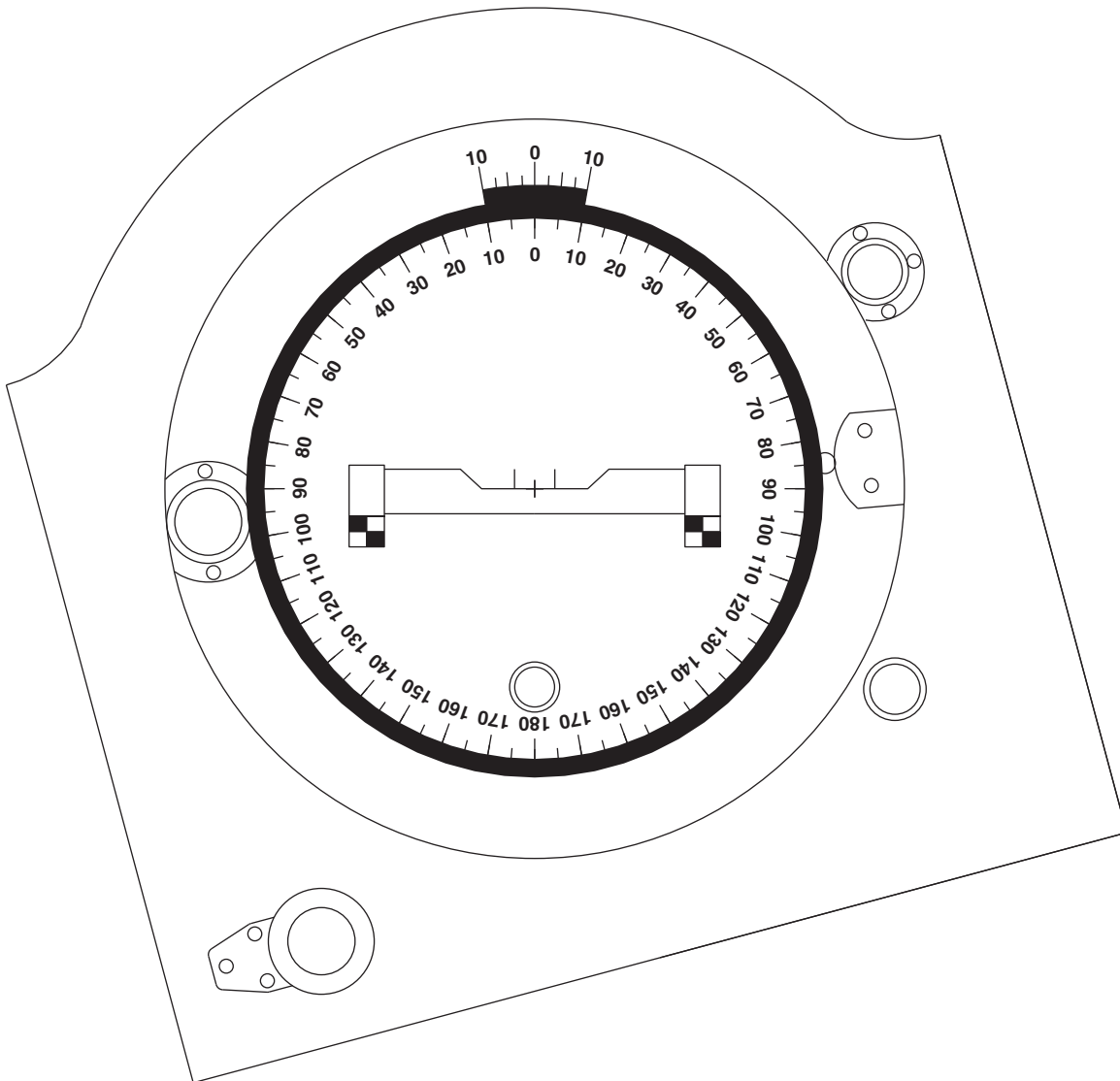
DESCRIPTION: The aircraft tire bead breaker is used to break the tire bead from the rim of a deflated wheel assembly. The aircraft tire bead breaker has a foot-operated hydraulic ram and release valve. It has a breaking capacity of 2000 pounds, on rims that range in size from 6–16–inches and tire widths of 9–inches. The aircraft tire bead breaker must be anchored to the floor.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FK1858B
SA

FIGURE C-82. AIRCRAFT TIRE BEAD BREAKER, 5033-010



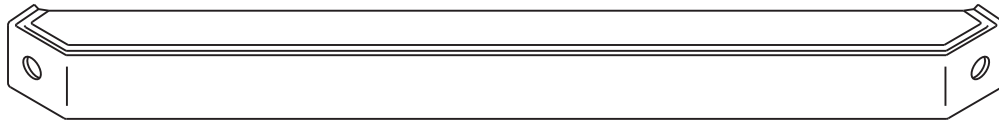
DESCRIPTION: The protractor may be used to measure propeller blade angle, control surface movement, or any other angle. It consists of an aluminum frame in which a steel ring and a disk are mounted. Spirit levels are mounted on both the frame and the disk, and locks are provided for locking the ring to the frame or the disk.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4081
SA

FIGURE C-83. PROTRACTOR



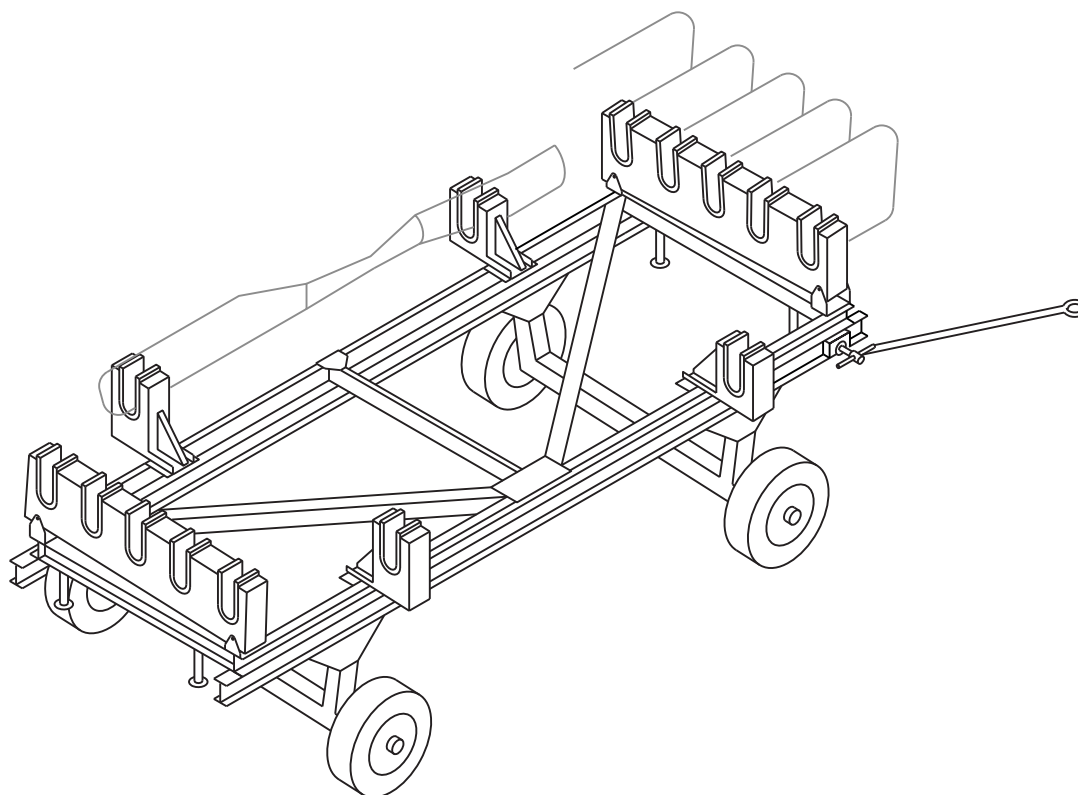
DESCRIPTION: The locking strut is used to secure open the tail rotor pylon during maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2482
SA

FIGURE C-84. MAINTENANCE LOCK STRUT, 70700-20385-041



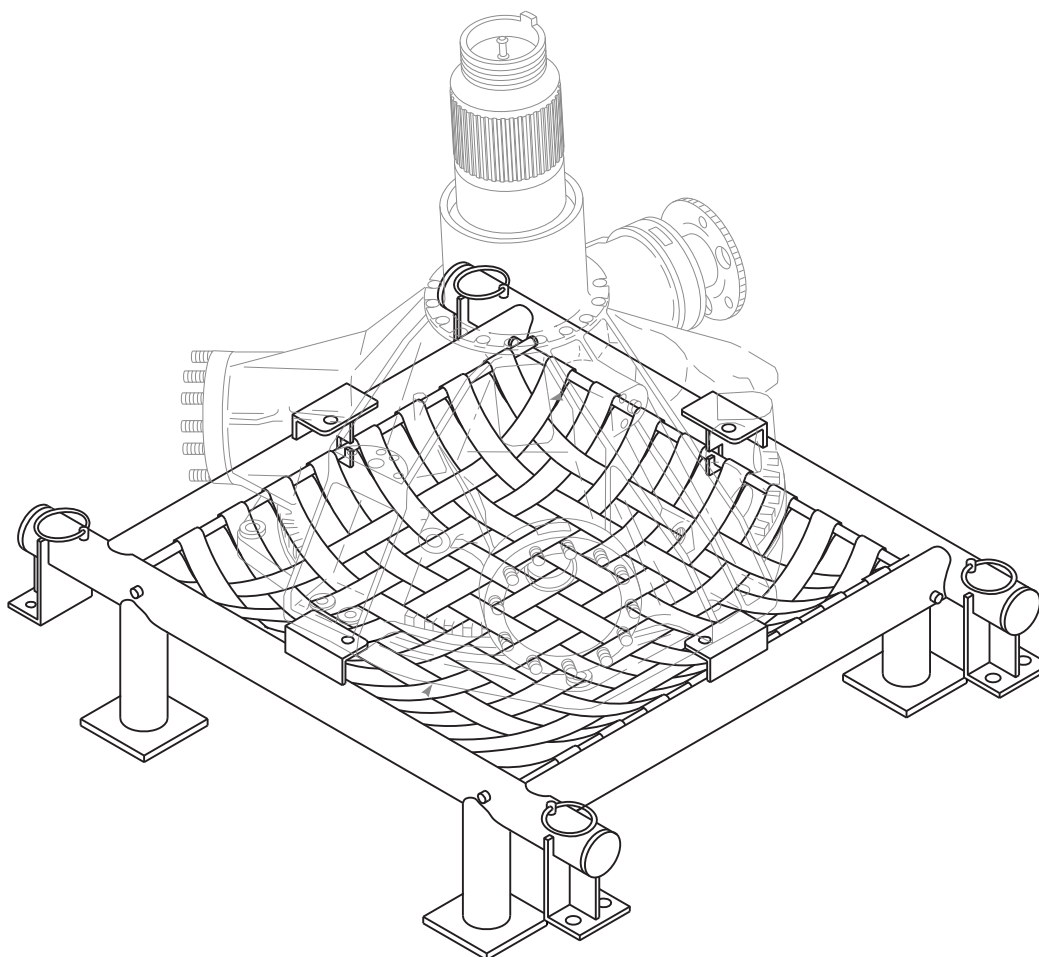
DESCRIPTION: The main and tail rotor blade adapter is used to secure the blades during maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2483A
SA

FIGURE C-85. MAIN AND TAIL ROTOR BLADE ADAPTER, 70700-20426-041



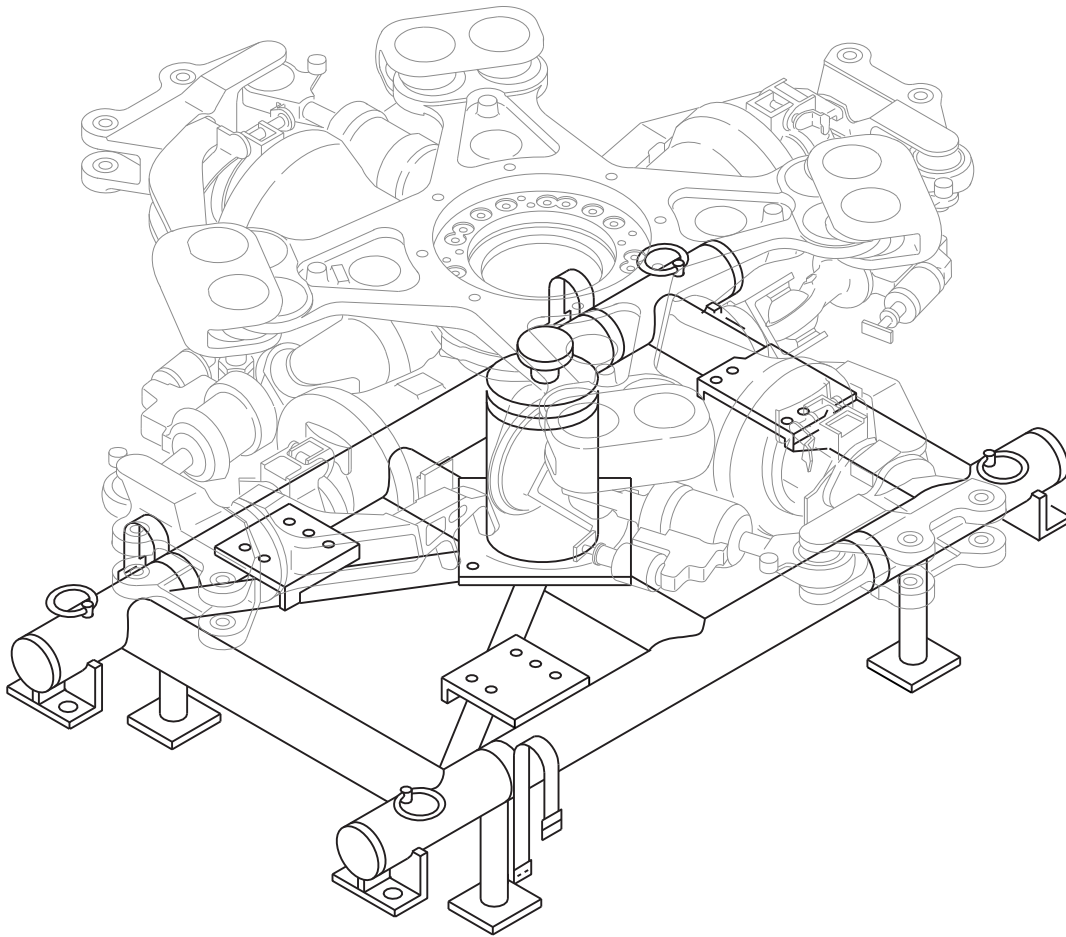
DESCRIPTION: The main gear box (MGB) adapter is used to support MGB during maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2484A
SA

FIGURE C-86. MAIN GEAR BOX ADAPTER, 70700-20420-041



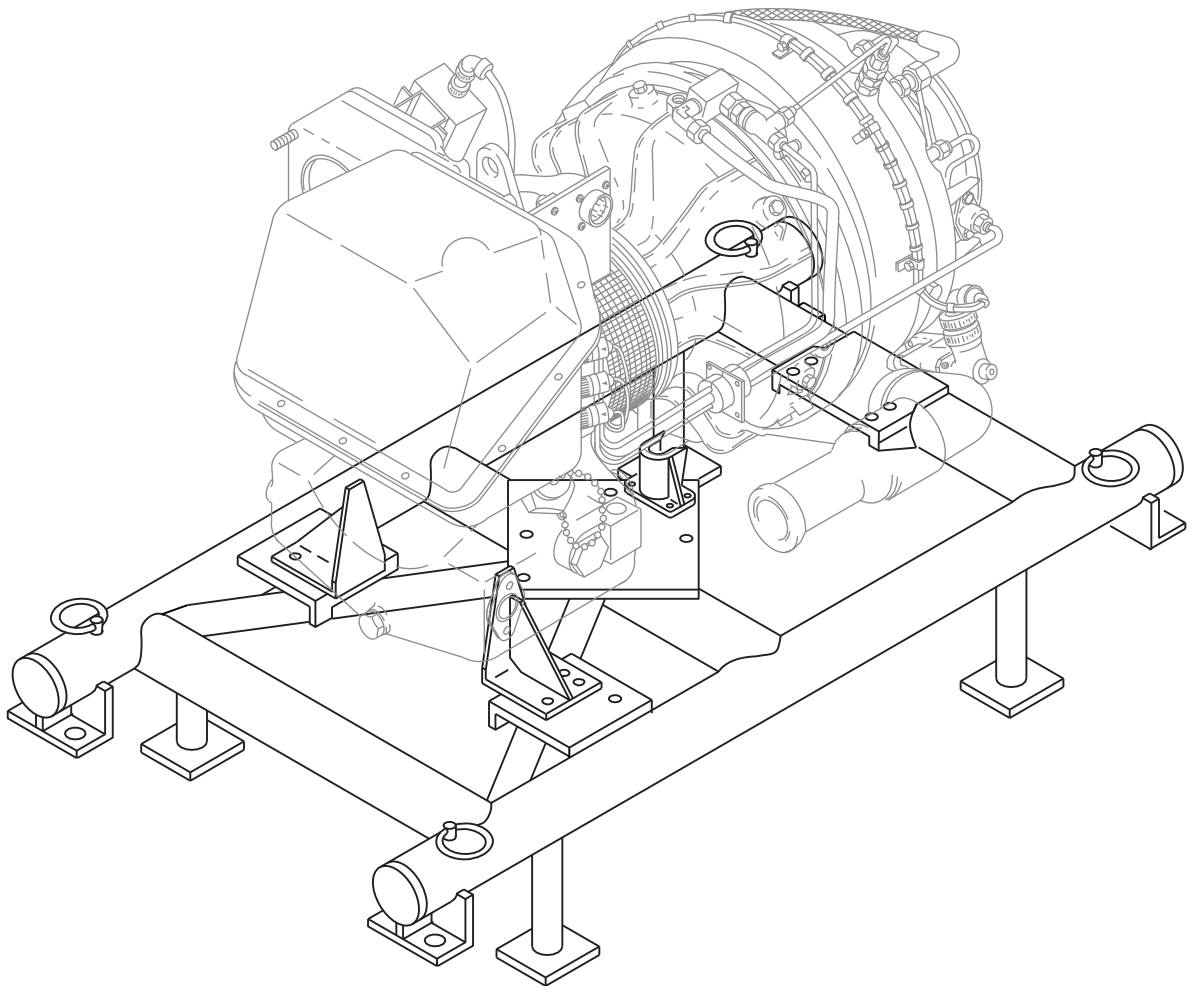
DESCRIPTION: The rotary wing head adapter is used to support the rotary wing head during maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2485A
SA

FIGURE C-87. ROTARY WING HEAD ADAPTER, 70700-20421-041



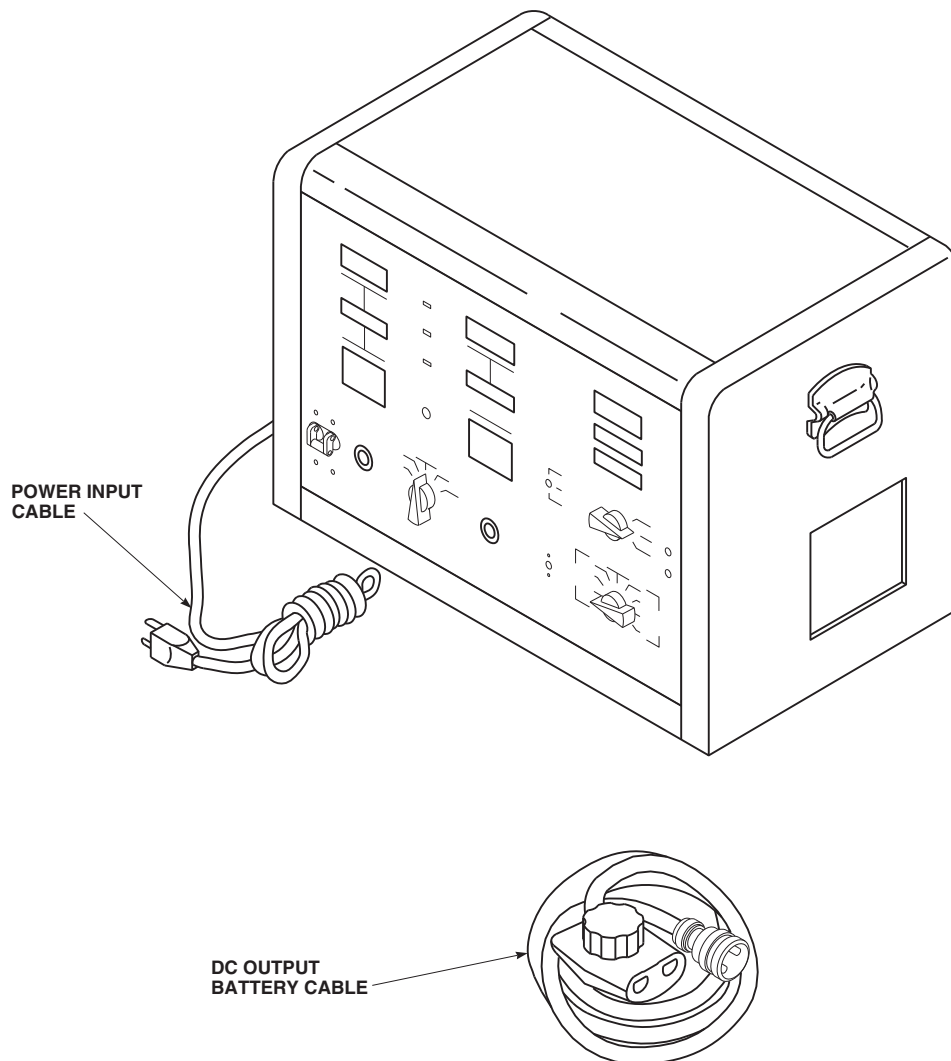
DESCRIPTION: The APU Adapter is used to support the APU during maintenance.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB2486
SA

FIGURE C-88. AUXILIARY POWER UNIT ADAPTER, 70700-20423-041



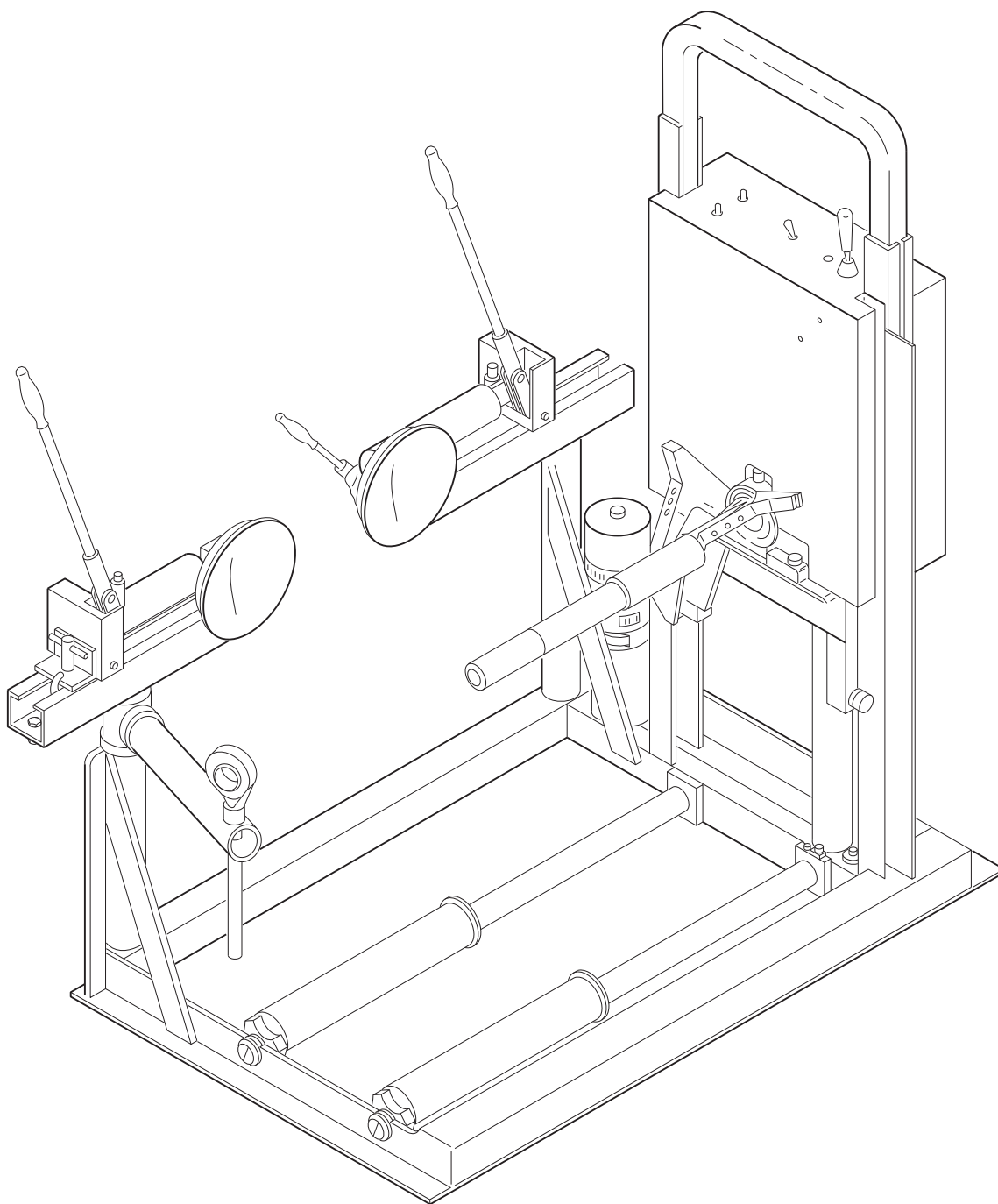
DESCRIPTION: The RF80-K Series Battery Charger / Analyzer consists of a transformer, rectifier, passing regulator circuit, timer programmer, and test discharge load bank. The unit is designed to charge and analyze aircraft batteries.

POWER REQUIREMENTS: This unit operates from a source of 187 to 265 VAC, 47-63 hertz, single phase.

CALIBRATION REQUIREMENTS: See manufacture manual.

FL0403
SA

FIGURE C-89. BATTERY CHARGER/ANALYZER, RF80-K



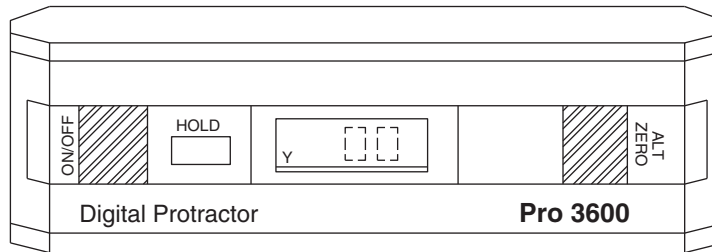
DESCRIPTION: The aircraft tire bead breaker is used to break the tire bead from the rim of a deflated wheel assembly. The aircraft tire bead breaker has two hydraulic hand pumps to break the bead. It has a breaking capacity of 12000 pounds.

POWER REQUIREMENTS: 115 / 230 Volts, 60 Hz, 1 PH, 15 Amps

CALIBRATION REQUIREMENTS: Not Applicable

FB4037B
SA

FIGURE C-90. AIRCRAFT TIRE BEAD BREAKER, LEE-1X



DESCRIPTION: The prop protractor is a digital instrument which provides maintenance personnel immediate digital indication of level, angle or tilt without the need to interpret a bubble or vernier scale. This instrument is used for rigging the flight controls and rotor blades.

POWER REQUIREMENTS: Standard 9 volt battery

CALIBRATION REQUIREMENTS:

FREQUENCY

Accuracy test shall be performed on digital protractor daily.
Digital protractor must be calibrated every 24 weeks regardless of being used or not.

EQUIPMENT REQUIRED

Precision Level
Precision Leveling Plate

CLEANING MATERIALS

Mild Detergent
Machinery Towel, Item 211, Appendix D

CALIBRATION PROCEDURES:

CLEANING / INSPECTION

Clean protractor with mild soap and a damp cloth. Never immerse in water.
Inspect for signs of damage, abuse, nicks, and dents.
Make sure a fresh battery is installed.

NOTE

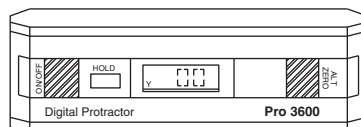
If digital protractor fails accuracy test, digital protractor must be recalibrated using Superset ® procedure.

FB4413_1A
SA

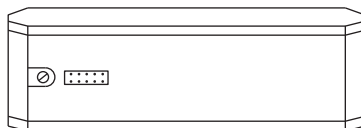
FIGURE C-91. DIGITAL PROPELLER PROTRACTOR, PRO 3600 (SHEET 1 OF 4)

CALIBRATION PROCEDURES: (CONTINUED)**ACCURACY TEST**

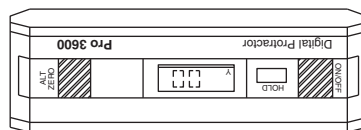
Position the digital protractor with the display facing the operator on a clean, precision leveling plate. Wait 10 seconds and note the angle on the display.



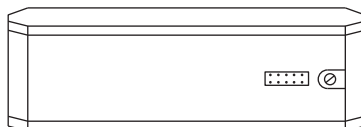
Rotate the digital protractor end-for-end so the display is facing away from operator. Wait 10 seconds and note the angle on the display.



Roll the digital protractor toward operator so the display is facing operator but the lettering on the digital protractor is upside down. Wait 10 seconds and note the angle on the display.



Rotate the digital protractor end-for-end so the display is facing away from operator (the lettering should still be upside down). Wait 10 seconds and note the angle on the display.



If any of the four measurements varies from any of the others by more than 0.1° , calibrate the digital protractor using Superset®.

 FB4413_2A
 SA

FIGURE C-91. DIGITAL PROPELLER PROTRACTOR, PRO 3600 (SHEET 2 OF 4)

CALIBRATION PROCEDURES: (CONTINUED)**SUPERSET®**

Turn the digital protractor on and place on a precision leveling plate.

NOTE

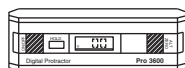
Each time the digital protractor is repositioned, wait a minimum of 10 seconds before pressing HOLD to advance to next step.

Press and hold the HOLD and ALT ZERO buttons simultaneously. Keep them pressed for approximately three seconds.

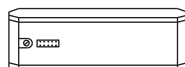
Release the buttons when the SUP appears. A 0 within flashing brackets will then appear. These brackets are composed of four horizontal and four vertical segments.

NOTE

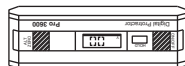
As you proceed through the following steps of Superset®, a new segment will hold steady after you complete each step.

Superset® – Horizontal Settings

Digital protractor faces operator and lettering on face plate is right side up.
Align with an edge or line – wait 10 seconds.
Press HOLD button until [1] appears.



Rotate digital protractor so it faces away from operator and lettering on face plate is still right side up.
Align with an edge or line – wait 10 seconds.
Press HOLD button until [2] appears.



Roll digital protractor so it faces operator and lettering on face plate is upside down.
Align with an edge or line – wait 10 seconds.
Press HOLD button until [3] appears.



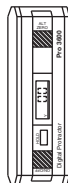
Rotate digital protractor so it faces away from operator and lettering on face plate is still upside down.
Align with an edge or line – wait 10 seconds.
Press HOLD button until [4] appears.

FB4413_3
SA

FIGURE C-91. DIGITAL PROPELLER PROTRACTOR, PRO 3600 (SHEET 3 OF 4)

CALIBRATION PROCEDURES: (CONTINUED)

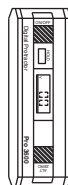
Superset® – Vertical Settings



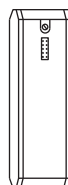
Digital protractor is placed against a vertical surface, faces operator and lettering reads from bottom to top. Align with an edge or line – wait 10 seconds. Press HOLD button until [5] appears.



Rotate digital protractor so it faces away from operator and lettering still reads from bottom to top. Align with an edge or line – wait 10 seconds. Press HOLD button until [6] appears.



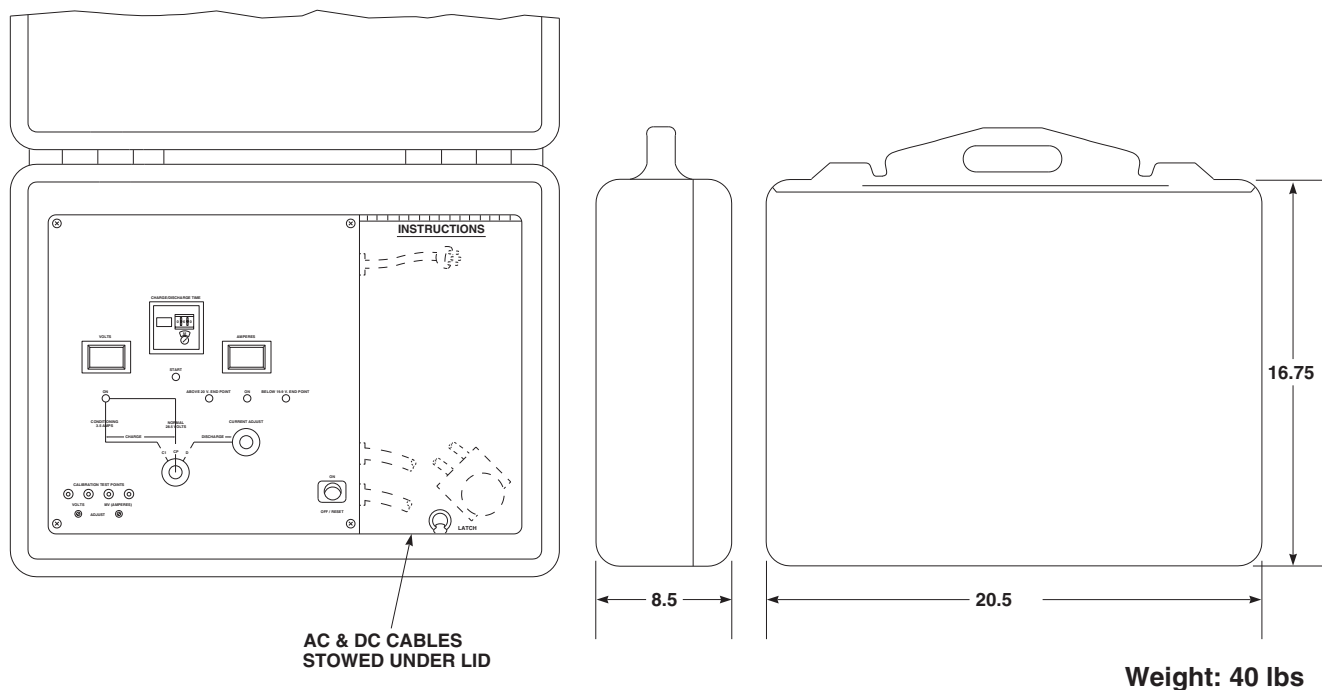
Roll digital protractor so it faces operator and lettering on face plate reads from top to bottom. Align with an edge or line – wait 10 seconds. Press HOLD button until [7] appears.



Rotate digital protractor so it faces away from operator and lettering still reads from top to bottom. Align with an edge or line – wait 10 seconds. Press HOLD button. [8] will very briefly appear, followed immediately by regular display.

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FIGURE C-91. DIGITAL PROPELLER PROTRACTOR, PRO 3600 (SHEET 4 OF 4)



DESCRIPTION: This Battery Charger / Analyzer is used for charging / discharging the Valve Regulated Sealed Lead Acid Battery (VRSLAB) at the intermediate level of maintenance.

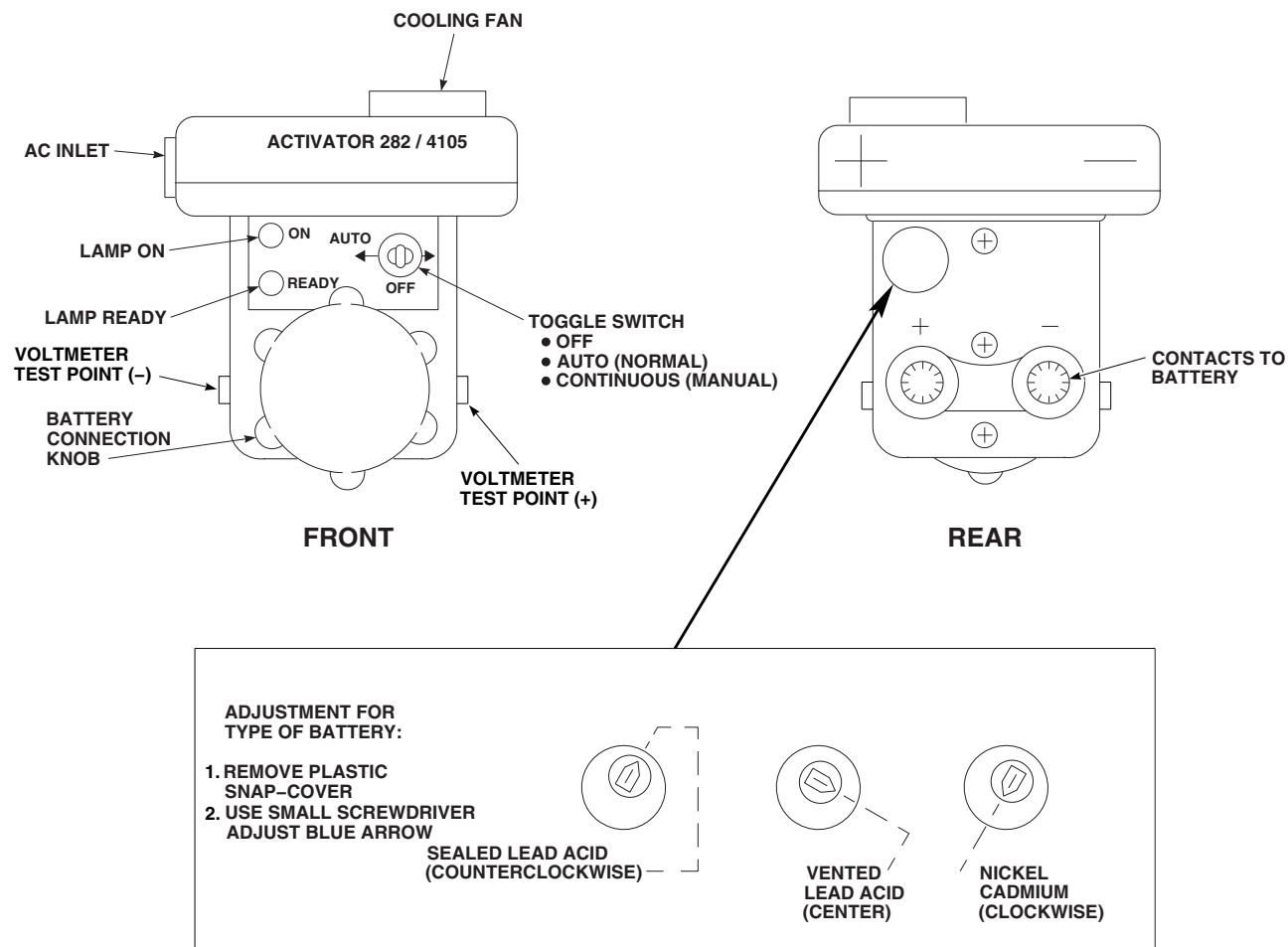
The Charger / Analyzer has full charging and discharging capabilities, an elapsed time counter, voltage and amperage displays, and fan cooled discharged resistors. The Charger / Analyzer is portable and has usage instructions contained within the case.

POWER REQUIREMENTS: This unit operates from a source of 95–135 VAC, 50/60 Hertz, 3 Amperes, or 190–207 VAC, 50/60 Hertz, 1.5 Amperes.

CALIBRATION REQUIREMENTS: TBD
Calibration Interval: TBD
Calibration Reference: TBD

FB4414B
SA

FIGURE C-92. VRSLAB - BATTERY CHARGER/ANALYZER, 4159



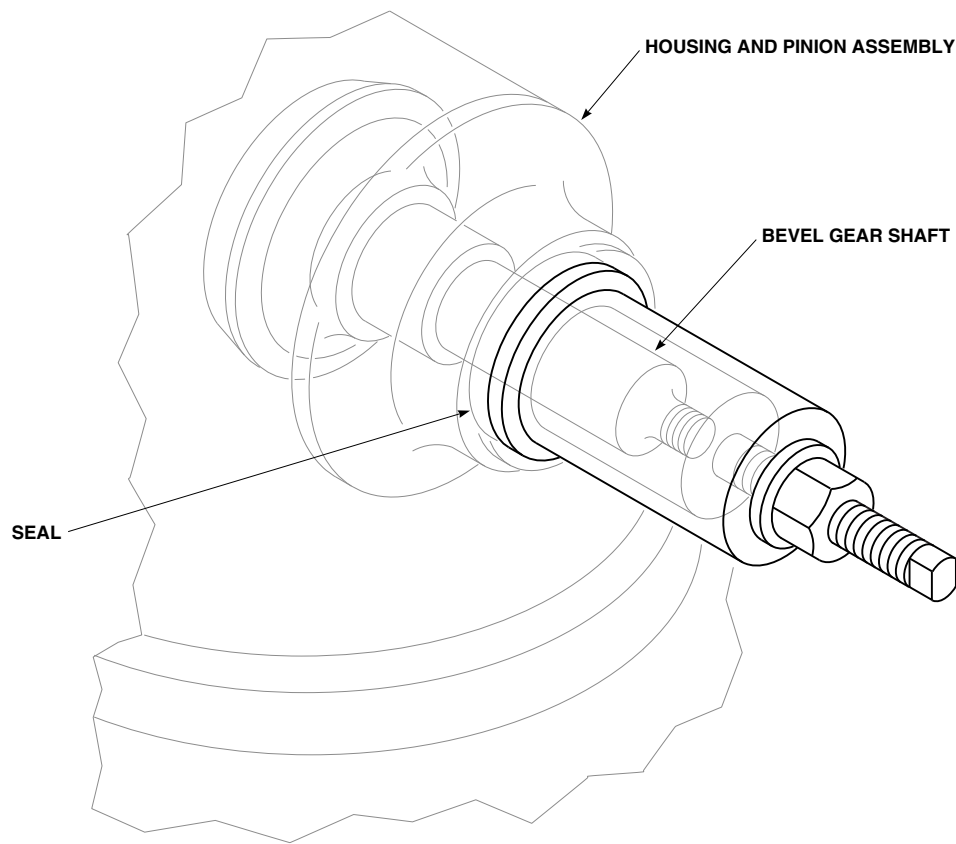
DESCRIPTION: The Battery Charger is a multi-purpose charger. It is capable of charging lead acid vented batteries, sealed lead acid batteries (including Valve Regulated Sealed Lead Acid Battery (VRLAB)), and Nickel Cadmium (NiCAD) batteries.

POWER REQUIREMENTS: TBD

CALIBRATION REQUIREMENTS: Refer to manufacturer's product manual for Activator 282 / 4105. Calibration interval: 12 months.

FB4418B
SA

FIGURE C-93. BATTERY CHARGER, 282/4105



DESCRIPTION: Used in replacement of the seal in the housing and pinion assembly of the main gear box. The seal is a 0.012" interference fit in the housing. The seal has an OD of 3.512". The bevel pinion shaft protrudes approximately three inches thru the seal.

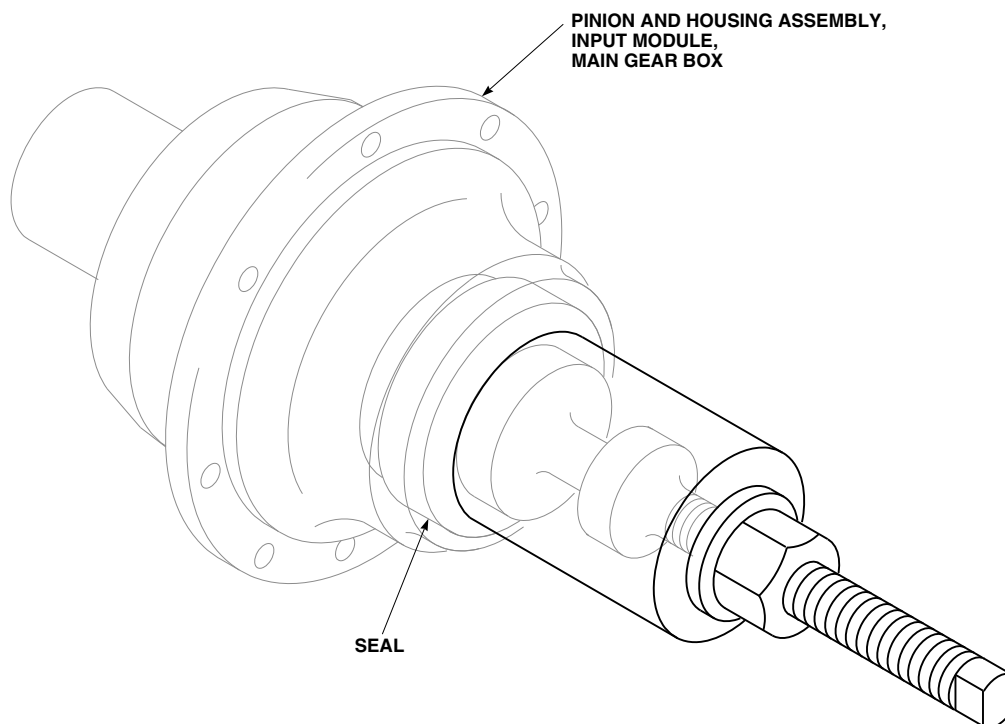
The Seal Installer consists of an outer cylinder and an inner shaft. The diameter of the outer cylinder is designed to mate the seal surface without touching the housing. One end of the inner shaft has internal threads to mate the external threads of the bevel pinion shaft. The opposite end is externally threaded. This tool is used by threading the inner shaft onto the bevel gear shaft then tightening a nut at the opposite end of the inner shaft against the top of the outer cylinder forcing the outer cylinder against the seal, pressing the seal in position in the housing.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4415A
SA

FIGURE C-94. SEAL INSTALLER, 70700-77308-041



DESCRIPTION: Used in the replacement of the seal in the input pinion and housing assembly of the main gear box input module. The seal is a line-to-line fit in the housing and must be pressed in to be seated.

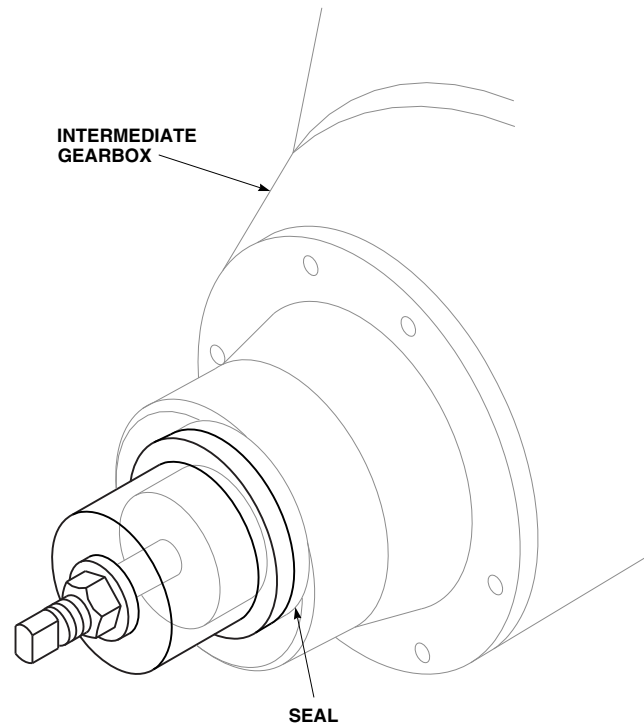
The Seal Installer consists of an outer cylinder and an inner shaft. The diameter of the outer cylinder is designed to mate the seal surface without touching the housing. The length of the cylinder will enable it to pass over the shaft and seat the seal without bottoming out against the end of the shaft. One end of the inner shaft has internal threads to mate the external threads of the end of the pinion shaft. The opposite end of the inner shaft is externally threaded. This tool is used by threading the inner shaft onto the pinion shaft then tightening a nut at the opposite end of the inner shaft against the top of the outer cylinder, forcing the outer cylinder against the seal, seating the seal in the housing.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4416A
SA

FIGURE C-95. SEAL INSTALLER, 70700-77309-041



DESCRIPTION: Gear box leakage, that requires the removal and replacement of the seal in the input and output bearings of the intermediate gear box and the input housings of the tail gear box requires a tool for installing these seals that are an interference fit.

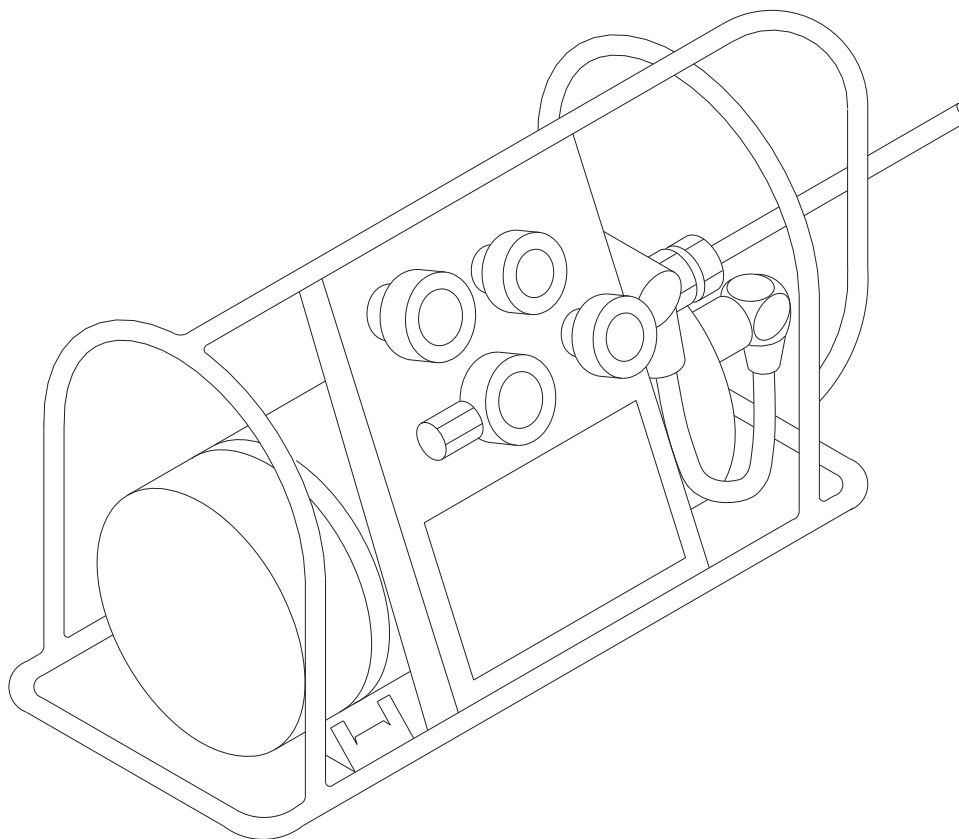
The Seal Installer consists of an outer cylinder and an inner shaft. The diameter of the outer cylinder is designed to mate to the seal face. The length of the cylinder enables it to pass over the longest shaft and seat the seal without bottoming out against the end of the shaft. One end of the inner shaft has internal threads to mate the external threads of the shafts. The opposite end of the inner shaft is externally threaded. This tool is used by threading the inner shaft on to the gear shaft, then tightening a nut at the opposite end of inner shaft against the top of the outer cylinder, forcing the outer cylinder against the seal seating the seal in the housing.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4417A
SA

FIGURE C-96. SEAL INSTALLER, 70700-77310-041



DESCRIPTION: The portable nitrogen cylinder is utilized for line maintenance servicing of the following components:

Main landing gear shock struts
Tail landing gear shock strut
Tail bumper shock strut
Main rotor accumulator
APU accumulator

The portable nitrogen cylinder is capable of pressure to 3000 PSI and comes complete with regulator, hose and fittings.

DIMENSIONS:

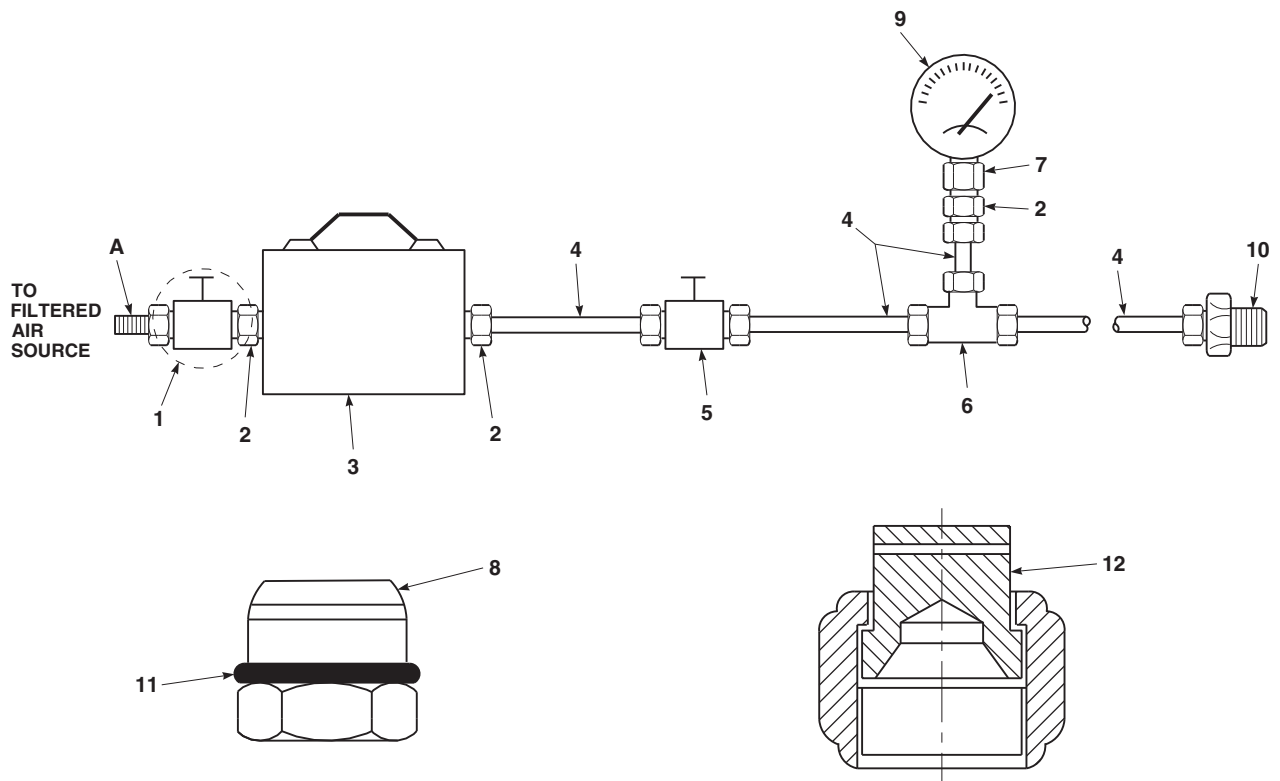
Length:	23"
Width:	17"
Height:	14.5"
Weight:	53 lbs

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4420A
SA

FIGURE C-97. NITROGEN SERVICING UNIT, 1428AS100-1



NO.	NOMENCLATURE	QTY REQUIRED
1	METERING VALVE	1
2	UNION	3
3	VACUUM PUMP	1
4	HOSE OR TUBE ASSEMBLY	5
5	SHUTOFF VALVE	1
6	TEE FITTING	1
7	ADAPTER	1
8	PLUG	5
9	VACUUM PRESSURE GAGE	1
10	REDUCER FITTING	2
11	PACKING	2
12	CAP	2

FB4419_1
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FIGURE C-98. FUEL INTEGRITY TEST SET, RJP001 (SHEET 1 OF 2)

NOTES

Item (1) is not required if air source is regulated. If item (1) is needed unscrew air valve (A) from vacuum pump (3) and thread (A) to (1) and connect (1) to (3) via union (2).

An 8" hose assembly (4) and reducer (10) is needed for bleeding crossfeed lines in addition to test setup illustrated.

Plug (8) to be used in sealing inlet adapter and engine fuel feed line.

Packing (11) is not required when plugging engine fuel feed line.

Cap (12) is used to cap prime shutoff valve during line purge and integrity test.

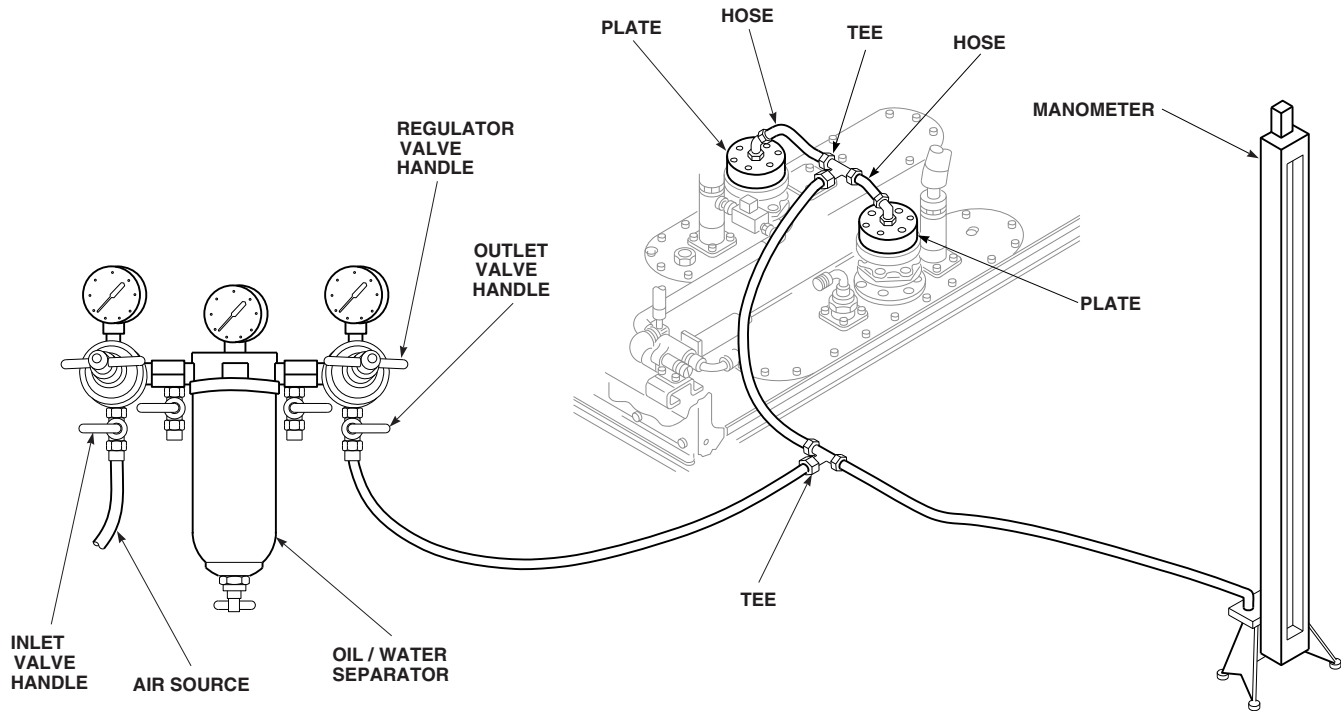
DESCRIPTION: The requirement exists at the organizational level of maintenance to test the integrity of the engine fuel feed system. This system must be subjected to both vacuum (-6 In Hg) and pressure (+30 PSIG) for a designated period of time (5 min) with no leakage. A means to create the vacuum, route the pressure, monitor the test (vac / press), and seal the system during test is needed.

The kit consists of a metering valve for regulating air pressure; a vacuum pump for creating the necessary vacuum; a shutoff valve for sealing the system during testing; a combination vacuum / pressure gage to monitor the test; as well as a variety of fittings, plugs, and hoses for routing and connecting the test equipment to the aircraft. A low pressure air source (50-100 PSI) will be connected to a metering valve and the metered air source to the vacuum pump. From the pump, a shutoff valve will be plumbed and from the shutoff valve a combination vacuum / pressure gage will be tapped into the test line. The test line will be adapted to the engine feed line(s). Various components of this system will be removed and their corresponding lines plugged.

Additionally, the main engine prime shutoff valves must be capped on the prime side of the valve during vacuum testing and fuel line purge. When plugging the crossfeed lines, an 8" hose assembly (low pressure 1/4" fittings) is connected to the opposite engine feed line via a reducer fitting and is used to drain purged fuel from the crossfeed lines into a suitable container. In addition, the discharge fuel hoses for the boost pump must be plugged.

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FIGURE C-98. FUEL INTEGRITY TEST SET, RJP001 (SHEET 2 OF 2)



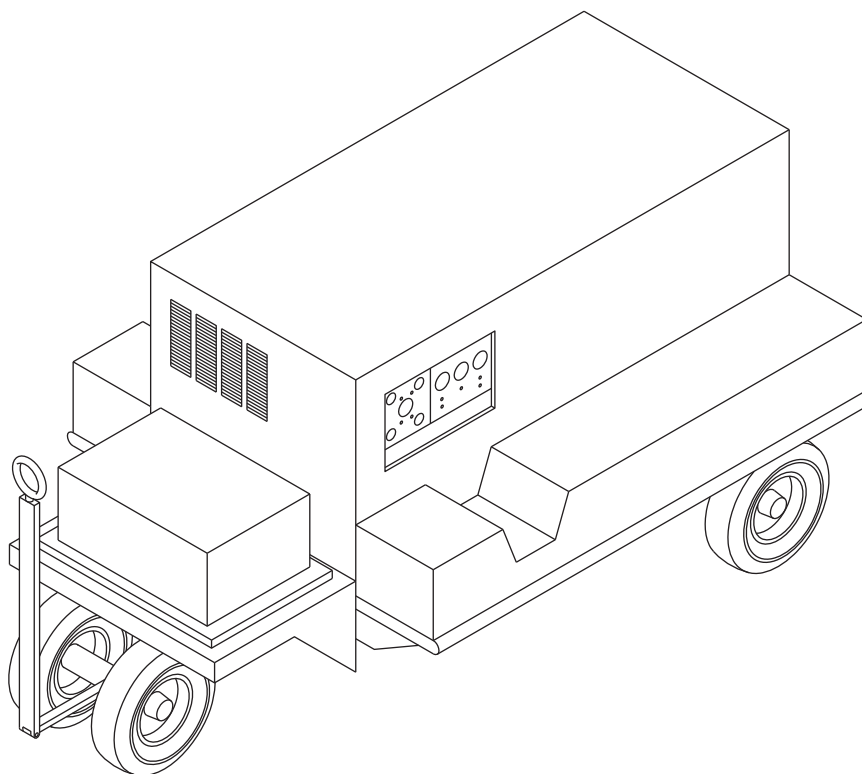
DESCRIPTION: A requirement exists at the 'O' level of maintenance to functionally test the main fuel cells by pressure. In order to accomplish this requirement, a method of capping off fuel cell is required with a way of applying pressure and measuring it.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4421A
SA

FIGURE C-99. FUEL CELL TEST KIT, RJP002-2



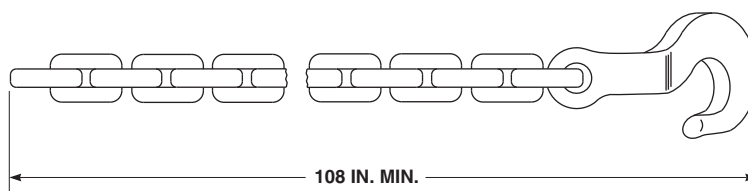
DESCRIPTION: The Generator unit will provide all AC electrical power needed to perform system checks, and troubleshooting of various electrical systems. The Generator unit is used in place of the helicopter APU in ground test.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FB4422
SA

FIGURE C-100. ELECTRICAL GENERATOR, DIESEL-POWERED, 90C24P5-S/A



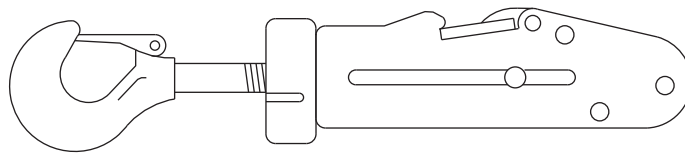
DESCRIPTION: The tiedown chain assembly is used for tying down the helicopter.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FC4111A
SA

FIGURE C-101. TIEDOWN CHAIN ASSEMBLY, 627728A



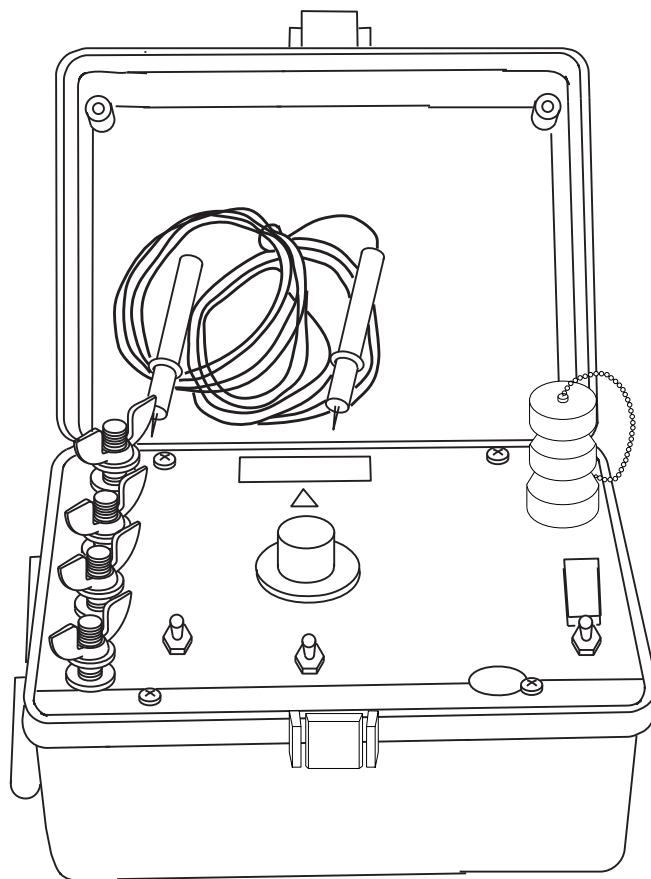
DESCRIPTION: The tiedown tensioner is used to adjust the length of chains in order to remove slack when tying down the helicopter.

POWER REQUIREMENTS: Not Applicable

CALIBRATION REQUIREMENTS: Not Applicable

FC4112A
SA

FIGURE C-102. TIEDOWN TENSIONER, CGU-4/E



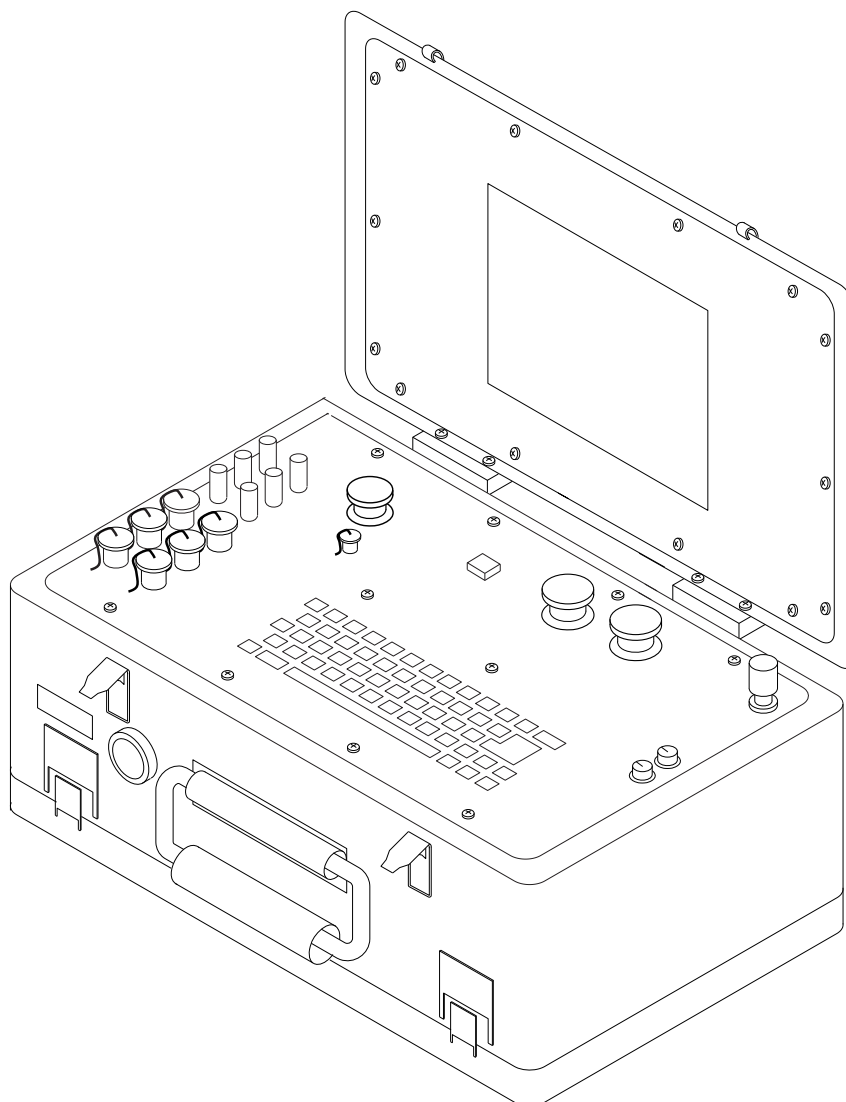
DESCRIPTION: The digital low-resistance ohmmeter is a portable instrument used for measuring resistance in milliohms.

POWER REQUIREMENTS: Rechargeable batteries (Self-contained)

CALIBRATION REQUIREMENTS: TBD

FB1359A
SA

FIGURE C-103. DIGITAL LOW-RESISTANCE OHMMETER, 247000



DESCRIPTION: The fuel quantity test set is an instrument that can be used on any aircraft with capacitance probe technology. The fuel quantity test set is a microprocessor or controlled test set that takes measurements and calculates pass/fail limits automatically. The fuel quantity test set is operated via menu driven software. The fuel quantity test set can also measure fuel, LOx and water systems.

POWER REQUIREMENTS: 18 Volt to 32 Volt DC Batteries

CALIBRATION REQUIREMENTS:

FREQUENCY

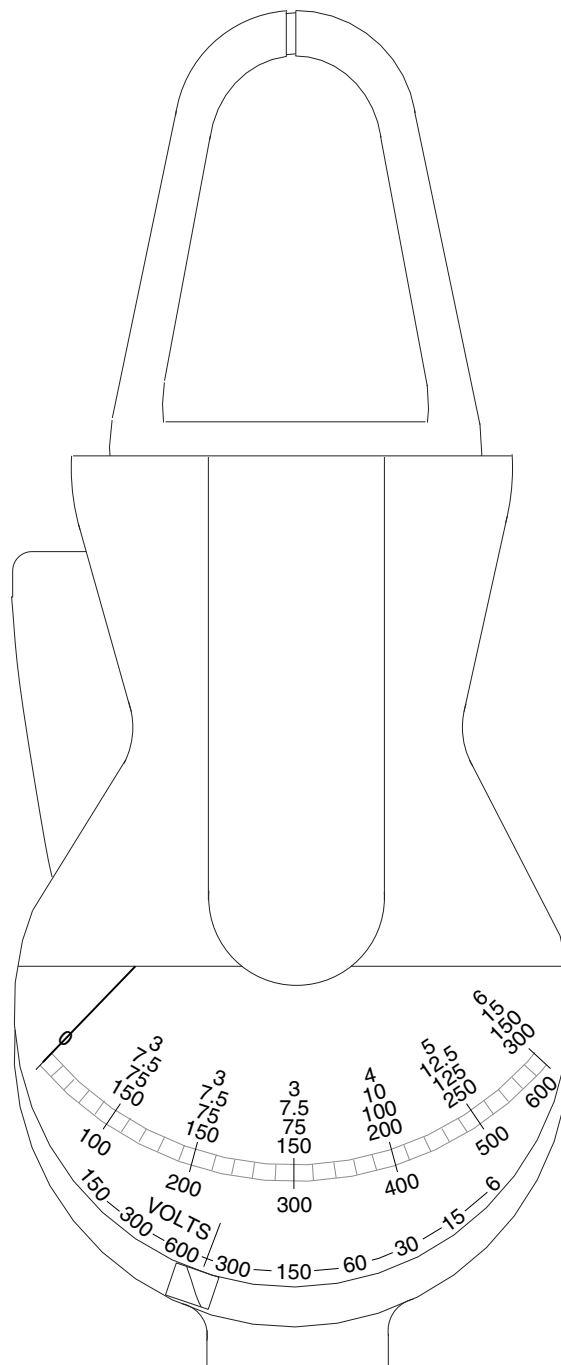
Continuous self calibration using very accurate internal reference.
1 year return period for full factory calibration.

EQUIPMENT REQUIRED

- Adapter Cable, BC899A8775-000
- Bonding Lead, BC899A8185-000
- Breakout Box Assembly, BC8998742-000

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SA

FIGURE C-104. FUEL QUANTITY TEST SET, BCF DE8491



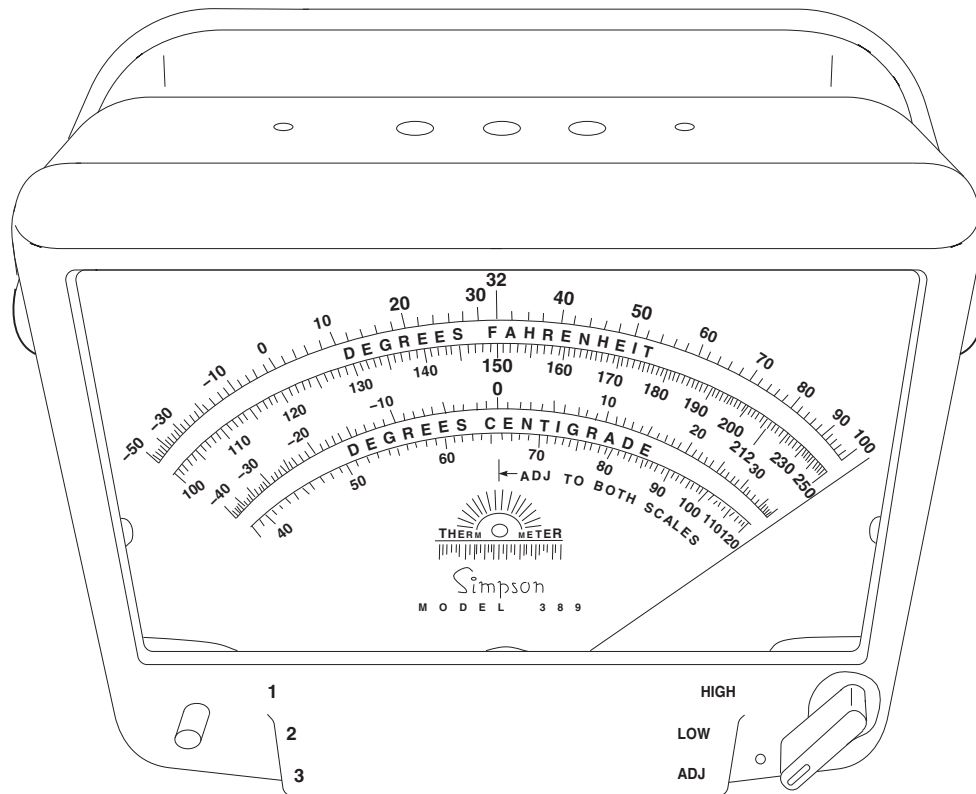
DESCRIPTION: The ammeter is used to measure the flow of electric current in a circuit.

POWER REQUIREMENTS: Not applicable

CALIBRATION REQUIREMENTS: Not applicable

FR0654
SA

FIGURE C-105. AMMETER, CLAMP-ON, 749



DESCRIPTION: The thermometer is used in conjunction with surface probe to measure windshield surface temperature.

POWER REQUIREMENTS: Not applicable

CALIBRATION REQUIREMENTS: Not applicable

FC2790
SA

FIGURE C-106. THERMOMETER, 389-3L

